



Terramechanics and Soil-Wheel Interactions for Road Vehicle Applications

by

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Abstract

The current research concerns the analysis and development of a soil-wheel interaction model intended for application in road vehicles, in order to support virtual vehicle development processes. As a first step, a review of the literature is conducted which reveals the absence of a reliable tyre model for off-road applications. In addition, it highlights two critical performance items for the soil-wheel interaction; tractive effort and rolling resistance. The rolling resistance is generated by soil compaction, horizontal soil displacement and tyre flexibility, while the tractive effort is generated by the soil shearing behaviour at the soil-wheel interface. Existing models for soil compaction (i.e. pressure-sinkage) are initially evaluated for their accuracy and applicability using literature data, but their performance is unsatisfactory. In addition, a large experimental campaign is conducted using two soil types and various experimental processes such as pressure-sinkage on flat and curved plates, shear tests, rolling wheel tests. An analysis confirms that collecting reliable soil data is a challenging task, and that measurements on soils with transient characteristics, such as moisture content, are affected the most. A new model for pressure-sinkage on flat plates is developed which greatly improves the correlation to experimental data. An extension of existing shear model is also created, which improves the applicability of the formulation to road vehicles. Using optimisation algorithms for model parameter identification is found advantageous compared to traditional methods. A dynamic soil-wheel interaction model is developed which combines the new models for pressure-sinkage and shearing behaviour. The dynamic model, along with a new formulation for the horizontal soil displacement, demonstrates good correlation to test data for dry sand. Overall conclusions are drawn and suggestions for future work are presented.

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Notation

Symbol	Description	Unit (SI)
A	a. Activity index b. Area	[-] [m ²]
A_0	Plate width and hardpan depth coefficient (Lyasko [31] model)	[-]
B	Smallest dimension of wheel loading area (or wheel width)	[m]
B_i	Bearing capacity (Lyasko [31] model)	[N/m ²]
D	Wheel diameter	[m]
E	Dynamic soil deformation modulus (Lyasko [31] model)	[N/m ²]
E_0	Steady-state soil deformation modulus (Lyasko [31] model)	[N/m ²]
F	Tractive Effort	[N]
$F_{v,T}$	Total vertical force (Bekker [1] model)	[N]
F_x	Tyre longitudinal force	[N]
F_y	Tyre side force	[N]
G_s	Specific gravity	[-]
H	Hardpan depth	[m]
J	Coefficient (Lyasko [31] model)	[-]
K	Parameter (Janosi and Hanamoto [24] model)	[m]
K_w	Strain at maximum shear stress	[m]
$K_{w,a}$	Strain at maximum shear stress coefficient (Extended Oida model)	[m ³ /N]
$K_{w,b}$	Strain at maximum shear stress coefficient (Extended Oida model)	[m]
L	Length of wheel loading area	[m]
LI	Liquidity index	[-]
N_c	Bearing capacity coefficient	[-]
N_q	Bearing capacity coefficient	[-]

N_γ	Bearing capacity coefficient	[-]
P	Pressure	[N/m ²]
PI	Plasticity index	[-]
R	a. Overall rolling resistance b. Radius	[N] [m]
R_c	Soil compaction resistance	[N]
V	Vehicle velocity	[m/s]
V_S	Volume of solids of a soil sample	[m ³]
V_T	Total volume of a soil sample	[m ³]
V_V	Volume of voids of a soil sample	[m ³]
V_x	Longitudinal velocity of the wheel centre	[m/s]
V_{sx}	Longitudinal slip velocity	[m/s]
W	Wheel vertical load	[N]
W_S	Weight of the dry solids of a soil sample	[N/m ³]
W_w	Weight of water of a soil sample	[N/m ³]
Y_0	Coefficient (Lyasko [31] model)	[-]
a_k	ADPS model coefficient	[N/m ⁿ⁺⁴]
a_n	ADPS model coefficient	[1/m ⁴]
b_k	ADPS model coefficient	[-]
b_n	ADPS model coefficient	[-]
c	Soil cohesion	[N/m ²]
e	Void ratio	[-]
i	a. Wheel slip (or skid) b. Hydraulic gradient	[-] [m/s]
j	Shear strain	[m]
j_l	Shear strain - wheel surface front region (Wong and Reece [34] model)	[m]

j_2	Shear strain - wheel surface rear region (Wong and Reece [34] model)	[m]
k	a. Coefficient of permeability b. Coefficient of proportionality c. Longitudinal slip ratio	[-] [-] [-]
k_c	Bekker pressure-sinkage coefficient	[N/m ⁽ⁿ⁺¹⁾]
k'_c	Reece pressure-sinkage coefficient	[-]
k_ϕ	Bekker pressure-sinkage coefficient	[N/m ⁽ⁿ⁺²⁾]
k'_ϕ	Reece pressure-sinkage coefficient	[-]
n	a. Porosity b. Pressure-sinkage exponent	[%] [-]
n_c	Bearing capacity plate shape factor (Lyasko [31] model)	[-]
n_q	Bearing capacity plate shape factor (Lyasko [31] model)	[-]
n_γ	Bearing capacity plate shape factor (Lyasko [31] model)	[-]
q	Fluid flow rate	[m ³ /s]
r_e	Wheel effective radius	[m]
w	Moisture content	[%]
w_l	Liquid limit (Consistency Limits)	[%]
w_p	Plastic limit (Consistency Limits)	[%]
w_s	Shrinkage limit (Consistency Limits)	[%]
z	Sinkage	[m]
z_b	Sinkage due to bulldozing	[m]
z_T	Total sinkage	[m]
Ω	Wheel angular velocity	[rad/s]
α	a. Tyre slip angle b. Total sinkage coefficient	[deg] [-]
β	Total sinkage coefficient	[1/m]
γ_s	Unit weight (solid portion of soil mass)	[N/m ³]
γ_d	Unit weight (dry soil)	[N/m ³]

γ_{sat}	Unit weight (saturated soil)	[N/m ³]
γ_w	unit weight of water at 4 °C	[N/m ³]
θ_1	Entry angle (measured from wheel vertical axis)	[deg]
θ_2	Exit angle	[deg]
θ_e	Entry angle (Gee-Clough [25] model)	[deg]
θ_M	Angle of maximum normal stress (Wong and Reece [34] model)	[deg]
v	Fluid velocity	[m ³ /s]
ζ	Contact pressure concentration coefficient (Lyasko [31] model)	[-]
ρ_d	Density (dry soil)	[kg/m ³]
ρ_{sat}	Density (saturated soil)	[kg/m ³]
σ	a. Normal stress b. Standard deviation	[N/m ²] [-]
σ_1	Normal stress - wheel surface front region (Wong and Reece [34] model)	[N/m ²]
σ_2	Normal stress - wheel surface rear region (Wong and Reece [34] model)	[N/m ²]
τ	Shear stress	[N/m ²]
τ_{max}	Maximum shear stress	[N/m ²]
τ_r	Shear residual stress	[N/m ²]
$\tau_{r,a}$	Shear residual stress coefficient (Extended Oida model)	[-]
$\tau_{r,b}$	Shear residual stress coefficient (Extended Oida model)	[N/m ²]
ϕ	Angle of shearing resistance.	[deg]
ω	Coefficient (Lyasko [31] model)	[-]

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Chapter 1: Introduction

1.1 Road Vehicles and the Role of Terramechanics

The automotive industry is currently facing an unprecedented transformation driven by stricter regulations on emissions and safety standards, an increasing market demand for sport utility vehicles (SUVs) and large off-road vehicles, as well as pressure to launch new models much more frequently than ever before. Since the pneumatic tyre is the sole element that comes into contact with the terrain (rigid or deformable), the understanding of the principles behind its behaviour and interactions is of great importance in the overall vehicle dynamic performance. However, the active lifestyle that SUVs are expected to satisfy, require the consideration of the tyre performance not only on rigid road surfaces (e.g. asphalt) but also on deformable terrains such as sand and clay.

Terramechanics is the branch of engineering dealing with the interaction between the machine and its environment, and in the specific case of road vehicles the soil-wheel interactions. The systematic study into the fundamentals of the terrain-machine interaction effectively begun in the 1950s by the works of M.G. Bekker [1], [2]. Since then, numerous researches and studies were conducted covering all aspects of Terramechanics for wheeled and tracked vehicles, focusing on agriculture, machinery, military, space exploration and road vehicle applications.

As the number of off-road and sport utility vehicles increases each year, and with the simultaneous toughening of emissions regulations, the study of vehicle performance in deformable terrains slowly becomes part of the standard development process. With tyre rolling resistance being one of the major contributors to overall vehicle emissions, a careful design specification is necessary. The laws of Terramechanics provide the foundation for analysing the vehicle performance in deformable terrains by means of prediction of tyre rolling resistance and traction, which directly influence the required vehicle power, and subsequently emissions. Additionally, advanced control systems responsible for the vehicle stability, such as traction control and anti-lock braking system, are also expected to perform equally well in both rigid and deformable terrains. Yet, an effective tuning of these systems for off-road conditions require an insight of the soil-wheel interaction mechanisms.

1.2 Problem Definition

Over the past few years there has been a market shift towards sport utility vehicles and cross-overs, with the requirements for these vehicles however being usually in conflict with each other. For instance, the higher mass and dimensions of an SUV compared to a regular car requires a larger power unit to provide the necessary propulsion and greater tyre dimensions to ensure safety during cornering, which ultimately results in significant emissions increase.

A common development practice in the automotive industry is the so-called *V-Shape development process*. In such process, as shown in Figure 1.1, the left part of the ‘V’ is dedicated to target cascading and specification setting, starting from the high-level vehicle requirements at the top-left, down to the component targets at the bottom. In the right part of the ‘V’ the implementation phase takes place, followed by validation at various system levels.

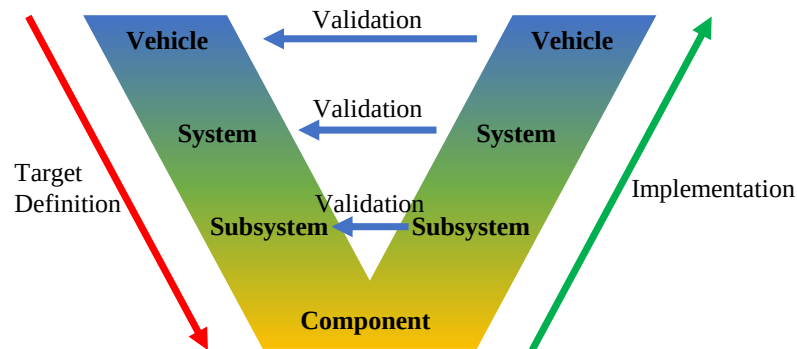


Figure 1.1 V-Shape development process.

Traditionally, both branches of the ‘V’ would be conducted using empirical methods or simple computational tools. Using such tools and methods require a large number of physical prototypes, which consequently increase the overall time and cost of the development cycle due to higher number of validation loops, and at the same time can lead to suboptimal final design. In recent years though, the advancements in simulation technology allowed the use of numerical methods in all stages of vehicle development process.

For the tyre development process, the automotive OEMs would normally specify high level requirements in terms of vehicle dynamics performance (handling, braking, etc.), and would either commission the development to tyre manufacturers, or would develop the tyre in a joint fashion. But, in

both cases a significant number of validation loops with physical prototypes is required. Today, even the most conservative automotive OEMs have recognised the merit and efficiency simulation methods can provide and have adopted these methods as part of the official development process.

Currently, there are several tyre models available, but the majority of them are designed to describe and predict the tyre behaviour on rigid road surfaces. However, there is a gap in the industry for a reliable tyre model for off-road vehicle applications, which means that evaluation of off-road capability and performance cannot be done using simulation methods. Such a condition can impact both the output product performance and the development cost.

1.3 Aim and Objectives

The current research aims at bridging the gap between physical prototypes and virtual development methods for off-road tyres, by investigating the necessary elements impacting their performance. Ultimate target is the development of a tyre model that can support all stages of a virtual vehicle development process. To achieve this aim, a number of objectives have been defined which are described below:

1. Conduct a literature research that provides a comprehensive overview of the fundamentals and principles concerning off-road tyres, and identify the key elements affecting the vehicle performance.
2. Evaluate the accuracy and robustness of existing methodologies, and their applicability to road vehicle applications and to virtual vehicle development process.
3. Perform in-house experimental procedures with different types of soils in order to gain an extensive understanding of the soil behaviour from a practical viewpoint, but also to support the validation of a tyre model.
4. Develop a longitudinal dynamics soil-wheel interaction tyre model for off-road vehicle applications, aiming at accurate prediction of key performance items as identified in objective (1).

1.4 Thesis Structure

This section is aimed at providing the reader an overview of the thesis structure which is organised in six chapters. A brief description of each chapter is discussed below.

Chapter 1: Introduction

The general demand for accurate simulation models is discussed. With pressure to reduce physical prototypes, the automotive industry is aiming at utilising simulation methods as much as possible. On that basis the aim and objectives are set which are focusing on the development of a soil-wheel interaction model for road vehicle applications.

Chapter 2: Literature review

In this chapter a review of the literature is conducted which starts with the fundamentals for soil mechanics, and transitions to the specific branch of Terramechanics which deals with the machine-terrain interactions. The literature review continues with a description of concepts regarding tyre modelling in rigid surfaces, and concludes with approaches to soil-wheel interaction modelling.

Chapter 3: Pressure-Sinkage Initial Study

As pressure-sinkage being one of the major components of soil-wheel performance, existing models are evaluated in this chapter using experimental data from the literature. Then, the most promising model is further analysed to confirm or reject its applicability to road vehicles.

Chapter 4: Soil Experiments and Modelling

A large experimental campaign is described, and the results are analysed in this chapter. The data from in-house pressure-sinkage tests support the development of a new model for flat plates. Analysis of the shear data takes place, and an existing model gets extended for allowing its applicability to road vehicles. Finally, the data from curved plates, that replicate vehicle wheels, are used for validating a dynamic pressure-sinkage model.

Chapter 5: Soil-Wheel Interaction

The finding from Chapter 4 are combined with additional knowledge from in-house rotating wheels test data to form the basis of a dynamic soil-wheel interaction model. The model is compared to test data and the results are discussed and analysed.

Chapter 6: Final Concussions

Discussion and summarisation of the findings and novelties of the current research takes place in Chapter 6. Furthermore, areas of future study are proposed.

Chapter 2: Literature Review

In this chapter a review of the literature is conducted, starting with some basic principles of soil mechanics in order to gain an understanding of the overall field of study. Then, moving into the field of terramechanics, with emphasis on concepts related to road vehicles such as rolling resistance. As the tyre is core element on the study of terrain-wheel interactions, a review of available tyre models is presented, followed by discussion of specific modelling approaches to soil-wheel interactions.

2.1 Fundamentals of Soil Mechanics

Soil Mechanics describe the properties and behaviour of soil, which is a non-homogeneous medium that can consist of various materials such as minerals, organic solids, liquids and gasses. As soil mechanics is a very wide field of engineering, only concepts and basic principles applicable to Terramechanics will be discussed in this text.

Soil characteristics can be categorised into two main groups; permanent and transient. Permanent characteristics refer to those soil parameters that are independent of the in-situ conditions such as particle size, particle shape and specific gravity. On the other hand, transient characteristics describe the soil parameters, such as moisture content and porosity, that vary with applied load and weather conditions. The subsequent paragraphs summarise the most dominant permanent and transient soil characteristics.

2.1.1 Permanent Characteristics of Soils

2.1.1.1 Particle Size Distribution

As most soils consist of a mixture of different types of materials and particles of various sizes, one of the methods for characterising a soil is by determining its particle size distribution, i.e. defining range of particle sizes as percentage by mass within a soil sample. There are two main methods for determining particle size distribution; Sieve analysis and Hydrometer analysis.

In sieve analysis, according to British Standard 1377-2 [3], a soil sample is shaken through a series of various opening size sieves, with the minimum sample mass being dependent on the largest significant particle size. The sieves are arranged in a column of increasing opening size sequence, with the largest aperture sieve of 75 mm placed at the top and the smallest aperture of 63 μm at the bottom. The sieve stack is then positioned on a mechanical shaker, and the soil sample is placed at the top sieve. After the soil sample is shaken for a minimum of 10 minutes, the weight of soil retained in each sieve is recorded and the results are plotted in a diagram with x-axis being the particle size in a logarithmic scale and y-axis the percentage of particles by weight.

After the sieve test is completed, if more than 10% of the material has passed through the 63 μm sieve, it is necessary to perform the hydrometer analysis in order to fully determine the particle size distribution of the soil sample [3]. In this test the soil sample that is passed through the finest sieve size is dispersed in water and let to settle in a 1000 mL cylinder. A thermometer and a hydrometer are placed in the cylinder, and as soil particles are settling the hydrometer readings as well as time and temperature are recorded. Stokes' equation describes the relationship between particle terminal velocity, particle diameter, fluid viscosity and other factors [4]. Using Stokes' equation the effective diameter of soil particles can be derived, and the hydrometer analysis results are presented similarly to the results from sieve test. Depending on the tested soil, in order to cover the whole range of particle sizes, both sieve and hydrometer may be required to determine the particle size distribution. Table 2.1 summarises the particle size ranges for various types of soils as found in the literature.

Types of material		Sizes (mm)
Boulders		Over 200
Cobbles		60-200
Gravel	Coarse	20-60
	Medium	6-20
	Fine	2-6
Sand	Coarse	0.6-2
	Medium	0.2-0.6
	Fine	0.06-0.2
Silt	Coarse	0.02-0.06
	Medium	0.006-0.02
	Fine	0.002-0.006
Clay		Less than 0.002

Table 2.1 Particle size distribution of different types of soils, data from [5].

2.1.1.2 Specific Gravity

Specific gravity (G_s) is defined as the ratio between the density of a substance and the density of a reference substance. Specifically in soil mechanics, the substance is the soil and usually the reference substance is distilled water at 4 °C [6]. The specific gravity is defined by Equation 2.1.

$$G_s = \frac{\gamma_s}{\gamma_w} \quad 2.1$$

where γ_s is the unit weight of the solid portion of the soil mass, and γ_w is the unit weight of water at 4 °C. The specific gravity is defined with different methods for coarse and fine soil particles, and the average of the two is used for analyses [4].

2.1.1.3 Consistency Limits

Consistency Limits or *Atterberg Limits* express the moisture content of cohesive soils as they are passing from one state to another. A schematic of consistency limits is presented in Figure 2.1. When considering a perfectly dry soil that then starts increasing its moisture content by adding water, *shrinkage limit* (w_s) is the moisture content where any increase in water will result in a volumetric change of the soil sample. At this point all voids are filled with water, and with further increase in moisture the soil behaves as a semi-solid. The transition point where the soil starts to behave like a plastic medium is called *plastic limit* (w_p), and it is defined as the lowest water content at which the soil can be rolled into 3 mm in diameter threads without the threads breaking into pieces. The transition between the plastic state of soil and the liquid state is called *liquid limit* (w_l), where at this state the soil has small but measurable shear strength; with moisture content beyond that limit the soils behave like viscous fluids.

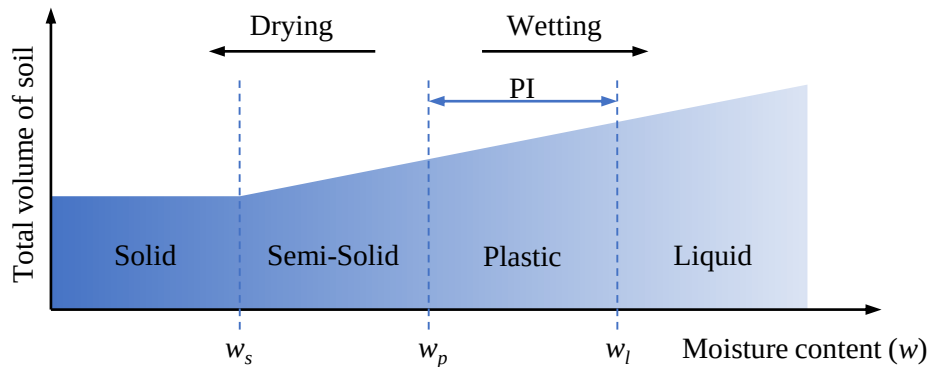


Figure 2.1 Consistency Limits schematic.

The consistency limits discussed in this section, which are derived using standardised tests and procedures [4], are particularly useful for soil identification and classification. Indeed, when they are used to derive indices, some fundamental soil properties can be defined [6]. The most widely accepted indices are the plasticity, the liquidity and the activity index. The consistency limits values of some typical types of soils, as found in the literature, are summarised in Table 2.2.

Type of Soil	Degree of Plasticity	Plasticity Index PI	Limits of Plasticity Indexes	Plastic Limit w_p	Liquid Limit w_l
Sand	Non-plastic	0	0	20	20
Silt	Low-plastic	5	<7	20	25
Silty Clay Clayey Silt	Medium-plastic	15	7-17	25	40
Clay	High-plastic	30	>17	40	70

Table 2.2 Consistency Limits for Some Typical Soil Types, data from [6].

Plasticity Index (PI) is the difference between the liquid and the plastic limit and it indicates the range of moisture content in the plastic soil state. Table 2.2 gives some typical values of PI for common soil types, where soils with high PI become soft and slippery in wet weather. In contrary, soils with low PI are expected to crack and become brittle in dry weather conditions.

Liquidity Index (LI) is the ratio of the difference between the moisture content of an in-situ soil and its liquid limit divided by the plasticity index [6].

$$LI = \frac{w - w_l}{PI} \quad 2.2$$

The liquidity index is used to indicate where the soil lies within the plastic range, and its value ranges between 0 and 1; $LI = 1$ if the in-situ moisture content is equal to the plastic limit, and $LI = 0$ if the moisture content is equal to the liquid limit.

Activity Index (A) is defined as the plasticity index divided by the percentage of mass of soil particles with effective diameter <0.002 mm.

$$A = \frac{PI}{\% \text{ by mass finer than } 0.002mm} \quad 2.3$$

The activity index is an indicator of soil plasticity. Soils with activity index between 0.75 and 1.25 are considered as ‘normal’ as the PI is practically equal to the clay percentage of soil mass, while soils with index of <0.75 are assumed ‘inactive’. ‘Active’ soils with activity index >1.25 tend to have high water holding and cation exchange capacity, they are more prone to swelling and volumetric change with water content change [4]. Therefore, they tend to have low permeability and shear strength.

2.1.2 Transient Characteristics of Soils

2.1.2.1 Void Ratio and Porosity

An important concept that distinguishes soil from other engineering materials is that it is a three-phase medium: solid, liquid and gas. The total volume of a soil sample consists of two main parts, the volume of solids (V_s) and the volume of voids (V_v). The volume of voids can be completely filled with liquid, completely filled with gas or a combination of both. A schematic of the three-phase soil is presented in Figure 2.2. Two parameters that describe the in-situ soil condition are the porosity and void ratio, with porosity (n) being defined as the ratio between the volume of voids and the total volume.

$$n = \frac{V_v}{V_T} \quad 2.4$$

However, porosity is a soil parameter difficult to measure because both the numerator and the denominator can vary independently with applied load. An easier way for deriving porosity is by using the void ratio (e), which is the ratio between the volume of voids and the volume of solids.

$$e = \frac{V_v}{V_s} \quad 2.5$$

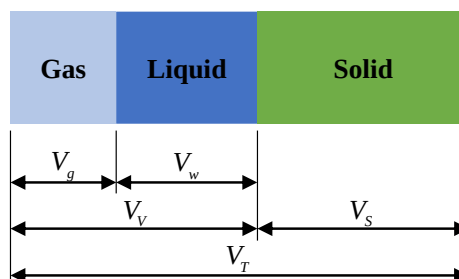


Figure 2.2 The Three-Phase soil.

2.1.2.2 Moisture Content

Moisture content (w) is defined as the percentage ratio of the weight of water of a soil sample and the weight of its dry solids.

$$w = \frac{W_w}{W_s} 100 \quad 2.6$$

where: W_w is the weight of water of a soil sample, and W_s is the weight of its dry solids.

Moisture content directly influences the shear strength of the soil [7], therefore it plays an important role in off-road vehicle mobility. There are variations in moisture content in different soil depths and these variations tend to be more significant near the soil surface due to weather fluctuations. Table 2.3 summarises some permanent and transient characteristics of typical soils in natural state.

Description	Porosity, n (%)	Void ratio, e (-)	Water content, w (%)	Density (Mg/m ³)		Unit Weight (kN/m ³)	
				ρ_d	ρ_{sat}	γ_d	γ_{sat}
Uniform sand, loose	46	0.85	32	1.43	1.89	14.0	18.5
Uniform sand, dense	34	0.51	19	1.75	2.09	17.2	20.5
Mixed-grained sand, loose	40	0.67	25	1.59	1.99	15.6	19.5
Mixed-grained sand, dense	30	0.43	16	1.86	2.16	18.2	21.2
Glacial till, very mixed-grained	20	0.25	9	2.12	2.32	20.8	22.7
Soft glacial clay	55	1.2	45		1.77	12.0	17.4
Stiff glacial clay	37	0.6	22		2.07	16.7	20.3
Soft slightly organic clay	66	1.9	70		1.58	9.1	15.5
Soft very organic clay	75	3.0	110		1.43	6.7	14.4
Soft bentonite	84	5.2	194		1.27	4.2	12.5

Table 2.3 Characteristics of Typical Soils in Natural State, data from [7].

2.1.2.3 Permeability

The ability of a soil to allow fluids (gas or liquids) to pass through its voids is called permeability. In general, a medium is considered as permeable if it permits a measurable quantity of fluid to pass through its voids in a finite time frame. On the other hand, if the rate of fluid through the medium is slow enough to be negligible then the medium is considered as impermeable. Darcy's law describes the flow of fluids through a fully saturated soil as:

$$v = ki \quad 2.7$$

or

$$q = Av = Aki \quad 2.8$$

where: v is the fluid velocity, k the coefficient of permeability, i the hydraulic gradient, q the flow rate and A the cross-sectional area of the soil corresponding to q . Darcy's law assumes that the hydraulic flow is laminar and that is usually the case in fine grained soils.

For the derivation of coefficient of permeability k , various laboratory methods are employed. The extraction of the soil sample to be tested must be careful to ensure that the sample is not disturbed, as it could lead to misleading results. There are also in-situ tests to determine the coefficient of permeability, such as the pumping test ([4], [7]), however they are expensive for off-road vehicle applications [6].

When considering a moving off-road vehicle, the running gear (wheels or grousers) will come into contact with a soil patch only temporarily, and the duration of the contact would vary depending on the vehicle speed. Therefore, the higher the coefficient of permeability the greater the soil compression under the effect of a moving vehicle, due to the soil's ability for quick reduction of its volume of voids. Contrarily, a soil with low permeability would compress less under the effect of the same vehicle and conditions, resulting in lower levels of rolling resistance.

2.1.3 Limit Analysis

The majority of problems in soil mechanics are divided into two main groups; the stability problems and the elasticity problems [8]. The elasticity problems deal with the stresses and strains of soils where no failure occurs, hence within the material's elastic region as described by the classical Mechanics of Materials. In such problems the relationship between stresses and strains is considered to be linear and the solution is provided by using the well-known Hooke's law of elasticity.

On the other hand, the stability problems deal with the condition of the material at ultimate failure. For the determination of loads that lead to soil ultimate failure the theory of perfect plasticity is employed, in which the soil is considered to remain within the elastic region only for small levels of stress. When the stress reaches the yield value, continuous plastic flow occurs with constant state of stress. Limit analysis is concerned with the applications and methods of determining the soil yield stress. The transition between the elastic and plastic deformation is called progressive failure and is not

considered by limit analysis. The law defining the limit of elastic behaviour under any possible combination of stresses is called *yield criterion* [9] and several yield criteria have been proposed over the years such as Von Mises [9], Tresca [9], Mohr-Coulomb [10] and Drucker-Prager [11].

2.1.3.1 Tresca and von Mises Yield Criteria

It is usually easier to describe the yield criteria in terms of principal stress space, where instead of Cartesian x-y-z axes, the orthogonal axes are represented by the principal stresses σ_1 , σ_2 and σ_3 . Using this construction, any stress can be plotted in the three-dimensional stress space [9].

In the example stress space of Figure 2.3, the line OH which represents the hydrostatic stress axis will be equally inclined to the three principal stress axes with direction cosines $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$. The stress vector **OQ**, can be resolved into two components; the **OG** along the OH line and **OP** perpendicular to OH. The plane perpendicular to OH that passes through the origin O is called the *deviatoric plane* or π -plane, and since the hydrostatic stress has no effect on yielding, the material yielding will be caused by the direction and magnitude of the deviatoric stress vector **OP**. This leads to a cylindrical yield surface of radius $\sqrt{2J_2}$, where J_2 is the second invariant of the deviatoric stress tensor.

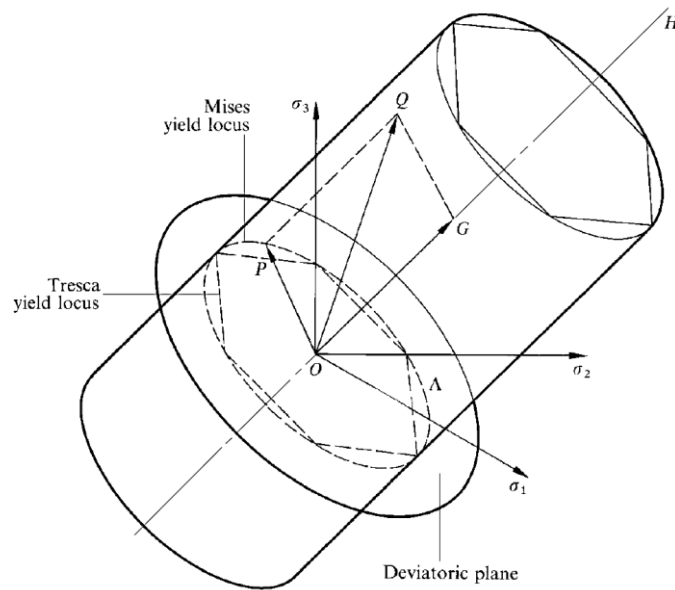


Figure 2.3 Graphical representation of the Tresca and von Mises yield criteria in the principal stress space [9]. Reprinted from Theory of Plasticity, 3rd ed., J. Chakrabarty, Copyright (2006), with permission from Elsevier.

The right cylinder of Figure 2.3 represents von Mises yield surface, and it defines the transition from the elastic state of stress (i.e. inside the cylinder) to the plastic state of stress (i.e. on the yield surface). On the other hand, the Tresca yield criterion states that material yielding will occur when the maximum shear stress reaches a critical value equal to the shearing yield in uniaxial tension test. As shown in Figure 2.3, the yield surface of Tresca criterion is a right hexagon, which when applied can lead to mathematical complexities due to singularities at the hexagon corners [9]. Both criteria predict quite well the plastic limit of ductile materials, however experimental data are in better agreement to von Mises yield criterion [9].

2.1.3.2 Mohr-Coulomb Yield Criterion

Tresca and von Mises yield criteria are used predominantly for the prediction of yield limit in ductile metals and they assume that the materials behave in the same manner in both tension and compression. Nonetheless, in problems concerning the soil plastic behaviour such assumption is not valid. In Mohr-Coulomb yield criterion, the maximum shear stress of a soil is a function of normal stress, and it is expressed by Equation 2.9 below:

$$\tau_{\max} = c + \sigma \tan(\phi) \quad 2.9$$

where c and ϕ are the soil cohesion and internal friction angle respectively.

To enable the treatment of the Mohr-Coulomb yield criterion in three-dimensional space, it has to be expressed in terms of principal stresses. Assuming $\sigma_1 > \sigma_2 > \sigma_3$, Equation 2.9 can be transformed into the form:

$$\sigma_1 - \sigma_3 = 2c \cos \phi + (\sigma_1 + \sigma_3) \sin \phi \quad 2.10$$

In the graphical representation of the criterion in Mohr diagram, as presented in Figure 2.4, the wedge-shaped envelope between the two straight lines starting from $(c \cot \phi, 0)$ and inclined at angles of amount ϕ to the positive σ -axis, denote the soil elastic region. As long as the state of stress $(\sigma_1, \sigma_2, \sigma_3)$ is such that the Mohr circles lie within the wedge-shaped region, the soil remains in the linear elastic range. Meaning that if the largest of the Mohr circles touches the two straight lines, plastic flow occurs. When

the Mohr-Coulomb yield curve is plotted in the principal stress space, the three-dimensional yield surface is derived, which is a right hexagonal pyramid equally inclined to the principal stress axes σ_1 , σ_2 and σ_3 as shown in Figure 2.5.

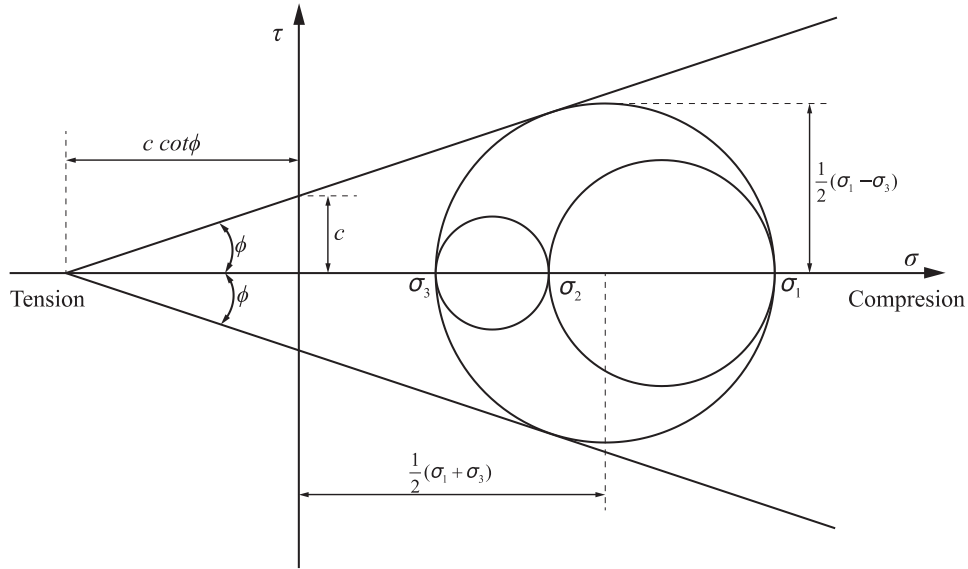


Figure 2.4 Mohr-Coulomb Yield Criterion in Mohr's diagram.

2.1.3.3 Drucker-Prager Yield Criterion

The Drucker-Prager yield criterion is the generalisation of a modified von Mises criterion for soils that has been proposed by Drucker and Prager [11]. In some references it can also be found as extended von Mises, but the criterion is effectively the von Mises criterion with the inclusion of the hydrostatic stress contribution to the yield limit. The Drucker-Prager yield surface in the principal stress space is a right circular cone equally inclined to the principal stress axes as shown in Figure 2.5 and it has the form:

$$aI_1 + \sqrt{J_2} = k \quad 2.11$$

where a and k are positive material constants, and if a is zero it reduces to von Mises criterion. In the case of plane strain, the criterion reduces to Mohr-Coulomb if:

$$a = \frac{3 \tan \phi}{\sqrt{9 + 12 \tan^2 \phi}} \quad k = \frac{3c}{\sqrt{9 + 12 \tan^2 \phi}} \quad 2.12$$

The Drucker-Prager yield criterion in various versions and modifications is widely used in geotechnical engineering studies [12-16]. Even though experimental data in granular soils have shown that it over-predicts the change in volume [9], it still remains one of the most popular theories for practical applications. The criterion's smooth yield surface (i.e. no edges) provides numerical stability and advances over the Mohr-Coulomb criterion which can cause singularities in certain cases when computer simulations are involved, for instance when performing Finite Element Analysis.

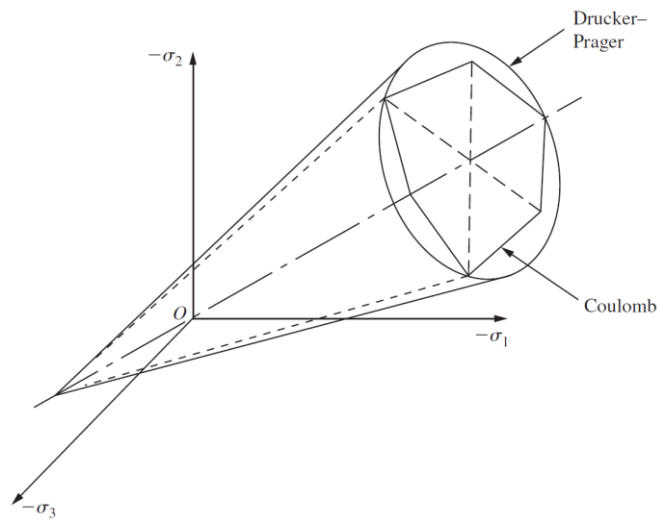


Figure 2.5 Drucker-Prager vs Mohr-Coulomb yield surface in the principal stress space [9]. Reprinted from Theory of Plasticity, 3rd ed., J. Chakrabarty, Copyright (2006), with permission from Elsevier.

2.2 Terramechanics

In the preceding paragraphs some fundamental concepts of soil mechanics are discussed, with emphasis on those more closely related to *Terramechanics* which is the branch of engineering dealing with the terrain-machine interaction; or more specifically, in the case of off-road vehicles, the soil-wheel interaction. There are two major concepts that concern Terramechanics for off-road vehicles: (a) the tractive effort which is generated at the soil-wheel interface enabling the vehicle longitudinal movement, and (b) the wheel rolling resistance which is a force caused by the soil vertical compaction and horizontal displacement. With the main aims of vehicle digital development being the early detection of design issues and physical prototype reduction, accurate simulation predictions for rolling resistance and tractive effort are crucial in off-road vehicle development process. These two concepts will be discussed in the following paragraphs.

2.2.1 Tractive Effort

As discussed in Section 2.1.3.2, the Mohr-Coulomb failure criterion describes the shear strength of a soil as a function of the normal stress. When a wheel is rolling on a deformable terrain forces arise in the normal and tangential directions of the wheel-soil interface, with the normal forces develop due to soil compaction and bulldozing effect while the tangential forces are the result of the relative shear strain between the two surfaces. The sum of tangential forces transformed in the wheel's longitudinal movement axis is called *thrust* or *tractive effort*, and for a mixed type of terrain is described by Equation 2.13 created based on the Mohr-Coulomb failure criterion [1].

$$F = \tau A = cA + W \tan \phi \quad 2.13$$

where A is the contact area between the wheel and the terrain, W is the wheel vertical load, c is the coefficient of cohesion in and ϕ is the internal friction angle. Equation 2.13 is graphically illustrated in graph (a) of Figure 2.6.

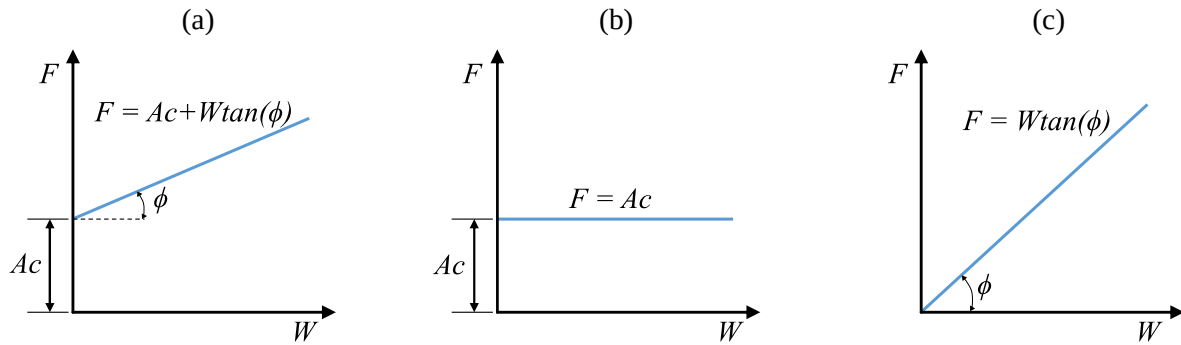


Figure 2.6 Tractive effort for (a) mixed, (b) cohesive and (c) frictional type of terrain.

Equation 2.13 provides the tractive effort for a mixed type of soil where both cohesive and frictional characteristics are present. In the case of a purely plastic medium (e.g. wet clay or snow), the cohesive forces between the soil particles, expressed by the coefficient of cohesion, do not change under the effect of pressure. Therefore, assuming that the running gear is a rigid wheel of certain dimensions, the tractive effort remains essentially constant and it is only a function of the coefficient of cohesion.

This type of soil is called purely cohesive soil [17-19], and its tractive effort can be expressed by a simplified version of Equation 2.13 as presented below:

$$F = cA \quad 2.14$$

If a purely frictional soil is considered, e.g. dry sand or cold snow, there are no cohesive forces holding the soil particles together [17-19]. Hence, when the soil is compressed under the load exerted by the running gear, due to frictional forces the soil particles stick together providing motion resistance. The forces developed are not constant and increase proportionally to vertical load. In this case, the tractive effort can be expressed as:

$$F = W \tan \phi \quad 2.15$$

In reality, most of terrains are mixture of both cohesive and frictional characteristics. However, for calculation simplifications reasons it is common to treat particular types of terrains as purely cohesive or purely frictional together [17-20], for instance clay-like soils as purely cohesive and sand-like soils as purely frictional.

Traditionally, the Terramechanics studies focus at the maximum tractive force a wheel can provide, and for the Mohr-Coulomb failure criterion, as described in paragraph 2.1.3.2, would correspond to the maximum shear stress at a given normal load. However, when considering the dynamic response of a wheel, the amount of tractive force generated by the wheel-soil interaction at any given time is a factor of wheel slip, i.e. shear strain at the wheel soil interface. Hence, capturing the complex shear stress-strain soil behaviour is essential for automotive applications, such as traction or ABS control tuning and optimisation.

Longitudinal slip is defined as the ratio of the longitudinal slip velocity V_{sx} and the longitudinal velocity of the wheel centre V_x :

$$k = -\frac{V_{sx}}{V_x} = -\frac{V_x - r_e \Omega}{V_x} \quad 2.16$$

where Ω is the angular velocity of the wheel spin axis, and r_e wheel effective radius. The sign of the longitudinal slip is set in way that driven wheels would result in positive values for k (i.e. slip), and

towed wheel at negative k values (i.e. negative slip or skid). Also, when the wheel is locked (i.e. $\Omega = 0$), then $k = -1$, while a stationary wheel (i.e. $V_x = 0$) the longitudinal slip would be $k = \infty$.

According to Wong [10], for “brittle” type of soils such as dry sand, it is expected that the shear stress-strain curves would display a peak at the maximum shear stress as illustrated by Figure 2.7 below. In the figure, the longitudinal axis corresponds to the shear strain j , which is the relative displacement at the shear plate-soil interface, while the vertical axis corresponds to the shear soil stress. For “brittle” soil types, after the peak value τ_{max} occurring at strain K_w , the shear stress drops until it converges to an asymptotic value called *residual stress*, τ_r .

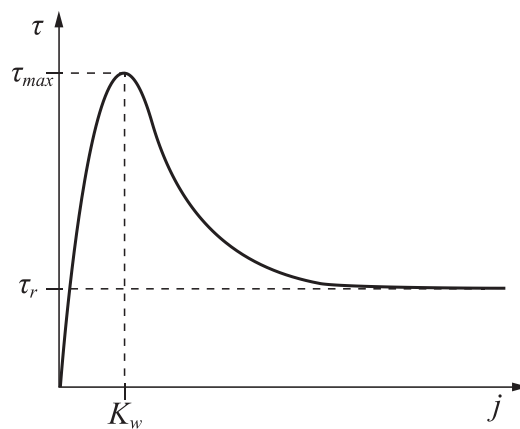


Figure 2.7 Schematic of a brittle-type soil shear response type.

After analysing a large sample of shear data from various types of soils, and comparing the ratios between the maximum and the residual shear stress values, Kacigin and Guskov [21] proposed the shear stress-strain formulation of Equation 2.17. Though, as highlighted by Oida [22], large discrepancies can occur between the desired (i.e. entered parameter) and resultant (i.e. curve output) values of maximum shear stress when the residual shear stress to normal stress ratio (τ_r/σ) approaches the value of 1. To overcome this issue, Oida [22] proposed the alternative formulation of Equation 2.18 which always produces stress-strain curves with the maximum shear stress matching the input parameter. According to Wong [23], the main drawback of Oida’s formulation is its complexity and that an iterative process is required for obtaining the necessary input parameters, so he proposed Equation 2.19 as an alternative formulation. Equation 2.19 however, is equally complicated with Equation 2.18, and it requires the same input parameters. In addition, the predicted maximum shear stress of Equation 2.19 cannot match the desired input parameter, with the error value usually being less than 3% [23]. Wong’s concerns could

be justified back in the days, but equation complexity and parameter identification process (which is required for all three formulations described in the current paragraph) can be easily handled by today's computational resources.

$$\text{Kacigin-Guskov [21]:} \quad \tau = \tau_r \left[1 + \frac{2.55 \left(\frac{\tau_{\max} - \tau_r}{\tau_r} \right)^{0.825}}{\cosh \left(\frac{j}{K_w} \right)} \right] \tanh \left(\frac{j}{K_w} \right) \quad 2.17$$

$$\text{Oida [22]:} \quad \tau = \tau_{\max} K_r \left[1 - \frac{\sqrt{1-K_r} \left(1 + \frac{\sqrt{1-K_r}-1}{K_r} \right)^{\frac{j}{K_w}}}{\sqrt{1-K_r} \left(1 - \frac{2}{K_r} \right) + \frac{2}{K_r} - 2} \right] \left[1 - \left(1 + \frac{\sqrt{1-K_r}-1}{K_r} \right)^{\frac{j}{K_w}} \right] \quad 2.18$$

$$\text{Wong [23]:} \quad \tau = \tau_{\max} K_r \left\{ 1 + \left[\frac{1}{K_r \left(1 - \frac{1}{\exp(1)} \right)} - 1 \right] \exp \left(1 - \frac{j}{K_w} \right) \right\} \left[1 - \exp \left(- \frac{j}{K_w} \right) \right] \quad 2.19$$

For the type of soils that do not exhibit a peak at the maximum shear stress, such as saturated clay, Janosi and Hanamoto [24] proposed the much simpler shear stress formulation of Equation 2.20.

$$\tau_{\max} = [c + \sigma \tan(\phi)] \left(1 - e^{-\frac{j}{K}} \right) \quad 2.20$$

where j is the shear strain, and K is the strain value where the slope near the origin intersects with the maximum shear stress value as shown in Figure 2.8.

Equation 2.20 is effectively the Mohr-Coulomb yield criterion with the addition of and exponential function for the strain dependency, and it reduces to Mohr-Coulomb as $j \rightarrow \infty$.

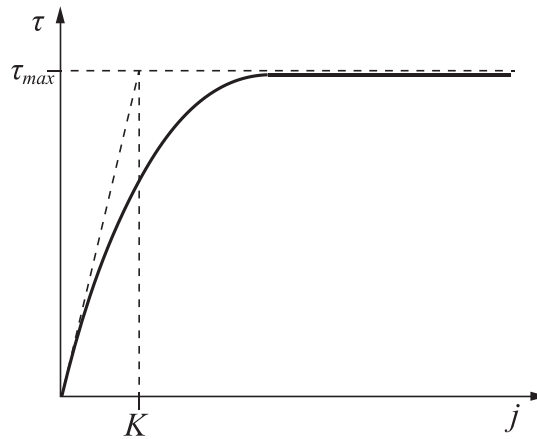


Figure 2.8 Typical shear stress-strain behaviour of cohesive soils.

2.2.2 Rolling Resistance

Typical off-road driving (i.e. excluding rallying/racing) would be done at low or moderate speeds where aerodynamic forces can be neglected, and so the forces resisting the vehicle motion would mostly be due to the tyre rolling resistance and road gradient. When a vehicle is traveling at constant velocity the sum of all applied forces is zero, which means that the sum of forces resisting its motion is equal to the tractive effort produced by the wheel/soil interface.

Rolling resistance is a force developing between the wheel and the terrain, and when considering the case of a rigid wheel rolling on a deformable terrain, the rolling resistance is caused mainly by two factors; (a) vertical soil compression, i.e. soil compaction, and (b) horizontal soil displacement, i.e. bulldozing. If instead of a rigid wheel a flexible tyre is considered, then there is an additional resistance caused by energy losses due to tyre deflection. The general concepts of pneumatic tyres and their interaction to rigid and deformable terrains will be explored in the following subsections.

2.2.2.1 Soil compaction

The pressure-sinkage relationship is of great importance for the prediction of rolling resistance due to soil compaction. As the stress-strain relationship of soils is very difficult to express, especially in the case of irregular loading areas such as wheels, some assumptions are normally made. One of these assumptions describes a linear relationship between pressure P and sinkage z , as if the soil was a purely elastic medium [1]:

$$P = kz \tag{2.21}$$

where k is the coefficient of proportionality. Equation 2.21 has been criticised of being overly simplistic and it never got widely accepted ([1],[10]), and an alternative form was proposed by Letoshnev, Goriatchkin, and others (as cited in [1], p.185):

$$P = kz^n \tag{2.22}$$

where n is an exponent depending on the soil type. Experiments indicate the range of values for n being between 0 and 2 [25].

The problem with the above Equation 2.22 is that k is a function of the size of loading area and cannot be considered as a “modulus” of deformation [1]. For that reason, Bekker [2] used some civil engineering soil mechanics and modified Equation 2.22 which transformed into the following form:

$$P = \left(\frac{k_c}{B} + k_\phi \right) z^n \tag{2.23}$$

where B is the smallest dimension of the loading area (or wheel width), and k_c and k_ϕ are coefficients linked to cohesion and friction respectively.

For the determination of the pressure-sinkage relationship and for obtaining the necessary parameters, an experimental apparatus called *Bevometer* is normally used. The Bevometer technique was first introduced by Bekker and it consists of two tests, the plate penetration test and the shear test. From the plate penetration test the pressure-sinkage relationship can be obtained. The plate is usually circular or oval-shaped, and its dimensions are equivalent to the contact area of the terrain-wheel or terrain-grouser interface to avoid uncertainties in extrapolating soil properties data. The applied vertical load is also similar to that exerted by a vehicle.

In the shear test a plate with size comparable to the terrain-running gear contact area is used. Various normal pressures are applied and the plate either translates perpendicular to the testing surface or rotates around the axis normal to the surface. The applied force (or torque) is recorded as well as the translational or rotational displacement and the shear stress-shear displacement can then be derived.

According to Bekker [2], experiments have shown that k_c and k_ϕ are independent of the size and form of the loading area and therefore being true moduli. However, that proved to be true only in homogeneous terrains. But since most types of soils are not homogeneous, the dimensions of the measuring penetration plate should be as close to those of a real contact area of vehicles as possible [2]. Table 2.4 summarises soil characteristics as well as Bekker parameters for various types of soils. As can be seen the Bekker parameters can vary significantly for the same soil type, even when the wet density and/or moisture content remains relatively constant. This confirms that Bekker parameters are not soil moduli.

Terrain Type	n	k_c (kN/m ⁽ⁿ⁺¹⁾)	k_ϕ (kN/m ⁽ⁿ⁺²⁾)	Wet density (kg/m ³)	Moisture content %
LETE sand	0.705	6.94	505.80	~1600	
	0.611	1.16	475.00		
	0.804	3.93	599.50		
	0.728		1348.00		
	0.578	9.08	2166.00		
	0.781	47.80	6076.00		
	0.806	155.90	4526.00		
Upland sandy loam	1.100	74.60	2080.00	1557	51.6
	0.970	65.50	1418.00	1542	49.2
	1.000	5.70	2293.00	1570	49.1
	0.740	26.80	1522.00	1519	44.3
	1.740	259.00	1643.00	1696	50.0
	0.850	3.30	2529.00	1471	28.6
	0.720	59.10	1856.00	1592	34.3
	0.770	58.40	2761.00	1559	35.1
	1.090	24.90	3573.00	1716	31.2
	0.700	70.60	1426.00	1470	27.3
	0.750	55.70	2464.00	1526	32.6
Rubicon sandy loam	0.660	6.90	752.00	1561	43.3
	0.650	10.50	880.00	1588	44.2
North Gower clayey loam	0.730	41.60	2471.00	1681	45.8
	0.850	6.80	1134.00	1597	52.0
Grenville loam	1.010	0.06	5880.00	1326	24.1
	1.020	66.00	4486.00	1339	18.2

Table 2.4 Parameters and Bekker coefficients for various types of terrains, data from [10].

For the part of tractive effort spent to overcome the rolling resistance due to soil compaction, Bekker proposed the following equation for rigid, towed wheels [2]:

$$R_c = \frac{1}{(n+1)(k_c + Bk_\phi)^{\frac{1}{n}}} \left(\frac{W}{L} \right)^{\frac{n+1}{n}} \quad 2.24$$

where W is the vertical load and L the length of the loading area.

The above Equation 2.24 was modified to take into account the curvature of the wheel which leads to the expression:

$$R_c = \frac{1}{(3-n)^{\frac{2n+2}{2n+1}} (n+1)(k_c + Bk_\phi)^{\frac{1}{2n+1}}} \left(\frac{3W}{\sqrt{D}} \right)^{\frac{2n+2}{2n+1}} \quad 2.25$$

where D is the wheel diameter.

Equation 2.25 was obtained considering the equilibrium of forces acting on the wheel in the vertical and horizontal direction. Also, an assumption was made which states that the normal (radial) pressure along the wheel-terrain interface is equal to the pressure exerted on a plate at the same sinkage level in a pressure-sinkage test [10]. In a similar fashion, the wheel sinkage is expressed as:

$$z = \left[\frac{3W}{(3-n)(k_c + Bk_\phi)\sqrt{D}} \right]^{\frac{2}{2n+1}} \quad 2.26$$

Bekker's approach does not consider the effect of slip or skid (i.e. negative slip) in rolling resistance and gives good prediction only for shallow sinkage. Gee-Clough [25] tried to extend Bekker's theory to include the effect of skid and deep sinkage of rigid-towed wheels. In his analysis he assumed that the wheel loses contact (exit angle) at $\theta_2 = 0^\circ$ (see Figure 2.9), which means he did not take into account the effect of rut recovery, z_r . He also assumed that the entry angle θ_1 does not change due to horizontal soil displacement, known as bulldozing. However, the work conducted by Wong [26] has shown that for rigid-towed wheels the horizontal soil displacement can be considerable and, therefore, the entry angle can increase significantly. Based on these assumptions, Gee-Clough proposed the following formulation for rolling resistance due to compaction:

$$R_c = \frac{Bk}{(n+1)(1+i)^{\frac{n}{2n+1}}} \left[\frac{3W}{Bk(3-n)\sqrt{2R}} \right]^{\frac{2n+2}{2n+1}} \quad 2.27$$

The above Equation 2.27 which takes onto account skid and deep sinkage, can be written in the alternative form of Equation 2.28, which differs from Bekker's Equation 2.25 only in the addition of the longitudinal slip ratio dependency factor.

$$R_c = \frac{1}{(3-n)^{\frac{2n+2}{2n+1}} (n+1) (k_c + Bk_\phi)^{\frac{1}{2n+1}}} \left(\frac{3W}{\sqrt{D}} \right)^{\frac{2n+2}{2n+1}} \frac{1}{(1+i)^{\frac{n}{2n+1}}} \quad 2.28$$

Two formulations for predicting the wheel skid have been obtained from Gee-Clough's analysis, one assuming purely cohesive soil (i.e. $\phi = 0$) and one assuming purely frictional soil (i.e. $c = 0$). For purely cohesive soil the wheel skid can be predicted using Equation 2.29 below:

$$i = \frac{\frac{\pi^2}{8} - \frac{\theta_e}{2} (\pi - \theta_e)}{\left(\frac{\pi}{2} - \theta_e \right) \cos \theta_e - (1 - \sin \theta_e)} - 1 \quad 2.29$$

The best fitting curve of Equation 2.29 is:

$$i = -0.03863 + 1.19976e^{-0.03832\theta_e} \quad 2.30$$

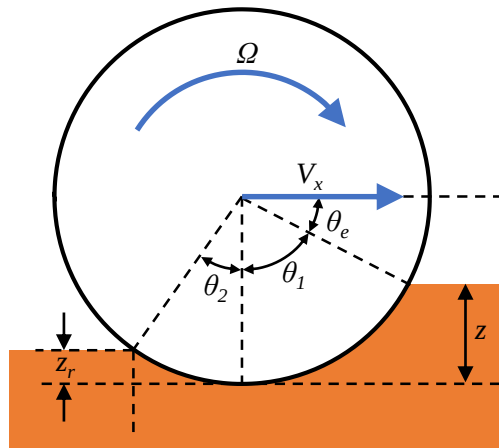


Figure 2.9 Rigid wheel rolling on soil.

For purely frictional soils, the skid can be predicted using Equation 2.31 and its best fitting curve of Equation 2.32. The skid level against entry angle for cohesive and frictional soils are illustrated in Figure 2.10 and Figure 2.11 respectively. As can be seen, for frictional soil (sand) the predicted level of skid seems reasonable but for clay is overestimated for small entry angles where the skid reaches almost 120%.

$$i = \frac{\sin \theta_e \left[\frac{\pi^2}{8} + 1 - \frac{\theta_e}{2} (\pi - \theta_e) \right] - 1}{\sin \theta_e \left[\left(\frac{\pi}{2} - \theta_e \right) \cos \theta_e + \frac{3}{2} \sin \theta_e - 1 \right] - \frac{1}{2}} - 1 \quad 2.31$$

$$i = -0.03908 + 1.0494e^{-0.03671\theta_e} \quad 2.32$$

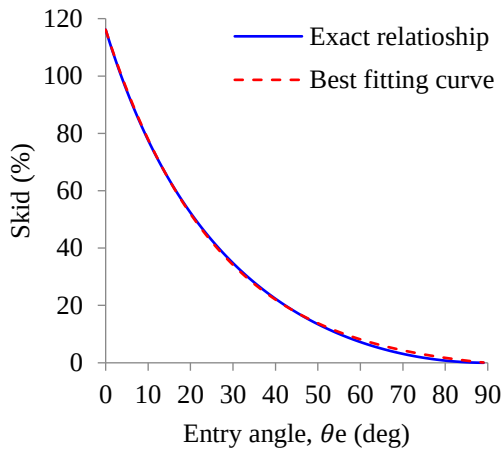


Figure 2.10 Skid vs. Entry angle for purely cohesive soils.

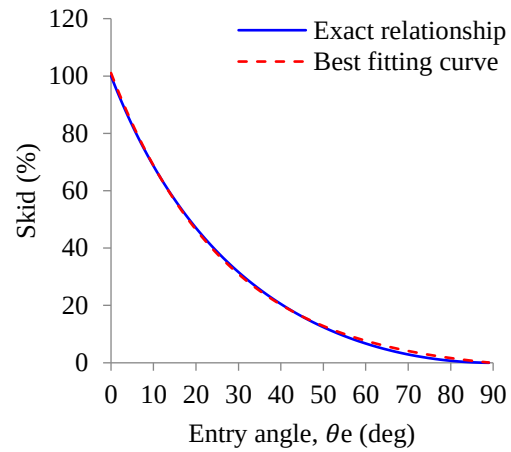


Figure 2.11 Skid vs. Entry angle for purely frictional soils.

Gee-Clough compared the predicted rolling resistance and sinkage derived from his formulations against Bekker's equations and experimental data from Wills, et al. [27]. The results showed good correlation with the experimental data on sand for both rolling resistance and sinkage but not so good correlation on clay mainly because of the overestimation of wheel skid.

Although Bekker's equation can predict with a reasonable accuracy the wheel sinkage for an applied pressure, it is however of an empirical nature. In addition, the constants of cohesion k_c and friction k_ϕ have variable dimensions which are depending on the value of soil exponent n . In order to

provide an alternative formulation for the pressure-sinkage relationship, Reece [28] started his analysis from the bearing capacity theory of classical soil mechanics (Equation 2.33). Bearing capacity is the maximum average pressure that a soil can support without producing shear failure. For a strip (or continuous) foundation footing, where the length of the loading area is much larger than its width, Terzaghi [29] proposed the following equation for shallow footing ($z / B \leq 1$) after an analysis of the general shear failure of soils (Figure 2.12):

$$P = cN_c + \gamma_s z N_q + 0.5\gamma B N_\gamma \quad 2.33$$

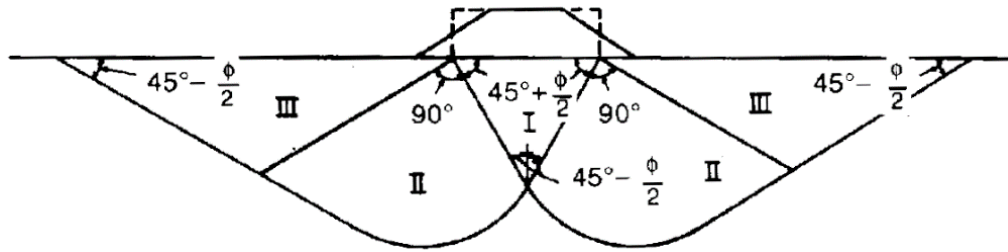


Figure 2.12 General Shear Failure under a strip footing [30]. Republished with permission of John Wiley & Sons Books, from Smith's Elements of Soil Mechanics, I. Smith, Oxford, 2006; permission conveyed through Copyright Clearance Centre, Inc.

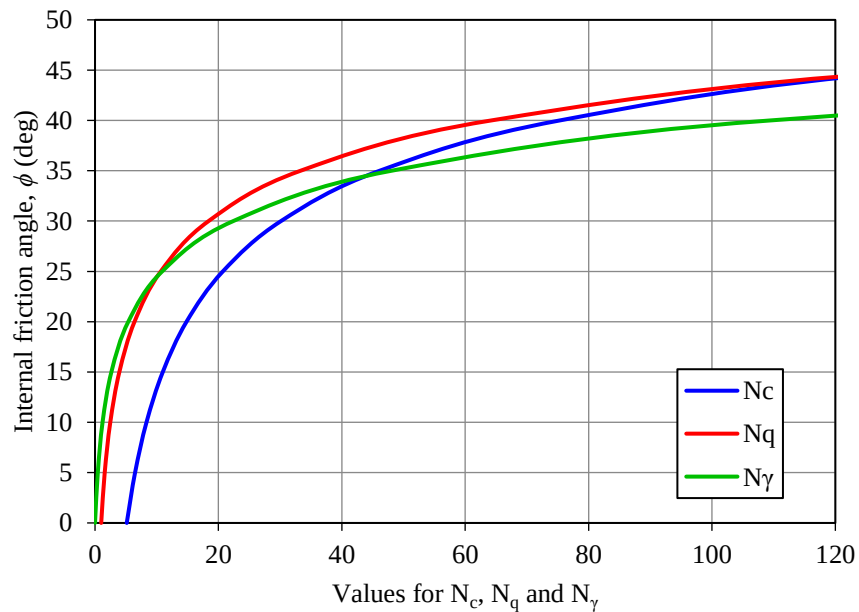


Figure 2.13 Terzaghi's bearing capacity coefficients [30].

In Equation 2.33, c is the soil cohesion (Pa), γ_s the unit weight (N/m³) and B the footing width (m). The empirical dimensionless coefficients N_c , N_q and N_γ depend on the soil internal friction angle and they allow the effect of soil cohesion, surcharge effect due to soil weight and the size of the footing respectively. In Figure 2.13 the bearing capacity coefficients are plotted against the internal friction angle and their numerical values are summarised in Table 2.5. Terzaghi also provided solutions for other types of footings which are presented in the following Equations 2.34 - 2.36.

ϕ	N_c	N_q	N_γ
0	5.14	1.00	0.00
5	6.49	1.57	0.45
10	8.34	2.47	1.22
15	10.98	3.94	2.65
20	14.83	6.40	5.39
25	20.72	10.66	10.88
30	30.14	18.40	22.40
35	46.12	33.30	48.03
40	75.31	64.20	109.41
45	133.87	134.87	271.75
50	266.88	319.06	762.86

Table 2.5 Bearing capacity factors in common use, data from [30].

Circular footing of diameter B :

$$P = 1.3cN_c + \gamma_s z N_q + 0.3\gamma B N_\gamma \quad 2.34$$

Square footing:

$$P = 1.3cN_c + \gamma_s z N_q + 0.4\gamma B N_\gamma \quad 2.35$$

Rectangular footing of width (or smaller dimension) B and length (or larger dimension) L :

$$P = cN_c \left(1 + 0.3 \frac{B}{L} \right) + \gamma_s z N_q + 0.5\gamma_s B N_\gamma \left(1 - 0.2 \frac{B}{L} \right) \quad 2.36$$

In his analysis, Reece assumed that there is no separate soil surcharge and, therefore, N_q in Equation 2.33 is zero because the soil weight is included in N_γ . Also, the remaining N factors are almost linear functions of z/B [28]. So, Equation 2.33 can be written in the following form:

$$P = ck'_c \left(\frac{z}{B} \right)^n + \gamma_s B k'_\phi \left(\frac{z}{B} \right)^n \quad 2.37$$

where k'_c and k'_ϕ are dimensionless parameters.

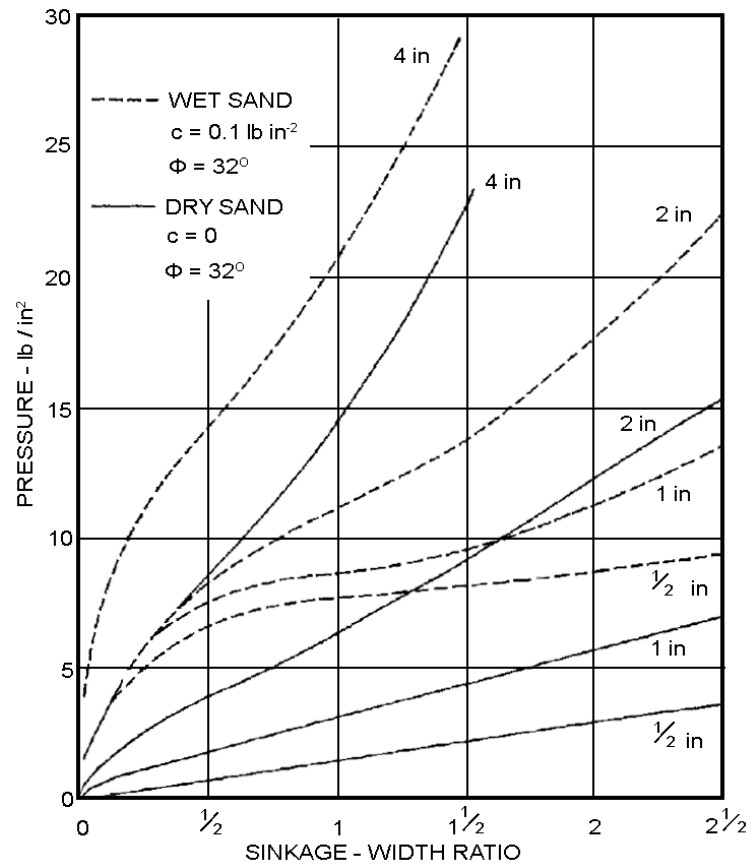


Figure 2.14 Experimental pressure-sinkage curves obtained using rectangular plates of 0.5, 1, 2 and 4 inches wide on dry and wet sand [28]. Reprinted from Proceedings of the Institution of Mechanical Engineers, A. R. Reece, Principles of Soil-Vehicle Mechanics, Copyright (1965), with permission from Elsevier.

To validate his model, Reece [28] performed a series of experiments with various types of soils (dry sand, wet sand and clay) using rectangular plates of aspect ratios up to 4.5. The above Equation 2.37 suggests that for frictionless saturated clays, the coefficient k'_ϕ should be negligible. That is

confirmed from the plot of Figure 2.15, where the clay pressure curves are plotted against z/B and the three curves for various plate widths almost overlay. In contrary, when dry cohesionless sand is considered, coefficient k'_c should be negligible and the plate width should contribute to the soil response. Again, this is confirmed by the measured data presented in Figure 2.14. Adding water to the sand increases the soil cohesion term but as expected the curves have similar dependency on z/B as the dry sand. Despite the claim that Reece's model has stronger theoretical background, it is very similar to Bekker's and it effectively only differs in the effect of plate width. Also, it requires the same experimental procedure to derive the dimensionless parameters k'_c , k'_ϕ and n which by no means are soil invariants.

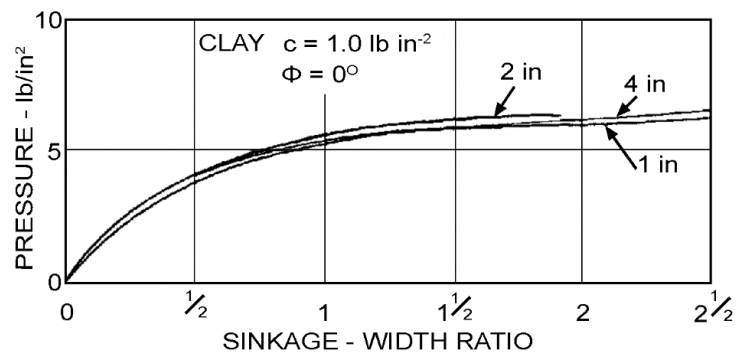


Figure 2.15 Experimental pressure-sinkage curves obtained using rectangular plates of 1, 2 and 4 inches wide on clay [28]. Reprinted from Proceedings of the Institution of Mechanical Engineers, A. R. Reece, Principles of Soil-Vehicle Mechanics, Copyright (1965), with permission from Elsevier.

Both models discussed in the preceding paragraphs rely on experimental data acquisition using large test equipment (Bevamer) to derive the soil parameters required. Furthermore, they are essentially empirical, and the results are difficult to extrapolate since they depend on the size of the penetration test plates. Hence, the experimental tests must be performed using plate size and vertical loads equivalent to those exerted by the vehicle of interest.

Lyasko's approach is to use only invariant soil parameters such as cohesion and internal friction angle to describe the pressure-sinkage relationship [31]. These invariant parameters can be derived from in-situ tests using small, hand-held test equipment; typically, a dynamic cone penetrometer. Lyasko proposed a complex equation for pressure-sinkage which includes soil invariant parameters as well as various empirical parameters and supporting equations. The main pressure-sinkage equation is:

$$P = \frac{1}{\frac{D_1}{B_i} + \frac{D_2}{E \cdot z} \cdot \omega \cdot B \cdot \xi} \quad 2.38$$

Where:

$$D_1 = \frac{2}{\pi} \arctan \left(\frac{\pi(H-z)}{2B} \right) \quad 2.39$$

$$D_1 = \arctan \left(\frac{H-z}{A_0 B} \right) \quad 2.40$$

The list below summarises the complete set of input parameters required by the model, and they are grouped by type.

Soil invariants:

C	Cohesion	(Pa)
ϕ	Internal friction angle	(deg)
γ	Bulk density	(kg/m ³)
E	Deformation modulus	(Pa)
H	Hardpan depth	(m)

Empirical soil parameters:

B_i	Bearing capacity	(Pa)
$\left. \begin{matrix} N_c \\ N_p \\ N_\gamma \end{matrix} \right\}$	Bearing capacity factors	(-)
$\left. \begin{matrix} n_c \\ n_p \\ n_\gamma \end{matrix} \right\}$	Bearing capacity plate shape factors	(-)
ξ	Contact pressure concentration coefficient	(-)
A_0	Plate width and hardpan depth coefficient	(-)
J	Coefficient	(-)
ω	Coefficient	(-)

From the above listed parameters, cohesion, internal friction angle, bulk density and hardpan depth can be measured directly. The remaining parameters are derived using empirical equations as presented below.

Soil bearing capacity:

$$B_l = cN_c n_c + \gamma_s z N_p n_p + \gamma_s B N_\gamma n_\gamma \quad 2.41$$

Where:

$$N_\gamma = \exp(6.65 \tan \phi - 1.75) \quad 2.42$$

$$N_p = \exp(4.76 \tan \phi + 0.15) \quad 2.43$$

$$N_c = \exp(3.32 \tan \phi - 1.5) \quad 2.44$$

$$n_\gamma = 1 + 0.25 \frac{B}{L} \quad 2.45$$

$$n_p = 1 + 1.5 \frac{B}{L} \quad 2.46$$

$$n_c = 1 + 0.3 \frac{B}{L} \quad 2.47$$

$$\gamma_s = \gamma g \quad 2.48$$

Contact pressure concentration coefficient:

$$\xi = 1 + 1.86 \left(\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{5} \right) \quad 2.49$$

$$x = 2.5 [1 - \exp(-3.74B)] \quad 2.50$$

Other coefficients:

$$J = \frac{0.03 + \frac{L}{B}}{0.6 + 0.43 \frac{L}{B}} \quad 2.51$$

$$A_0 = 0.64 \left(1 + \frac{B}{H} \right) \quad 2.52$$

$$\omega = A_0 J \quad 2.53$$

Soil deformation modulus E_0 for steady-state loading with negligible velocity:

$$E_0 = 100 [2000 + c \exp(0.03\phi - 2.5)] \exp(0.011\phi^{1.5}) \quad 2.54$$

Dynamic soil deformation modulus E depends on velocity:

$$E = E_0 (1 + Y_0 \sqrt{V}) \quad 2.55$$

where V is the vehicle ground speed in m/s and Y_0 is the dimensionless coefficient depending on soil cohesion.

$$Y_0 = 0.114 + c3.37 \times 10^{-6} \quad 2.56$$

Lyasko also describes the methodology required to derive the soil invariant parameters (i.e. cohesion, internal friction angle and deformation modulus) using the dynamic cone penetrometer as well as some additional supporting equations. However, these will not be discussed in the current text. An analysis and evaluation of the pressure-sinkage formulations discussed in the section is conducted in Chapter 3.

2.2.2.2 Bulldozing

The part of rolling resistance caused by the horizontal soil displacement is called *bulldozing*. When a wheel is rolling on soil, part of the soil compresses vertically (compaction) and part of it accumulates in front of the wheel causing additional resistance. Narrow wheels suffer less from bulldozing resistance as most of the soil particles are forced aside [32]. However, for wider wheels a considerable amount of soil would build-up in front of them causing high resistance ([2], [32]) which would further increase with the increase of sinkage [2]. Additionally, based on the research conducted by Wong & Reece [33-34], it can be concluded that the towed wheels generate higher bulldozing resistance than driven wheels due to differences in the flow patterns between the two.

The maximum normal and shear stresses beneath a rigid wheel vary depending on the wheel slip. In addition, there are two failure zones which move the material forward and backward respectively, and they meet at the location of maximum normal stress. In Figure 2.16, a rigid-towed wheel is running on a soil surface at 23% skid (i.e. negative slip), with point A at the location of the maximum normal stress and where the two failure zones meet. Point A is quite close to the wheel bottom-dead-centre which leads to a significant volume of soil been moved towards the front of the wheel. In contrast, the

maximum normal stress of a rigid driven wheel presented in Figure 2.17 has moved further forward and away from the bottom-dead-centre. This results in reduction of material flow of the front failure zone and consequently less soil build-up in front of the wheel.

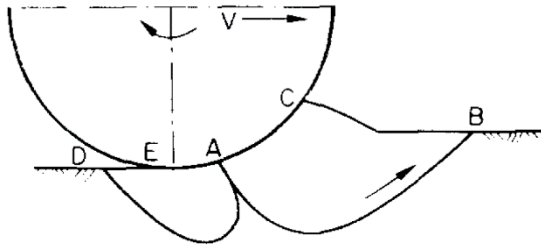


Figure 2.16 Soil material flow of towed wheel. Skid 23%, [34]. †

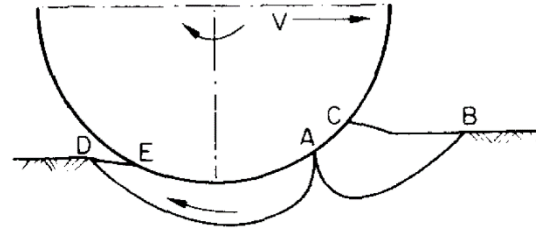


Figure 2.17 Soil material flow of driven wheel. Slip 63%, [34]. †

† Reprinted from Journal of Terramechanics, vol. 4, no. 2, J. Y. Wong and A. R. Reece, Prediction of rigid wheel performance based on the analysis of soil-wheel stresses. Part II: Performance of towed rigid wheels., pp. 7-25, Copyright (1967), with permission from Elsevier.

Gee-Clough [35] tried to investigate how exactly the wheel width affects the rolling resistance. He conducted a series of experiments with different combinations of wheel diameters and widths in sand. His findings confirmed that the rolling resistance and wheel skid for towed wheels, increase rapidly with increasing width at a particular sinkage. With the industry trending towards heavier vehicles with wider tyres, the bulldozing effect would have greater impact on their off-road performance. The phenomenon of bulldozing is complicated and difficult to express mathematically. Due to this reason, it is often neglected from the rolling resistance computations.

Apart from experimental and analytical investigations, numerical models are also used for the prediction of rolling resistance, and more specifically the bulldozing effect. Typically, two numerical approaches are used to model the soil behaviour: (a) the finite element method (FEM) where the material is considered as continuous ([18], [19], [36], [37]), (b) the discrete element method (DEM) where the soil is represented as an assembly of individual particles, usually spheres, with contact stiffness and damping in the normal direction and friction in the tangential direction [38].

There are certain disadvantages by using the FEM/DEM in Terramechanics such as: (a) the soil is assumed to be homogeneous, (b) very high computational effort as the number of elements used for the simulation can be several thousands, (c) the particle size used in simulations (DEM) are at least one

order of magnitude larger than the actual soil particles in order to reduce the computational effort, (d) a lack of widely accepted method for deriving the parameters required for soil modelling.

2.3 Tyre Modelling on Rigid Surface

In the previous subsections 2.1 and 2.2 of the literature review the fundamentals of Soil Mechanics and Terramechanics are discussed, with emphasis on the concepts applicable to road vehicles. The current section concerns the general principles of tyres and discusses existing approaches to tyre modelling.

The tyre is one of the most important parameters of a road vehicle. The forces due to friction generated between the road surface and the contact patch provide the necessary traction to enable the vehicle to accelerate, brake and change direction. In Figure 2.18, the cross-section of a passenger car tyre is presented and as it can be seen, the tyre has a complex structure consisting of various materials such as nylon, steel and rubber. Each one of these materials has specific structural properties and behaves differently under loading, temperature and weather conditions. Creating a mathematical model that takes all these factors into account and that can accurately represent the tyre overall behaviour under accelerating, braking and cornering conditions is a challenging task. However, several tyre models have been created over the years. These models can be grouped into two main categories; (a) empirical models, (b) physical models. In the subsequent sections the most widely accepted models from each group will be discussed.

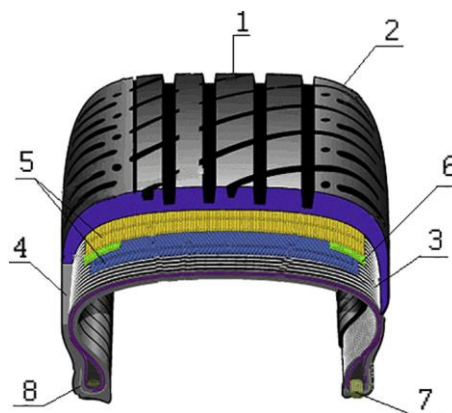


Figure 2.18 Cross-section of a passenger car tyre (1 & 2 – tread, 3 – carcass, 4 – sidewall, 5 & 6 – belt, 7 – bead rope, 8 – bead), [39]. Reprinted from Engineering Failure Analysis, no. 104, G. Fedorko, et al., Failure analysis of irreversible changes in the construction of car tyres, pp. 399-408, Copyright (2019), with permission from Elsevier.

Figure 2.19 illustrates the characteristic curves of side force (left graph), brake forces (middle graph) and the combined condition (right graph). The tyre lateral forces arise at the contact patch during steering and cornering as a result of slip angle [41], which is the angle between the tyre plane and the actual tyre traveling direction. Under braking conditions, negative longitudinal slip (skid) occurs which produces longitudinal (braking) forces at the contact patch. The resulting situation of the side and brake force is called the combined slip condition. The shape and peak value of the characteristic curve and surfaces depend on various parameters such as tyre pressure, wheel camber angle, friction coefficient, tyre temperature and others. Forces due to pure lateral, pure longitudinal and combined slip conditions are usually at the start of any discussion concerning the tyre behaviour under steady-state conditions.

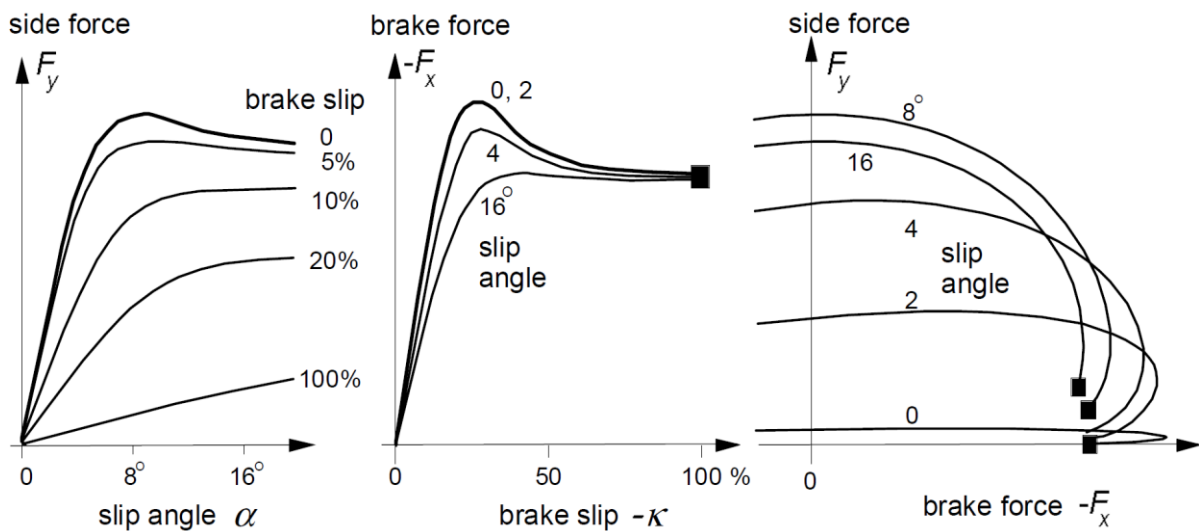


Figure 2.19 Tyre characteristic curves on rigid surface [42]. This article was published in *Tire and Vehicle Dynamics*, 3rd ed., H. Pacejka, Copyright Elsevier (2012).

2.3.1 Empirical models

Empirical tyre models attempt to capture the tyre behaviour through empirical equations derived from large libraries of experimental data. Two examples of empirical models, the Magic Formula and Similarity method, are discussed in the following paragraphs.

2.3.1.1 Magic Formula

Empirical tyre models are trying to replicate the tyre behaviour in terms of forces and moments based on processing and manipulation of physical test data. The magic formula is an empirical tyre model that has been developed in the mid-eighties as the result of a collaboration between TU-Delft and Volvo. Various versions of the model have been created ([43], [44]) in which the combined slip was initially described from a physical viewpoint, but later on, a purely empirical approach was adopted. Over the years the model has been evolved and expanded to include representation of pneumatic trail for the calculation of self-aligning moment, pressure dependency, relaxation length for transient behaviour and others.

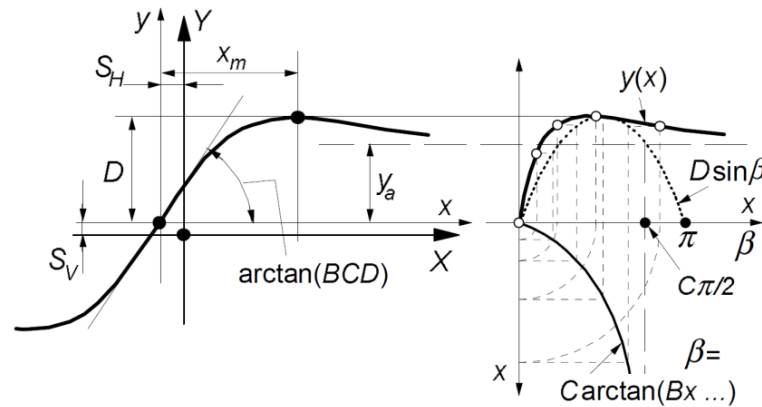


Figure 2.20 The curve produced by the Magic Formula equation [42]. This article was published in *Tire and Vehicle Dynamics*, 3rd ed., H. Pacejka, Copyright Elsevier (2012).

All factors used in the formula are derived from the characteristic curves at pure slip. When the factors are identified from the raw data and applied to the formula, a curve that fits the measured data for pure slip conditions is obtained. Usually, the resultant curves fit quite well to measured data which enables successful and accurate complete vehicle simulations to be conducted. Since the Magic Formula is essentially a curve fitting tool, it doesn't provide an insight into the tyre's complex behaviour. Nevertheless, because the tyre characteristic curves can be well captured, Magic Formula is still widely used for overall vehicle performance evaluation and development by the automotive industry.

2.3.1.2 Similarity Method

The observation that the tyre characteristic curves at pure slip, under conditions different from the reference condition, have very similar shape to each-other has led to the development of the similarity method [42]. Reference condition would typically be a tyre with a nominal vertical load, at zero camber angle, with either zero longitudinal or lateral slip at a specific road surface. In order to derive the characteristic curves for various conditions, for example at various vertical loads or camber angles, first the characteristic curves at reference conditions have to be derived by processing the raw data. The Magic Formula would normally be employed as a fitting tool. Then, the curves are normalised and shifted so they become unitless and pass through the origin. After the normalised curves are derived, by working backwards the curves at different conditions can be obtained. In Figure 2.21, a diagram of the normalised lateral force vs. normalised slip angle at various vertical loads is presented. As it can be seen, the various curves have been reduced to effectively one curve after normalising the lateral force and slip angle.

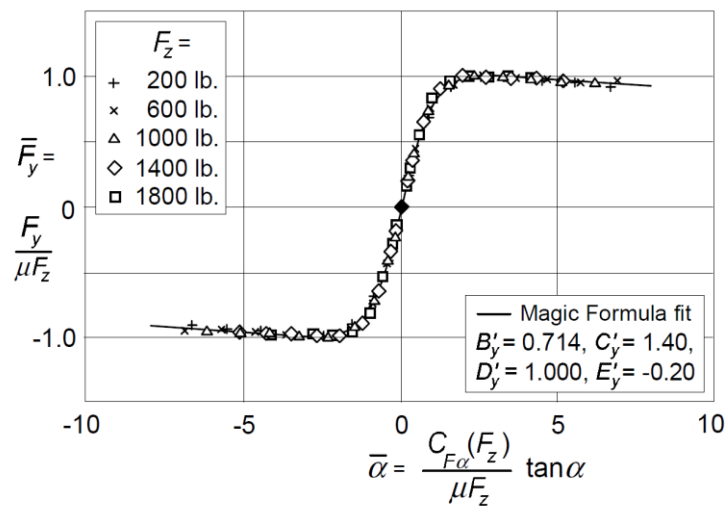


Figure 2.21 Normalised lateral force vs slip angle for several vertical loads [42]. This article was published in *Tire and Vehicle Dynamics*, 3rd ed., H. Pacejka, Copyright Elsevier (2012).

2.3.2 Physical models

In contrary to the empirical models, physical tyre models consider detailed tyre characteristic such as materials properties, tyre construction and shape, non-linear friction and other. The physical tyre models aim at allowing the engineers to conduct parameter sensitivity studies using tyre design parameters instead of fitting coefficients which have no physical meaning. Some of the most common advanced tyre models commercially available are described below.

2.3.2.1 Tyre Brush model

A physical tyre model which is capable of describing quite well the pure lateral and longitudinal as well as the combined slip at steady-state conditions is the Brush model [42]. The model consists of a row of elastic bristles normal to the tyre circumference which represent the overall tyre compliance including the carcass, belt and tread elements.

When the tyre is considered at free rolling (i.e. perpendicular to the road surface with zero camber angle and slip), as the wheel rotates the bristles come into contact with the road and move from the leading edge to the trailing edge without generating lateral or longitudinal forces. Due to friction, the tips of the bristles stick to the road surface and they are always vertical since there is no longitudinal or side slip. During vehicle cornering, lateral slip arises, which causes the bristles to deform in the lateral direction as shown by the top-right graph of Figure 2.22. Similarly, when longitudinal slip occurs, the bristles would deform in the longitudinal direction (middle-right, Figure 2.22). Finally, the bottom-right diagram of Figure 2.22 illustrates the case of a combined cornering and braking condition.

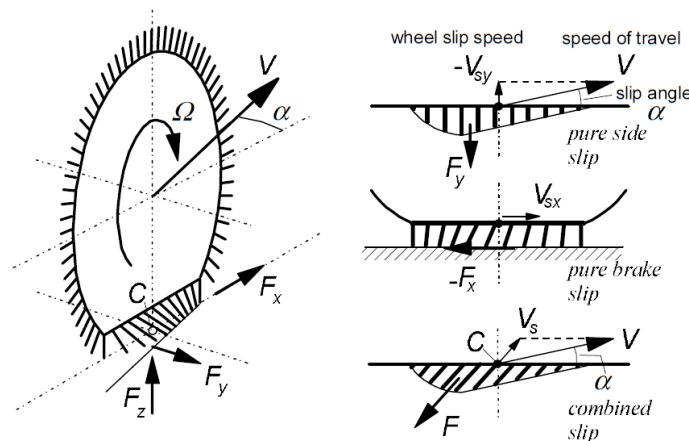


Figure 2.22 Tyre Brush model under pure and combined slip conditions [42]. This article was published in *Tire and Vehicle Dynamics*, 3rd ed., H. Pacejka, Copyright Elsevier (2012).

Due to the high accuracy of lateral and longitudinal forces predicted by the brush model, it is often used as part of other advanced tyre models ([45], [46]).

2.3.2.2 Ftire

Ftire is a physical tire model, very popular within the automotive industry, with high level of fidelity but below of that of FEA based models. It is mainly used for ride comfort and durability studies and it consists of two main components; the structural model and the tread model. In the structural model, the tyre belt is represented by a set of flexible bodies called belt segments. Each segment has 4 rigid body degrees of freedom with respect to the rim; longitudinal, lateral and vertical displacement of the centre point (Figure 2.23.a) and angular displacement of the centre point about the circumferential axis (Figure 2.23.b). In addition, the segments can elastically bend as shown in Figure 2.23.c. The rim itself can be assumed as perfectly rigid, elastic or viscoelastic. The number of belt segments varies between 50 and 500 and they are connected with each other and with the rim via several inflation-pressure dependent, non-linear force elements.

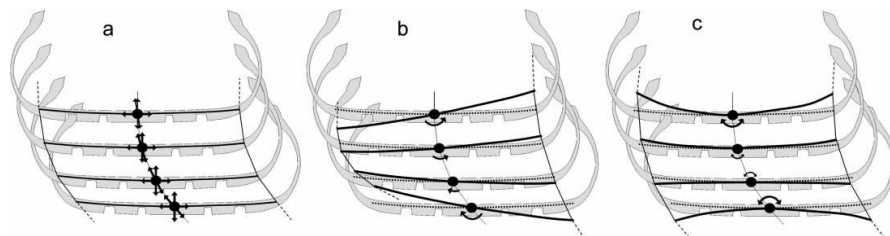


Figure 2.23 Belt segments degrees of freedom. (a) translation along x/y/z, (b) rotation about circumferential axis, (c) lateral bending [47]. Reprinted from Vehicle System Dynamics, Vol. 43 Supplement, M. Gipser, FTire: a physically based application-oriented tyre model for use with detailed MBS and finite-element suspension models, pp. 76-91, Copyright (2005), with permission from Taylor & Francis.

In Figure 2.24 a set of force elements acting between the rim and a belt segment in the radial direction is presented. Similarly, force elements are placed for the circumferential and lateral directions but are not presented in the figure. Parallel spring and damper elements are used for the tyre vertical stiffness, Maxwell elements (spring-damper in series) are used for the dynamic tyre stiffening at high rolling velocities and dry friction elements to represent rubber hysteresis. An optional feature is the thermal model which varies the force element properties appropriately during the simulation.

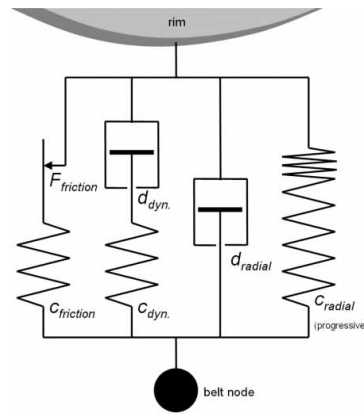


Figure 2.24 Structure model: radial force elements [47]. Reprinted from Vehicle System Dynamics, Vol. 43 Supplement, M. Gipser, FTire: a physically based application-oriented tyre model for use with detailed MBS and finite-element suspension models, pp. 76-91, Copyright (2005), with permission from Taylor & Francis.

For the tread model, several contact and friction elements between the belt segments are used. The number of elements can be determined by the user and it depends on the road surface resolution. The normal force of each contact element is a function of radial deflection and deflection velocity, describing tread rubber compression stiffness and damping [47]. The coefficient of friction, which is a function of position, ground pressure, sliding velocity and tread temperature, as well as the normal force are used to calculate the friction forces in the tangential direction.

2.4 Soil-Wheel Interaction

2.4.1 Normal and Shear Stresses under rigid wheels

As already discussed in previous paragraphs, the shear strength on the wheel-soil interface is a function of normal stress. Therefore, the determination of normal stress is essential for deriving the maximum shear strength and consequently the maximum tractive effort of the soil-wheel interface.

In the work conducted by Onafeko and Reece [48] the distribution of stresses (normal and shear) beneath a rigid wheel was investigated. They concluded that in sand the maximum stresses of slipping or skidding wheels are forward to the bottom dead centre and would move further forward with increasing slip or skid. That is in contrast to Bekker's theory which states that the maximum stresses

would occur at the bottom dead centre and would be equivalent to the one generated by a penetration plate. Wong [26] who investigated the soil flow patterns beneath rigid wheels confirms Onafeko & Reece's conclusion. An example of the findings from Onafeko and Reece is presented in Figure 2.25.

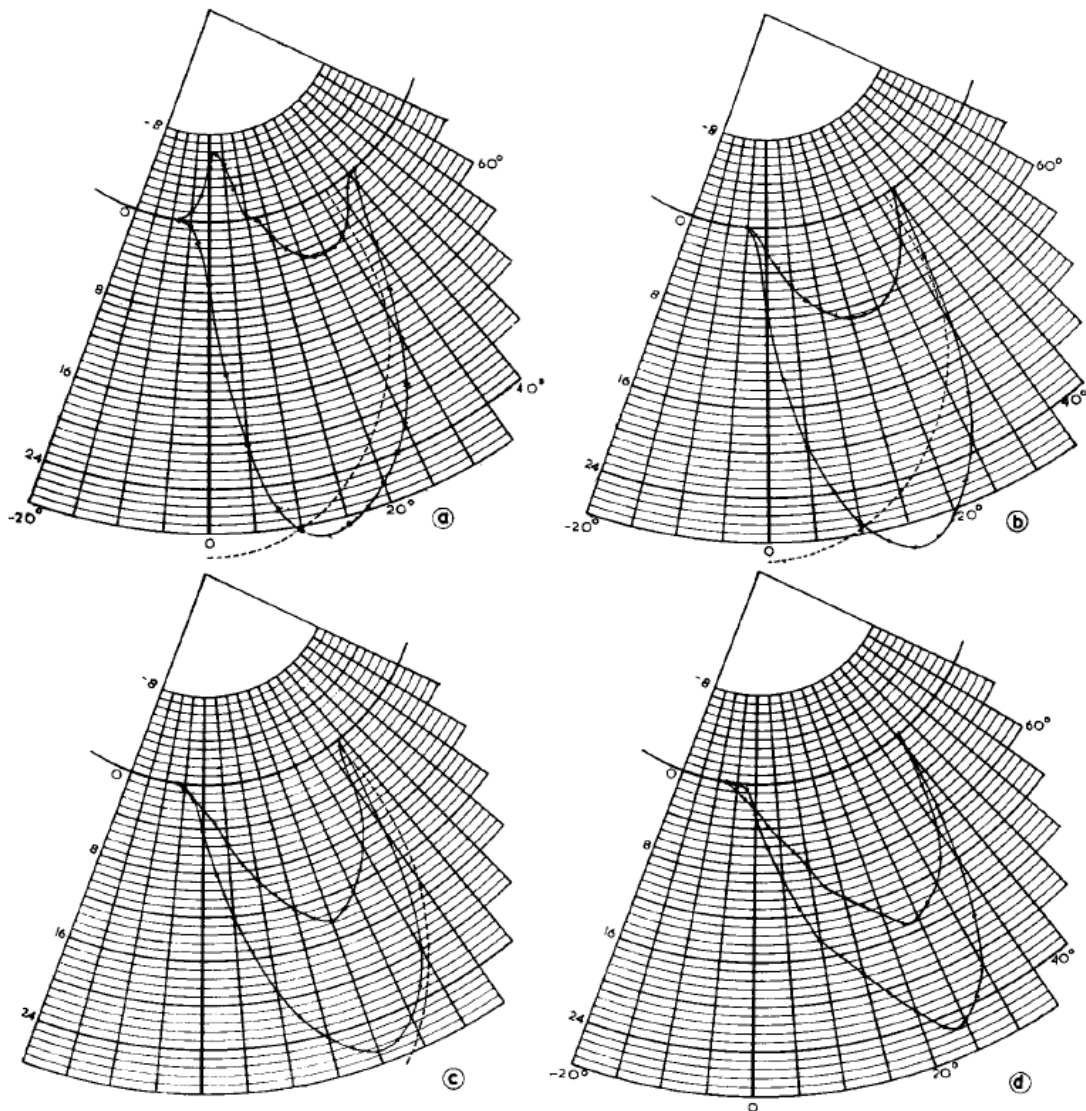


Figure 2.25 Normal and Shear Stress distribution on wheel of diameter = 49" and width = 6" in compact sand. a) 44.5% skid, b) 3.1% slip, c) 22.1% slip, d) 41.5% slip (Radial stresses in lb/m²) [48]. Reprinted from Journal of Terramechanics, vol. 4, no. 1, O. Onafeko and A. R. Reece, Soil stresses and deformations beneath rigid wheels, pp. 59-80, Copyright (1967), with permission from Elsevier.

Onafeko and Reece's work has been further validated [49], with various modified and extended versions of the Wong and Reece's original models for towed [34] and driven [33] rigid wheels currently existing in the literature ([16], [49-52]).

2.4.2 Elastic wheels on deformable terrain

All formulations of wheel-soil interaction discussed up to this point consider the wheel as a rigid body. However, most off-road wheeled vehicles are equipped with tyres. Sometimes, if the terrain deflection is orders of magnitude softer than the tyre's deflection, then the rigid wheel approach can still be valid.

A model developed by Harnisch, et al. [53] uses Bekker's formulations in combination with an elastic tyre. The Bekker's equations are used for the determination of the pressure-sinkage relationship and the Mohr-Coulomb theory for the calculation of the maximum shear strength of soil. The relationship between the shear stress and shear displacement is given by the empirical approach of Janosi and Hanamoto [24] discussed in subsection 2.2.1.

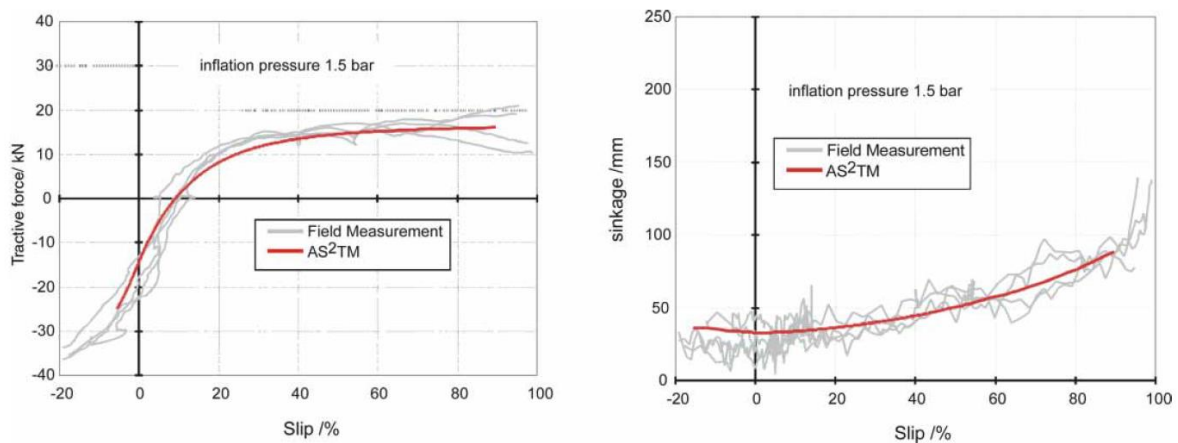


Figure 2.26 Tractive force and sinkage vs. slip. Correlation between measured data and AS²TM simulation results [53]. Reprinted from Vehicle System Dynamics, vol. 43 Supplement, C. Harnisch et al, A new tyre–soil interaction model for vehicle simulation on deformable ground, pp. 384-394, Copyright (2005), with permission from Taylor & Francis.

For the elastic tyre, the model considers a larger wheel diameter to capture the increase of contact area between the wheel and soil due to tyre deflection. Various parameters are used for the determination of the wheel deflection among them the inflation pressure. The effects of slip sinkage and multi-pass [10] as well as the influence of tyre-soil friction and tread are also included in the model. The exact mathematical formulation for the tyre is not provided, however good correlation is achieved for the tractive effort and sinkage as function of wheel slip (Figure 2.26).

The increase of computational capacity of modern computers along with the numerical efficiency improvement of FEA software like ABAQUS has allowed the development of a number of FEA-based tyre models for soil [54-56] and snow ([14], [16]). The majority of these models use the Drucker-Prager/Cap model (a modification of the traditional Drucker-Prager formulation) for the physical representation of the terrain compaction. Furthermore, usually the tread pattern is neglected from the FEA models. However, in the work done by Li and Schindler [37] the tread pattern of tractor tyres is considered and very good correlation with experimental data is presented.

2.5 Summary

In this chapter a review of the literature was conducted, which underlined the lack of a reliable tyre model for off-road vehicle applications. There are various available methods and modelling approaches for replicating different phenomena, however they all tend to be studied in isolation and decoupled. The rolling resistance and tractive effort have been identified as critical components in the overall tyre and vehicle performance. Furthermore, a review of existing tyre models for rigid terrains has been summarised, in order to gain an understanding of the modelling techniques in use.

Based on the above remarks, a study into the rolling resistance and tractive effort, individually and in combination, will be analysed in the following chapters supported by in-house and literature derived experimental data. In the light of the above, objective 1 has been fulfilled.

Chapter 3: Pressure-Sinkage Initial Study

As highlighted by the literature review of the previous chapter, a core soil behaviour related to both rolling resistance and tractive effort is the pressure-sinkage relationship, with three main models being discussed. In this chapter, the pressure-sinkage models proposed by Bekker, Reece and Lyasko are evaluated using test data found in the literature, with the aim of establishing whether these models can form appropriate candidates for a soil-wheel interaction model.

3.1 Primary Considerations

Aim of the current research is the development of an off-road tyre model for application in passenger cars; typically sport utility vehicles (SUVs). Therefore, some specific vehicle design aspects, such as corner weight and tyre dimensions, need to be considered. Corner weight is the main parameter affecting the wheel sinkage. Starting from the vehicle weight, the region of interest should be between the lightest corner weight of a small SUV and the heaviest corner weight of a large off-road vehicle.

A small SUV weighs approximately 1500 kg. Assuming a front/rear axle distribution of 60/40, then the lightest corner would weigh 300 kg $\approx$ 3000 N. Following similar assumptions for large SUVs with curb weight of around 3000 kg, the heaviest corner weight would be 900 kg $\approx$ 9000 N. Therefore, when investigating soil sinkage the main focus would be on the predictions between 3 kN and 9 kN. The soil response outside of this region will still be considered for assessing overall validity of the methodology applied as it is sometimes the case that vehicles travelling over rough terrain can have one or more wheels leave the ground for a short duration, e.g. ruts across opposing wheels.

The tyre dimension is the other vehicle characteristic which has a significant effect on the performance on deformable terrain. After performing a market research and summarising specifications of more than 60 vehicle variants of various manufacturers, the author has found that the tyre widths of off-road passenger cars currently vary between 215mm and 295mm with various aspect ratios and rim diameters. It is known from literature ([1], [10]) that for wheeled vehicles the tyre/soil contact area is oval shaped with the smallest dimension being the wheel width and the largest the length of the contact

area. Hence, it is valid to assume that the minimum contact area dimension would be equal to the wheel width. In a rolling tyre, the maximum contact length of the front region would be the wheel radius, and if the soil elastic recovery at the rear part of the contact area is also considered, an increase of the contact length by a further 20% is a fair assumption. Therefore, the maximum contact length can be estimated as $L_{max} = 1.2 \times R_{tyre}$ (see Figure 3.1). The following Table 3.1 provides a comprehensive summary of the wheel dimensions for off-road vehicles available in the UK market in 2016. The min and max diameters are derived from the tyre aspect ratio and rim diameters, and are used for defining the width/length and length/width ratios. The values of these ratios are later considered for the dimensions of flat penetration plates in pressure-sinkage, as described and analysed in Chapter 4.

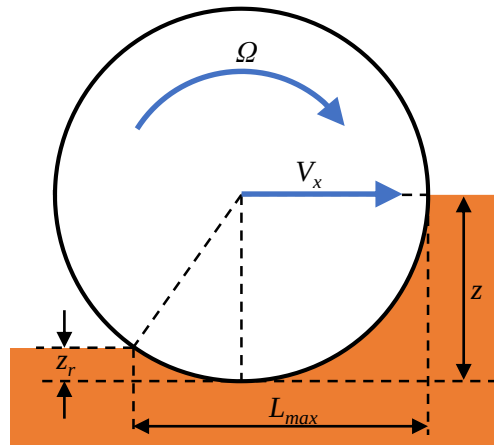


Figure 3.1 Maximum length of the soil/wheel contact area.

B (mm)	D (mm)			L (mm)			B/L (-)			L/B (-)		
	Min	Max	AVG	Min	Max	AVG	L Min	L Max	AVG	L Min	L Max	AVG
215	668	707	688	215	413	314	1.00	0.52	0.76	1.00	1.92	1.46
225	660	744	702	225	421	323	1.00	0.53	0.77	1.00	1.87	1.44
235	690	806	748	235	449	342	1.00	0.52	0.76	1.00	1.91	1.46
245	703	799	751	245	451	348	1.00	0.54	0.77	1.00	1.84	1.42
255	687	814	750	255	450	353	1.00	0.57	0.78	1.00	1.77	1.38
265	744	776	760	265	456	361	1.00	0.58	0.79	1.00	1.72	1.36
275	728	758	743	275	446	360	1.00	0.62	0.81	1.00	1.62	1.31
285	771	799	785	285	471	378	1.00	0.61	0.80	1.00	1.65	1.33
295	736	744	740	295	444	369	1.00	0.67	0.83	1.00	1.51	1.25

Table 3.1 Summary of tyre data for a number of SUVs available on the UK market in 2016.

3.2 Soil Data Processing

In Section 2.2.2.1 an overview of three pressure-sinkage models was presented, all claiming they can predict with reasonable accuracy the wheel response on a soil under loading. In order to evaluate the accuracy of each model, they will be tested against measured data found in the literature [57]. All three models (Bekker [2], Reece [28] and Lyasko [31]) require a particular set of parameters which are derived from experimental data. For clarity, an overview of the required input parameters for the three models is presented in Table 3.2.

Description	Symbol	Unit	Bekker [2]	Reece [28]	Lyasko [31]
Soil exponent	n	-	○	○	×
Cohesion parameter	k_c	N/m^{n+1}	○	×	×
Friction parameter	k_ϕ	N/m^{n+2}	○	×	×
Cohesion parameter	k'_c	-	×	○	×
Friction parameter	k'_ϕ	-	×	○	×
Cohesion	c	N/m^2	×	○	○
Internal friction angle	ϕ	deg	×	○	○
Moisture content	w	%	×	×	○
Unit weight	γ_s	N/m^3	×	○	○
Hardpan depth	H	m	×	×	○
○: Required ×: Not required					

Table 3.2 Required input parameters for Bekker, Reece and Lyasko models.

Bekker and Reece models as described by Equations 2.23 and 2.37 respectively require a set of empirical parameters that can be obtained from the Bevameter penetration test. At least two plates of different size must be tested to allow data interpolation and extrapolation. After the pressure versus sinkage data for two plates are collected, they are processed as described below. Considering the Bekker formulation and taking the logarithm on both sides, Equation 2.23 can be written as:

$$\log P = \log \left(\frac{k_c}{B} + k_\phi \right) + n \log z \quad 3.1$$

As it can be seen, Equation 3.1 is a linear equation of the form $y = a + bx$ at log scale. Obtaining the linear equation coefficients requires plotting the measured data (pressure and sinkage) in a logarithmic scale (see Figure 3.2). The linear regression coefficients of the log-log plot are then easily found. In the past, the best fitting line would be drawn by hand and would rely on the engineer's skills to ensure consistency and fitting quality. Today, the best fitting line can be found in a consistent and accurate way via computer algorithms. The most common algorithms calculate the regression line by minimising the sum of the squared distance between each data point and the regression line.

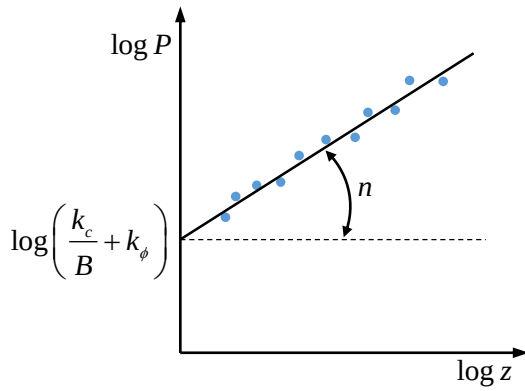


Figure 3.2 Schematic of the soil data processing for Bekker.

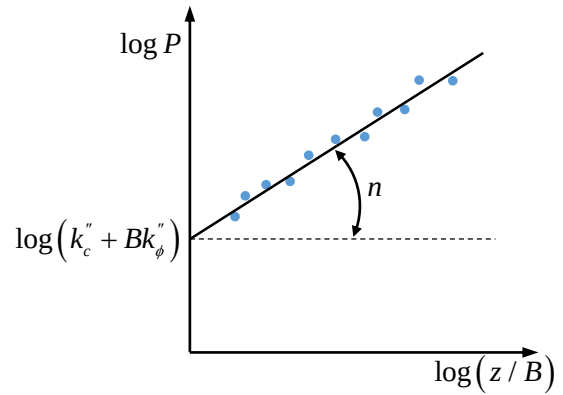


Figure 3.3 Schematic of the soil data processing for Reece.

The same methodology described above for the Bekker model can be followed to derive the pressure-sinkage parameters for the Reece formulation. According to Wong [10], Equation 2.37 may be written as:

$$P = (k_c'' + Bk_\phi'') \left(\frac{z}{B} \right)^n \quad 3.2$$

where: $k_c'' = ck_c'$ and $k_\phi'' = \gamma_s k_\phi'$

Taking the logarithm of Equation 3.2 leads to:

$$\log P = \log(k_c'' + Bk_\phi'') + n \log(z/B) \quad 3.3$$

In order to perform a fair comparison between the Bekker, Reece and Lyasko models, a common set of soil data is required. An abundance of Bekker parameters is available in the literature, and reasonable number of parameters are available for Reece as well. Unfortunately, it was not possible to find a single source that – under the same controlled conditions, using the same test equipment and measured at the same time – would provide all soil parameters required for the three models as summarised in Table 3.2.

In the work conducted by Reece [57], the pressure-sinkage raw data for three soils (wet sand, dry sand and saturated clay) as well as the soil cohesion, internal friction angle, bulk density and moisture content are provided. The tests were conducted in a range of rectangular and circular plates as summarised in Table 3.3. Since for obtaining the parameters for the Bekker and Reece models, two plates of different dimensions are required, the author selected the largest and smallest plate for each case as denoted with (*) in Table 3.3. Then, the linear fitting in logarithmic scale was performed, as described above and illustrated in Figures 3.4-3.10. There are many different and more advance alternative fitting methods, for example through optimisation or machine learning, however it was deemed useful to assess the models using the methods proposed in the literature.

Plate ID	B (mm)	L (mm)	D (mm)	Soil		
				Saturated Clay	Wet Sand	Dry Sand
1	12.7	457.2	-	○ (*)	○ (*)	○ (*)
2	25.4	457.2	-	×	○	○
3	50.8	457.2	-	○	○	○
4	76.2	457.2	-	○	○	○
5	101.6	457.2	-	○ (*)	○ (*)	○ (*)
6	25.4	114.3	-	○ (*)	○ (*)	○ (*)
7	50.8	228.6	-	○	×	×
8	76.2	342.9	-	○	×	×
9	101.6	457.2	-	○ (*)	○ (*)	○ (*)
10	-	-	25.4	○ (*)	○ (*)	○ (*)
11	-	-	50.8	○	○	○
12	-	-	76.2	×	○	○
13	-	-	101.6	○	○	○
14	-	-	127	×	○	○
15	-	-	152.4	○ (*)	○ (*)	○ (*)
○: Data available				(*): Used for parameter identification		
×: No Data						

Table 3.3 Summary of plates used by Reece [57].

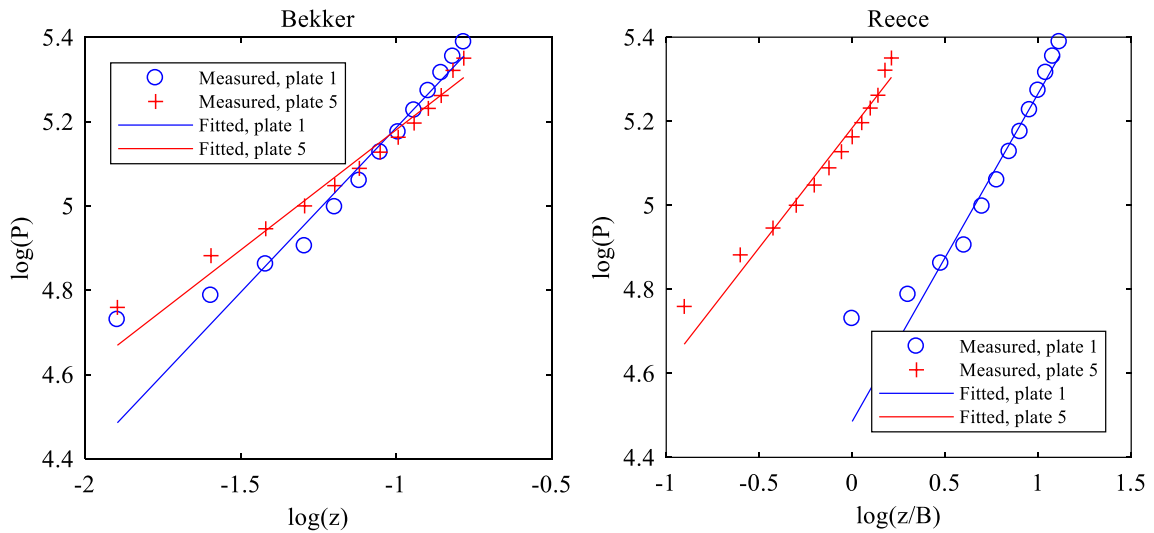


Figure 3.4 Rectangular plates set 1 in wet sand, fitted Bekker (left) and Reece (right).

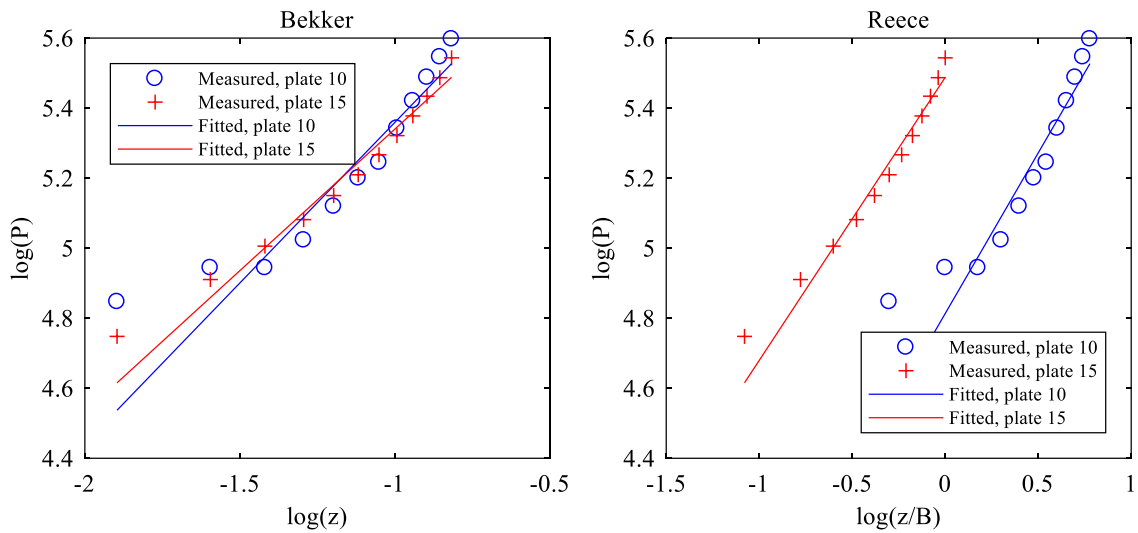


Figure 3.5 Circular plates in wet sand, fitted Bekker (left) and Reece (right).

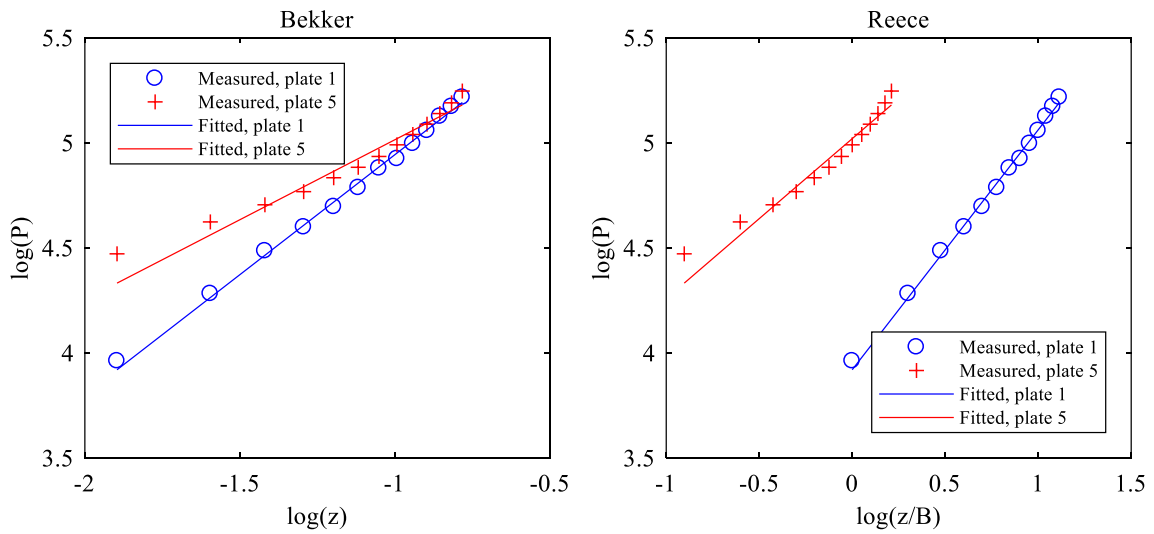


Figure 3.6 Rectangular plates set 1 in dry sand, fitted Bekker (left) and Reece (right).

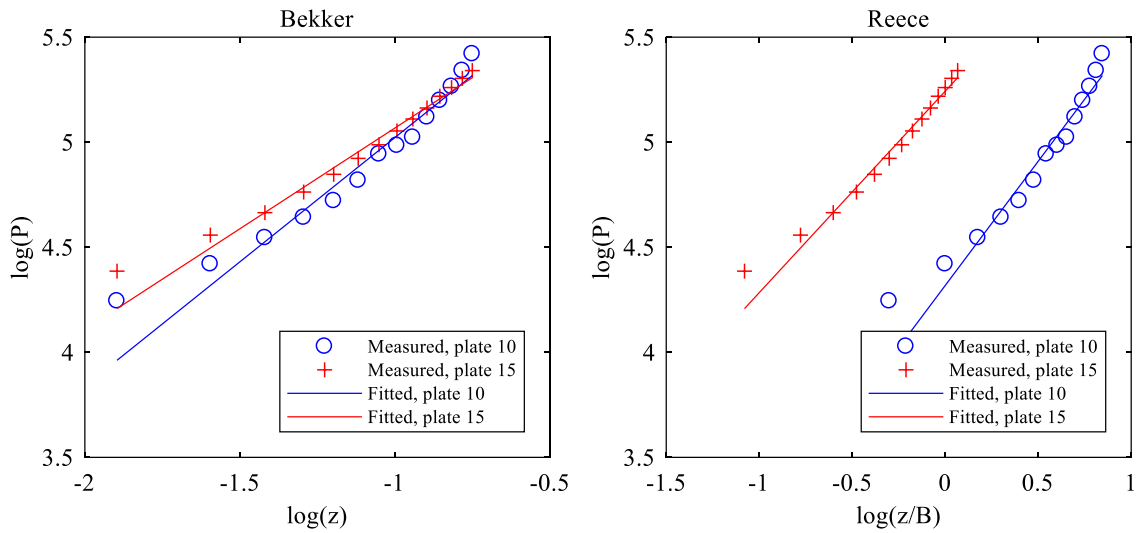


Figure 3.7 Circular plates in dry sand, fitted Bekker (left) and Reece (right).

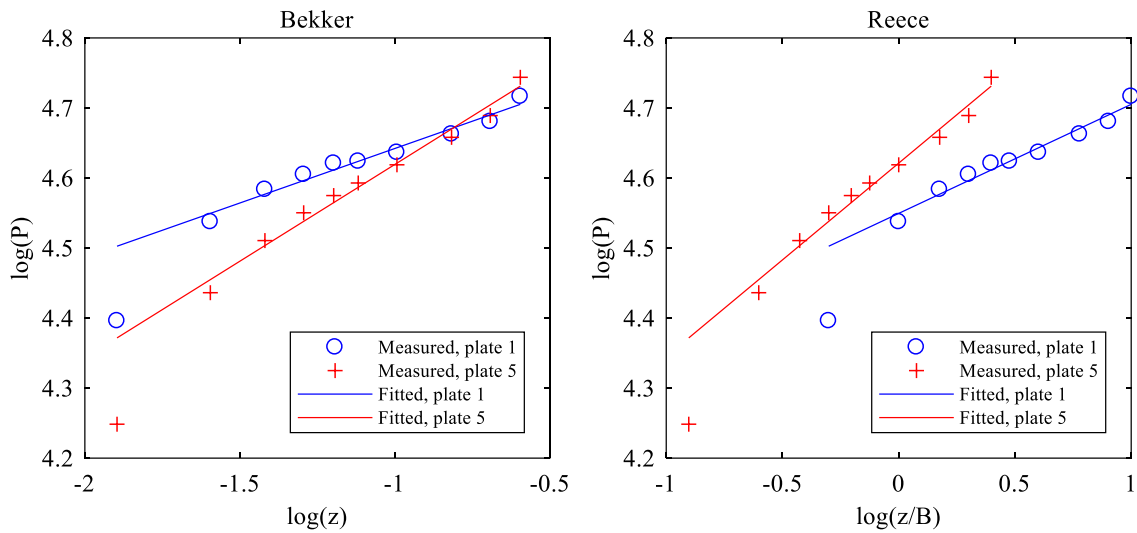


Figure 3.8 Rectangular plates set 1 in saturated clay, fitted Bekker (left) and Reece (right).

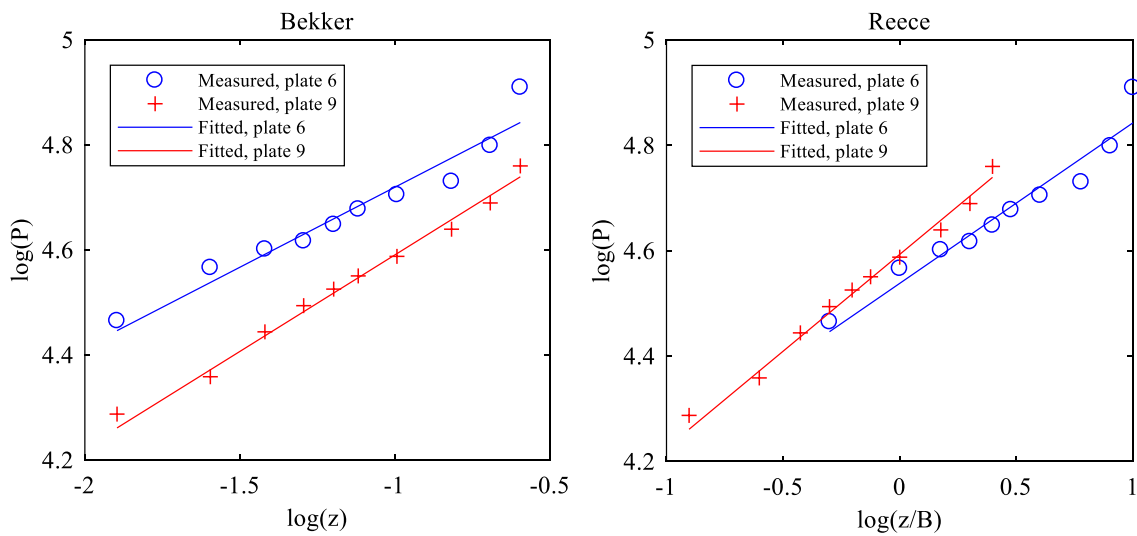


Figure 3.9 Rectangular plates set 2 in saturated clay, fitted Bekker (left) and Reece (right).

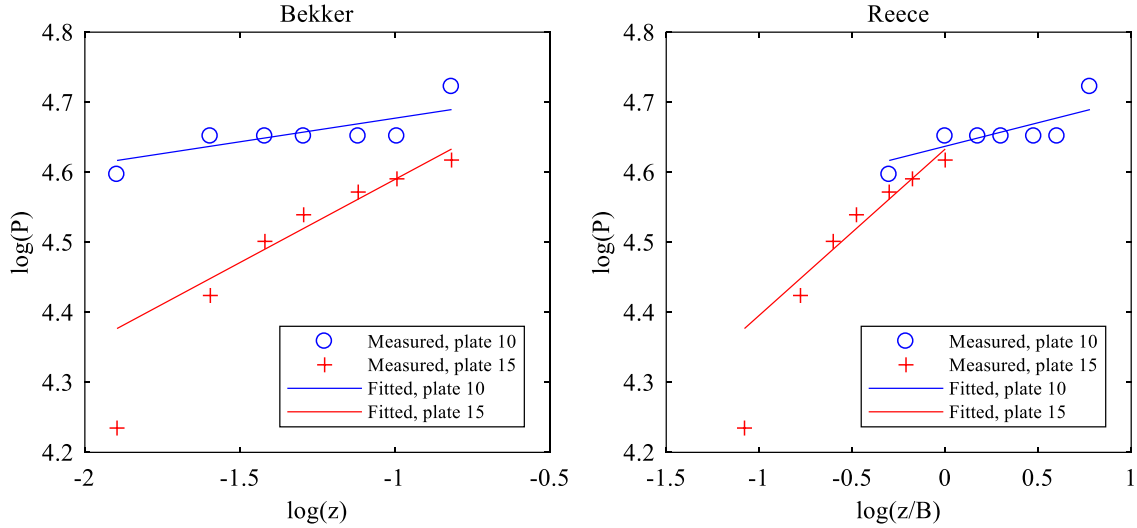


Figure 3.10 Circular plates in saturated clay, fitted Bekker (left) and Reece (right).

An alternative way for obtaining the terrain parameters for the Bekker and Reece models has been developed by Wong [10] which uses a weighted least squares method. The proposed method uses a weighing factor to minimise the effect of low pressures which are the least significant.

Similarly to the traditional method, the logarithms are taken and after a detailed analysis, the following equations are derived for the terrain parameters for the Bekker model.

$$\ln P = \ln k_{eq} + n \ln z \quad 3.4$$

where $k_{eq} = k_c/B + k_\phi$

The best fitting curves of the soil exponent n and the logarithm of coefficient of proportionality $\ln k_{eq}$ for a particular set of measured data can be obtained by using the following equations.

$$n = \frac{\sum P^2 \sum P^2 \ln P \ln z - \sum P^2 \ln P \sum P^2 \ln z}{\sum P^2 \sum P^2 (\ln z)^2 - (\sum P^2 \ln z)^2} \quad 3.5$$

$$\ln k_{eq} = \frac{\sum P^2 \ln P - n \sum P^2 \ln z}{\sum P^2} \quad 3.6$$

Normally the penetration test is performed using two different sized plates. The fitted values of n for these two sets of data are rarely identical [10]. So, in order to derive a unique value for n , the average $\bar{n}$ is taken.

$$\bar{n} = \frac{(n)_{B=B_1} + (n)_{B=B_2}}{2} \quad 3.7$$

where B_1 and B_2 are the plate widths.

Using the average exponent, Eq. 3.6 becomes:

$$\ln k_{eq} = \frac{\sum P^2 \ln P - \bar{n} \sum P^2 \ln z}{\sum P^2} \quad 3.8$$

After the coefficient of proportionality is found, the parameters for cohesion k_c and friction k_ϕ can be calculated using the following equations.

$$k_c = \frac{(k_{eq})_{B=B_1} - (k_{eq})_{B=B_2}}{B_2 - B_1} B_1 B_2 \quad 3.9$$

$$k_\phi = (k_{eq})_{B=B_1} - \frac{(k_{eq})_{B=B_1} - (k_{eq})_{B=B_2}}{B_2 - B_1} B_2 \quad 3.10$$

Following a similar approach, parameters for the Reece model can be calculated using the following equations.

$$\ln P = \ln k_{eq}'' + n \ln(z/B) \quad 3.11$$

where $k_{eq}'' = k_c'' + Bk_\phi''$

Soil exponent for each plate:

$$n = \frac{\sum P^2 \sum P^2 \ln P \ln(z/B) - \sum P^2 \ln P \sum P^2 \ln(z/B)}{\sum P^2 \sum P^2 [\ln(z/B)]^2 - [\sum P^2 \ln(z/B)]^2} \quad 3.12$$

Average exponent:

$$\bar{n} = \frac{(n)_{B=B_1} + (n)_{B=B_2}}{2} \quad 3.13$$

Logarithm of the coefficient of proportionality:

$$\ln k_{eq}'' = \frac{\sum P^2 \ln P - \bar{n} \sum P^2 \ln(z/B)}{\sum P^2} \quad 3.14$$

Coefficient of cohesion:

$$k_c'' = (k_{eq}'')_{B=B_1} - \frac{(k_{eq}'')_{B=B_1} - (k_{eq}'')_{B=B_2}}{B_1 - B_2} B_1 \quad 3.15$$

Coefficient of friction:

$$k_\phi'' = \frac{(k_{eq}'')_{B=B_1} - (k_{eq}'')_{B=B_2}}{B_1 - B_2} \quad 3.16$$

In order to establish which of the two methods for processing raw soil data is superior, a Matlab code for the Wong's method was created. The soil parameters derived using both methods are summarised in Table 3.4, and as it can be seen there isn't much difference between the Standard and Wong's method for the soil exponent. However, for cohesion and friction parameters, there are significant differences for both Bekker and Reece models.

The two sets of parameters for the Bekker model – using Standard and Wong's parameters – were compared against the measured data for all available soils and plate sizes. From this comparison it was difficult to properly assess which of the two methods performs best, as in some plots the curves using Standard parameters are closer to measurements and in others the Wong's parameters are closer, and the sum of least squares provided mixed results too. So, Wong's weighted squares method did not show a considerable improvement in fitting accuracy to test data, the Standard least squares method will be used in the rest of the chapter.

	n (-)			k_c (N/m ⁿ⁺¹)			k_ϕ (N/m ⁿ⁺²)			k''_c			k''_ϕ		
	Least squares	Wong	Diff (%)	Least squares	Wong	Diff (%)	Least squares	Wong	Diff (%)	Least squares	Wong	Diff (%)	Least squares	Wong	Diff (%)
Dry Sand - Rect Plates	0.95	1.04	8.7	9013	10855	17.0	508646	633211	19.7	1862	-7403	125.2	1084547	1126906	3.8
Dry Sand - Circ Plates	1.07	1.25	14.4	16420	53499	69.3	954438	928090	-2.8	-3447	-19592	82.4	1213239	1318603	8.0
Wet Sand - Rect Plates	0.68	0.69	1.4	5246	4883	-7.4	511362	566575	9.7	24091	13637	-76.7	1375099	1394172	1.4
Wet Sand - Circ Plates	0.86	0.93	7.5	14699	18101	18.8	1310838	1597007	17.9	31758	11113	-185.8	1909404	2057663	7.2
Sat Clay - Rect Plates 1	0.22	0.24	8.3	-537	-519	-3.5	83937	86812	3.3	33305	31454	-5.9	83583	99612	16.1
Sat Clay - Rect Plates 2	0.34	0.35	2.9	505	830	39.2	85750	82510	-3.9	32317	31809	-1.6	61089	73928	17.4
Sat Clay - Circ Plates	0.15	0.18	16.7	-355	-448	20.8	69493	76263	8.9	41590	42467	2.1	-3085	8465	136.4

Table 3.4 Soil parameters derived by the standard least squares method of linear fit and Wong's weighted least squares method.

3.3 Review of Results

After obtaining the soil parameters for Bekker and Reece using the least square method described in the previous subsection, the pressure sinkage curves could be drawn. The following figures illustrate the comparison between Bekker, Reece and Lyasko models against numerical data from [57]. As data for the unit weight of saturated clay and hardpan depth in all cases were not provided, assumptions were made based on the dimensions of the in-house sand box and data from [31], in order to include the Lyasko model in the comparison. In the following pages, a series of figures are presented illustrating, comparing test data from Reece [57] and the corresponding fitted curves using the Bekker, Reece and Lyasko models. The figures are followed by an analysis of the results and discussion.

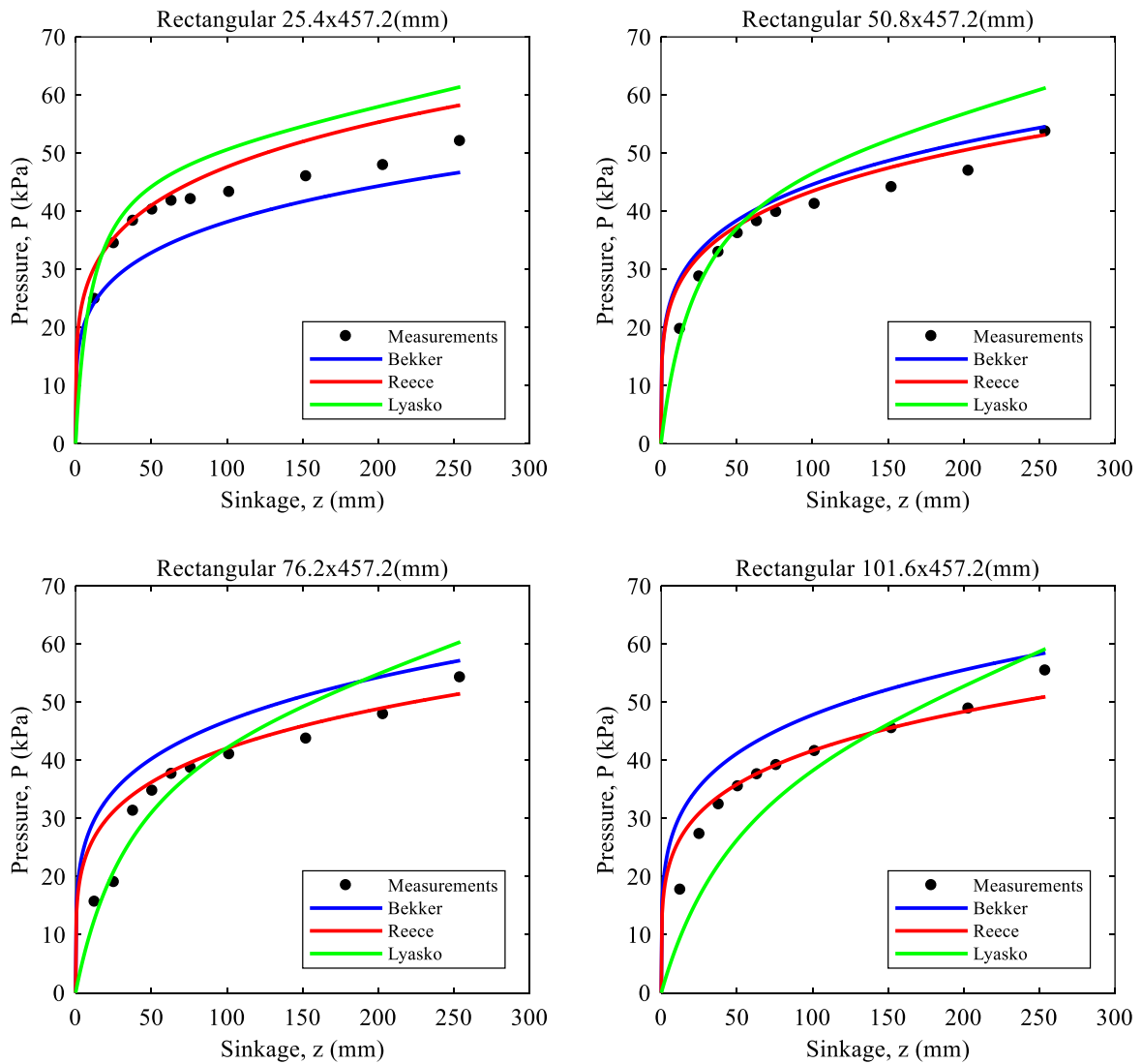


Figure 3.11 Pressure vs sinkage in saturated clay, rectangular plates of 457.2 mm (18in) length and various widths. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

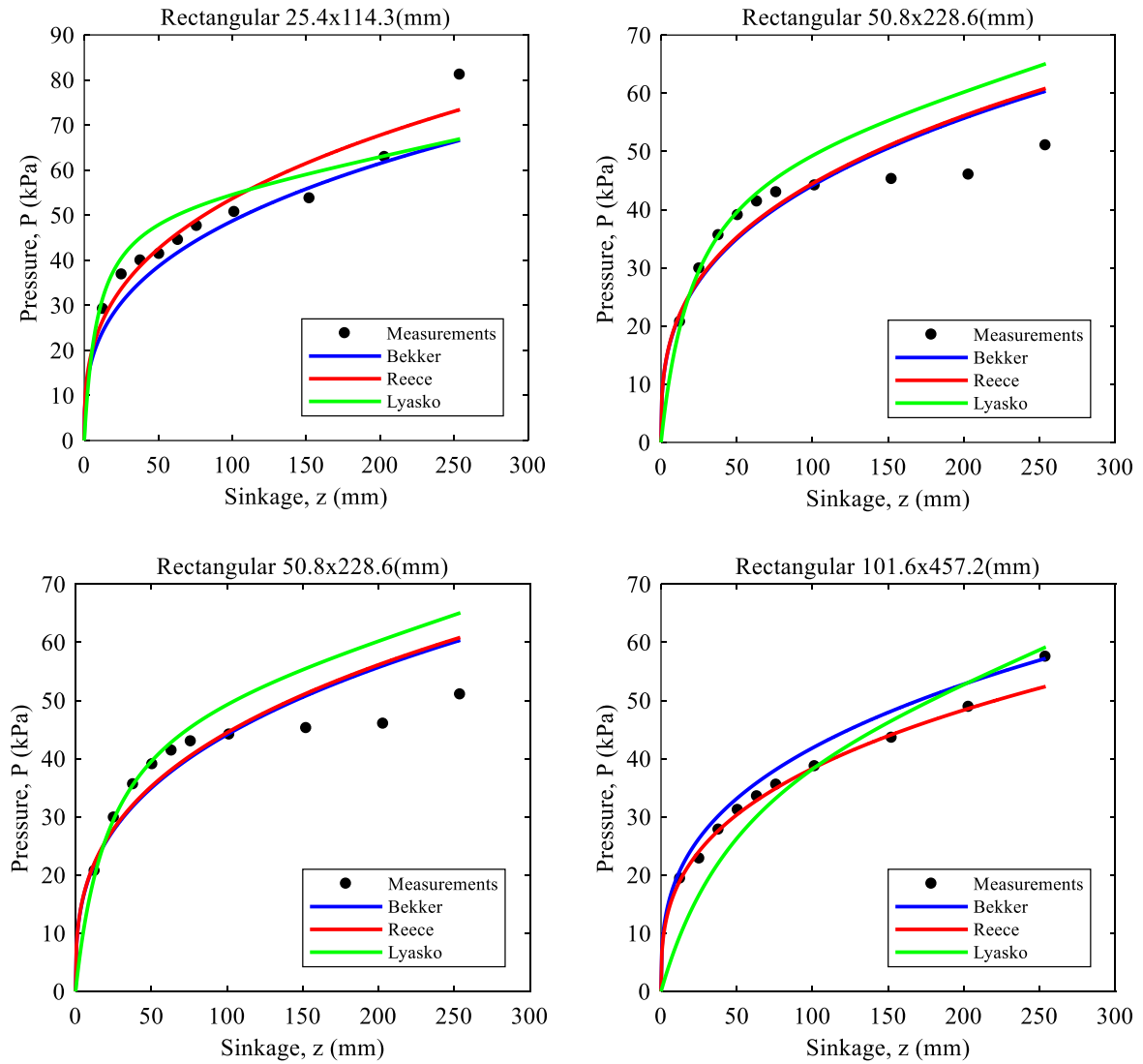


Figure 3.12 Pressure vs sinkage in saturated clay, rectangular plates of various length and widths. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

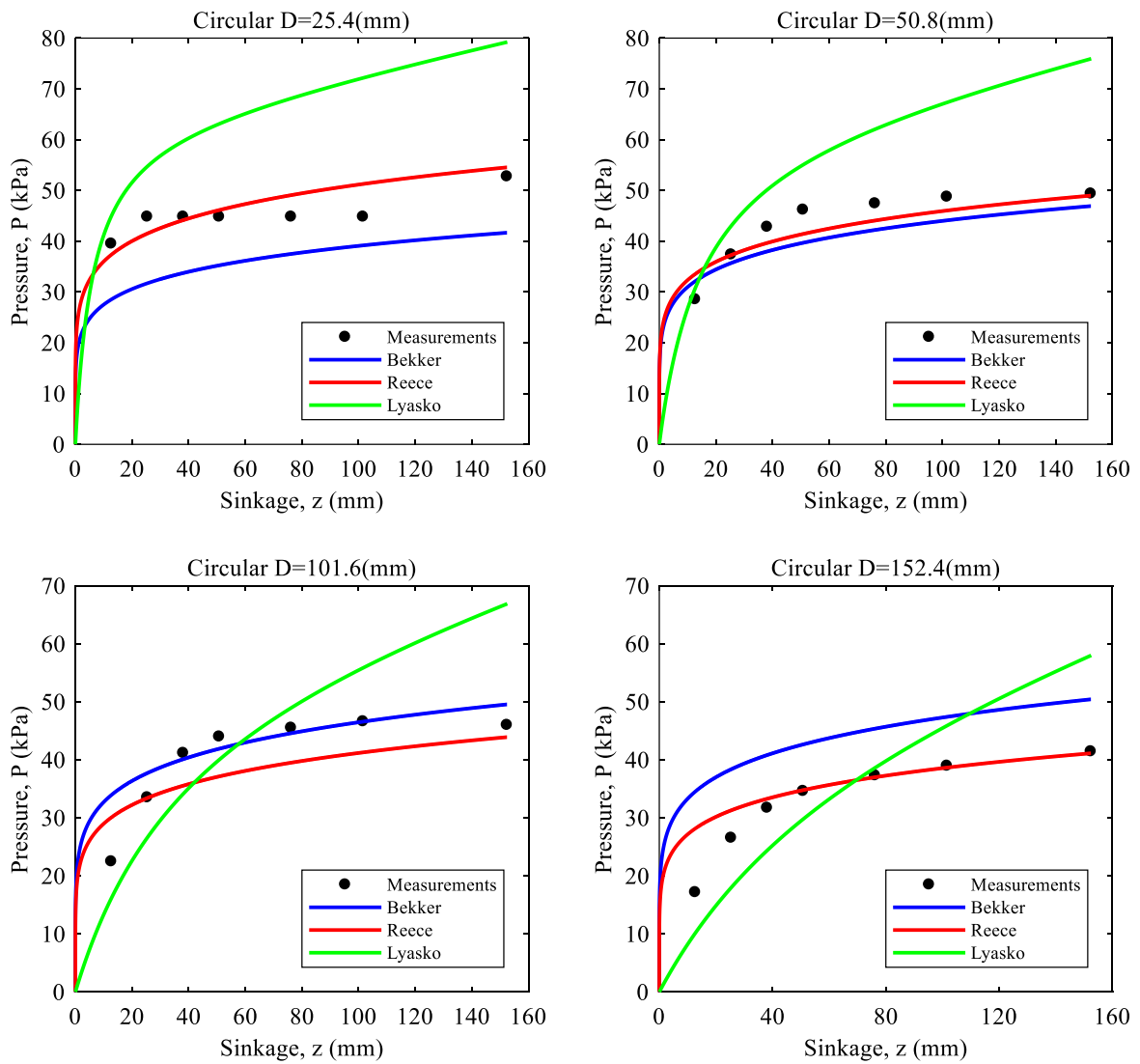


Figure 3.13 Pressure vs sinkage in saturated clay, circular plates of various diameters. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

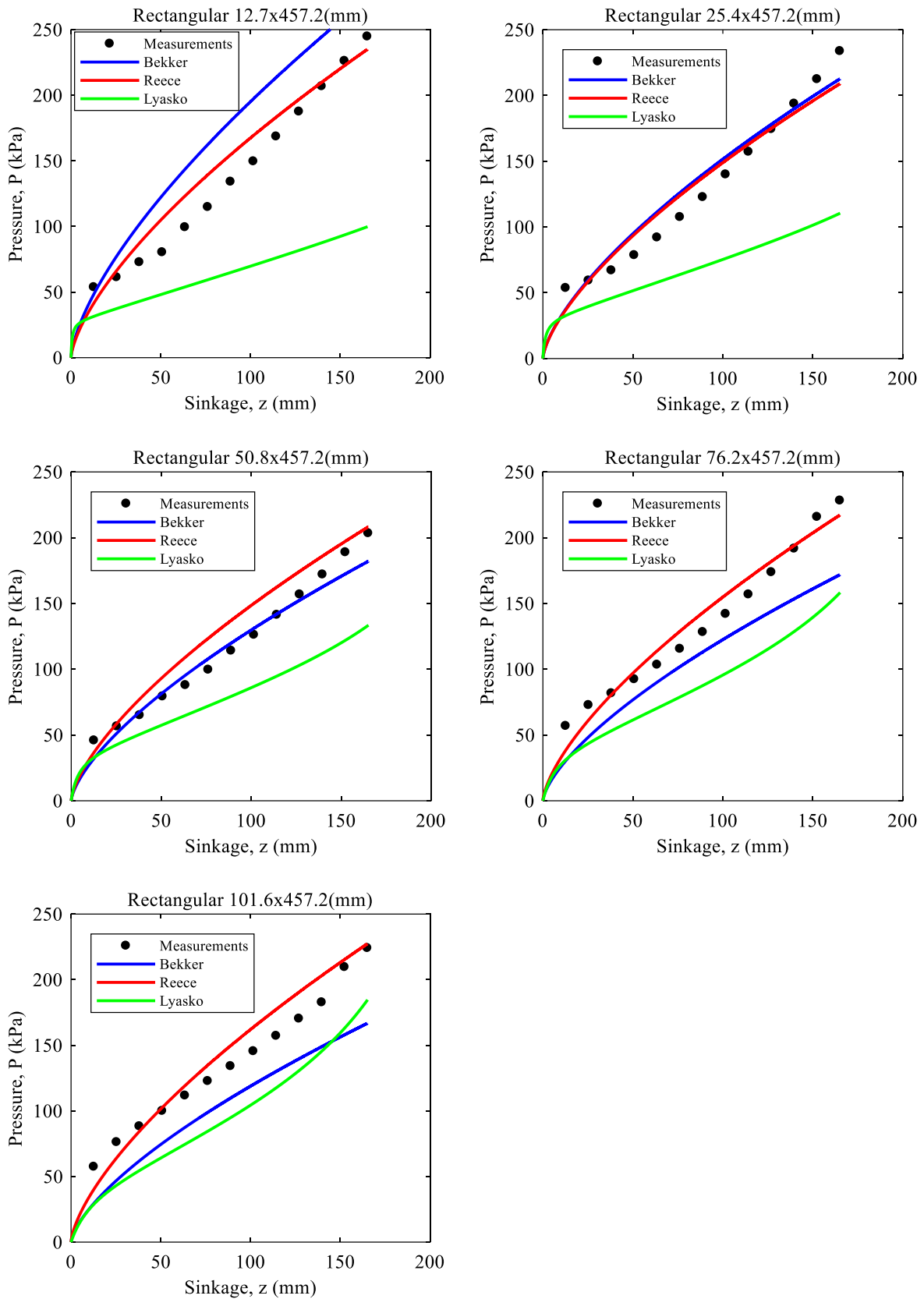


Figure 3.14 Pressure vs sinkage in wet sand, rectangular plates of 457.2 mm (18 in) length and various widths. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

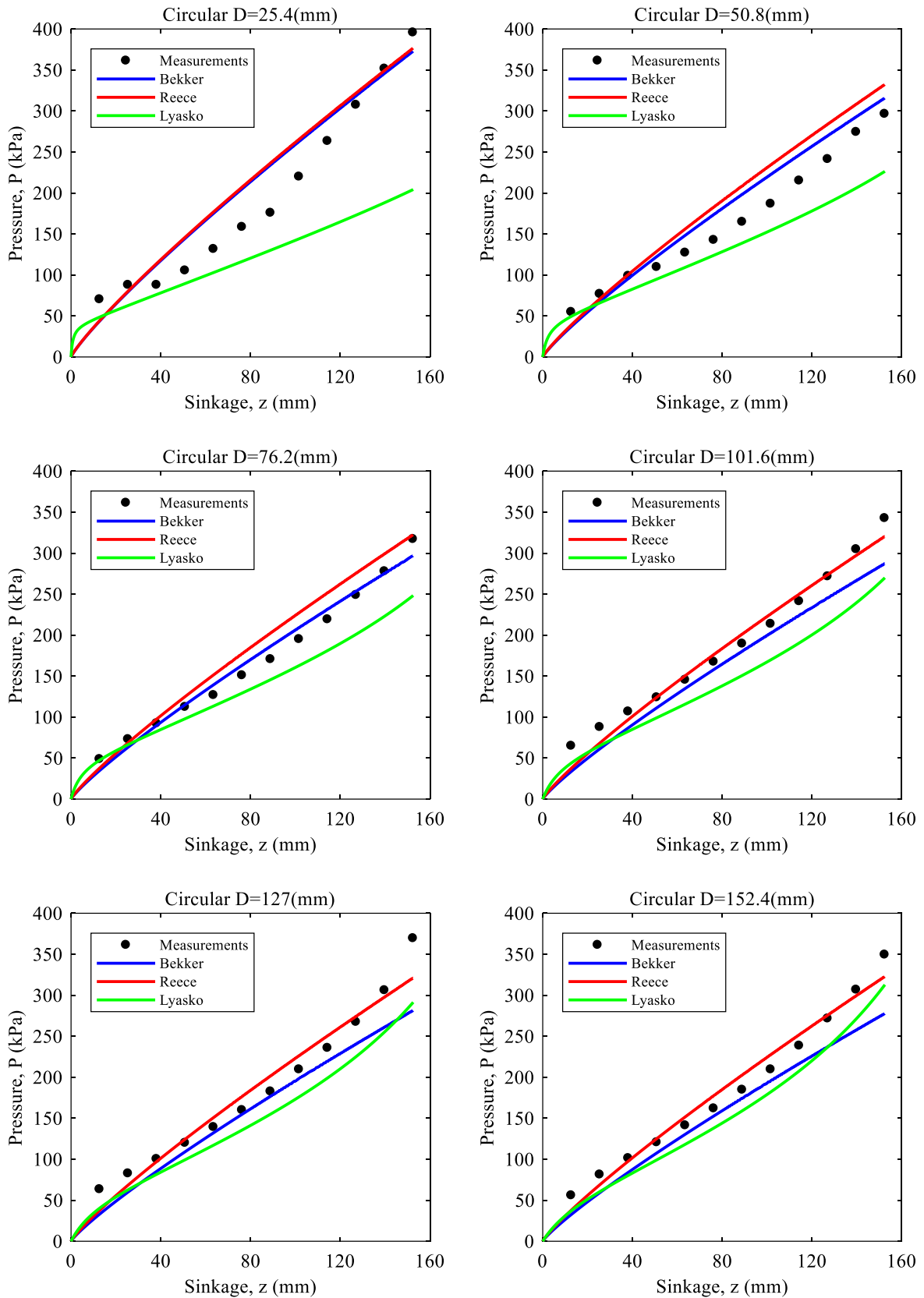


Figure 3.15 Pressure vs sinkage in wet sand, circular plates of various diameters. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

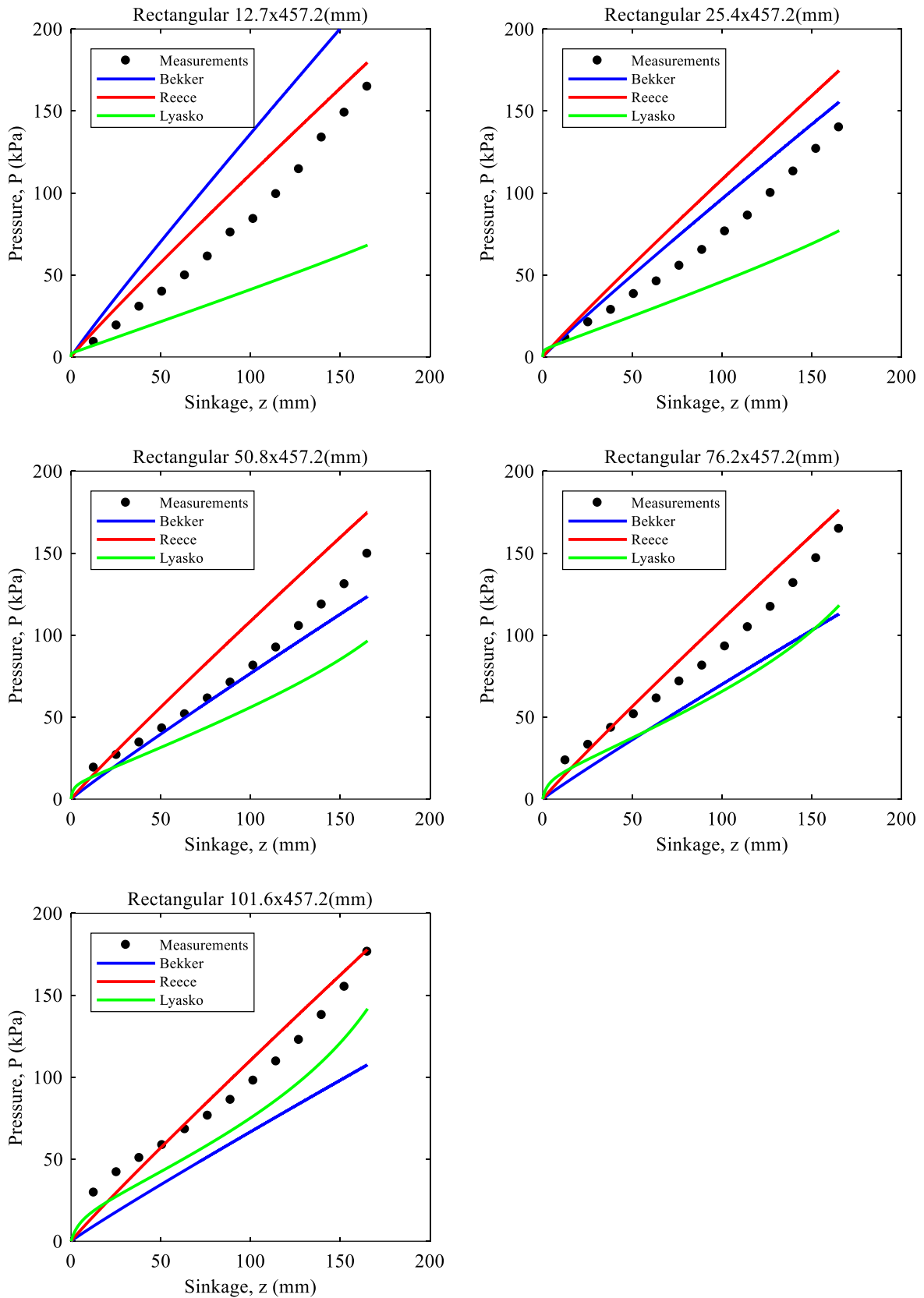


Figure 3.16 Pressure vs sinkage in dry sand, rectangular plates of 457.2 mm (18 in) length and various widths. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

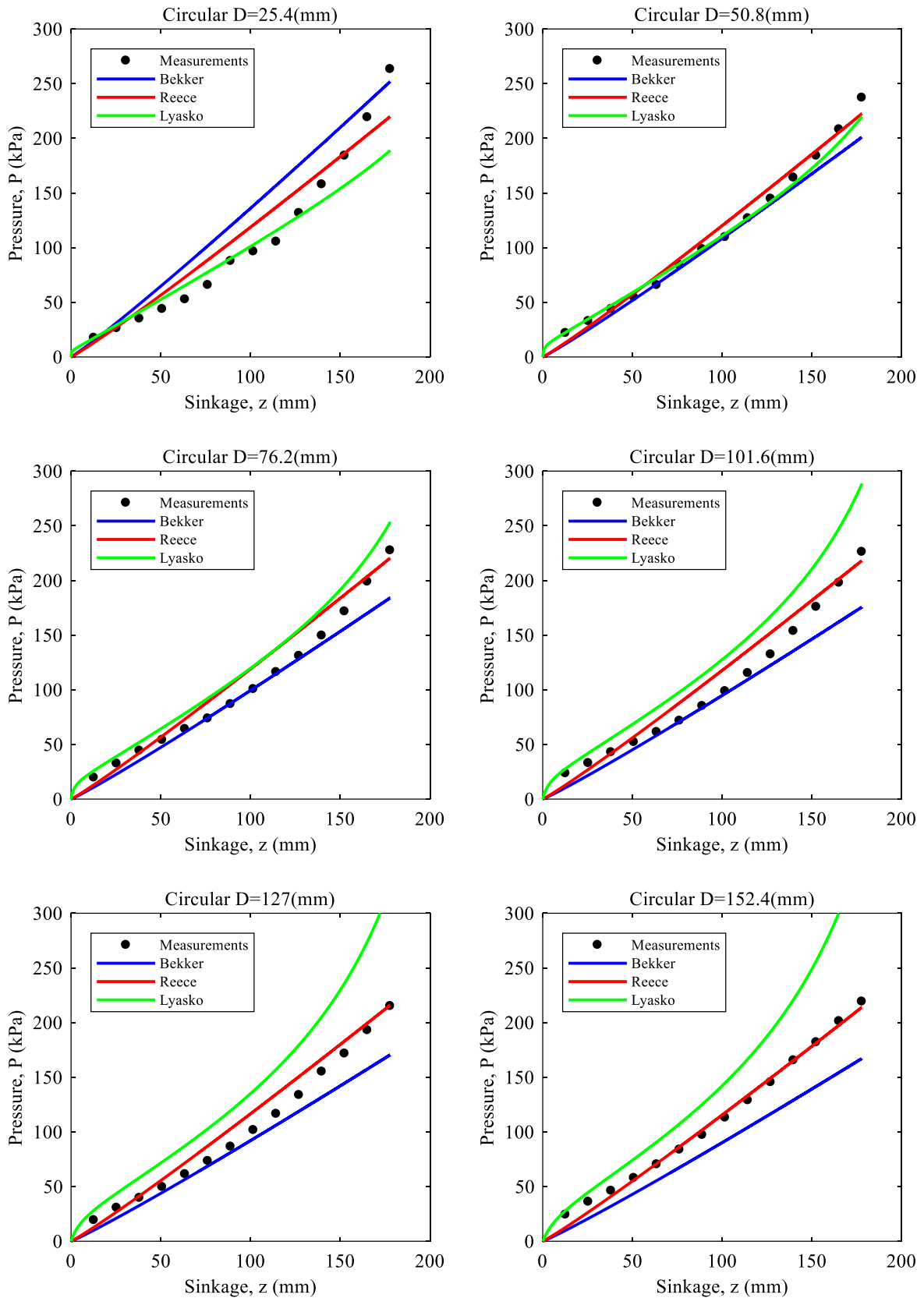


Figure 3.17 Pressure vs sinkage in dry sand, circular plates of various diameters. Comparison between test data from [57] and Bekker, Reece and Lyasko models.

Before summarising the conclusions from the model comparison, it is necessary to make some general remarks which will help in interpretation and understanding of the comments to follow.

- As described earlier in the text, the current research concerns tyre widths between 215mm and 295mm. However, the smallest test plates compared here have about 1/10 and the largest about 1/2 the width of interest. Although the comparisons of all test plates are presented in order to assess the correlation of the three methods, the main focus when analysing the results will be on the largest plates.
- In almost all rectangular plate tests, the plate length is fixed to 0.475m which is approximately the maximum contact area length of interest in as summarised in Table 3.1.
- Similarly, the length/width ratio of the test plates (L/B) is between 4.5 and 36, when the ratios of interest are approximately between 0.5 and 1 (see Table 3.1).
- The width/length ratio of the test plates (B/L) is between 0.0278 and 0.2222, when the ratios of interest are approximately between 1 and 1.9 (see Table 3.1).
- Vertical loads between 3000N and 9000N are mostly concerned.

Comparing the results of the three analysed model against the test results for three types of soil (wet sand, dry sand and saturated clay) from [57], the following conclusions can be made:

- Through visual inspection, all models provide reasonable accuracy (especially at large plate sizes) as the curves in most cases lie near the test data.
- Lyasko is not as accurate at small plates in sand. Nonetheless, this was expected since the model, according to the author [31], is validated for agricultural and earthmoving vehicles where the vertical loads and contact areas are much larger than the ones from the test results.
- In wet sand with rectangular plates, Bekker and Reece are more consistent than Lyasko with Reece also being the most accurate.
- In wet sand with circular plates, Lyasko model captures very well the soil response and the change in gradient at high sinkage for the two largest plates. It is also the most accurate on the largest plate.

- In dry sand, all models are inaccurate at small plates. In large plate plots, Bekker and Lyasko are closer to the measure data with Lyasko, like in wet sand, capturing very well the change in gradient at high sinkage values.
- In dry sand with circular plates, Reece is the most accurate all across the plate sizes. At large plates, Lyasko and Bekker over- and underestimate respectively the vertical load for specific sinkage values.
- For saturated clay the measured data only go up to 2500N of vertical load; outside the region of interest. Therefore, it is difficult to conclude which model is more appropriate for the desired load level.

Despite the above observations and commentary, it is challenging to quantify what ‘good’ or ‘accurate’ correlation really is. The test data did not provide any statistical information, such as standard deviation, so it is difficult to judge whether the three models compared in this section can really capturing the soil response. There are, however, many graphs where one or more models completely fail to match the experimental data as the magnitude of the error is several kilo newton (e.g. Figure 3.15, $D=25.4$ mm). Additionally, none of the three models demonstrated consistency across all test data, which could mean that either they are not suitable for all soil types and could selectively be used in certain cases, or that their formulations are inherently problematic. In any case though, there is an obvious gap that requires further attention and will be discussed later in the text.

3.4 Lyasko Model – Further Analysis

From the comparison between Bekker, Reece and Lyasko models the general conclusion is that all three demonstrate similar accuracy, but further improvements are needed. The main advantage of the Lyasko model though is that depends on soil invariants; physical properties of soil, and not just fitting parameters. This can provide an opportunity to gain an insight into soil behaviour and the effect of certain physical parameters in the response under particular loading conditions. For instance, the effect of internal friction angle can be investigated without requiring tests of various types of soil with different values of ϕ . This kind of studies cannot be performed using the Bekker or Reece models. In the following paragraphs the results of a sensitivity study in various soil parameters will be discussed. Also, the effect

of different empirical Lyasko model parameters is investigated and analysed in an attempt to simplify the formulation if possible.

3.4.1 Contact Pressure Concentration Coefficient ζ

One of the empirical parameters of the Lyasko model is the contact pressure concentration coefficient of Equation 2.49 and 2.50. This coefficient depends on plate width and it has been derived from typical soils in tractor operating conditions [31]. As it can be seen from in Figure 3.18 the curve of coefficient ζ is very non-linear with values varying between 1 and 2.5. Yet, this research concerns plates with width much narrower than the one presented in Figure 3.18. In Figure 3.19 the parameter ζ is presented for the range of plate widths of SUVs as per Table 3.1. From Figure 3.19 it is clear that the variation of ζ is much smaller for that width range, with the parameter values spanning between 2.49 and 2.55 with average value of 2.54.

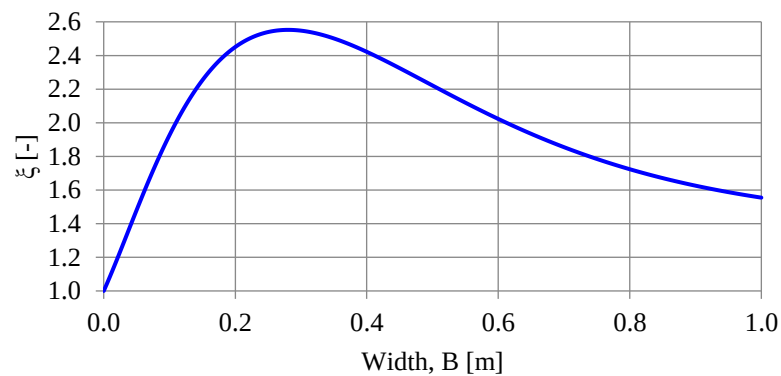


Figure 3.18 Contact pressure concentration coefficient against plate width.

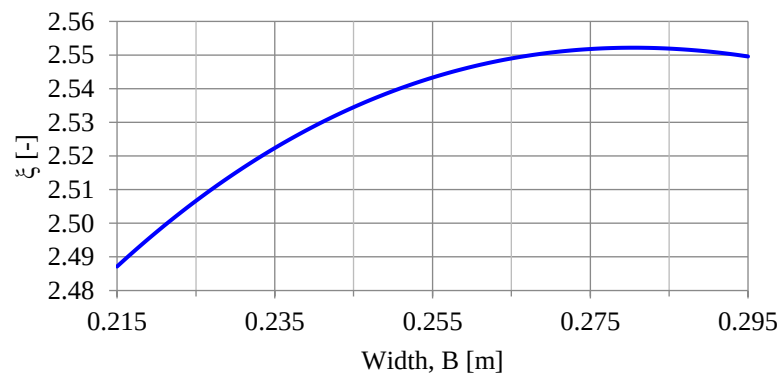


Figure 3.19 Parameter ζ for plate width limits of SUVs.

In order to check the effect of parameter ζ in the overall pressure-sinkage equation, the Lyasko model was run for the three soils reviewed earlier and for the plate widths of Table 3.1. The model was run using both $\zeta = 2.54$ (average value) and $\zeta = \text{calculated}$ using Equation 2.49. It was found that the difference between the calculated and fixed value of ζ , at vertical loads between 3 kN and 9 kN, is negligible with maximum value of difference equal to 73 N. Hence, Equation 2.49 can be replaced by the constant value of $\zeta = 2.54$.

3.4.2 Soil deformation modulus E_0

Very limited information is provided by Lyasko on how the soil deformation modulus is derived or its theoretical basis. The Equation 2.54, which is repeated here, is a function of both soil cohesion and internal friction angle. In this paragraph it will be investigated the effect these two parameters have on E_0 .

$$E_0 = 100[2000 + c \exp(0.03\phi - 2.5)] \exp(0.011\phi^{1.5}) \quad 2.54$$

Re-writing Equation 2.54 with the exponential functions being separate factors, it takes the form:

$$E_0 = 100[2000 + cE_A]E_B$$

where: $E_A = \exp(0.03\phi - 2.5)$ and $E_B = \exp(0.011\phi^{1.5})$

The reason for re-writing Equation 2.54 is to enable the analysis of exponential functions E_A and E_B factors separately. In the following Figures 3.20 and 3.21 the exponential factors are plotted against the internal friction angle with soil cohesion equal to zero (assuming dry sand). As it can be seen in Figure 3.20, factor E_A is steadily increasing with an increase of internal friction angle in an almost linear manner. On the other hand, E_B in Figure 3.21 demonstrates a much more non-linear behaviour with a rapid increase after $\sim 30^\circ$. Furthermore, it is multiplied with the rest of the soil deformation modulus equation, which means that it greatly impacts the shape of E_0 . The last statement is confirmed when looking at Figure 3.22 where the soil deformation modulus is plotted against the internal friction angle; as expected the shape of E_0 is almost the same with E_B but different in magnitude.

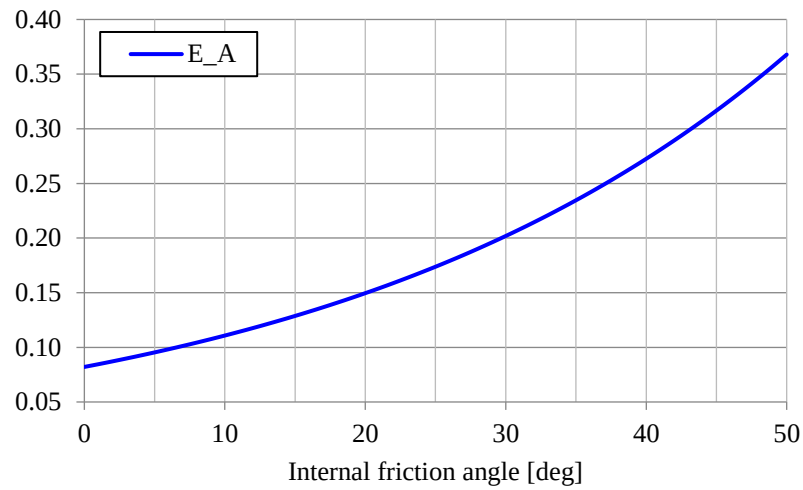


Figure 3.20 Factor E_A against internal friction angle.

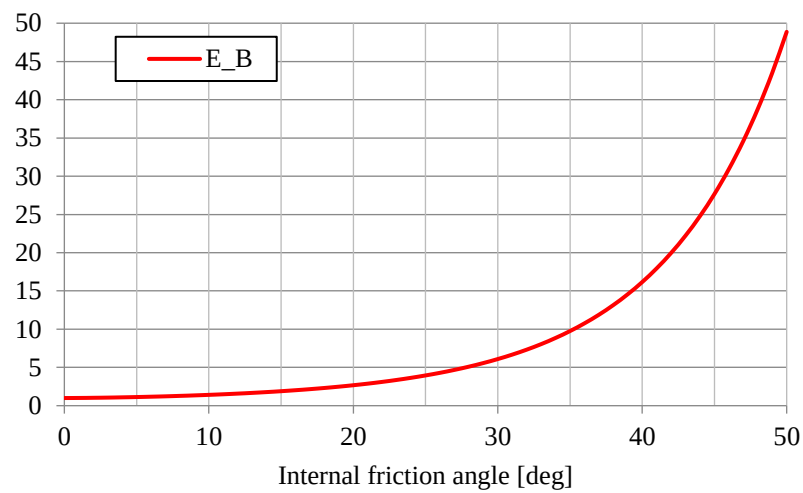


Figure 3.21 Factor E_B against internal friction angle.

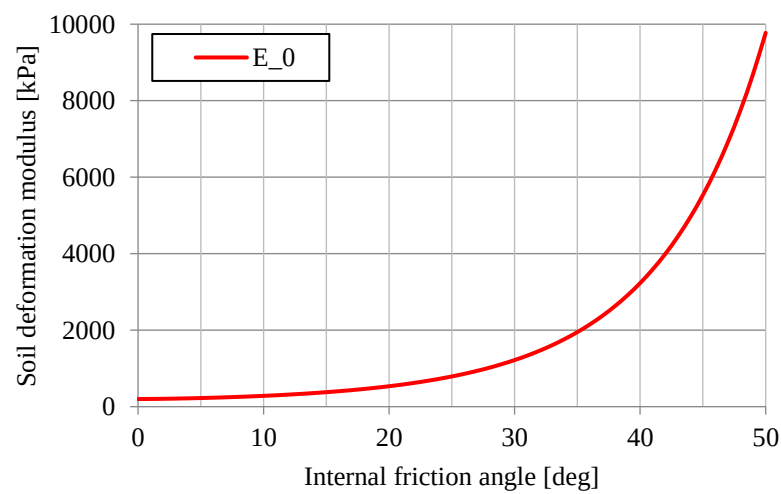


Figure 3.22 Soil deformation modulus against internal friction angle for cohesion $c = 0$.

On the other hand, the increase of cohesion doesn't seem to have as significant impact as the internal friction angle. In Figure 3.23 the deformation modulus is plotted against soil cohesion for three discrete internal friction angles; 10° , 20° and 30° . As the figure illustrates, the increase of soil cohesion is followed by an almost linear, low gradient, increase in deformation modulus. Also, the higher the internal friction angle, the higher the curve gradient due to the effect of E_B at high friction angles.

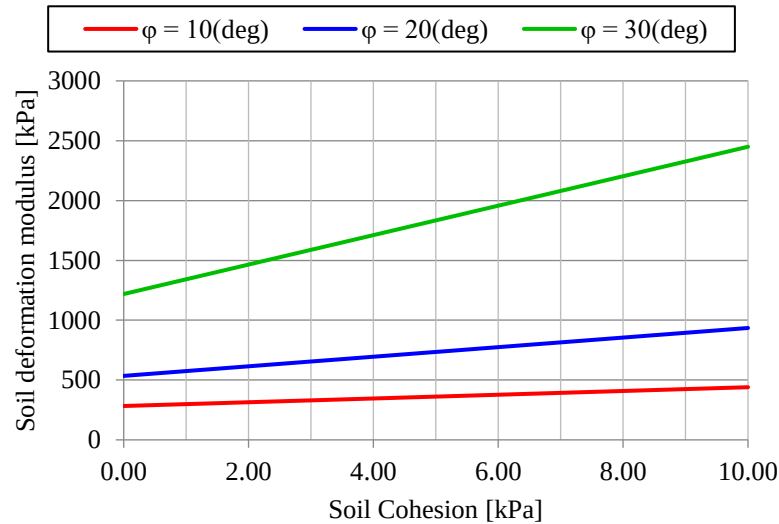


Figure 3.23 Soil deformation modulus against cohesion for 10° , 20° and 30° of internal friction angle.

The above findings highlight that the internal friction angle is the critical parameter mostly affect the deformation modulus, and that soil cohesion has much smaller impact. Additionally, for frictionless types of soil, such as clay with low internal friction angle (e.g. less than 10°), the deformation modulus could be assumed constant for practical applications. It is though still unclear the role of the deformation modulus, and how it affects the total soil compaction.

3.4.3 Soil bearing capacity B_i

As discussed in Chapter 2, in classical soil mechanics bearing capacity is the maximum average pressure that a soil can support without producing shear failure. For a strip (or continuous) foundation footing, where the length of the loading area is much larger than its width, Terzaghi [29] proposed the following Equation 2.33 for shallow footing ($z / B \leq 1$) after an analysis of the general shear failure of

soils. Terzaghi also proposed bearing capacity equations for different types of footing; circular, square and rectangular. Equation 2.36 for rectangular plates is repeated below.

$$\text{Terzaghi:} \quad P = cN_c \left(1 + 0.3 \frac{B}{L} \right) + \gamma_s z N_q + 0.5 \gamma_s B N_\gamma \left(1 - 0.2 \frac{B}{L} \right) \quad 2.36$$

$$\text{Lyasko:} \quad B_i = cN_c \left(1 + 0.25 \frac{B}{L} \right) + \gamma_s z N_p \left(1 + 1.5 \frac{B}{L} \right) + \gamma_s B N_\gamma \left(1 + 0.3 \frac{B}{L} \right) \quad 2.41$$

Comparing Terzaghi and Lyasko's formulations some similarities can be observed. As it can be seen the plate shape coefficients are very similar between the two with main difference being the absence in Terzaghi of plate shape coefficient connection to the surcharge effect due to soil weight. Lyasko claims that his bearing capacity equation has been validated for applications involving agricultural and earthmoving vehicles [31]. In Table 3.5 the numerical values of bearing capacity factors for Terzaghi and Lyasko are summarised, with Figure 3.24, Figure 3.25 and Figure 3.26 presenting plots of variables N_c , N_q and N_γ respectively. The differences between Terzaghi and Lyasko for the N_c and N_q variables are very small, however N_γ is significantly different. Looking more closely into the equations, one can notice that in Terzaghi's Equation 4.23 only half of N_γ values are considered as it's multiplied with a scaling factor of 0.5. When considering this scaling factor, a better agreement is reached.

ϕ (deg)	N_c			N_q			N_γ			
	Terzaghi	Lyasko	Diff (%)	Terzaghi	Lyasko	Diff (%)	Terzaghi	Terzaghi*0.5	Lyasko	Diff (%)
0	5.14	4.5	12.8	1.00	1.2	-16.2	0.00	0.0	0.2	100.0
5	6.49	6.0	7.7	1.57	1.8	-12.2	0.45	0.2	0.3	27.6
10	8.34	8.0	3.5	2.47	2.7	-8.9	1.22	0.6	0.6	-8.7
15	10.98	10.9	0.6	3.94	4.2	-5.6	2.65	1.3	1.0	-28.3
20	14.83	15.0	-1.2	6.40	6.6	-2.7	5.39	2.7	2.0	-37.9
25	20.72	21.1	-1.7	10.66	10.7	-0.3	10.88	5.4	3.9	-40.9
30	30.14	30.5	-1.1	18.40	18.1	1.4	22.40	11.2	8.1	-38.6
35	46.12	45.8	0.7	33.30	32.6	2.2	48.03	24.0	18.3	-31.3
40	75.31	72.7	3.5	64.20	63.1	1.8	109.41	54.7	46.1	-18.8
45	133.87	124.0	7.4	134.87	135.6	-0.6	271.75	135.9	134.3	-1.2

Table 3.5 Comparison of bearing capacity factors between Terzaghi and Lyasko.

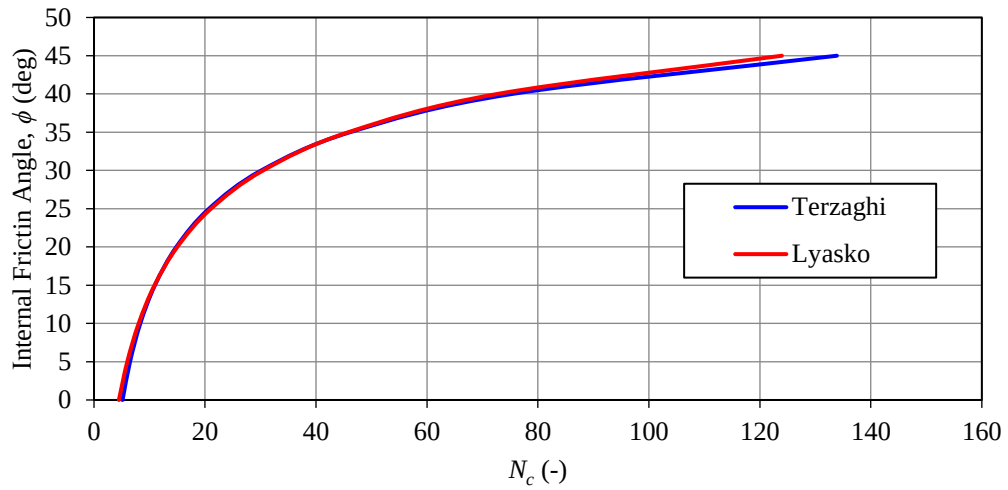


Figure 3.24. Comparison of bearing capacity factor N_c between Terzaghi and Lyasko.

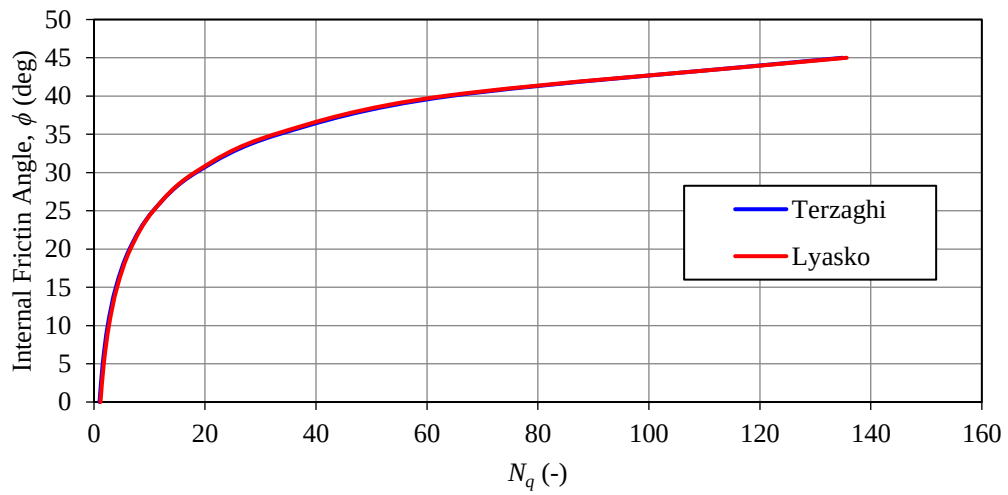


Figure 3.25. Comparison of bearing capacity factor N_q between Terzaghi and Lyasko.

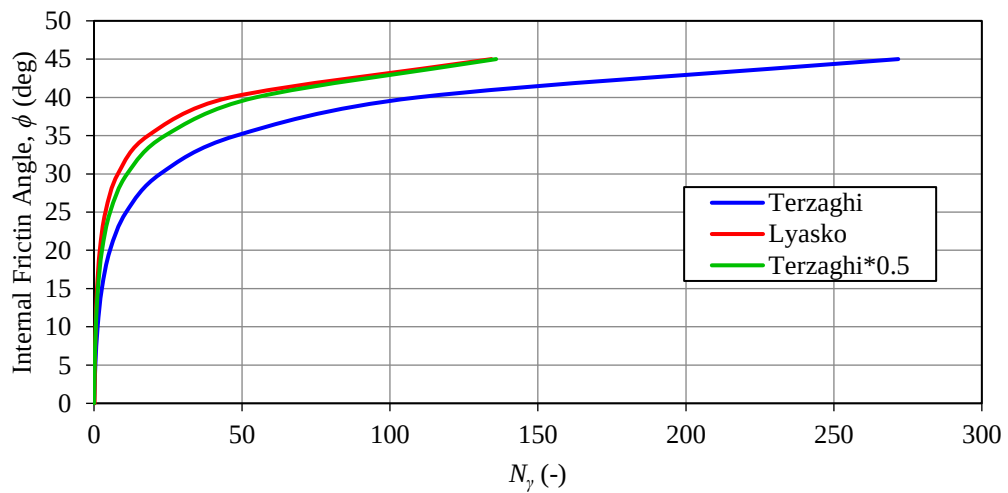


Figure 3.26 Comparison of bearing capacity factor N_γ between Terzaghi and Lyasko.

From the above it can be concluded that the bearing capacity equation used by Lyasko, is a modification of Terzaghi's work for rectangular plates. However, no clear indication as to why the particular modifications are given, a particular weakness of empirical methods as they cannot necessarily be directly tied to physically meaningful occurrences.

3.4.4 Analysis conclusions

In this Chapter an analysis of various empirical parameters of the Lyasko model was conducted. The analysis aimed at gaining an understanding of the contribution and function of these parameters in the overall soil response, since not enough information is provided in the literature. An overall conclusion about the model is that, although requires soil invariants as inputs, it is still heavily dependent on many empirical equations, the derivation of which is not well documented at present. Furthermore, in terms of correlation to test data, it is not particularly advantageous when compared to Bekker and Reece. In view of the facts above, it has been decided to not proceed with Lyasko's method for pressure-sinkage in the future modelling development. Instead, the well documented, simple and widely adopted Bekker model will consist the starting point.

3.5 Chapter Conclusions

Aim of Chapter 3 was the evaluation of existing pressure-sinkage models, as part of the second objective, in order to establish whether they can accurately predict the soil behaviour. The analysis has highlighted that neither of the three assessed models are consistent in terms of correlation to experimental data. Therefore, a development of a new or extension of an existing model is required. The Lyasko model was initially considered as a candidate for further development. However, despite the claim of being based on soil invariant parameters, the Lyasko model was eventually rejected based on the complexity and the lack of clarity regarding the origin of the vast set of empirical parameters. It was therefore decided that the most established and simple model by Bekker would be used for further investigation. In addition, it has been confirmed the need for in-house soil data in order to perform a meaningful evaluation of any newly proposed formulation. In Chapter 4, in-house experimental data are analysed and discussed, and various models (based on Bekker and others) are developed.

Chapter 4: Soil Experiments and Modelling

In Chapter 3 an analysis was conducted considering three pressure-sinkage models currently available in the literature. The analysis conclusion was that the model proposed by Bekker has certain advantages over Reece and Lyasko and it would be considered as a starting point for further research. The lack of widely available soil numerical data in the literature raised the necessity to conduct in-house soil measurements, which would support the investigation of an appropriate pressure-sinkage model. This chapter concerns the soil data acquisition setup and process, analysis of the measurement results and a proposal for a new pressure-sinkage model which provides a significant improvement over the classical Bekker formulation.

4.1 Terramechanics Scale Rig

Loughborough University is equipped with a test rig aimed for Terramechanics related studies on scaled wheels and plates (see Figure 4.1). The rig's two degrees of freedom – vertical and horizontal displacement – can be applied through manual input from an operator. Three load cells are installed for measuring forces in vertical, horizontal and lateral direction. Furthermore, two string pots are in place for acquiring vertical and horizontal displacements.

Two types of soil testing are described in the current chapter; pressure-sinkage and shear tests. For the pressure-sinkage tests, which are essential for characterising the compaction soil behaviour, only forces and displacement in the vertical direction are required. On the other hand, the shear tests are necessary for obtaining the soil physical characteristics of cohesion and internal friction angle. As discussed in preceding paragraphs, all failure criteria require information about the soil behaviour in both normal and shear direction. Therefore, for the shear tests forces and displacements in both vertical (normal) and longitudinal (shear) directions are acquired.

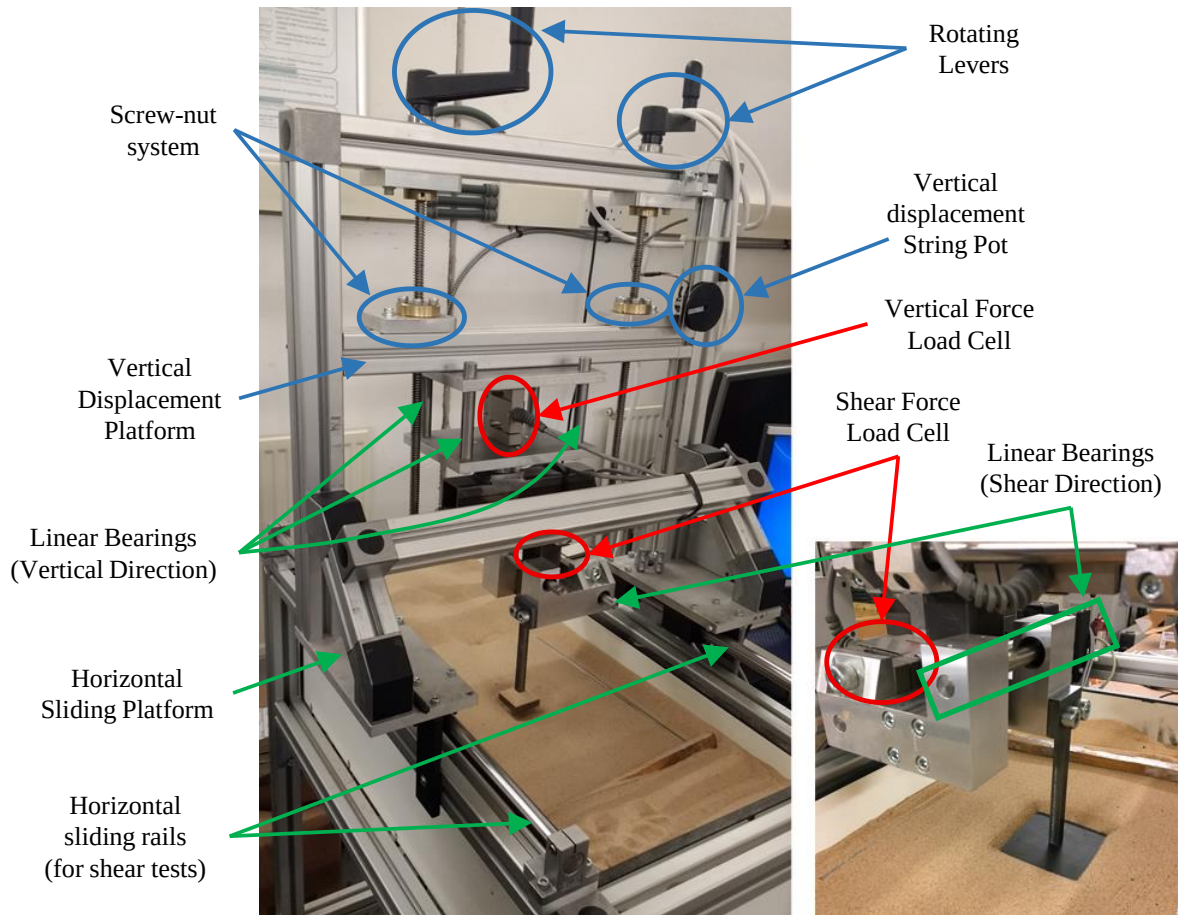


Figure 4.1 Loughborough University Terramechanics scale rig.

Terramechanics Scale Rig technical specifications:

- Internal sand box dimensions (Width, Length, Depth): 300 mm, 760 mm, 240 mm
- Maximum vertical travel of the screws: 300 mm
- Overall rig dimensions (Height, Width, Length): 1650 mm, 900 mm, 900 mm
- Vertical load cell capacity: 100 kg
- Horizontal load cell capacity: 50 kg
- Strain gauge amplifier box: RaceLogic RLVBSGA02
- Strain gauge amplifier USB: DCSUSB 17064836 0-5V
- Vertical travel is via a linear potentiometer 0-5V (in-house)
- Horizontal travel is via a strong potentiometer MicroEpsilon WPS-1250-MK46-CR-U10. Serial number 9612, 1250mm range, 8.004mV/mm, with < 10% error.

4.2 Pressure vs Sinkage – Flat Plates

As part of the current research, a series of penetration tests have been performed by the author for measuring pressure and sinkage on a number of flat plates of various dimensions. Aim of these tests is the acquisition of numerical data for assessing the accuracy of existing models and validating newly proposed alternatives. The list of flat plates manufactured for this purpose is summarised in Table 4.1, which consists of 18 square and rectangular plates of width (smallest dimension) between 30 mm and 80 mm, and length (largest dimension) between 40 mm and 140 mm, at various length/width ratio values spanning between 1 (square plates) and 3. All plates are made of steel of 10 mm thickness in order to ensure no plate deformation occurs during the pressure-sinkage tests. As presented in Table 3.1, the length to width ratio (L/B) for the SUVs tyres span between 1 and 2, so the 2/3 of the flat test plates of Table 4.1 are within that range. The remaining 1/3 of the plates have L/B ratios between 2 and 3, and they have been chosen for allowing potential extrapolation studies to take place.

ID	Type	Width, B (mm)	Length, L (mm)	L/B (-)	Area, A (mm ²)
1	Rectangular	30	60	2.0	1,800
2	Rectangular	30	80	2.7	2,400
3	Rectangular	30	90	3.0	2,700
4	Rectangular	40	60	1.5	2,400
5	Rectangular	40	100	2.5	4,000
6	Rectangular	40	120	3.0	4,800
7	Rectangular	50	75	1.5	3,750
8	Rectangular	50	80	1.6	4,000
9	Rectangular	50	120	2.4	6,000
10	Rectangular	50	140	2.8	7,000
11	Rectangular	60	90	1.5	5,400
12	Rectangular	70	105	1.5	7,350
13	Rectangular	80	120	1.5	9,600
14	Square	40	40	1.0	1,600
15	Square	50	50	1.0	2,500
16	Square	60	60	1.0	3,600
17	Square	70	70	1.0	4,900
18	Square	80	80	1.0	6,400

Table 4.1 List of flat penetration plates.

Two soil types have been selected for conducting the experiments; fine sand and sharp sand. Fine sand's moisture content is 0 % and it consists of small-sized particles with quite narrow distribution, i.e. small particle size variation. Therefore, the soil is considered dry and fairly homogenous. In contrary, sharp sand as received from the supplier contained an inherit amount of water, with measured moisture content of approximately 4%. Also, the particle size distribution spans wider than the fine sand with small and larger particles mixed together, characterising sharp sand as less homogenous. The difference in characteristics of the two tested soil types will allow a wider validation of the proposed pressure-sinkage model.

Figure 4.2 presents a schematic of the rig to assist describing the operation during the pressure-sinkage test. As presented in Figure 4.2, the vertical displacement is applied by rotating two levers at the top part of the rig. As the moving platform translates downwards and the penetration plate sinks into the soil, the string pot and load cell mounted on the vertical moving platform acquire displacement and force data against time. Finally, the time-domain data are filtered and cut through Matlab scripts for deriving the pressure-sinkage diagrams and the corresponding numerical data. Four linear bearings, between the parts connecting the two ends of the load cell, are in place for ensuring that no torque is applied to the load cell.

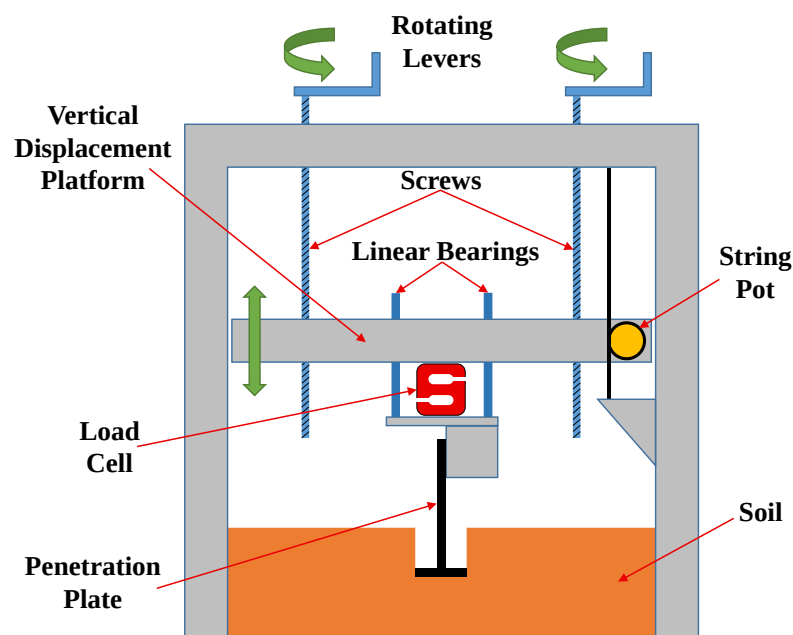


Figure 4.2 Schematic of Loughborough University Terramechanics scale rig.

4.2.1 Acquisition and Processing Methodology

Ensuring high level of measurement accuracy and repeatability requires a carefully designed test procedure, with the first attention item being the soil state. As described in [20], the soil response varies significantly between undisturbed and compacted condition. So, prior to each penetration test the soil must be as close to a ‘nominal’ state as possible. To achieve this, the sand must be well stirred inside the box to break-down any high density areas and the surface should be flattened out prior to each penetration test. Already mentioned that the plate vertical displacement is applied through manual rotation of rig’s levers. During this manual process, great attention was paid by the author in applying the vertical displacement consistently at a rate of approximately 3 mm/s. This rate was chosen as it corresponds to one lever full rotation per second, a pace that can be reasonably maintained by the rig operator. Figure 4.3 presents two examples of pressure-sinkage tests for fine and sharp sand.

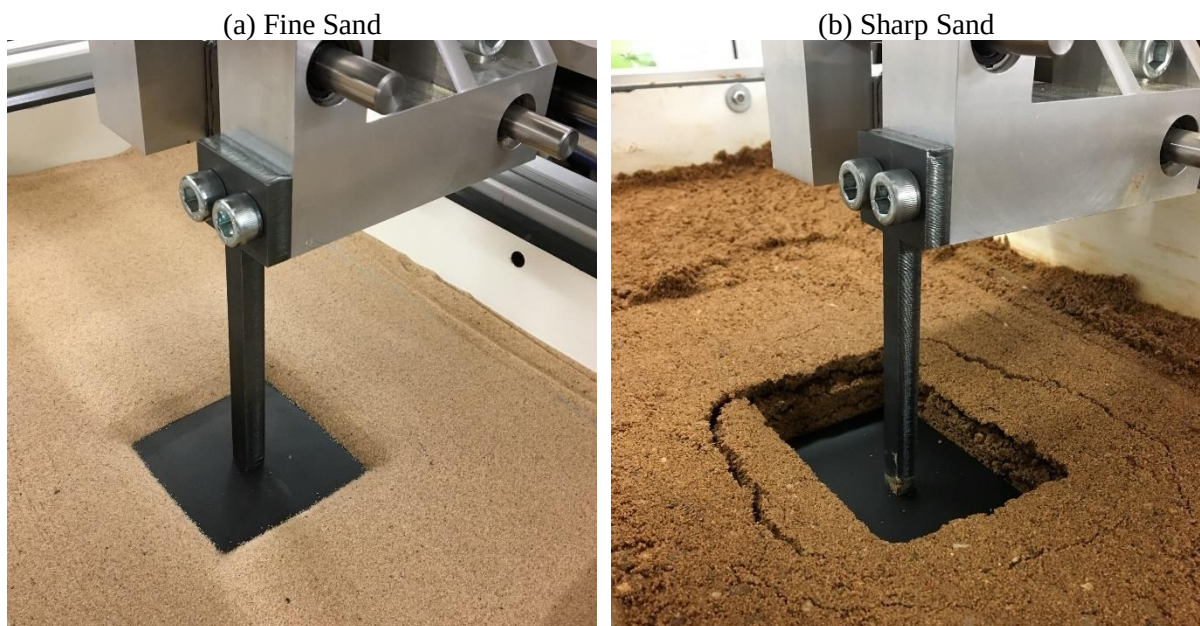


Figure 4.3 Examples from the in-house experiments of flat plates during pressure-sinkage tests.

It is generally difficult to retrieve consistent and repeatable soil measurements despite the efforts in generating a controlled test environment. There are always some discrepancies on the measured data caused by various factors related to both the tested material and/or the testing equipment/process. Such factors could entail friction of the rig’s sliding parts (e.g. linear bearings), soil homogeneity, soil preparation, room temperature and humidity, etc. Therefore, collecting a reasonable sample of data on

the same test plate and soil would aid in evaluating repeatability and in setting acceptable error boundaries. For this reason, the pressure-sinkage tests were repeated at least five times on each penetration plate and soil type. An example of the result outcome of the 30×60 mm plate is presented in Figure 4.4, where each grey curve corresponds to one test repetition on fine sand, with the red curve being the mean value.

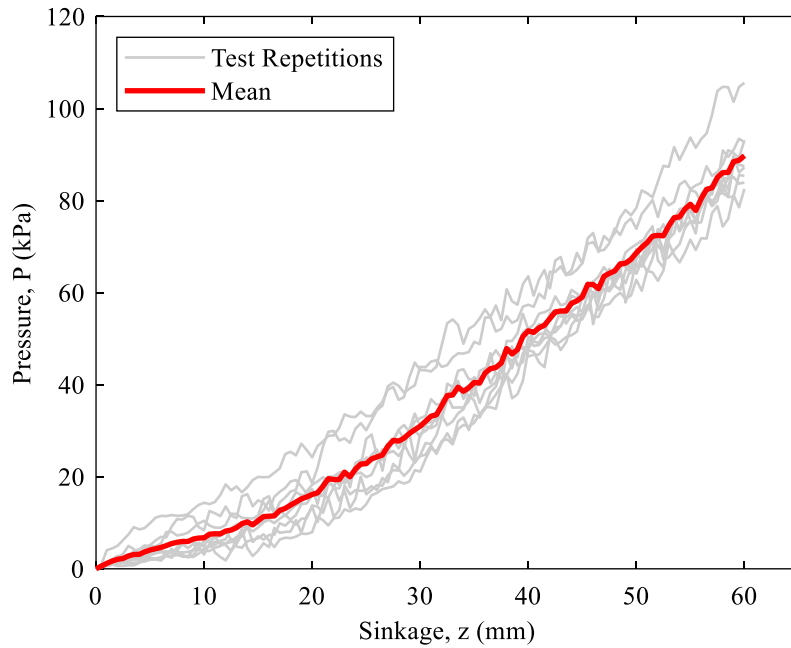


Figure 4.4 Fine sand response on a 30x60mm flat plate. Each curve denotes one test repetition.

The level of spread of data around an average value is called *dispersion* or *variation* [58]. One of the measures of dispersion is the *standard deviation*, which is denoted by the Greek letter σ and is defined by Equation 4.1. Generally, a large standard deviation indicates data spread far from the mean value, while with small standard deviation the data are closer to the mean value. Standard deviation is particularly useful as it defines a statistical probability for future measurement data based on analysis of an existing data sample. For example, if the mean value of a sample is equal to $\bar{x} = 10$ with a standard deviation $\sigma = 1.5$ then a future data sample is expected to be between 8.5 and 11.5 with a probability of 68.27 % ($\pm 1\sigma$ from mean) or between 7 and 13 with a probability of 95.45 % ($\pm 2\sigma$ from mean). An example of normal distribution plot from the literature is shown in Figure 4.5.

$$\sigma = \sqrt{\frac{1}{N} \sum_{j=1}^N (x_j - \bar{x})^2} \quad 4.1$$

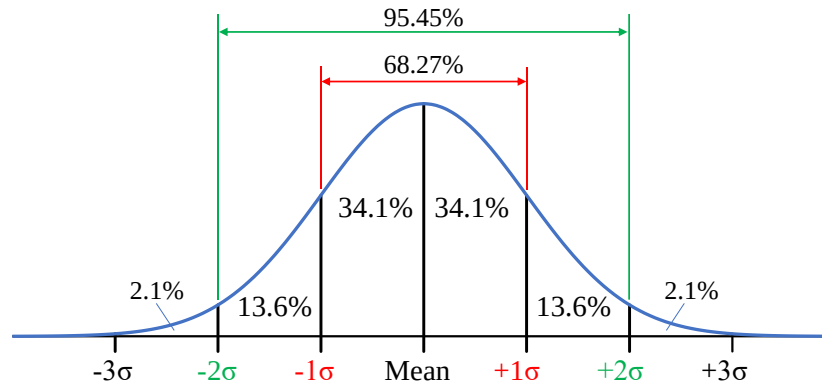


Figure 4.5 Example of normal distribution.

After repeating the penetration tests multiple times for each plate and soil, the standard deviation value was calculated for every 10 mm of sinkage. In the example plot of Figure 4.6, the soil response mean values are plotted including the error bars indicating the limits of $\pm 2\sigma$. In this case, apart from providing an indication of the test repeatability, standard deviation is also useful for defining acceptable limits when comparing measured data with modelling results. A good correlation would be assumed if the simulation results lie within the of $\pm 2\sigma$ error limits, as it would lie within the 95.45% probability range.

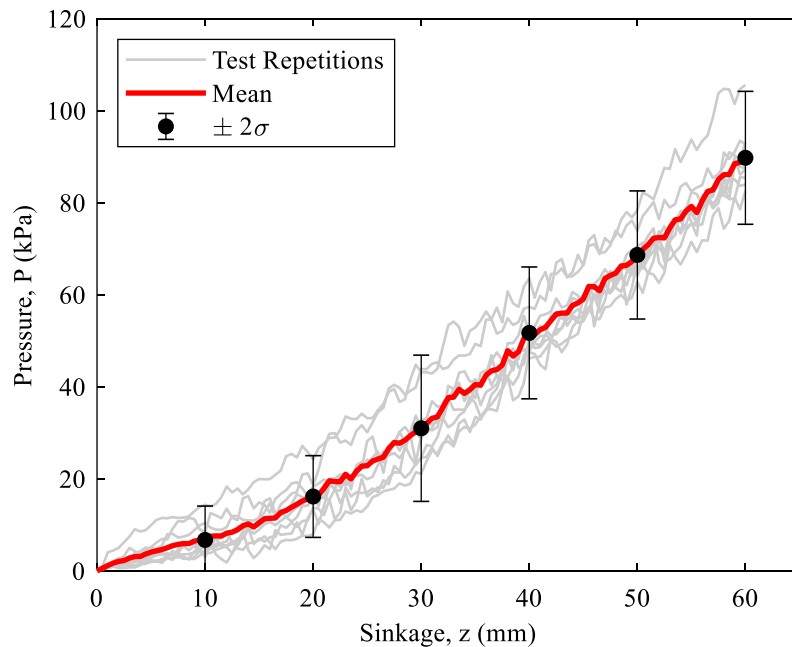


Figure 4.6 Fine sand response on a 30x60mm flat plate. Error bars denote ± 2 standard deviations, solid line is the mean curve.

Standard deviation expresses the spread of a data set with respect to the mean value in absolute terms. However, when comparing different data sets where the mean values vary significantly (e.g. two

different soil types), the standard deviation value could be misleading. In this case, a useful metric for relative data variation is used called *coefficient of variation* (CV) or *coefficient of dispersion* [59]. Coefficient of variation is commonly used in geosciences [59] and it is defined as the ratio of standard deviation σ to the mean $\bar{x}$.

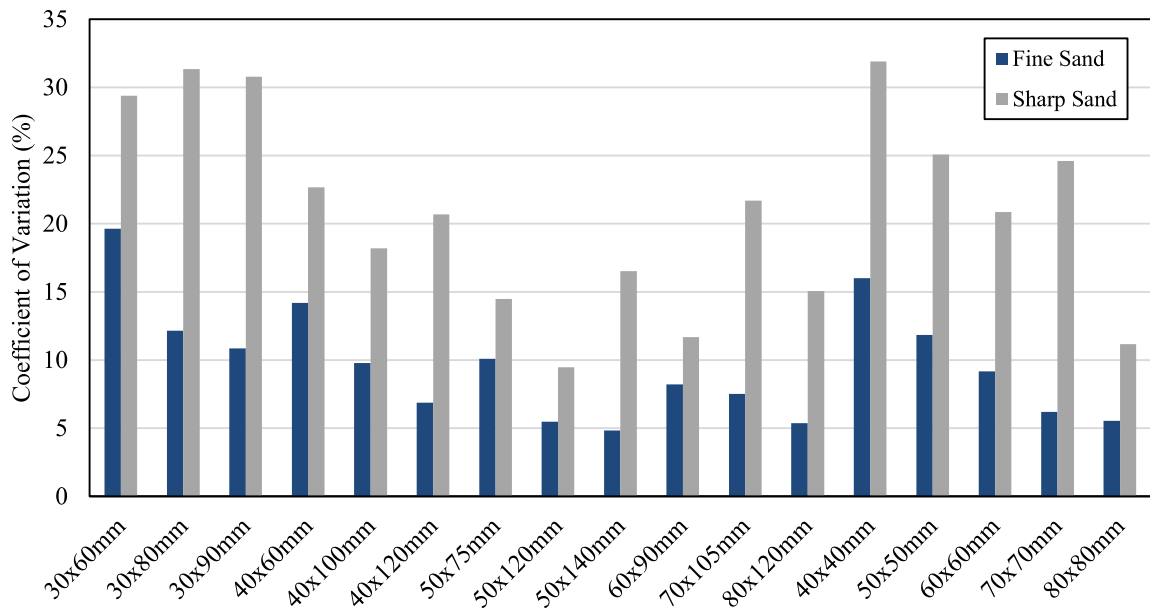


Figure 4.7 Comparison of the coefficient of variation between fine and sharp sand for all plates.

The coefficient of variation is used in the current research to compare the test repeatability for the two tested soils, with each bar in Figure 4.7 representing the average CV value across all controlled sinkage levels (i.e. between 10 mm and 60 mm of sinkage at 10 mm intervals) for each tested plate. From the bar chart of Figure 4.7, it can be clearly noted that the coefficient of variation for sharp sand is significantly higher than for fine sand. Furthermore, dividing the sharp sand CV ratios with the corresponding fine sand CV ratios, the calculated values are spanning between 1.4 (60×90 mm plate) and 4.0 (70×70 mm plate), with 2.3 as an average ratio across all flat plates. Which means that the coefficient of variation in sharp sand is significantly higher (up to 4 times) compared to fine sand, denoting lower level of test repeatability for sharp sand.

Another way of looking at the CV data is illustrated in Figure 4.8, where each dot is the average CV value across all tested plates at the discrete controlled sinkage levels, and the dashed lines are the best fitted negative exponential functions. As can be seen, for both soils the relative dispersion is higher

at shallow sinkage and it decreases in a negative exponential fashion at deeper sinkage levels. This observation is as expected for two reasons; firstly, because the unavoidable experimental errors (caused for instance by frictional forces on the rig's moving parts), would account for greater portion on the total measuring forces when these are still small. As the forces build up at deeper sinkage, the measurement errors have less impact on the overall force magnitude.

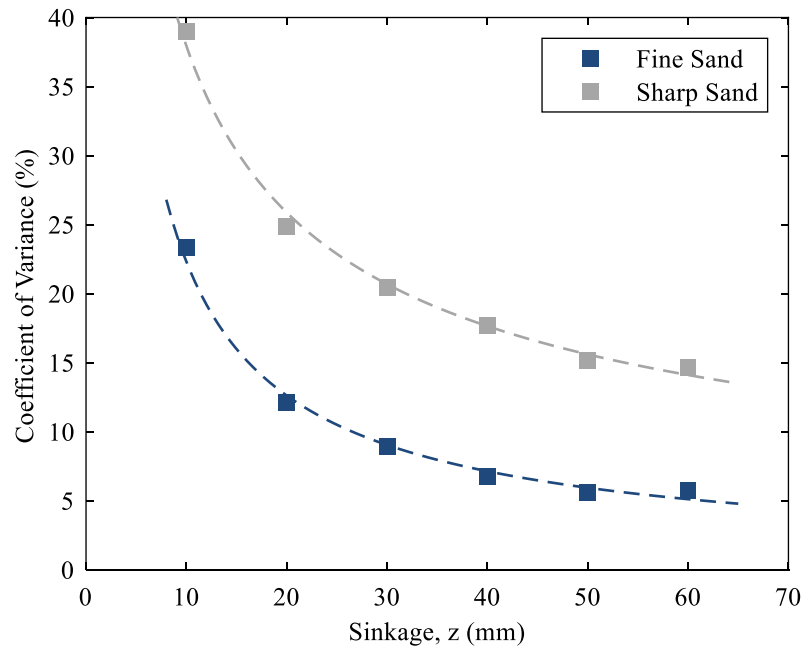


Figure 4.8 Comparison of the coefficient of variation between fine and sharp sand at different sinkage levels.

Secondly, the data dispersion caused by the soil state (e.g. void ratio, moisture content, etc.) would be more pronounced at shallow sinkage where the forces are relatively small. Again, this effect should fade as the total force magnitude increases. However, despite the author's efforts during the experiments, it is practically impossible to attain the same soil conditions at each experimental repetition, especially considering that soils are non-homogeneous materials. As described earlier in this chapter, sharp sand is less homogeneous compared to fine sand, and with inherent moisture content. Due to the latter, the particles of sharp sand tend to stick together under the effect of adhesion, leading to the formation 'pockets' of air and larger void ratio within the soil specimen. The randomness of such behaviour is believed to lead to higher relative dispersion of sharp sand compared to fine sand, with the former being approximately double in magnitude with respect to the latter.

4.2.2 Platewise Fitting

After performing the pressure-sinkage test in all plates of Table 4.1 with the required number of repetitions, the Bekker model discussed in preceding sections was employed to fit the mean values for each individual plate. For the Bekker model in its simple form $P = kz^n$ (Equation 2.22), there are two unknown parameters: the coefficient of proportionality k , and soil exponent n . As described in Section 3.2, to derive these two parameters the test data are plotted in a log-log scale. A linear regression is then fitted where the slope and y-intercept represent the exponent n and coefficient of proportionality k respectively. This method was conceived in the 1960s and the best fitting curve was initially drawn by hand [10]. But in today's plethora of computational resources such methodology seems rather dated.

An alternative approach that continuously gains popularity is to perform data fitting through an optimisation algorithm. Ultimate aim of an optimisation algorithm is to find the 'optimum' solution to a problem through maximising or minimising a function called *objective function* [60]. Usually, the objective function depends on a set of variables called *control variables*. The optimisation algorithm iterates on the values of the control variables until the objective function converges to a maximum or minimum. When an optimisation algorithm is used for a curve fitting problem, such objective function requires two elements: (a) the fitting equation for evaluation and (b) the goodness-of-fit criterion. In the fitting problem concerning this paragraph, the fitting equation is the simple form of Bekker's pressure-sinkage formulation with two control variables (k and n). The least squares method is employed as the goodness-of-fit criterion, where the best fitting curve is obtained when the value of the sum of squares of offsets between the fitting curve and the experimental points is minimum [61].

For the optimisation process, it was decided to use one of the Matlab's build-in algorithms. The commonly used 'fminsearch' function finds the local minimum of an unconstrained multivariable objective function using the Nelder-Mead simplex algorithm described in [62]. The algorithm follows a specific procedure and step sequence for modifying the simplex and converging to the local minimum, with the detailed algorithm procedure described in Matlab documentation [63]. In the context of the platewise fitting concerning this paragraph, the objective function for minimisation is presented by Equation 4.2 below.

$$SSE = \sum_{i=1}^m (P_i - kz_i^n)^2 \quad 4.2$$

The optimisation algorithm was used to fit each individual plate data by finding the appropriate values for Bekker variables k and n . Through the platewise fitting-via-optimisation process, a set of k and n values has been obtained for both fine and sharp sand, and the sum of least squares was used for assessing the fitting quality of each method on the available plate and soils. The radar chart of Figure 4.9 presents a comparison – in terms of sum of least squares – between the classical fitting method (dashed line) and the fitting-via-optimisation (solid line) for each individual plate on fine sand.

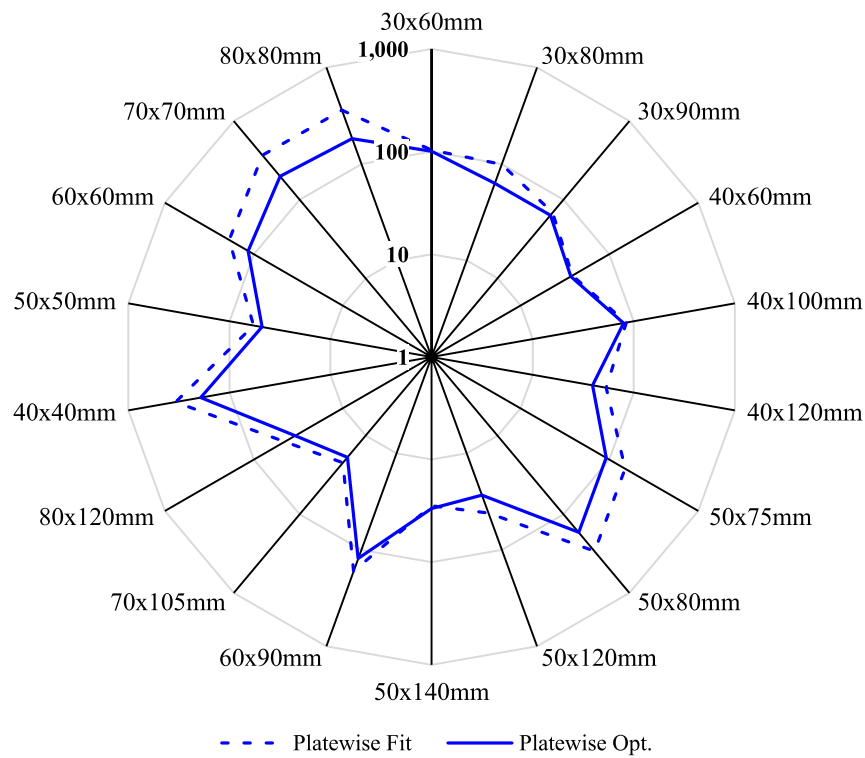


Figure 4.9 Bekker platewise classical fitting vs fitting-via-optimisation. Sum of least squares on fine sand.

Although the fitting-via-optimisation leads to either smaller or equivalent sum of least squares on fine sand, the true merit of fitting-via-optimisation method is better demonstrated when looking at the results of sharp sand. As presented in Figure 4.10, a vast fitting error reduction is observed when the optimisation algorithm is employed. With the sharp sand response being much less linear compared to fine sand, obtaining high quality fitting via the classical method proved problematic.

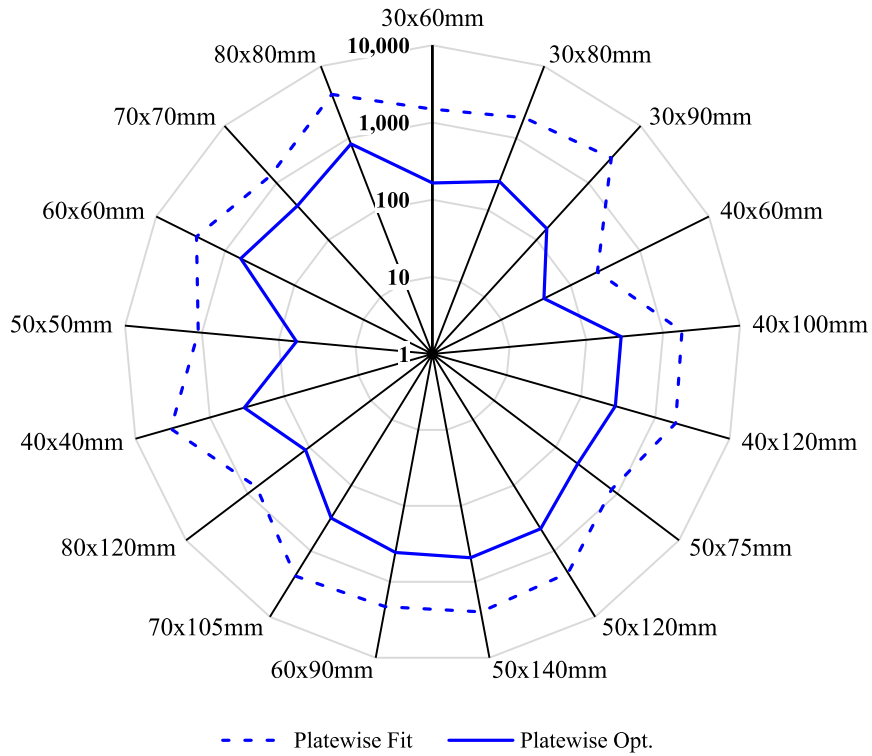


Figure 4.10 Bekker platewise classical fitting vs fitting-via-optimisation. Sum of least squares on sharp sand.

To further highlight the benefit of the optimisation method, two examples are used for visually inspecting the fitting quality. Figures 4.11 and 4.12 present the results of a rectangular (30×80 mm) and a square plate (50×50 mm) respectively; graphs on left-hand-side correspond to fine sand and on right-hand-side to sharp sand. A visual check confirms a good fit to test data on fine sand through both methods as the soil response is relatively linear (soil exponent $n \cong 1.4$). Contrarily, the sharp sand's higher non-linearity (soil exponent $n > 2$) leads to poor fitting with the classical method but very accurate results with optimisation. In all cases the derived curves are within the error bars, however for platewise parameter identification an almost perfect fitting is targeted. The error bars are intended mainly for global process outputs (i.e. common parameters for all plates) which will be discussed in subsequent paragraphs.

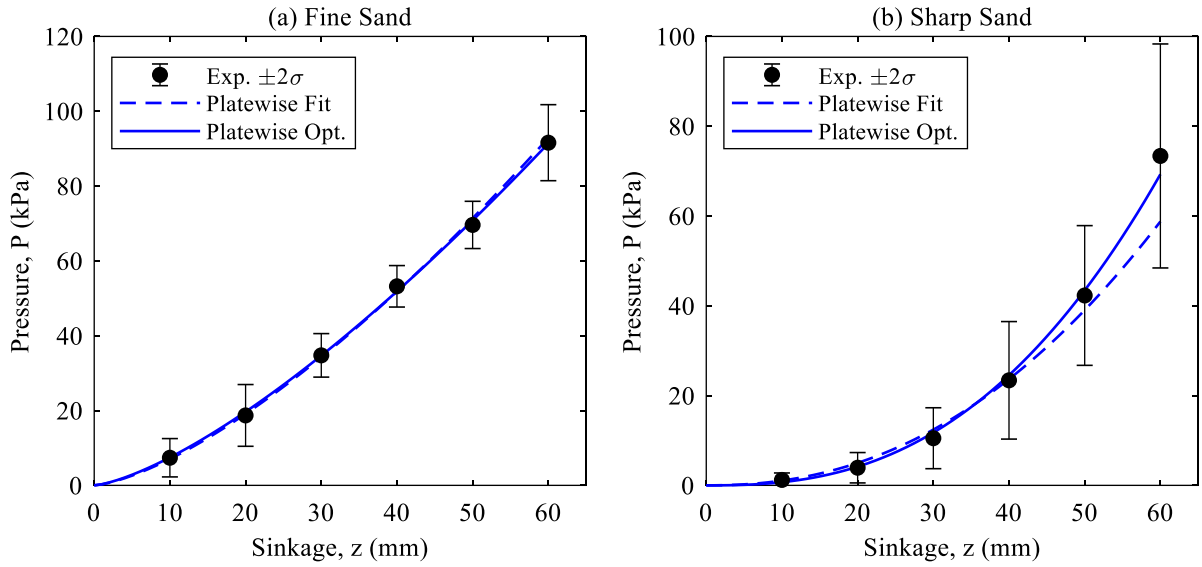


Figure 4.11 Pressure-Sinkage plots for 30×80 mm rectangular plate, comparison between classic fitting and fitting through optimisation on fine and sharp sand.

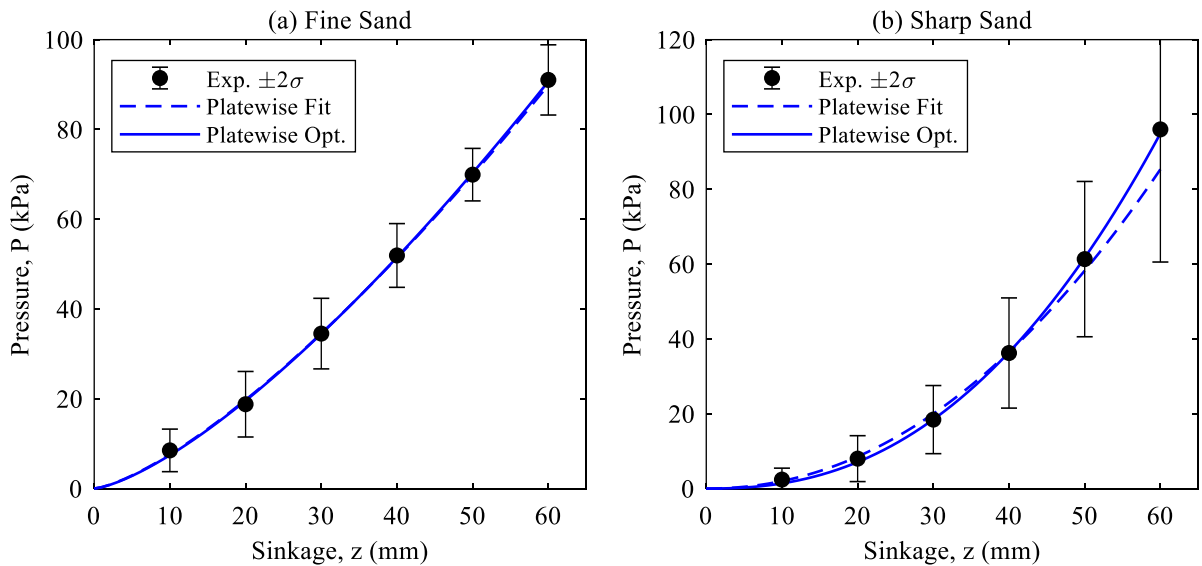


Figure 4.12 Pressure-Sinkage plots for 50×50 mm square plate, comparison between classic fitting and fitting through optimisation on fine and sharp sand.

In conclusion, the fitting-via-optimisation method is proved more accurate and robust in fitting various types of soil responses. As of this point and for the remaining of the current text, unless otherwise stated, when referring to ‘platewise fitting’ it will always correspond to fitting-via-optimisation results on individual plates. Furthermore, the platewise fitting results will be used as a benchmark for accuracy when alternative pressure-sinkage models are evaluated.

4.2.3 Global Fitting

The pressure-sinkage results presented in the previous Section 4.2.2 demonstrated high quality correlation with respect to test data (especially via optimisation) when the fitting is performed for each individual plate. However, such approach is impractical in real-life Terramechanics studies, as it would require to each time measure a plate of approximately the same dimensions as the intended vehicle running gear. In the fully expanded Bekker formulation for pressure-sinkage, which is discussed in Section 2.2.2.1 and repeated here, the coefficient of proportionality is a function of plate width B and two soil constants k_c and k_ϕ . From this point and for the remaining of this text, referring to ‘Bekker global fitting parameters’ would correspond to k_c , k_ϕ and n which are common for each tested soil.

Bekker:

$$P = \left(\frac{k_c}{B} + k_\phi \right) z^n \quad 2.23$$

As discussed in Section 3.2, in the classical method to identify parameters k_c and k_ϕ , it is necessary to perform measurements on two penetration plates of different widths. Then, after identifying the coefficient of proportionality for each of the two plates with widths B_1 and B_2 , a two-unknown two-equation problem is solved [10].

The above methodology assumes a linear interpolation and extrapolation between the two tested plates for the coefficient of cohesion. Additionally, it assumes that soil exponent is constant for a tested soil, and therefore just a simple average between the values of the two tested plates is enough to eliminate measurement and/or fitting errors. However, the in-house experiments and fitting exercise resulted in soil exponent values spanning between 0.9 and 1.5 for fine sand and between 1.8 and 2.8 for sharp sand. In the subsequent paragraphs, the validity of the above assumptions will be investigated by analysing the in-house soil data.

Since a large number of various plate sizes and width/length ratios have been tested by the author, a selection process is required for nominating two of the test plates to serve as the ‘largest’ and ‘smallest’ for calculating the Bekker global parameters. Furthermore, the two plates must meet all criteria listed below:

- a) Their size should allow the evaluation of both interpolation and extrapolation, i.e. can be neither the absolute largest nor absolute smallest plate.

- b) They must have different width in order to apply Equations 3.9 and 3.10.
- c) They must contain complete data set. Some plates due to their size, could not undergo large sinkage due to exceeding the Terramechanics scale rig load cell capacity.

Since multiple plates of equal width have been measured, the selection process would be based on the total area. In Figure 4.13 the plate widths (x-axis) are plotted against their areas (y-axis), where each black dot representing a plate. As per above requirement (c), the three largest area plates have been eliminated as they contain incomplete data sets (crossed out in Figure 4.13). The desired level of sinkage for all tested plates has been set to 60 mm depth. However, during test measurements on these plates, the load cell capacity was reached before achieving the desired sinkage level.

As a first step to satisfy condition (a), the limits in terms of minimum and maximum total plate area are defined and highlighted with the grey dashed lines in Figure 4.13 (the three plates with incomplete datasets are excluded). Regression models are normally applied to the complete dataset (i.e. interpolation only) in order to achieve as good correlation as possible. Extrapolating regression models always involves a level of risk, but in Terramechanics studies it is often impossible to conduct pressure-sinkage measurements to the exact dimensions of the vehicle's running gear due to testing equipment and other practical limitations. Hence, a pressure-sinkage model that can offer an extended extrapolation accuracy would be advantageous. In order to evaluate Bekker's formulation in such conditions, it's been decided to exploit an ambitious target of 40 % of total extrapolation in terms of plate area. The green and blue lines in Figure 4.13 consist the borders between interpolation and extrapolation (ratio of 60/40). Plates with area around the values defined by the green and blue lines will be selected to serve as the 'largest' and 'smallest' plate in Bekker's model. For the 'largest' plate, the 60×90 mm is selected as the only one matching the desired area. Four candidates are available for the 'smallest' plate which all satisfy condition (b), i.e. being different width than the 'largest' plate. The 30×80 mm was selected however as it satisfies two further conditions: (d) its width is not too close to the width of 'largest' plate avoiding potential risk of – due to measurement dispersion – the test data values between the two plates being similar, (e) its width is smaller than the width of 'mean' plate. The purpose of 'mean' plate will be discussed later in the text.

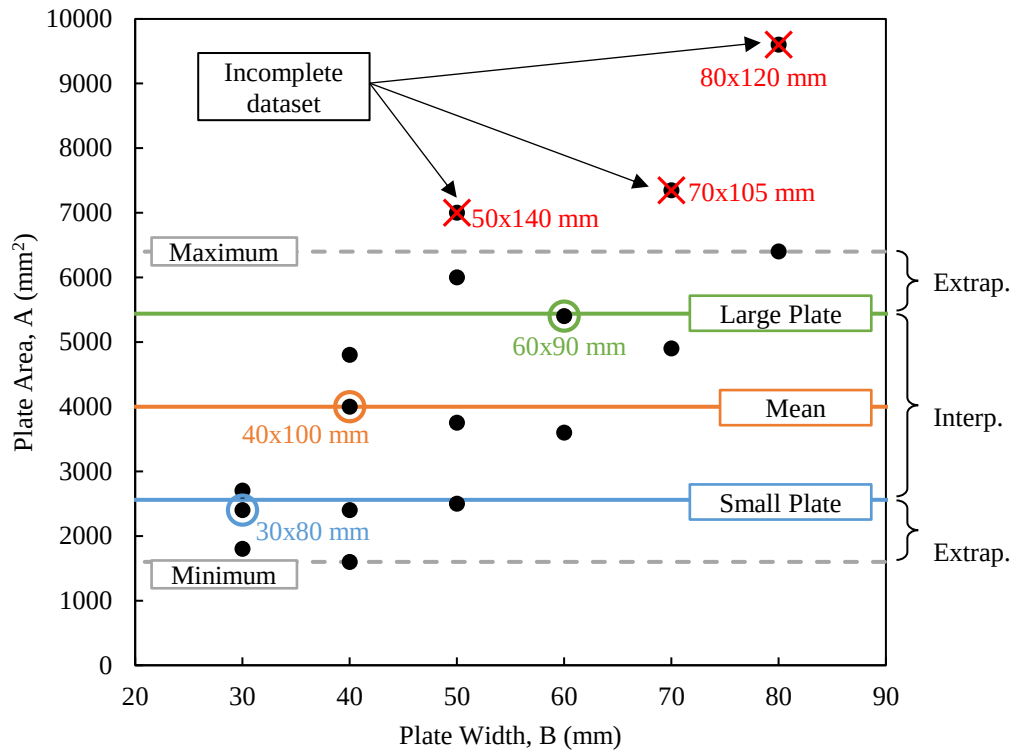


Figure 4.13 Plate selection for Bekker global fitting.

After selecting plates #2 (30×80 mm) and #11 (60×90 mm) of Table 4.1 as the ‘smallest’ and ‘largest’ plates respectively, the Equations 3.9, 3.10 and 3.13 applied for obtaining the global soil parameters for fine and sharp sand. Since the platewise optimisation resulted in more accurate fitting, the coefficients of proportionality k derived via this method are used for calculating k_c and k_ϕ . Similarly, the global soil exponent n was calculated as a simple average of the platewise n values. A summary of the Bekker global fitting parameters for fine and sharp sand is presented in Table 4.2.

	Classical Fitting			Optimisation		
	k_c (kN/m^{n+1})	k_ϕ (kN/m^{n+2})	n (-)	k_c (kN/m^{n+1})	k_ϕ (kN/m^{n+2})	n (-)
Fine sand	95.10	1,501.60	1.27	-70.34	5,134.74	1.23
Sharp sand	4,046.33	-41,861.63	2.25	-1,202.05	64,756.78	2.12

Table 4.2 Bekker global pressure-sinkage parameters for two tested soils.

In addition to the classical global fitting, the same optimisation method applied for the platewise parameter identification can be employed for deriving the global parameters as an alternative approach. While the classical global fitting approach is performed in two steps (1. derive model parameters for two plates of different width, 2. calculate global soil parameters using Equations 3.9, 3.10 and 3.13), the optimisation algorithm can manage to simultaneously fit both plates. In such case, the algorithm iterates on the three parameters (k_c , k_ϕ and n) until the total sum of least squares (for both plates) converges to a minimum. After deploying the optimisation algorithm to the same plates #2 and #11 as the classical fitting, the soil parameters in the rightmost part of Table 4.2 were attained.

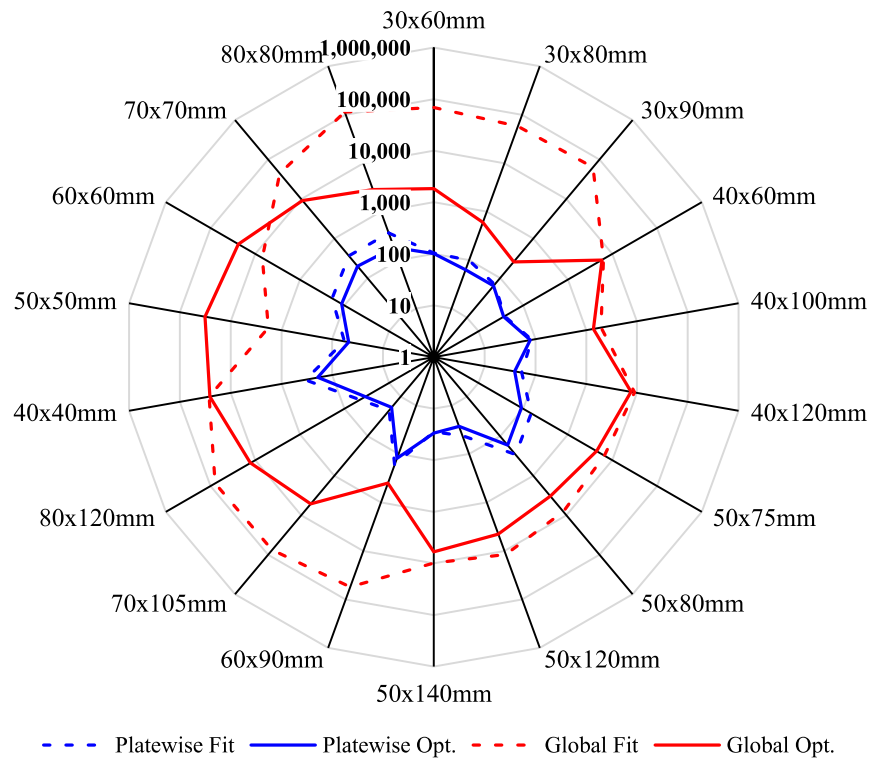


Figure 4.14 Sum of least squares in log axis between experimental data and fitting methods for all tested plates on fine sand.

To assess the global fitting quality of both methods, radar charts are used once again. In Figure 4.14, the sum of least squares for each plate on fine sand are presented, comparing all four methods discussed so far. It has been already established that platewise fitting via optimisation is the most accurate method and therefore is used as a benchmark. Looking into Figure 4.14, it is apparent that global parameters – derived either by classical fitting or optimisation algorithm – lead to far less accurate

correlation in terms of sum of least squares, especially considering the sum of least squares axis is in logarithmic scale. Furthermore, the global fitting via optimisation again shows better correlation compared to the classical approach for the majority of plates.

The sums of least squares is a useful indicator for assessing the fitting quality of a regression model, however it does not consider the statistical variation of test data. In the radar chart of Figure 4.15 the black solid line represents the $\pm 2\sigma$ as an average value across the whole dataset of each plate, i.e. from 10 mm to 60 mm sinkage. The axis scale is constructed in a way to represent a relative error between the fitted curve and the $\pm 2\sigma$ of test data, with 100 % corresponding to $\pm 2\sigma$. In essence, the relative fitting error indicates where the fitted curve lies within the defined acceptable envelope of two standard deviations; a value of 100 % or less means that (on average) the fitted curve lies within the upper and lower envelope of two standard deviations from mean. Consequently, relative error values above 100 % is outside the two standard deviations area. Of course, the smallest the relative error, the more accurate and closest to mean values the regression model is. An important remark about this fitting quality indicator is that it considers an average standard deviation across the evaluation range (i.e. 10 mm to 60 mm sinkage). The measured data have shown that standard deviation can vary with sinkage, with larger sinkage levels having the tendency to demonstrate higher standard deviation values, especially on sharp sand. Relative error to standard deviation is a useful indicator as it provides a fitting quality assessment using a single number, though a visual check of the actual pressure-sinkage graph should always be conducted for confirming the correlation quality.

While evaluating the results of Figure 4.15, it is observed that for fine sand the global fitting through the classical method is in most cases far outside the acceptable limits for standard deviation, and in some cases the relative fitting error approaching 1000 %, i.e. ten times outside the two standard deviations envelope. On the other hand, the global fitting-via-optimisation shows an overall better correlation, with approximately half of the plates being within the two standard deviations range. Similar output is observed on the results of sharp sand as demonstrated in Figure 4.16 for the sum of least squares and Figure 4.17 for the relative fitting error.

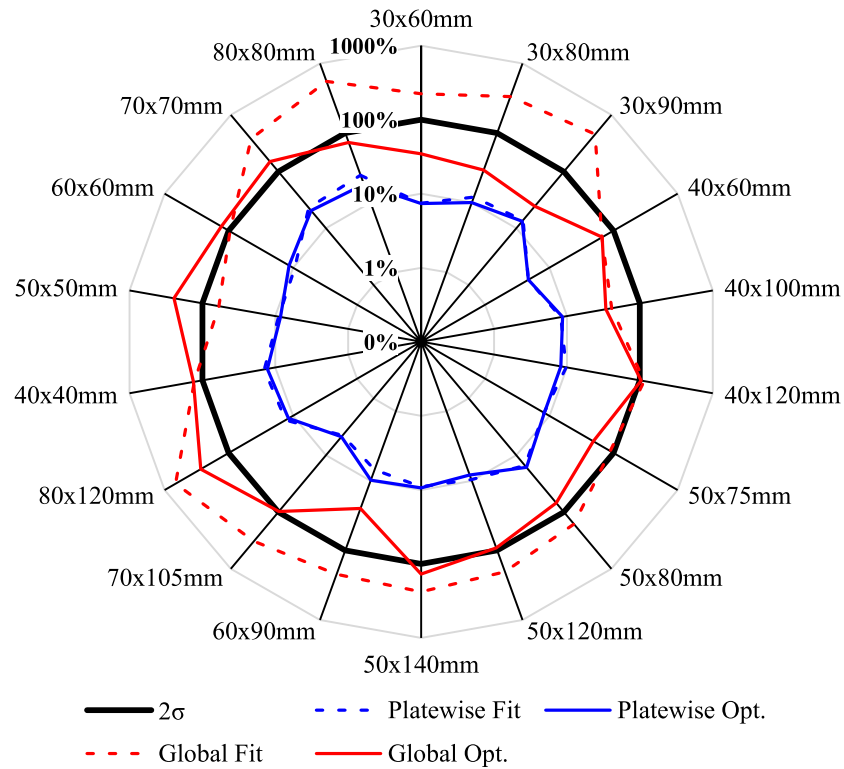


Figure 4.15 Relative error to standard deviation between fine sand experimental data and fitting methods.

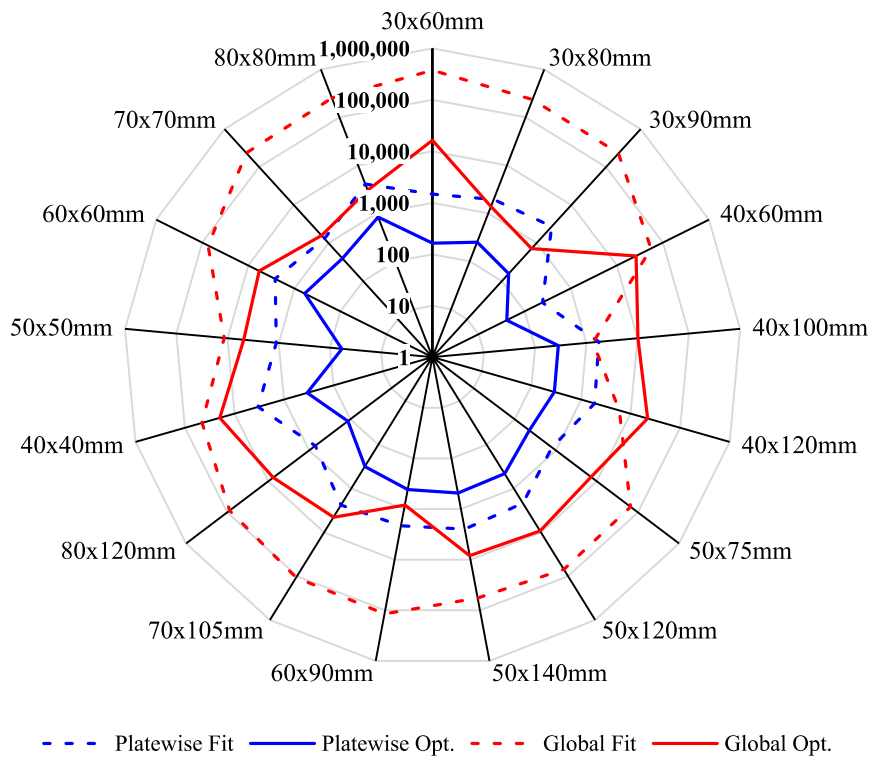


Figure 4.16 Sum of least squares in log axis between experimental data and fitting methods for all tested plates on sharp sand.

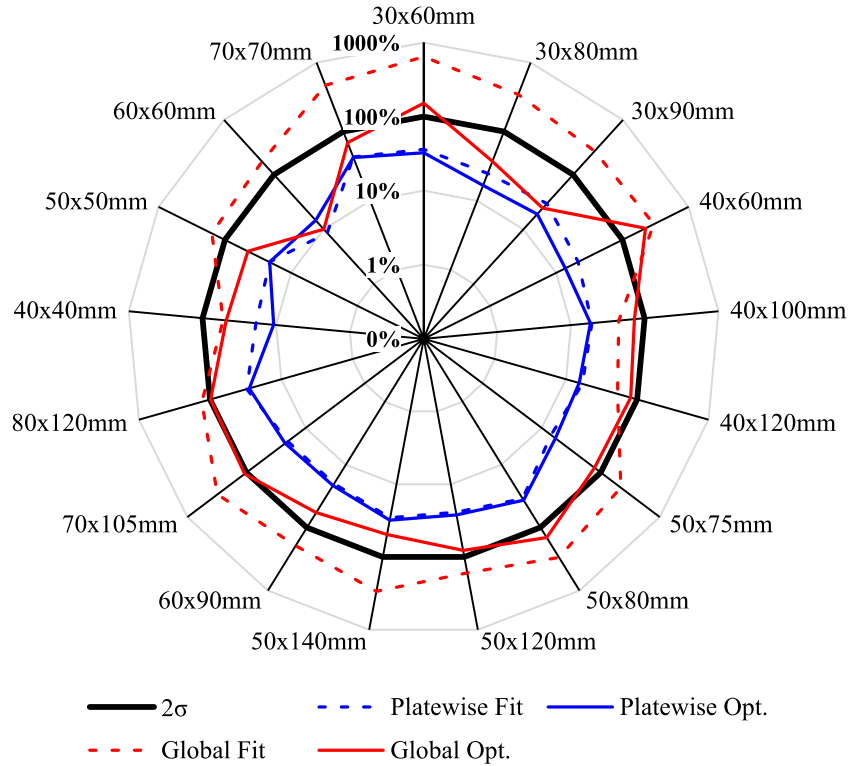


Figure 4.17 Relative error to standard deviation between sharp sand experimental data and fitting methods.

In order to understand the reasons behind the lack of accuracy of classical global fitting, the in-house experimental data are further analysed. In previous paragraphs three plates were selected namely ‘small’, ‘medium’ and ‘large’ corresponding to 30×80 mm, 40×100 mm and 60×90 mm plate dimensions respectively, with the ‘small’ and ‘large’ plates being used for identifying Bekker’s global parameters. In Figure 4.18 graphs (a) & (b) the pressure-sinkage experimental data are plotted for the three plates and, not surprisingly, the data indicate an increase of pressure level with increasing plate size for both fine and sharp sand. It is logical to expect that at a constant pressure value – for instance – a large plate would sink less into the soil compared to a small plate. Equally, at a given sinkage level a large plate would experience higher pressure values compared to a small one.

If the Bekker’s pressure-sinkage model is considered ($P = kz^n$, where $k = k_c/B + k_\phi$), then capturing the experimental data tendency would require the coefficient of proportionality to increase with increased plate dimensions. For this to happen, the following conditions must be met:

- (I) $k_c < 0$ and $k_\phi > 0$ in order for $k \uparrow$ when $B \uparrow$
- (II) $|k_c| \ll |k_\phi|$ in order for $k > 0$ at all times, even when $B \rightarrow 0$

However, as summarised in Table 4.2, the values for k_c and k_ϕ derived from the classical global fitting methodology do not satisfy condition (I). For fine sand, both parameters are positive numbers, while for sharp sand they have the opposite sign from what is required (i.e. $k_c > 0$ and $k_\phi < 0$). Using these parameter values results in coefficient k which is decreasing with increasing plate dimensions as shown in Figure 4.18 graphs (c) & (d); the exact opposite direction to experimental data. A range of values for k_c and k_ϕ are found in the literature ([2], [10], [64]), in the vast majority of which both parameters are usually positive numbers, or in rare occasions $k_c > 0$ and $k_\phi < 0$ [64]. Therefore, they do not meet condition (I), while condition (II) is satisfied in the majority of cases. Hence, there is an alignment between the literature and the author regarding the general direction for $k_c > 0$ and $k_\phi < 0$ values when the classical global fitting is used. Based on these findings the following conclusions can be drawn: (i) the classical global fitting approach cannot be successfully applied into test plates with large dimensional variation (i.e. only small relative size difference between the ‘small’ and ‘large’ test plates is possible), (ii) the derived output parameters cannot accurately predict the soil response under the effect of plates outside the tested range (i.e. no accurate extrapolation). The classical global fitting methodology imposes constraints and limits the flexibility and applicability of real-life Terramechanics studies, as it requires to perform pressure-sinkage measurements with test plates very close to the intended running gear.

On the other hand, the values for k_c and k_ϕ derived through the optimisation algorithm (Table 4.2) satisfy both conditions (I) and (II), and result in coefficients k which are increasing with increasing plate width. As shown in Figure 4.18 graphs (e) and (f) the pressure direction is aligned to the experimental data, and it is worth mentioning that this output was achieved using the ‘fminsearch’ Matlab function which does not impose any type of constraints to the control variables. This acts as a confirmation that the optimisation algorithm can successfully converge to outputs which make physical sense when the mathematical problem is well defined. Furthermore, a greater level of interpolation and extrapolation accuracy can be achieved through global fitting-via-optimisation, as demonstrated by the radio charts previously shown.

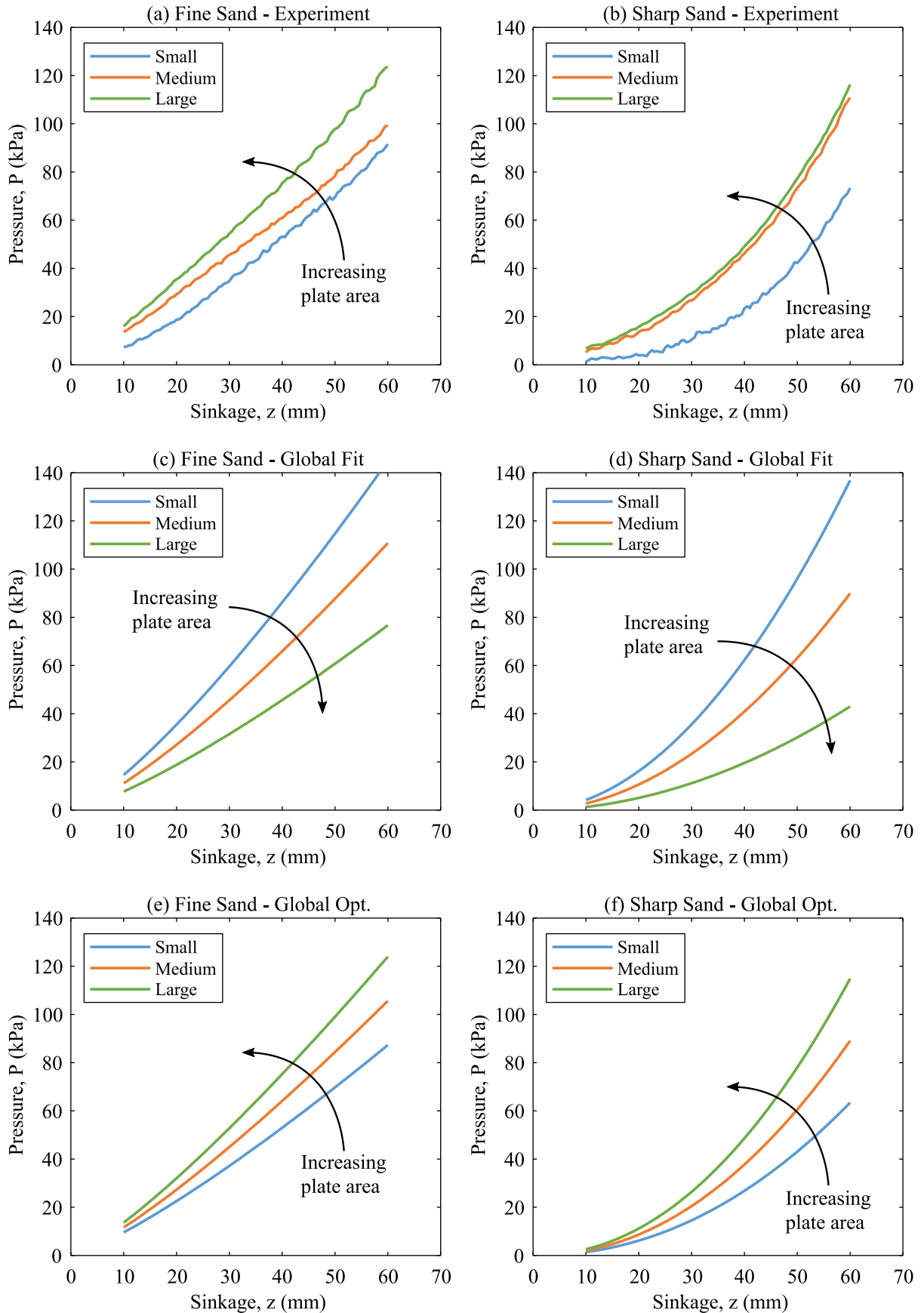


Figure 4.18 Pressure-sinkage comparison between three different sized plates of in-house experimental data on fine and sharp sand (a) & (b), global classical fitting (c) & (d), global optimisation (e) & (f).

In addition to radio charts, a sample of pressure-sinkage diagrams as shown by the Figures 4.19 and 4.20 provide a visual confirmation to the above conclusions for fine and sharp sand respectively. In both Figures 4.19 and 4.20, graphs (a) and (b) illustrate the fitting quality to experimental data for the ‘smallest’ and ‘largest’ plates used for the global fitting and global optimisation. Graphs (c) and (d) present two samples of interpolated plate results, and graphs (e) and (f) two samples of extrapolated plate results. For both soil types the global fitting leads to poor correlation even with the ‘smallest’ and ‘largest’ plates, which were originally used to perform the fitting, while the error further increases when looking at the extrapolated plates. The global fitting-via-optimisation results provide a reasonable overall correlation, while the platewise correlations (especially via optimisation) capture very accurately the soil behaviour of all plates.

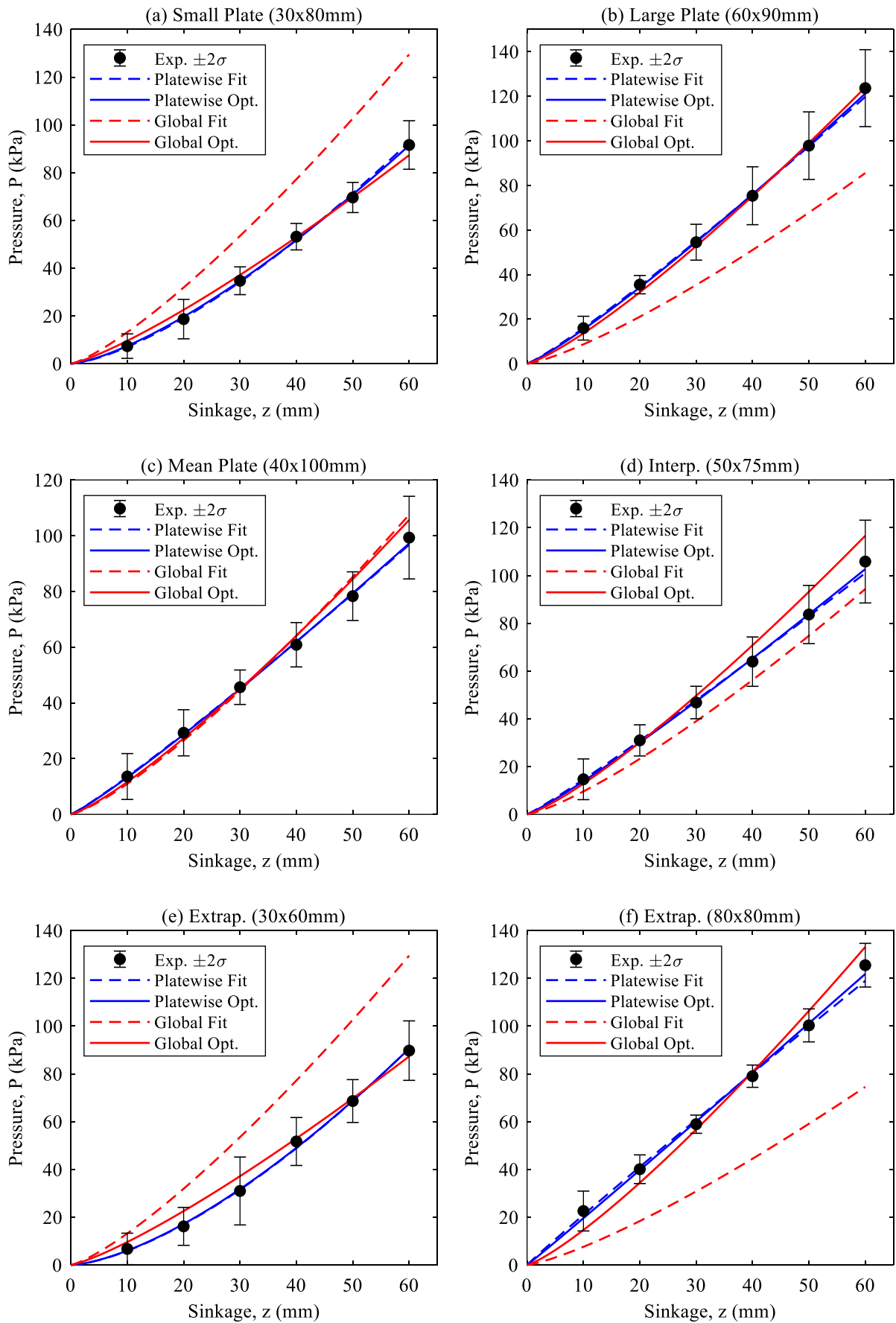


Figure 4.19 Pressure-sinkage fitting comparison for various plates on fine sand.

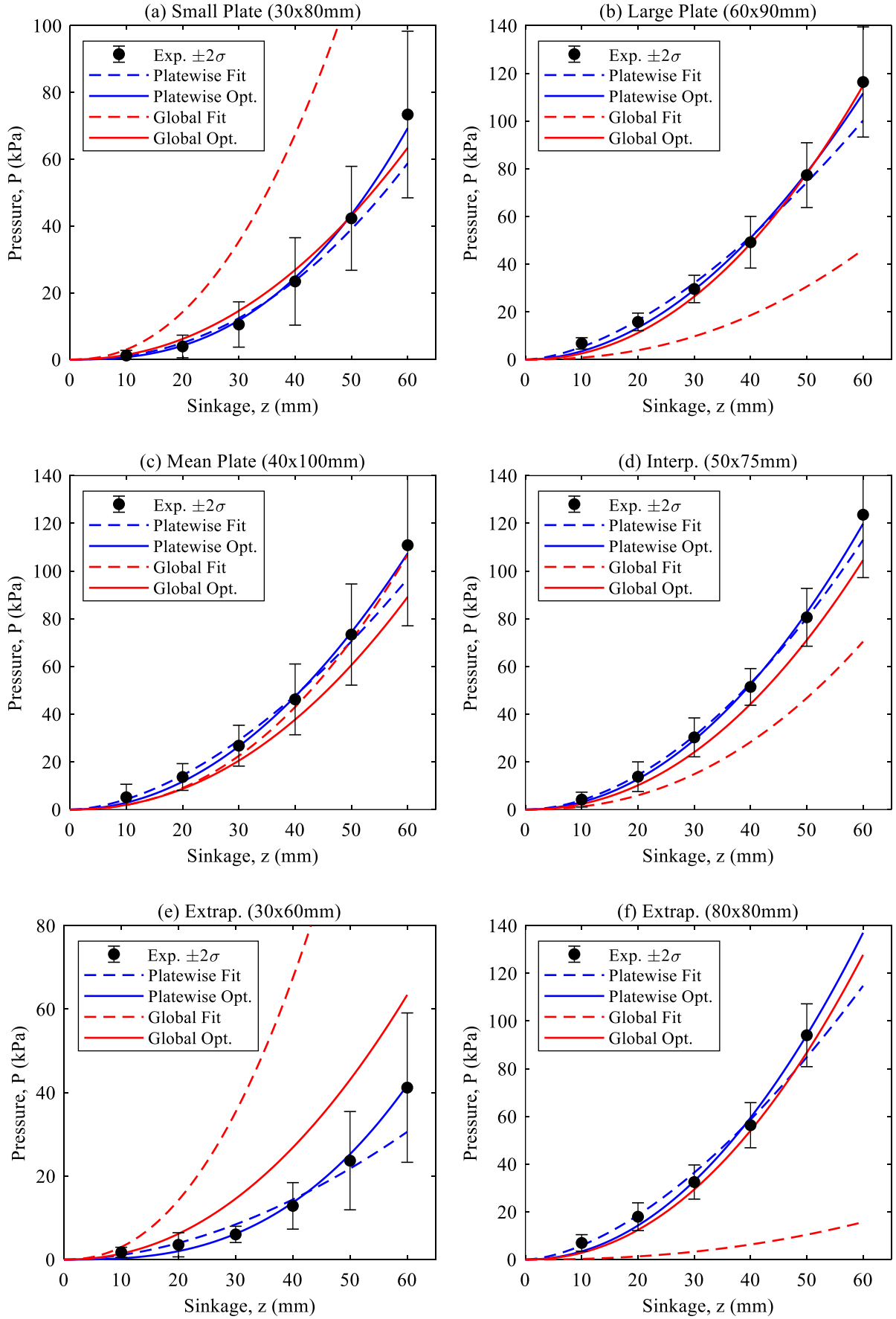


Figure 4.20 Pressure-sinkage fitting comparison for various plates on sharp sand.

4.2.4 New Pressure-Sinkage Model

As presented in the previous section, attempting to derive the Bekker global parameters through the classical fitting methodology leads to poor pressure-sinkage correlation. Consequently, this methodology is unsuitable for Terramechanics studies when relatively large interpolation and/or extrapolation is expected. On the other hand, a reasonable correlation can be achieved when utilising an optimisation algorithm for deriving the global parameters. In spite that, Bekker's assumption that plates of equal width but different length will undergo the same level of sinkage under the effect of the same vertical load is counterintuitive.

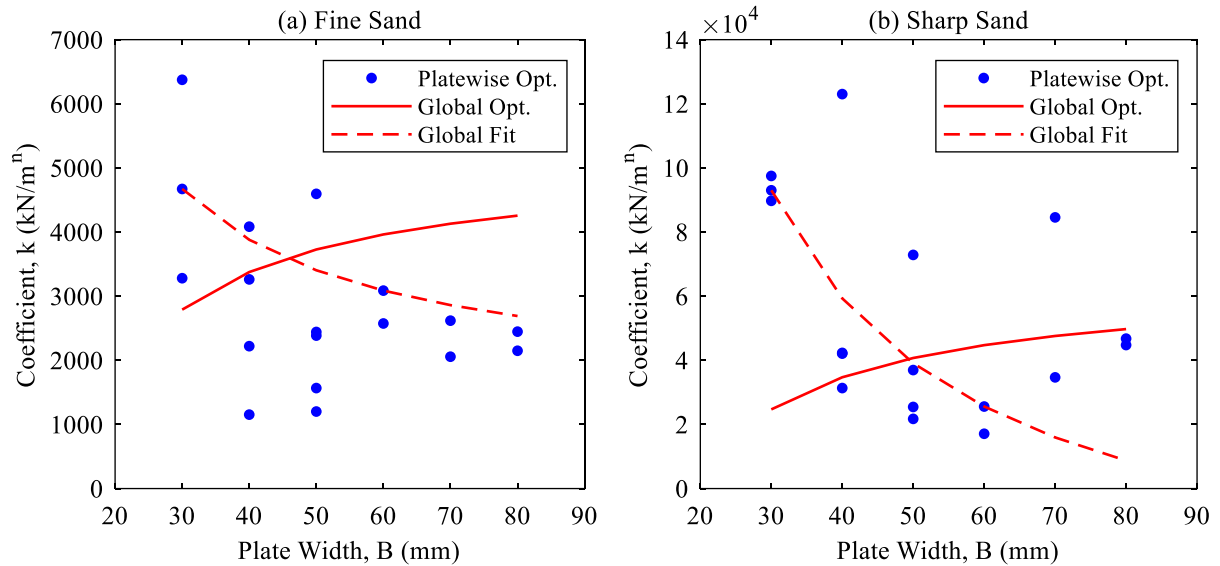


Figure 4.21 Bekker's coefficient of proportionality calculated through various methods for fine (a) and sharp sand (b).

In Figure 4.21 the coefficient of proportionality values for each plate are presented as blue dots for both fine and sharp sand. These values are derived through the platewise optimisation algorithm, with the red lines relate to the equivalent k when the global parameters of Table 4.2 are used. Observing graphs (a) and (b) in Figure 4.21, no resemblance can be seen between any of the three data sources. Similar results are presented in Figure 4.22 for the soil exponent, but in addition, large dispersion on values of n appears for plates of even the same width which leads to the conclusion that n is not a constant, as suggested by Bekker, but perhaps a variable related to plate dimensions and/or other factors.

Even though no recognisable pattern can be identified when the platewise scatter values are plotted against plate width, there is no doubt that other factors – in addition to plate width – are contributing to the overall soil response. An investigation and analysis of such factors is conducted in the succeeding paragraphs along with a proposal of a new pressure-sinkage formulation which provides an improvement over Bekker.

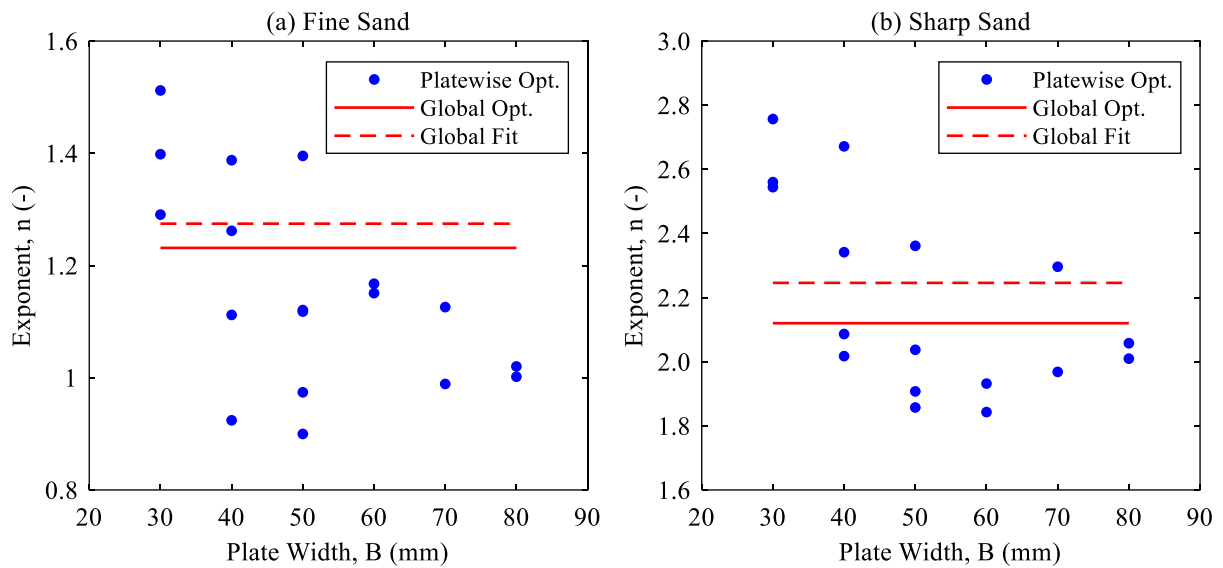


Figure 4.22 Bekker's soil exponent calculated through various methods for fine (a) and sharp sand (b).

A first candidate for further investigating the platewise parameter dispersion is the plate length. When plates with equal width but different length are sunk into the soil, it is logical to assume that – due to difference in the total contact area – the pressure-sinkage response would vary. In order to explore the effect of plate length, groups of equally wide plates are required. From the list of tested plates, three groups of 30 mm, 40 mm and 50 mm width are formed, with each group containing at least three plates in order to allow the study of potentially non-linear relationships.

When looking into Figures 4.21 and 4.22 for the coefficient of proportionality and soil exponent of the two tested soils, no clear indication on the influence of the plate length can be observed. Interestingly however, when plotting the coefficient of proportionality normalised with plate length versus the plate length, a clear parameter linearity is present. In Figure 4.23 below, the same colour dots represent the normalised coefficients of proportionality (k/L) for fine sand of equally wide plates, where

the dashed lines of the same colour correspond to the linear regressions. So, for each plate width there is a linear relationship between k/L and L in the form of $y = ax + b$. By substituting y with k/L and x with L , the author can express the coefficient of proportionality by Equation 4.3. Similar output is observed for sharp sand as shown by Figure 4.24.

$$k = aL^2 + bL \quad 4.3$$

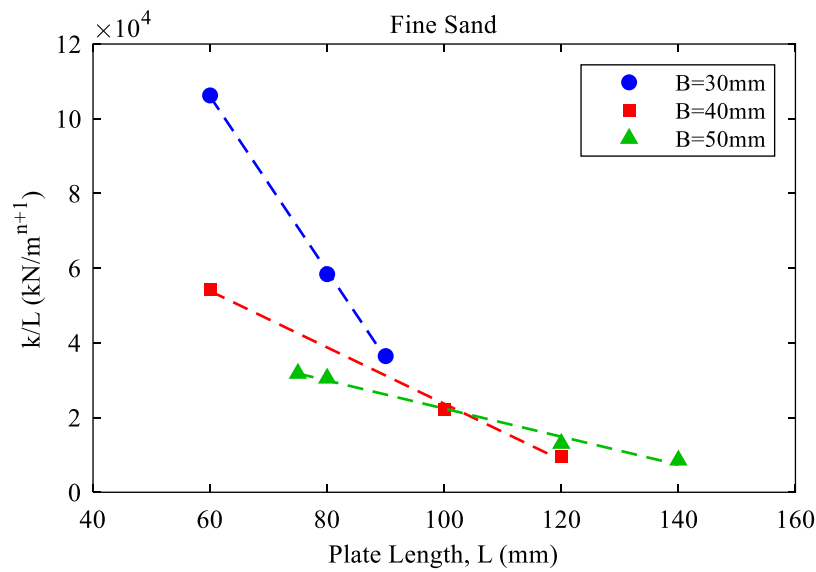


Figure 4.23 Normalised coefficient of proportionality with length vs plate length on fine sand.

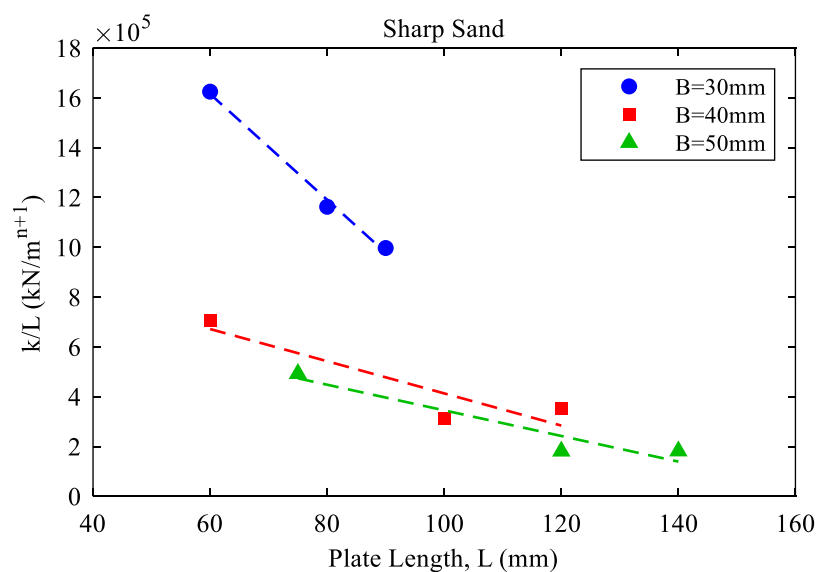


Figure 4.24 Normalised coefficient of proportionality with length vs plate length on sharp sand.

Equation 4.3 expresses the relationship between parameter k and length L for a specific plate width and has the form of a second order polynomial. It is also apparent that for each plate width, the linear regression results in different slope and y-intercept, which means that coefficients a and b are not constant but dependent on plate width. These observations confirm the dependency of k on both plate length and width.

In addition to the effect of plate length, the influence of the total plate area to the pressure-sinkage relationship is investigated. In the plots presented in Figures 4.25 and 4.26, where the experimental pressure (y-axis) is plotted against plate area (x-axis). Each data point represents the pressure at a particular sinkage level for a specific plate area. The data points of the same colour and shape correspond to the pressure exerted on the test plates at a particular sinkage level, e.g. each blue circle corresponds to the pressure at 10 mm depth for each plate plotted versus its total area.

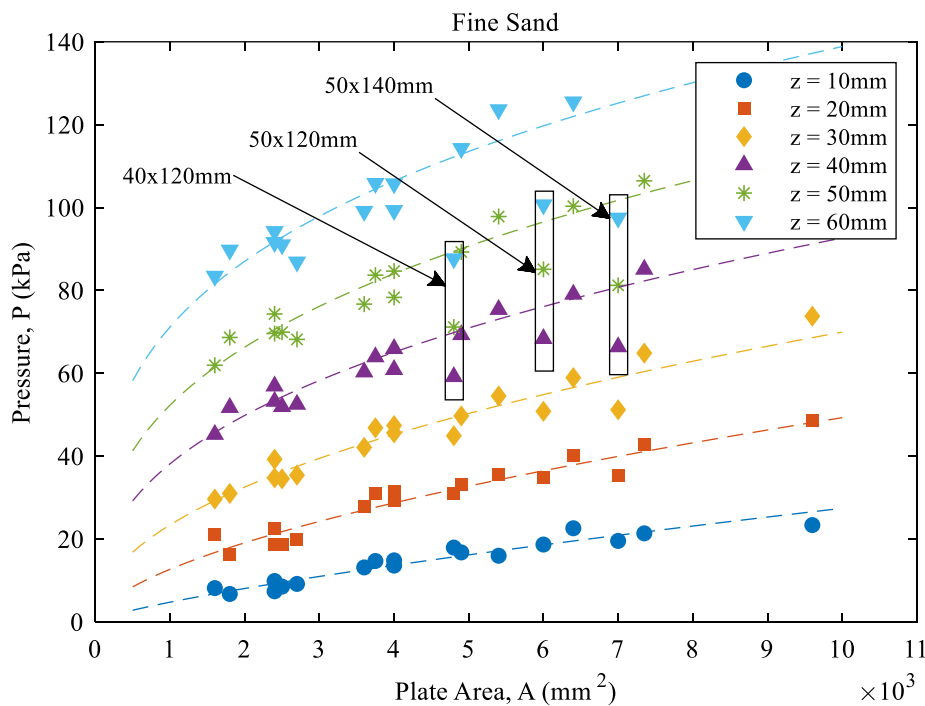


Figure 4.25 Pressure vs plate area at various sinkage levels on fine sand.

As shown by the dashed lines, there is a clear trend between pressure and total plate area in both tested soils. Such observation suggests that there is a dependency between pressure and plate area. There are three exceptions to this trend on fine sand, where the plates with dimensions 40×120 mm,

50×120 mm and 50×140 mm are outliers at sinkage levels beyond 30 mm (highlighted by the rectangular boxes in Figure 4.25). The responses of these plates follows a different pattern as they sink deeper into the soil. While the remaining of the plates follow an ascending pressure curve ($n > 1$), the three plates follow a descending shape ($n < 1$). The reasons for such behaviour, which occurs only on fine sand data, are unclear. A few possibilities however are considered: a) data acquisition errors during the experiments, b) plate/soil interactions with the rig due to large plate lengths, c) actual physical phenomenon linked to the length/width ratios which, for the three plates, are the largest among all tested plates. A combination of the above potential reasons, or an alternative explanation is also possible.

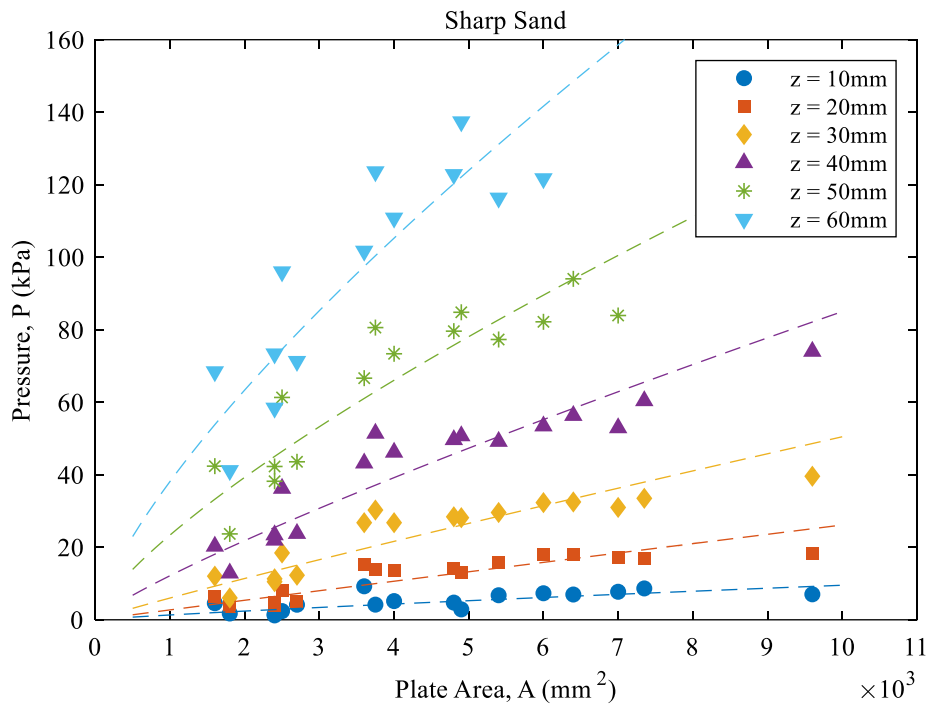


Figure 4.26 Pressure vs plate area at various sinkage levels on sharp sand.

Considering the plate dependency findings the alternative plot of Figure 4.27 was created, where k is normalised and plotted against plate area for the same three groups of plates presented in Figure 4.23. The results of Figure 4.27 (a) are similar to the ones in Figure 4.23, i.e. a linear relationship between k/A and A for a fixed plate width. However, looking more closely it becomes apparent that a common exponential function would fit reasonably well the data of all plate sizes simultaneously. In Figure 4.27 (b), the k/A data of measured plates with $B = 30$ mm, $B = 40$ mm and $B = 50$ mm are plotted

against A (dotted data) on fine sand. One can see that an exponential function of the form $y = ax^b$, illustrated by the black dashed line, fits well the fine sand data across the whole range. Likewise, the results of sharp sand shown in Figure 4.28 exhibit the same tendency. When substituting y with k/A and x with A , an area-dependent formulation is created for the coefficient of proportionality as expressed by Equation 4.4 below, where a_k and b_k are soil constants.

$$k = a_k A^{b_k + 1}$$

4.4

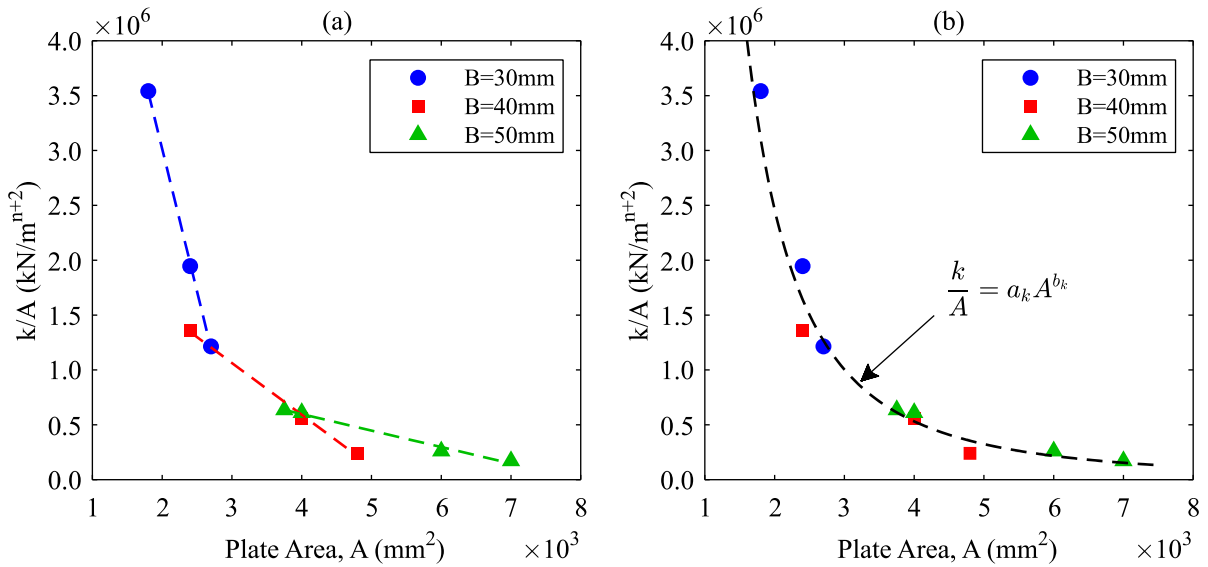


Figure 4.27 Coefficient of proportionality normalised and plotted against plate area for plates of 30 mm, 40 mm, and 50 mm width on fine sand.

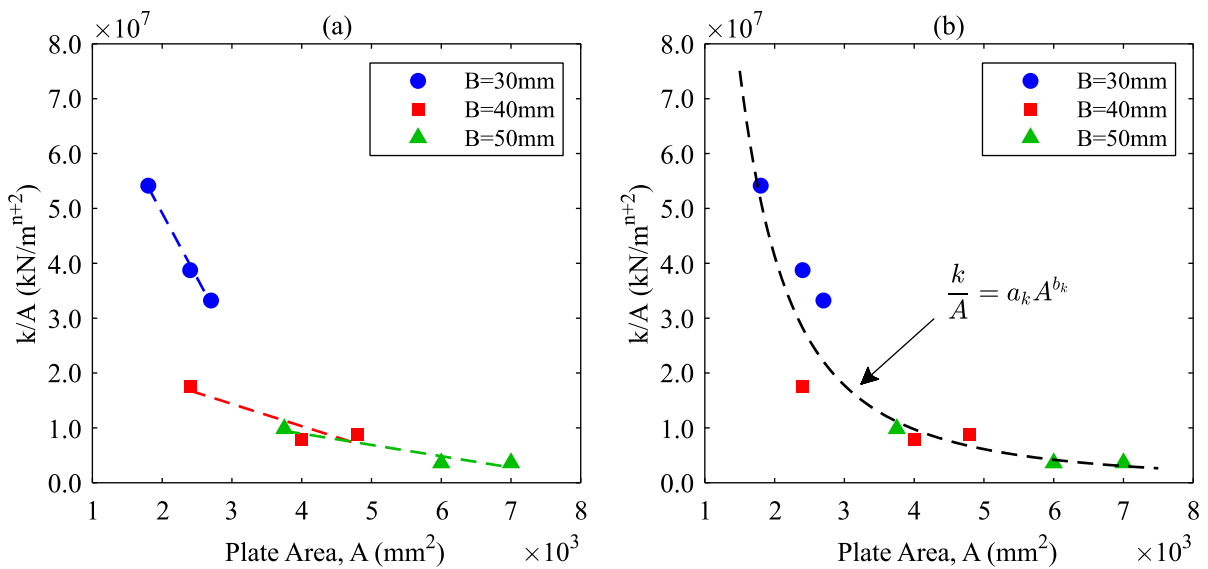


Figure 4.28 Coefficient of proportionality normalised and plotted against plate area for plates of 30 mm, 40 mm, and 50 mm width on sharp sand.

In a similar fashion, soil exponent n is normalised and plotted against plate areas, with the platewise parameters of $B = 30$ mm, $B = 40$ mm and $B = 50$ mm plates presented in Figure 4.29. Likewise to coefficient of proportionality, an exponential function fits well all data simultaneously as shown by the black dashed line of. Figure 4.29 (b) Therefore, the soil exponent can also be expressed as a function of plate area by an exponential equation with two coefficients (Equation 4.5).

$$n = a_n A^{b_n+1}$$

4.5

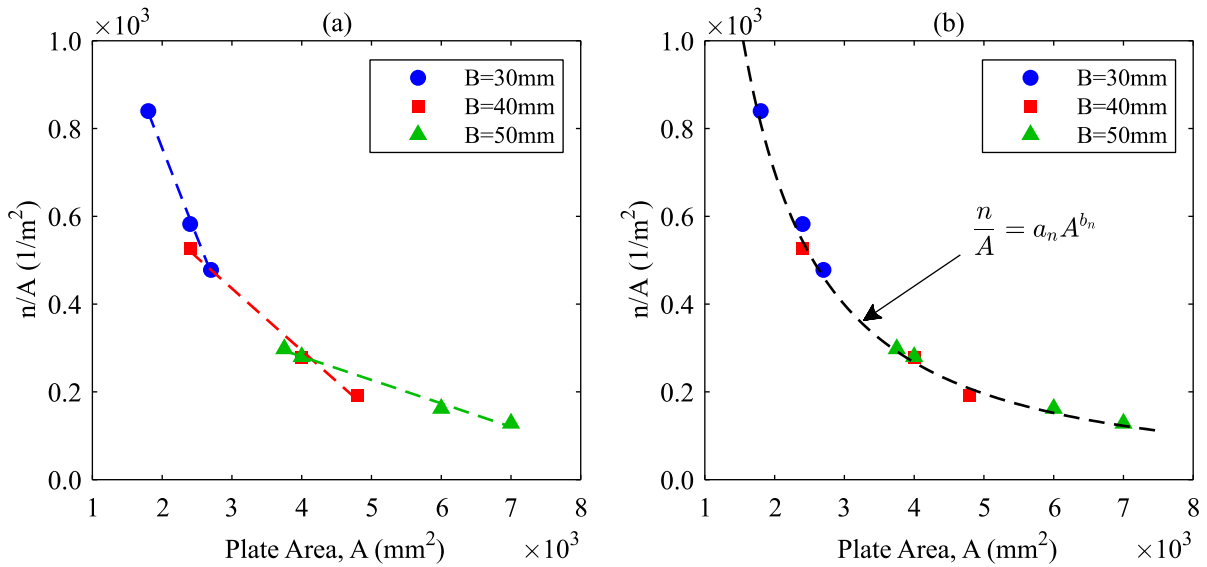


Figure 4.29 Soil exponent normalised and plotted against plate area for plates of 30 mm, 40 mm, and 50 mm width (fine sand).

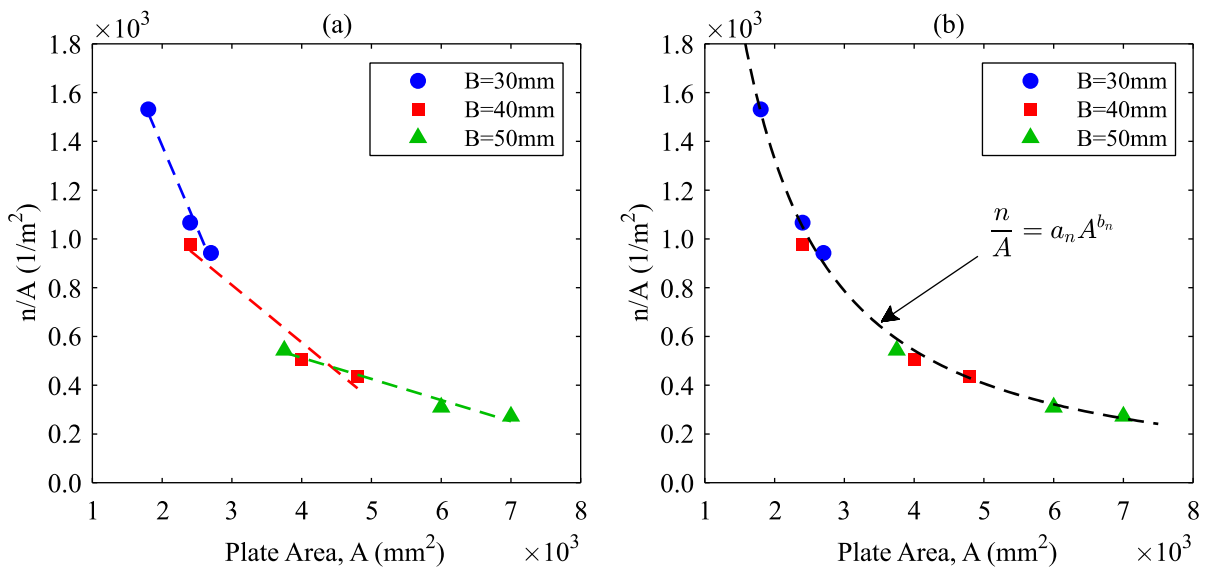


Figure 4.30 Soil exponent normalised and plotted against plate area for plates of 30 mm, 40 mm, and 50 mm width (sharp sand).

Combining the simple form of Bekker's pressure-sinkage from Equation 2.22 with the exponential relationships of Equations 4.4 and 4.5 for coefficient of proportionality and soil exponent respectively, an alternative Area-Dependent Pressure-Sinkage (ADPS) formulation is obtained. In this new model, which is expressed by Equation 4.6, the pressure-sinkage relationship is a function of plate area irrespective of length/width ratio.

$$\text{Bekker} \quad P = \left(\frac{k_c}{B} + k_\phi \right) z^n \quad 2.23$$

$$\text{ADPS model:} \quad P = \left[a_k A^{(b_k+1)} \right] z^{\left[a_n A^{(b_n+1)} \right]} \quad 4.6$$

In order to identify the soil parameters for the proposed pressure-sinkage model of Equation 4.6 for both fine and sharp sand, the optimisation algorithm is utilised once again. In preceding paragraphs, three plates were selected to serve as 'small', 'medium' and 'large' for the fitting activities. For the Bekker formulation only the 'small' and 'large' plates are needed. For the new model, due to the observed non-linear relationship between pressure and plate area, all three plates are necessary. As a goodness-of-fit criterion for the algorithm, the sums of least squares for all three plates is used, with its mathematical expression presented below by Equation 4.7. The standard unconstrained Matlab algorithm 'fminsearch' is used here as well for the parameter identification. Running the algorithm for the available soil data, two sets of parameters derived and they are summarised in Table 4.3.

$$SSE = \sum_{j=1}^3 \left\{ \sum_{i=1}^m \left[P_{i,j} - \left(a_k A_j^{b_k+1} \right) z_{i,j}^{(a_n A_j^{b_n+1})} \right]^2 \right\} \quad 4.7$$

	a_k (kN/m ⁿ⁺⁴)	b_k (-)	a_n (1/m ⁴)	b_n (-)
Fine Sand	2,376.7	-1.044	0.587	-1.129
Sharp Sand	1,273.5	-1.615	0.686	-1.204

Table 4.3 Area-Dependent Pressure-Sinkage (ADPS) model parameters derived through optimisation algorithm for two soil types.

After completing the optimisation for the three plates on ADPS, the same parameters are applied to all in-house measurements for both fine and sharp sand. In the bar chart of Figure 4.31 all pressure-sinkage methods discussed in the current chapter are summarised in terms of sum of square errors. It has already been established that the best Bekker global parameters are derived through the optimisation method. However, the correlation of newly developed ADPS to test data demonstrates a significant improvement over Bekker's global optimisation results, with the sum of square errors (mean across all plates) of ADPS being less than half compared to Bekker. In addition, looking at the bar chart Figure 4.32, the mean relative error between ADPS and two standard deviations is 58 % and 48 % for fine and sharp sand respectively (100 % corresponds to two standard deviations), while the equivalent values for Bekker global optimisation is almost double at 98 % and 81 %.

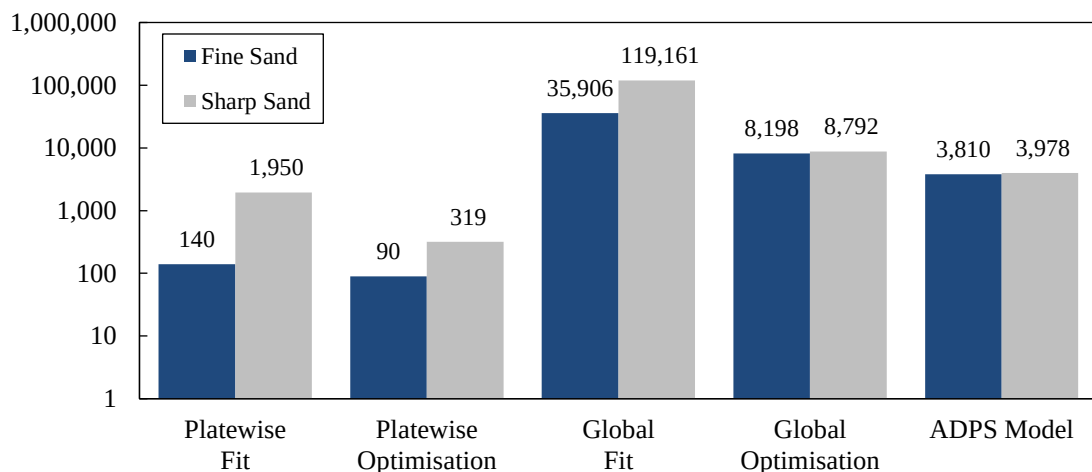


Figure 4.31 Sum of square errors, mean value across plates for each fitting method and soil type.

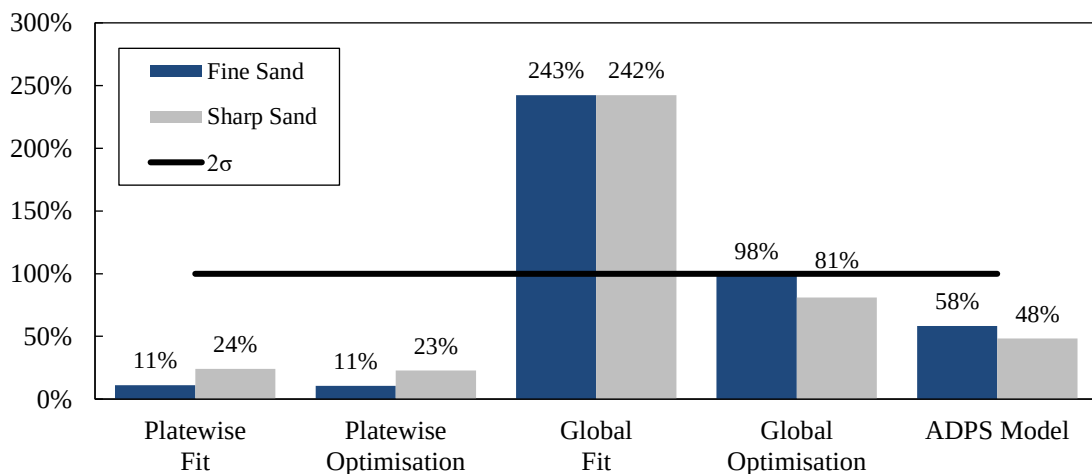


Figure 4.32 Relative difference between the fitting method and the test data 2σ .

The bar charts of Figures 4.31 and 4.32 present the average correlation across all tested plates. To further demonstrate the merit of ADPS and its greater correlation level, radar charts have been created which compare the three optimisation methods (Bekker platewise, Bekker global, ADPS) against test data. Figure 4.33 illustrates the sum of least squares of the three models for each penetration plate on fine sand. As it can be seen, the sum of square error values of ADPS (green line) are smaller than Bekker's global optimisation results (red line) for the majority of cases; especially on the square plates where the gap between the two is substantial. Furthermore, when examining the relative error to standard deviation as presented by the radar chart of Figure 4.34, it is again observed that ADPS is closer to the pressure-sinkage mean values and in all but three cases within the two standard deviation limits. The three outliers correspond to the same plates discussed earlier in the text that they do not follow the same ascending force behaviour at higher sinkage levels. Similar results can be observed for sharp sand when analysing the radar charts of Figures 4.35 and 4.36 for the sum of least squares and relative error to standard deviation respectively. So, in conclusion, the newly developed ADPS model provides improved correlation to test data compared to Bekker's formulation even at largely extrapolated plate sizes.

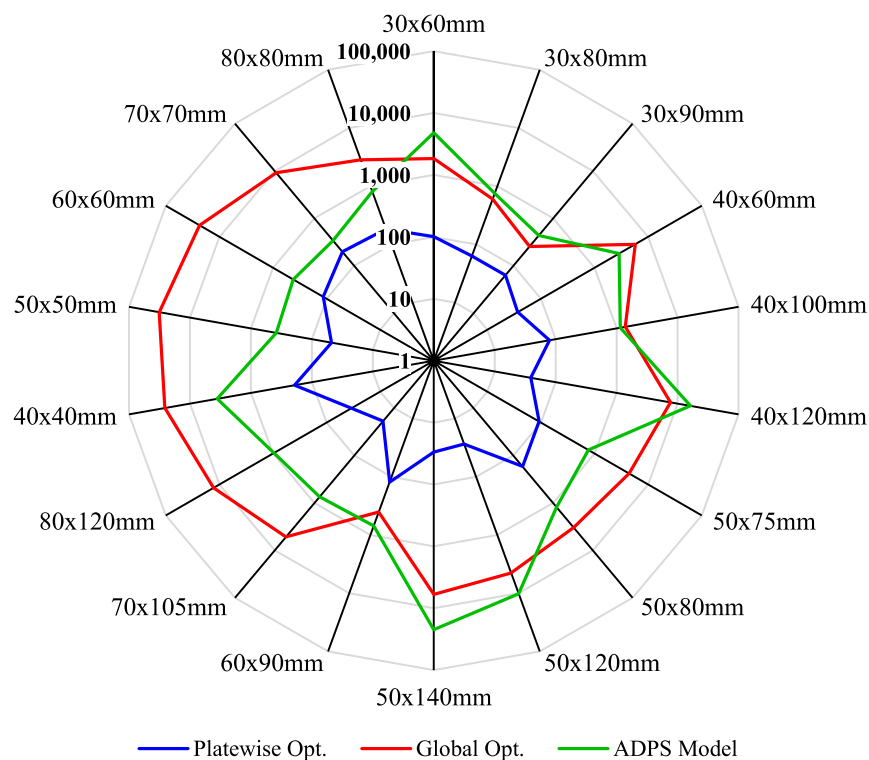


Figure 4.33 Sum of least squares between test data and fitting methods for all tested plates on fine sand.

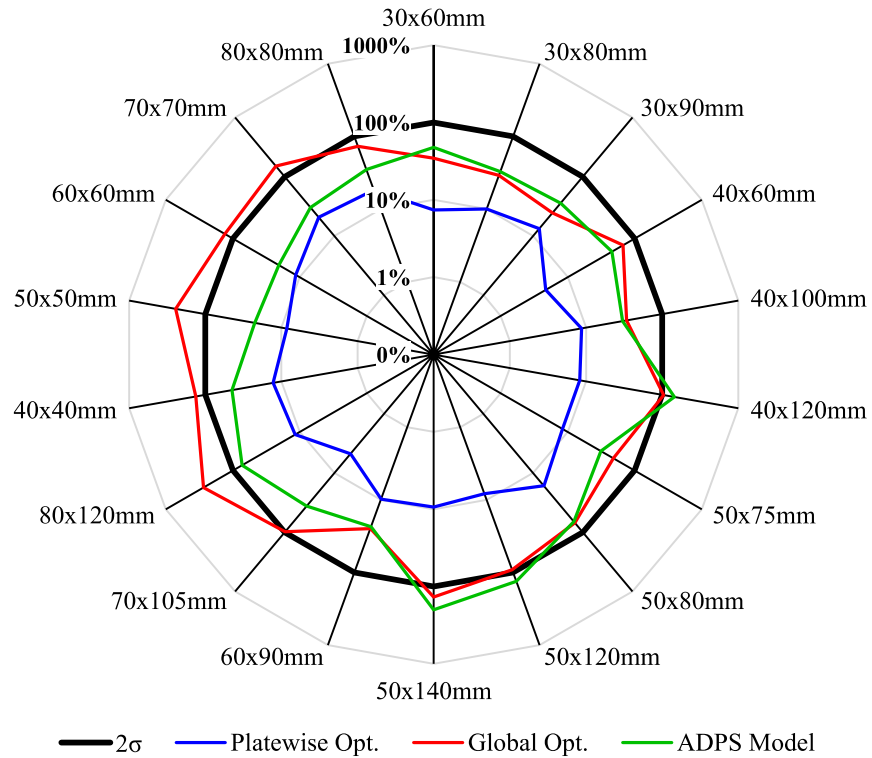


Figure 4.34 Relative error to standard deviation between fine sand test data and fitting methods.

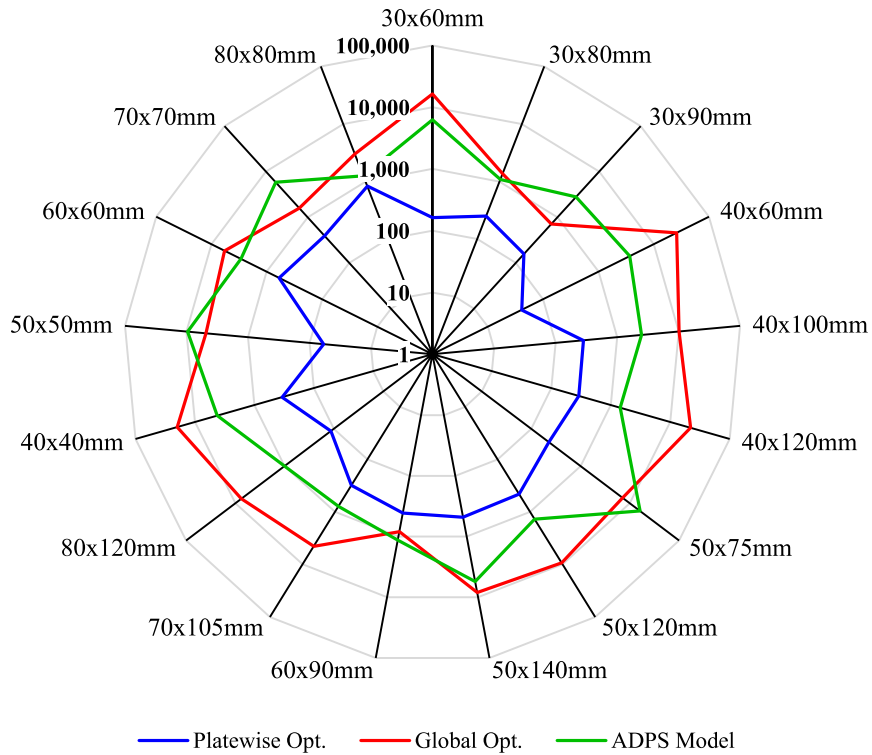


Figure 4.35 Sum of least squares between test data and fitting methods for all tested plates on sharp sand.

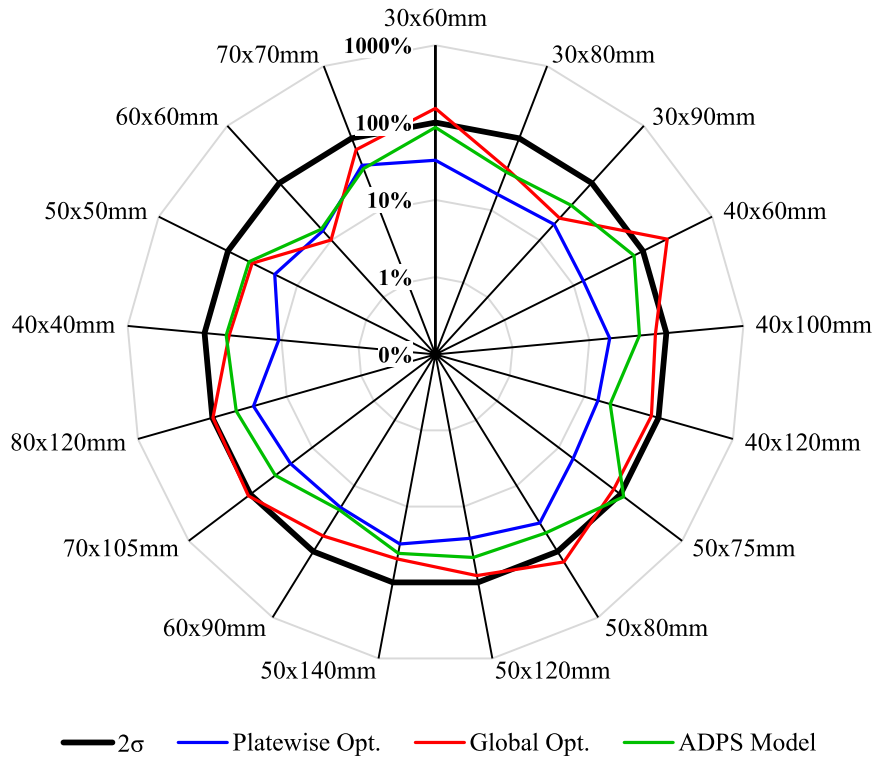


Figure 4.36 Relative error to standard deviation between sharp sand test data and fitting methods.

4.3 Shear Strength

Deriving soil cohesion and internal friction angle requires a different experimental procedure compared to the pressure-sinkage tests. The following paragraphs describe the experimental setup and data acquisition methodology used by the author for the shear test measurements and an analysis of the obtained data. Furthermore, three shear stress–strain empirical models are evaluated in order to establish their suitability to in-house test data. Finally, a new methodology for determining the shear model parameters is described, followed by results discussion.

4.3.1 Acquisition and Processing Methodology

The shear tests start by pressing the grooved plate of Figure 4.37 onto the soil until adequate normal pressure is recorded and the plate grooves have fully penetrated the soil surface. Then, a slow horizontal displacement is applied for deriving the shear measurements. In this test configuration, the initial sinkage should not be too deep as the resultant horizontal forces may include not only the soil shear but also bulldozing forces due to horizontal soil displacement. During each test repetition four

channels are continuously recording forces and displacements in the vertical and horizontal directions. The same process is repeated multiple times at different normal loads for both tested soils; fine and sharp sand. Similarly to pressure-sinkage tests, the time-domain acquired data are filtered and cut through Matlab scripts in order to produce the necessary shear stress-strain curves.



Figure 4.37 Shear test plate.

4.3.2 Data Analysis and Discussion

According to Wong [10], for ‘brittle’ type of soils such as dry sands, it is expected that the stress-strain curves would display a peak at the maximum shear stress. Figure 4.38 presents three examples of shear stress-strain test data, at different normal stress, for each tested soil. The results of both in-house tested soils confirm the literature, as indeed the curves display a peak at the maximum shear stress followed by an asymptotic convergence to a lower stress value called *residual stress*, τ_r .

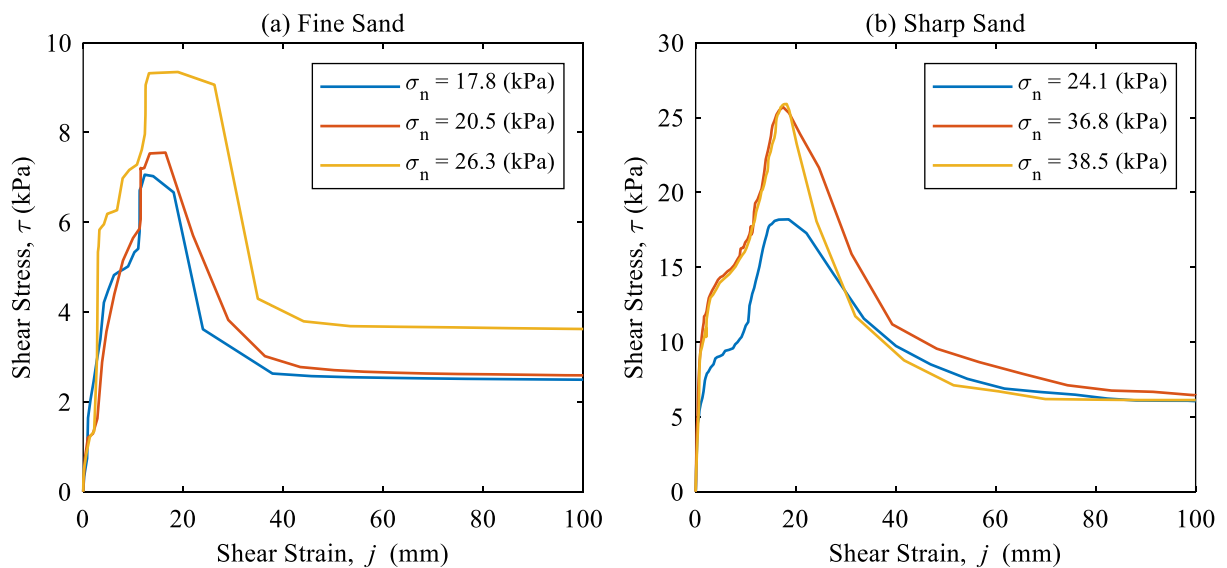


Figure 4.38 Shear stress vs horizontal displacement for two soil types.

In the graphs of Figure 4.38, the shearing forces for both tested soils vary as a function of strain. Three formulations of shear stress-strain are found in the literature [21-23] and in this chapter they are evaluated against the in-house test data. For all three formulations, which are defined by Equations 2.17, 2.18 and 2.19 respectively, the following parameters are required which are extracted from the shear stress-strain response as shown in the schematic of Figure 4.39: maximum shear stress τ_{max} , residual shear stress τ_r , and shear strain at the maximum shear stress K_w . In Oida [22] and Wong [23] models, the additional parameter K_r is also used which is the ratio of the residual shear stress τ_r to the maximum shear stress τ_{max} (i.e. $K_r = \tau_r / \tau_{max}$).

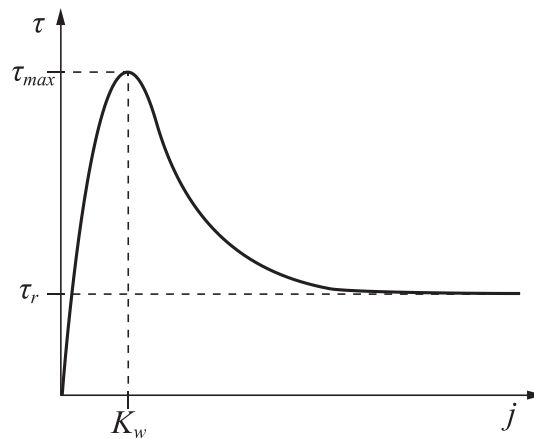


Figure 4.39 Schematic of a brittle-type soil shear response type.

Identifying the above parameters from test data is not always an easy task. The most trivial parameter to identify is perhaps the residual stress as the soil tends to find an asymptote, so taking the shear stress end value would suffice. However, as highlighted by the curves of fine sand in Figure 4.38, spotting the location of the maximum shear stress as well as the corresponding strain at the maximum shear stress can be tricky and open to interpretations. For this reason, it has been decided to identify all parameters by performing a curve fitting through optimisation. So, instead of identifying the necessary parameters directly from the test data, an optimisation algorithm would iterate on the three control variables (i.e. τ_{max} , τ_r and K_w) until the sum of squared errors between the fitted model and test data converges to a minimum. Similarly to the pressure-sinkage tests discussed in Section 4.2, the ‘fminsearch’ Matlab algorithm was used here to derive the appropriate values for τ_{max} , τ_r and K_w .

However, before proceeding with the optimisation process, a measurement error related to the experimental setup had to be addressed. In Figure 4.40 the solid line represents the experimental data of shear stress-strain of sharp sand at a 24.1 kPa normal load. As one can see, there is a step-like response between 2 mm and 10 mm of strain, where the slope of shear stress reduces before recovering again beyond 10 mm of strain. Such response is abnormal for a soil with a hump-type shearing behaviour, with the dashed line representing an approximate shear line of the expected behaviour for the first few millimetres of strain. Therefore, the true starting point of the shear stress-strain curve is not at the zero value of x-axis but at an offset j_{err} as shown in Figure 4.40. Analysing all shear test data from both fine and sharp sand, it is estimated that $j_{err} \cong 8$ mm, i.e. the actual ‘zero’ of the soil response is at an 8 mm offset (origin of dashed line).

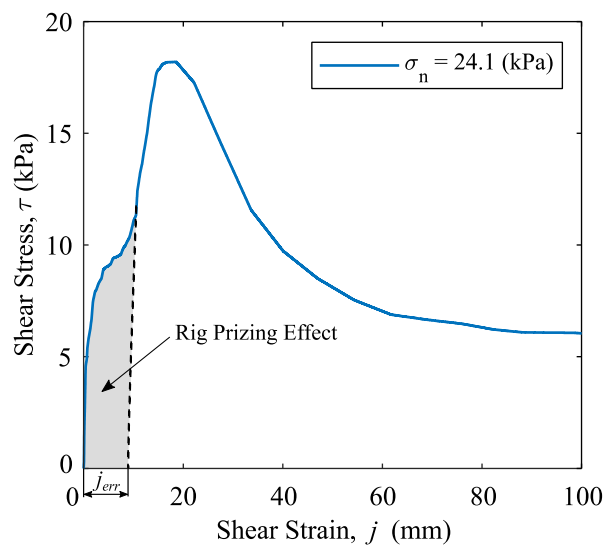


Figure 4.40 Rig prizing effect on sharp sand shear stress-strain curve.

The above described measurement error is introduced by the Terramechanics scale rig construction and setup, which will be described with the aid of the schematic of Figure 4.41. The Terramechanics scale rig consists of a sliding platform which can only translate horizontally across one direction, and a vertical displacement platform that can travel only vertically with respect to the sliding platform through the action of a screw-nut system. All forces (vertical and horizontal) are measured through two load cells installed on the vertical displacement platform. However, the shear strain is

measured via a string pot between the rig's fixed base and the horizontal sliding platform (not directly the shear plate). In an ideal situation, where no tolerances between parts exist and the rig is infinitely stiff, there would be no substantial measurement issue. In reality though, when the horizontal translation starts, a slight rotation of the vertical displacement platform with respect to the sliding platform takes place; the centre of rotation is the lower screw-nut system on the vertical sliding platform. Hence, due to the string pot not being attached directly to the shear plate, such prizing movement creates a misalignment between the shear force and shear strain as shown in Figure 4.40.

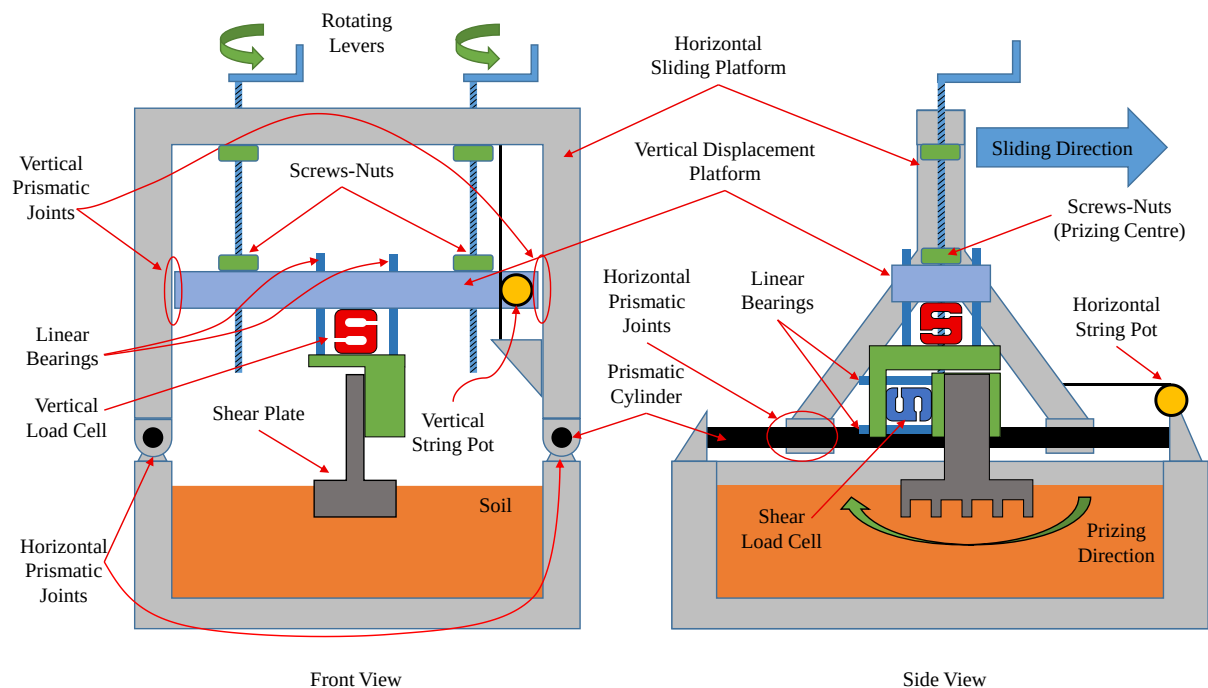


Figure 4.41 Schematic of Terramechanics scale rig in shear test mode.

Since the shear plate can be tested at various depths, attaching the string pot directly on the shear plate would create the additional issue of not acquiring purely horizontal displacement measurements. A potential solution to the problem could have been the use of a laser sensor pointing directly to the shear plate. In any case, the author would suggest a torsional system instead of the existing translational one. In a torsional system, a circular shear plate with radial grooves is used, where after preloading the plate, a rotational movement is applied measuring application angle and resultant torque. There are two main advantages of a torsional system. Firstly, the rig stiffness and tolerances would have less impact

on the measurements due to everything being operated along a common vertical axis. Secondly, in a torsional system there are less restrictions regarding the preload (i.e. depth) of the shearing plate, because no bulldozing effects can occur, and therefore the plate can be sunk deeper into the soil acquiring shear measurements at significantly greater normal loads.

As a consequence of the above described issue, the experimental data had to be modified prior to the fitting process. For each test repetition, the first 10 mm of strain were truncated and the overall curves shifted horizontally by 8 mm to the left, so essentially the first 2 mm of stress-strain data were excluded. The optimisation process was then applied for each test repetition and soil type for the Oida [22], Kacigin-Guskov [21], and Wong [23] formulations. The following Figures 4.42 and 4.43 illustrate the fitted curves (coloured) in comparison to experimental data (black curves) for fine and sharp sand respectively.

Visually, all three models demonstrate reasonable to very good correlation to experimental data, with Oida and Wong curves appearing quite similar in most cases. On the other hand, the KG model in some occasions overestimates the maximum shear stress, though this could be a result of the optimisation algorithm converging to a local instead of the global minimum. Thus, any of the three could be confidently used to characterise the shearing behaviour of the in-house tested soils.

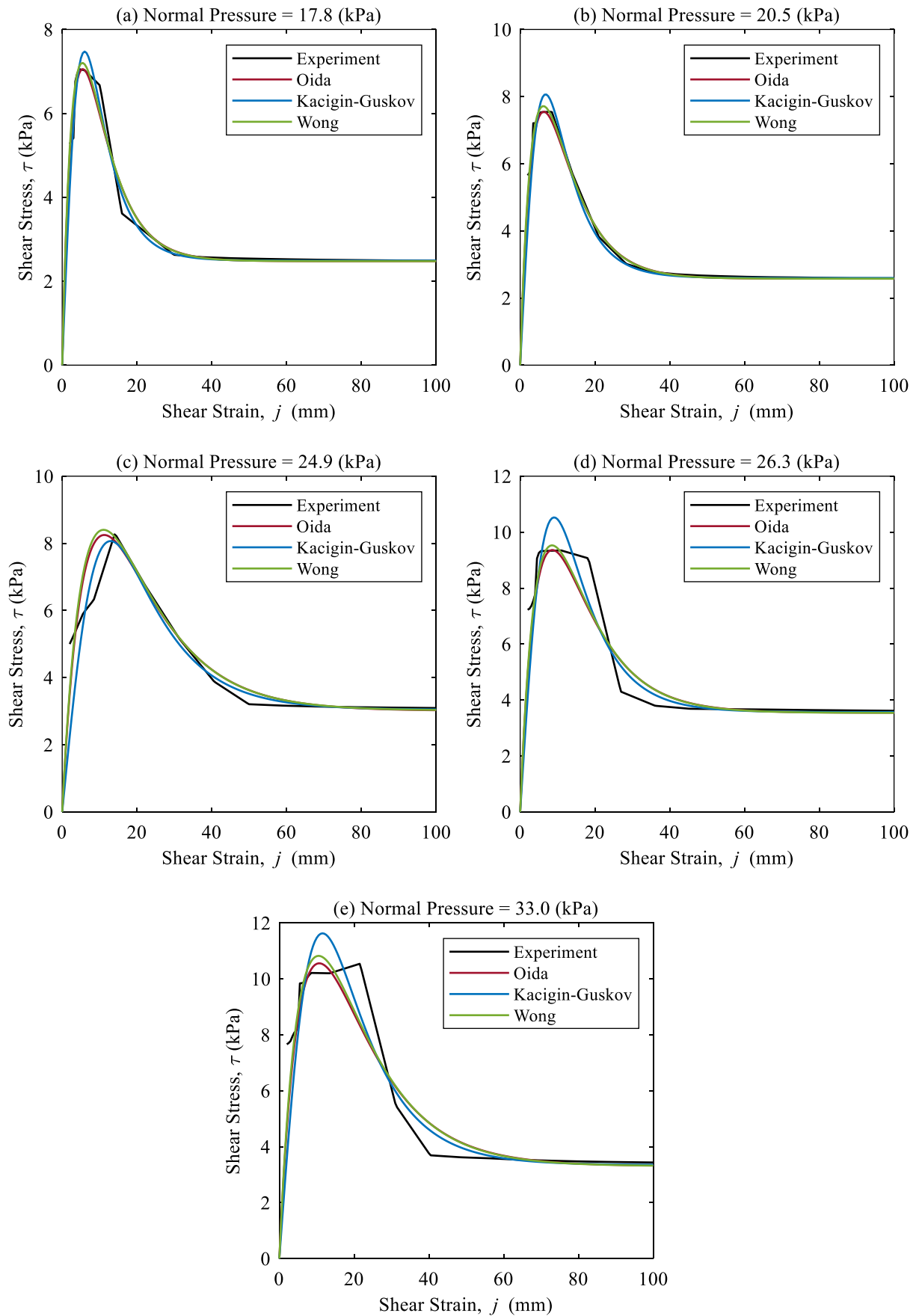


Figure 4.42 Shear stress vs shear strain experimental data fitted with Oida, Kacigin-Guskov and Wong models on fine sand.

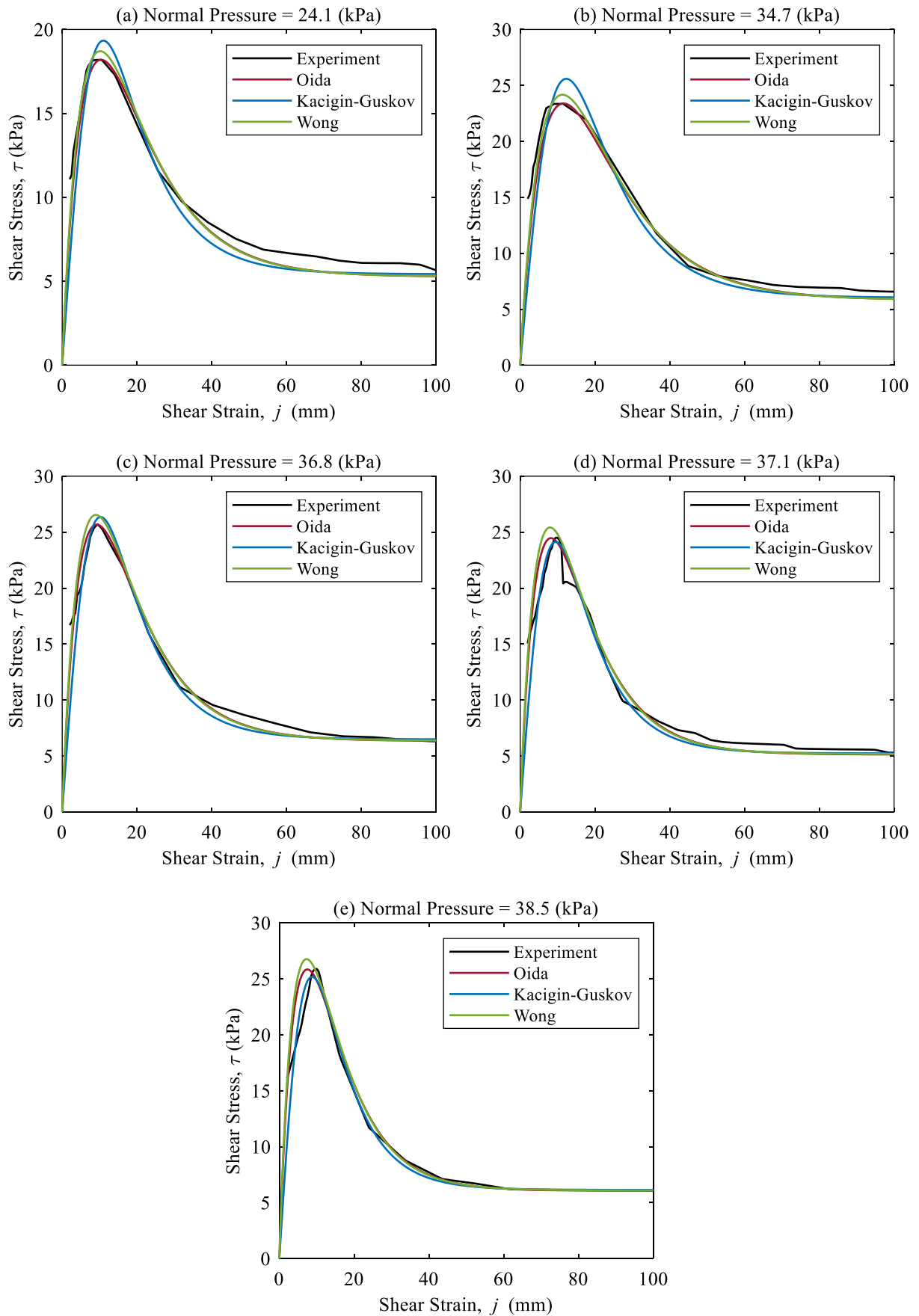


Figure 4.43 Shear stress vs shear strain experimental data fitted with Oida, Kacigin-Guskov and Wong models on sharp sand.

In each of the three models the maximum shear stress is an input parameter, so one may expect the peak of the resultant curves to match exactly the τ_{max} input value. However, as shown by the bar charts of Figures 4.44 and 4.45, an error exists between the input τ_{max} and the peak curve values for KG and Wong. According to Wong [10], such error for the Wong model is usually less than 3 % while the Oida model is error free. For the two tested soils, the calculated error for Wong is 2.2 % and 3.6 % for fine and sharp sand respectively, yet the calculated error for Oida is indeed 0 % for both soils. Therefore the literature is in agreement with the findings of the current research. No reference was found in the literature about a potential error between τ_{max} and peak curve values for the KG model, but using the in-house test data the calculated error was found between 0.2 % and 5.2 %.

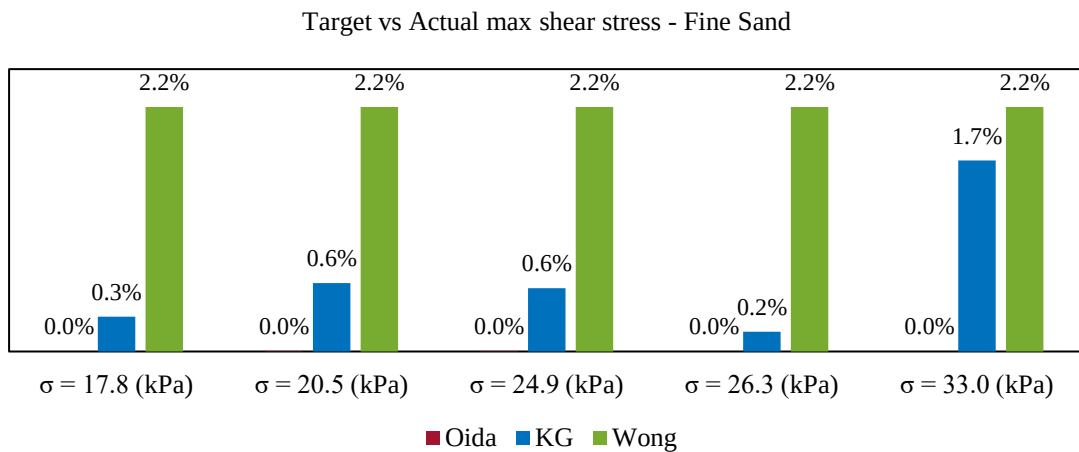


Figure 4.44 Error between the target maximum shear stress and the achieved values for Oida, Kacigin-Guskov (KG) and Wong models at various normal loads on fine sand.

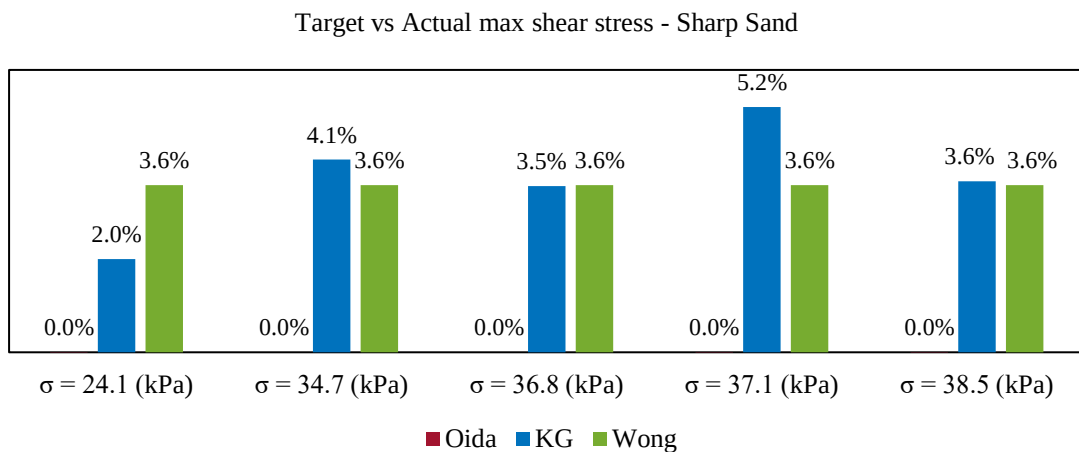


Figure 4.45 Error between the target maximum shear stress and the achieved values for Oida, Kacigin-Guskov (KG) and Wong models at various normal loads on sharp sand.

Since all three models characterise fairly well the tested soils' shear stress-strain behaviour, the decisive factor for choosing the Oida model for further investigation over KG or Wong is solely due to the error between the input and resultant τ_{max} . The τ_{max} value corresponds to the Mohr-Coulomb line at a particular normal pressure, where the Mohr-Coulomb (MC) line defines the maximum shear strength of a soil at a particular normal pressure; in other words it determines the maximum tractive effort that can be generated at the wheel-soil interface. Therefore, an accurate estimation of the MC line, as expressed by Equation 2.9, is imperative in Terramechanics studies. Using either KG or Wong models, an error to the MC prediction of up to ~5 % is expected. Such error level is of course fairly small and perhaps smaller than the test dispersion. However, the Oida model is completely error free, which is the sole reason for being selected.

For both fine and sharp sand, shear test data at five different normal loads were recorded. For each individual test and soil the optimisation algorithm was employed using the Oida model, and the values of the three control variables (i.e. τ_{max} , τ_r and K_w) were obtained. In Table 4.4 a complete summary of the optimisation process output is presented, with the numerical data of each parameter.

Two soil constants are necessary to define the Mohr-Coulomb failure criterion, namely cohesion and internal friction angle. Obtaining values for these parameters requires to perform a linear regression of the maximum shear stresses τ_{max} versus the corresponding normal stress σ . In Figures 4.46 and 4.47 for fine and sharp sand respectively, the red dots represent the τ_{max} values from the Oida optimisation process (as summarised in Table 4.4), while the blue line is the linear regression (i.e. the Mohr-Coulomb line). The y-intercept and slope of the regression line correspond to the soil cohesion and tangent of the internal friction angle values respectively.

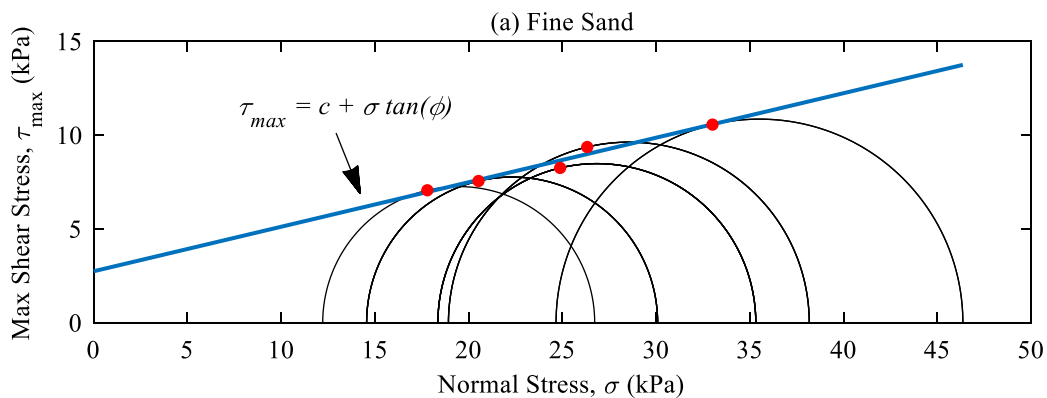


Figure 4.46 Calculated Mohr-Coulomb line for fine sand.

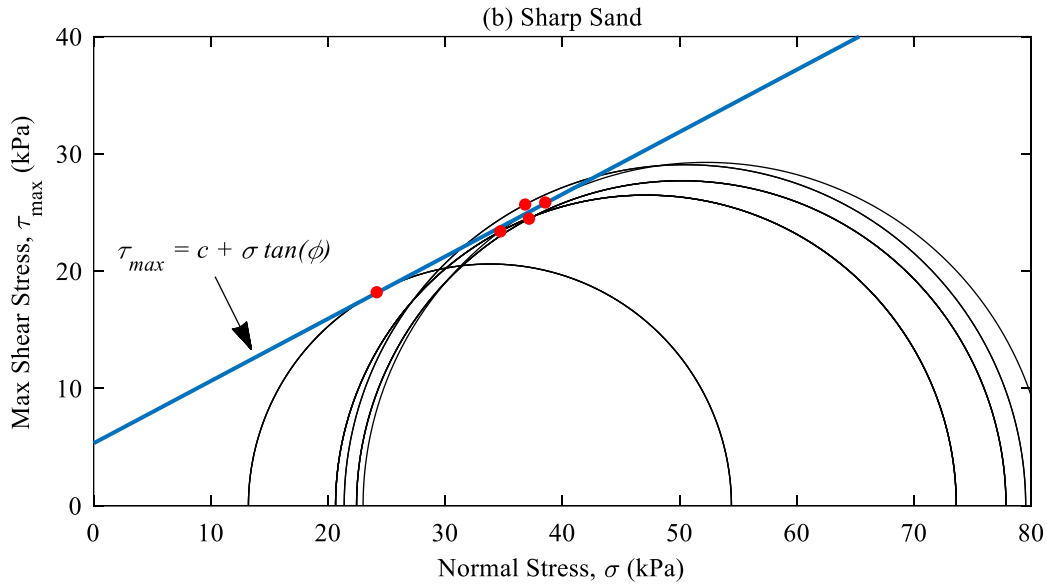


Figure 4.47 Calculated Mohr-Coulomb line for sharp sand.

		Oida Optimisation		
	σ (kPa)	τ_{max} (kPa)	τ_r (kPa)	K_w (mm)
Fine Sand	17.792	7.057	2.478	5.491
	20.524	7.552	2.576	6.304
	24.878	8.249	3.011	11.244
	26.321	9.363	3.544	8.610
	33.005	10.557	3.310	10.669
Sharp Sand	24.147	18.199	5.270	10.384
	34.701	23.388	5.821	11.581
	36.815	25.678	6.339	9.221
	37.144	24.469	5.123	8.226
	38.531	25.852	6.062	7.487

Table 4.4 Soil shear parameters for Oida obtained through optimisation, separately for each individual normal pressure.

Experimental data from Zapolsky (as cited in [21]) indicate that the strain K_w at the maximum shear stress is not constant but it decreases with increasing normal pressure. Similarly, the residual stress τ_r also shows a normal pressure dependency with its values increasing along with normal pressure [21]. In Figure 4.48 the in-house τ_r data of Table 4.4 for the two soils are plotted against their corresponding normal stress. Observing the two plots, it is apparent that the residual stress increases with increase in normal pressure, similarly to the observations cited in [21], with the relationship between the two

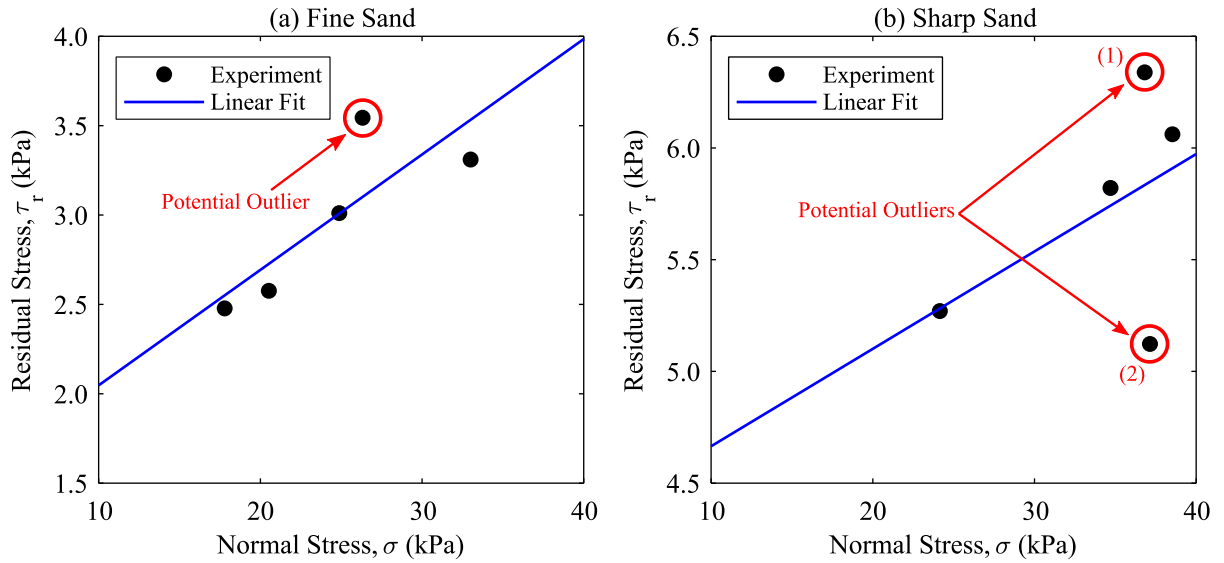


Figure 4.48 Residual shear stress vs normal stress for fine (a) and sharp sand (b).

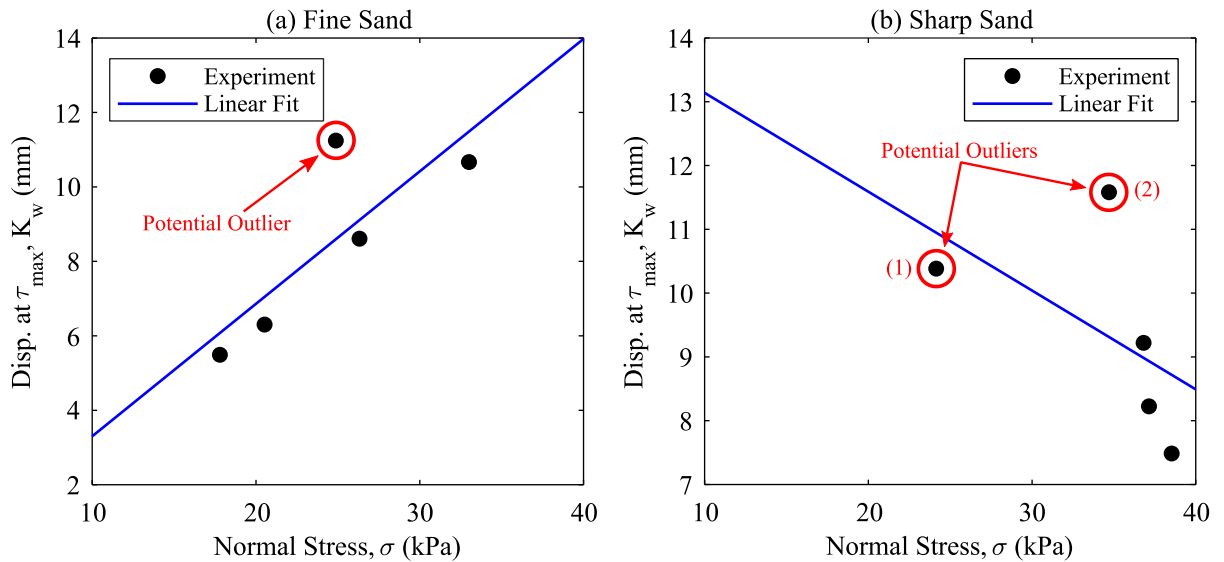


Figure 4.49 Shear displacement at maximum shear stress vs normal stress for fine (a) and sharp sand (b).

described reasonably well by a linear function with positive slope. In both graphs of Figure 4.48 there are potential data outliers, with more obvious example the two points in graph (b) for sharp sand.

Likewise, the in-house values of parameter K_w are also plotted against normal pressure as shown in Figure 4.49. For fine sand in Figure 4.49 (a), the values of K_w are also increasing with increased normal stress, which contradicts with the findings of the observations cited in [21]. On the other hand, the sharp sand data show a decrease of K_w values with increasing normal stress, which is in alignment

with the literature observations. Furthermore, two of the data points in graph (b) could be considered as potential outliers (highlighted with red circles), with point (2) being the most possible candidate. Yet, for both soil types a linear regression seems a reasonable approximation to characterise the K_w dependency to normal load.

4.3.3 Extended Oida model

The Oida model described in preceding paragraphs and mathematically expressed by Equation 2.18 characterises the soil shearing behaviour for a single normal pressure only. However, as found by the author, and supported by literature, all three Oida model parameters are dependent on normal pressure. Such finding raises the necessity to extend the current Oida formulation to a third dimension (i.e. normal pressure), for application in vehicle Terramechanics studies. Since the Mohr-Coulomb failure criterion defines the maximum shear stress of a soil, then the scalar parameter τ_{max} in the Oida model can be replaced by the MC Equation 2.9. The remaining two scalar parameters (τ_r and K_w) were found to also reasonably correlate to linear functions with normal pressure as the independent variable. Therefore, the extended Oida formulation can take the form of Equation 4.8.

$$\tau = \tau_{max} K'_r \left[1 - \frac{\sqrt{1-K'_r} \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}}}{\sqrt{1-K'_r} \left(1 - \frac{2}{K'_r} \right) + \frac{2}{K'_r} - 2} \right] \times \left[1 - \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}} \right] \quad 4.8$$

Where:

$$\tau_{max} = c + \sigma \tan(\phi) \quad 2.9$$

$$\tau'_r = \tau_{r,a} \sigma + \tau_{r,b} \quad 4.9$$

$$K'_w = K_{w,a} \sigma + K_{w,b} \quad 4.10$$

$$K'_r = \frac{\tau'_r}{\tau_{max}} = \frac{\tau_{r,a} \sigma + \tau_{r,b}}{c + \sigma \tan(\phi)} \quad 4.11$$

The expended Oida formulation requires in total six input parameters (c , ϕ , $\tau_{r,a}$, $\tau_{r,b}$, $K_{w,a}$, $K_{w,b}$), and for the identification process an optimisation method was used once again. In contrary to previous paragraphs, instead of trying to identify all control variables simultaneously, the optimisation was performed in three main phases as illustrated by the flowchart of Figure 4.50.

In first phase (flowchart processes 1-5), the optimisation Matlab algorithm ‘fminsearch’ is run on each individual shear stress-strain measurement for the original Oida model. Out of the three parameters obtained for each normal pressure (i.e. τ_{max} , τ_r and K_w), only τ_{max} is used in the second phase (flowchart processes 6-8), where the $\sigma - \tau_{max}$ data pairs are used to fit the Mohr-Coulomb failure criterion and obtaining the soil cohesion and internal friction angle. In final phase, ‘fminsearch’ is called on all shear stress-strain data with the Mohr-Coulomb parameters from the previous phase being fixed. So, the optimisation iterates on the values of the remaining four parameters ($\tau_{r,a}$, $\tau_{r,b}$, $K_{w,a}$, $K_{w,b}$).

The reason for performing the parameter identification in multiple phases is related to results accuracy. When attempting to obtain all six model parameters at once, the optimisation algorithm cannot always converge. Although convergence could eventually be achieved by tweaking the algorithm settings, the Mohr-Coulomb parameters could vary significantly depending on the setting used. But since the Mohr-Coulomb parameters are the most critical, the multi-phase optimisation process was selected in order improve robustness. The final soil parameters for the extended Oida model are summarised in Table 4.5

	Cohesion, c (kPa)	Int. Friction Angle, ϕ (deg)	Residual stress		Strain at τ_{max}	
			$\tau_{r,a}$ (-)	$\tau_{r,b}$ (kPa)	$K_{w,a}$ (mm/kPa)	$K_{w,b}$ (mm)
Fine Sand	2.740	13.350	6.655×10^{-2}	1.366	0.333	0.137
Sharp Sand	5.323	27.966	2.010×10^{-2}	5.091	0.000	9.085

Table 4.5 Final soil parameters for the extended Oida model.

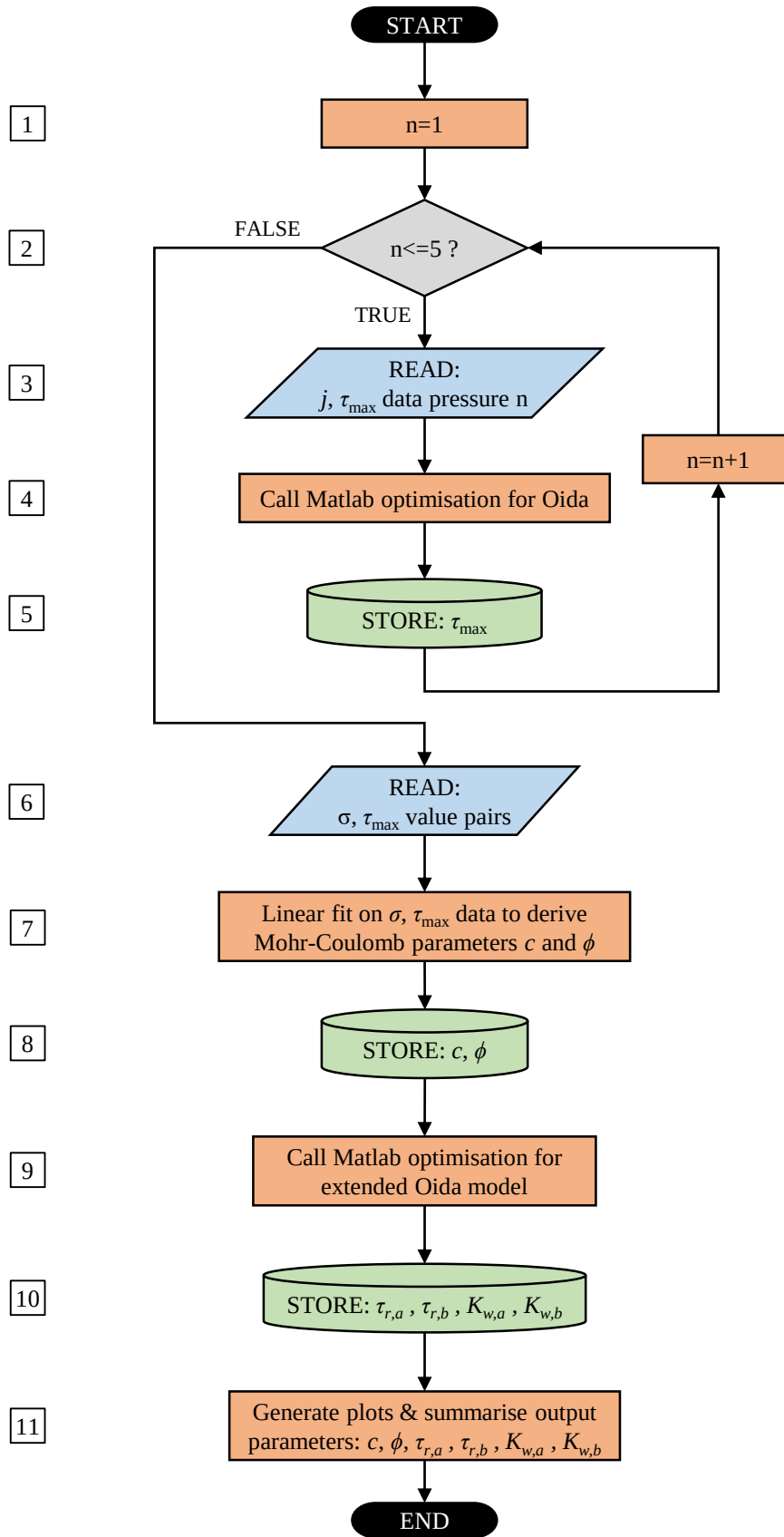


Figure 4.50 Flowchart of optimisation process for deriving the extended Oida model parameters.

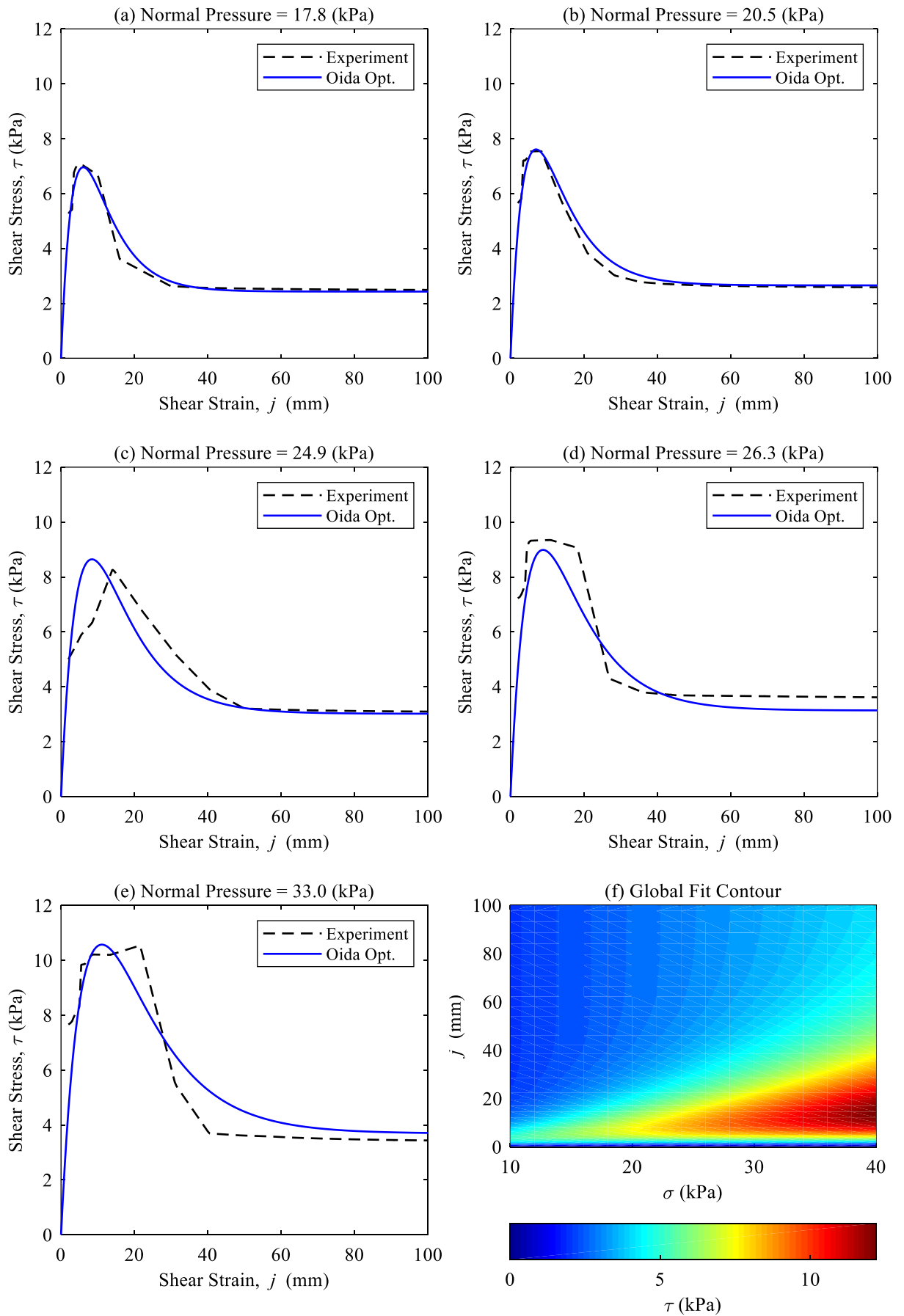


Figure 4.51 Shear stress-strain global optimisation fitting, fine sand.

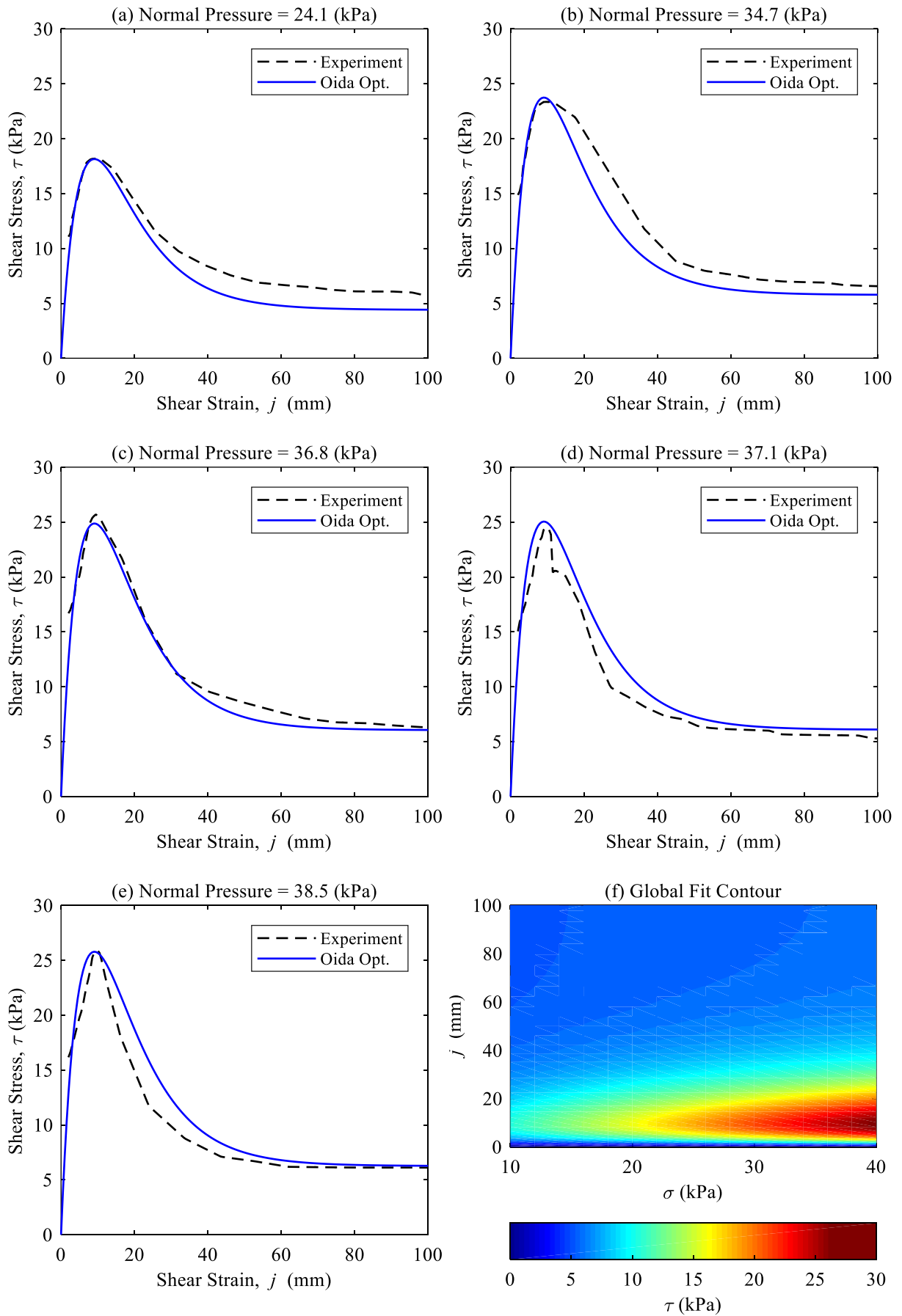


Figure 4.52 Shear stress-strain global optimisation fitting, sharp sand.

4.4 Pressure vs Sinkage – Curved Plates

One of the requirements of vehicle Terramechanics studies is the ability to accurately predict the wheel sinkage under the effect of an applied vertical load. When considering wheeled vehicles, the contact area can be approximated as cylindrical (assuming very little to none tyre deformation), which means that flat plate pressure-sinkage models need to be adapted for curved contact areas. For this reason, a set of curved plates have been manufactured and tested for pressure-sinkage, and the results are described in the subsequent paragraphs. Three groups of curved plates have been manufactured, with constant width for each group of 30 mm, 40 mm and 50 mm at various fixed radii as summarised in Table 4.6. The curved plates have been tested for pressure-sinkage using the same experimental methodology and soils as described in Section 4.2.

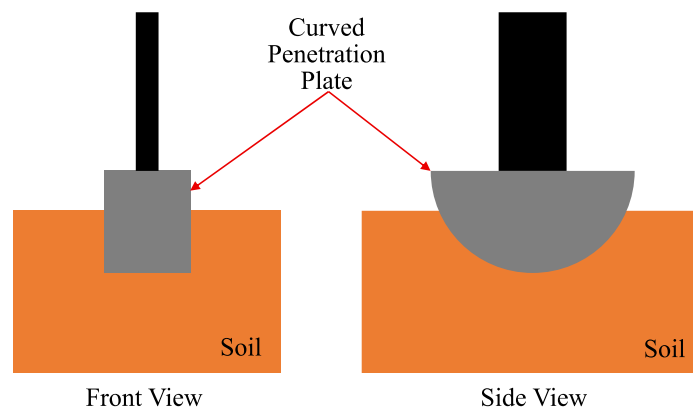


Figure 4.53 Schematic of pressure-sinkage for curved plates.

ID	Width, B (mm)	Radius, R (mm)	R/B (-)
1	30	42	1.4
2	30	45	1.5
3	30	48	1.6
4	40	56	1.4
5	40	60	1.5
6	40	64	1.6
7	50	70	1.4
8	50	75	1.5
9	50	80	1.6

Table 4.6 List of tested curved plates.

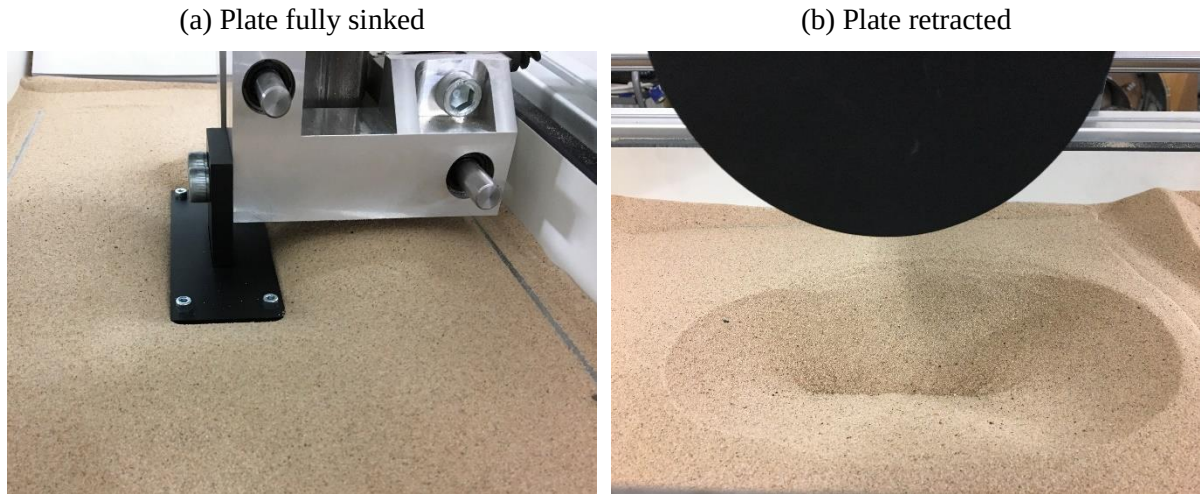


Figure 4.54 Pressure-sinkage test with curved plate on fine sand. (a) plate fully sunk, (b) footprint after removing the plate.

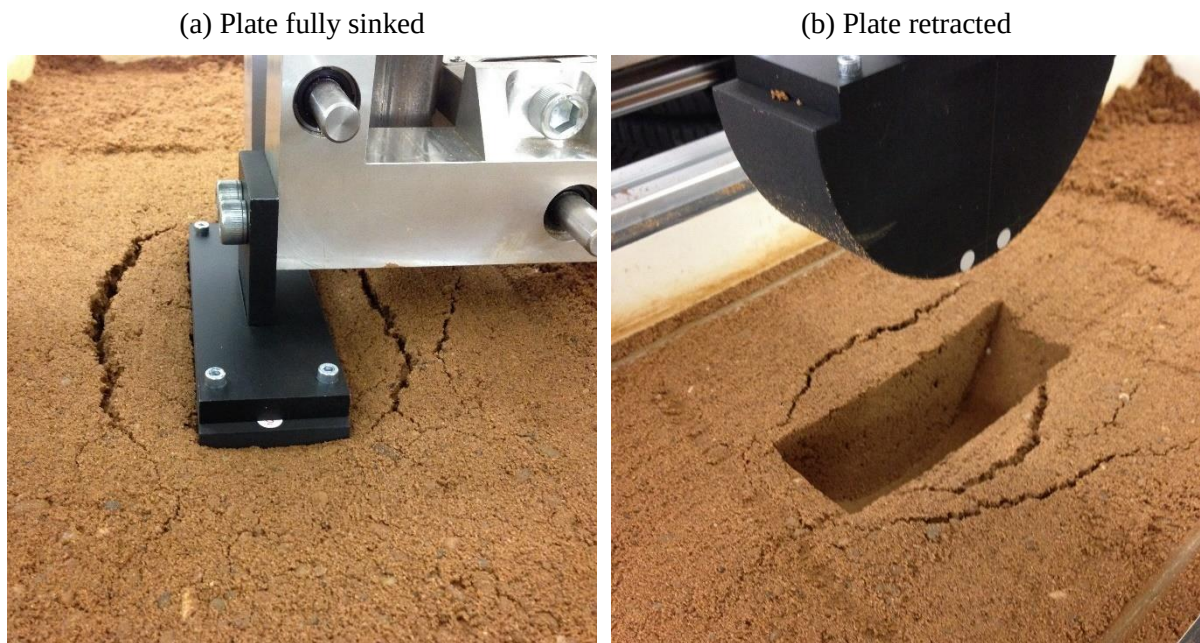


Figure 4.55 Pressure-sinkage test with curved plate on sharp sand. (a) plate fully sunk, (b) footprint after removing the plate.

According to Bekker [1], for a rigid wheel on deformable terrain, as shown in Figure 4.56, under the effect of vertical load W and horizontal force R_c , the total soil reaction force N would be the sum of the radial forces dN . Furthermore, the radial force dN at a particular sinkage is assumed to be equal to the force on a flat plate at the same depth. Considering these assumptions, the force equilibrium resolves to the following Equations 4.12 and 4.13:

$$R_c = BR \int_0^{\theta_0} P \sin \theta d\theta \quad 4.12$$

$$W = BR \int_0^{\theta_0} P \cos \theta d\theta \quad 4.13$$

where B is the wheel width, R the wheel radius, $P = dN$ is the radial pressure as defined by Bekker Equation 2.23 discussed earlier in the text, and θ_0 the contact angle as shown in Figure 4.56. After a geometrical and mathematical analysis, Bekker arrived to the analytical Equation 4.14 below:

$$W = \frac{Bk\sqrt{Dz_0}}{3} z_0^n (3-n) \quad 4.14$$

where k and n are the coefficient of proportionality and soil exponent respectively from of flat plates pressure-sinkage model, and D the wheel diameter.

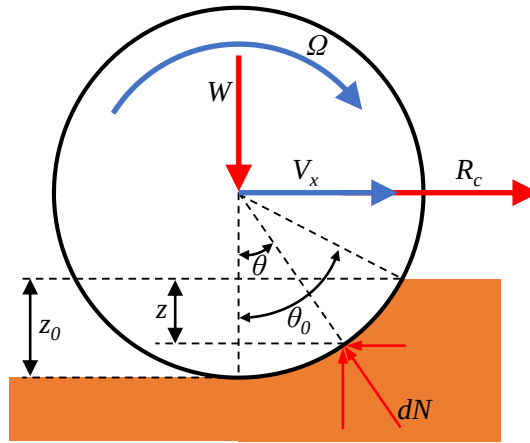


Figure 4.56 Forces on rigid wheel on deformable terrain as proposed by Bekker [1].

However, to arrive to Equation 4.14, Bekker performed some simplification in the mathematical formulation as described in Bekker [1]. These simplifications limit the validity of Equation 4.14 to a narrow range of sinkage levels [1] [10] where the sinkage-to-wheel diameter ratio is small enough to be assumed almost flat [65]. Yet the literature does not provide a clear and tangible definition of ‘shallow’ or ‘deep’ sinkage, it relies therefore to the researcher’s discretion and rationale.

In the case of non-rotating wheel, both front and rear face of the wheel is in contact with the soil. Furthermore, by combining Equations 4.13 and 2.23, the total vertical soil reaction force can be expressed as:

$$F_{V,T} = 2W = 2BR \int_0^{\theta_0} \left(\frac{k_c}{B} + k_\phi \right) (R \cos \theta - R \cos \theta_0)^n \cos \theta d\theta \quad 4.15$$

where: $z = (R \cos \theta - R \cos \theta_0)$

Similarly, combining Equations 4.14 and 2.23 and assuming front and rear contact areas, the Bekker's analytical form of vertical force would be:

$$F_{V,T} = 2W = \frac{B \left(\frac{k_c}{B} + k_\phi \right) \sqrt{Dz_0}}{3} z_0^n (3 - n) \quad 4.16$$

Equation 4.15 is solved through a numerical integration and Equation 4.16 by standard analytical approach for the plates of Table 4.6, using the fine sand global optimisation Bekker parameters from Table 4.2. In addition, platewise fitting via optimisation was performed using equation $P = kz^n$ to be used as a benchmark for result accuracy. The bar chart of Figure 4.57 illustrates the results of the average (across all plates) sum of lease squares of vertical forces between the test mean data and the three methods. The results are evaluated at various sinkage levels as percentage of the wheel radius, as well as a combined error from 0 % to 90 % of wheel radius.

As expected from the literature, the results illustrate that the force prediction error using the analytical Equation 4.16 increases at deeper sinkage levels. Furthermore, for shallow sinkage (which the author defines as up to 10 % of the wheel radius), the analytical equation (pink bars) provides reasonable force prediction. Although, the error to test data from the analytical equation is higher by a factor of 3 when compared to the platewise (blue bars) or numerical equation (red bars) results.

For moderate sinkage level of up to 30 % of the wheel radius, the numerical Bekker equation is reasonably close to the platewise results with approximately double average sum of lease squares error, while the analytical equation deteriorates further reaching an error which is higher by factor of 33 when

compared to platewise results. For sinkage levels beyond 30 % of the wheel radius, both Bekker methods (numerical and analytical) demonstrate a significant error increase. Especially the analytical method, when looking at a cumulative sum of least squares for up to 90 % of the wheel radius, the imposed error factor is larger than 1100 when compared to platewise results, requiring a log-scale vertical axis to allow proper visualisation of all data.

From the above observations, three conclusions can be drawn. Firstly, it is confirmed that the Bekker analytical equation can be applicable to shallow sinkage only as suggested by the literature. Secondly, when numerically integrating Equations 4.15 the results accuracy greatly improves, however at deep sinkage the overall error level is still quite significant. Finally, the original assumption that the overall vertical force is the sum of individual component forces along the wheel-soil contact surface, equivalent to the forces of flat plates at the same sinkage seems valid. In Figure 4.58 two force-sinkage examples are presented for visualising the fitting results discussed so far.

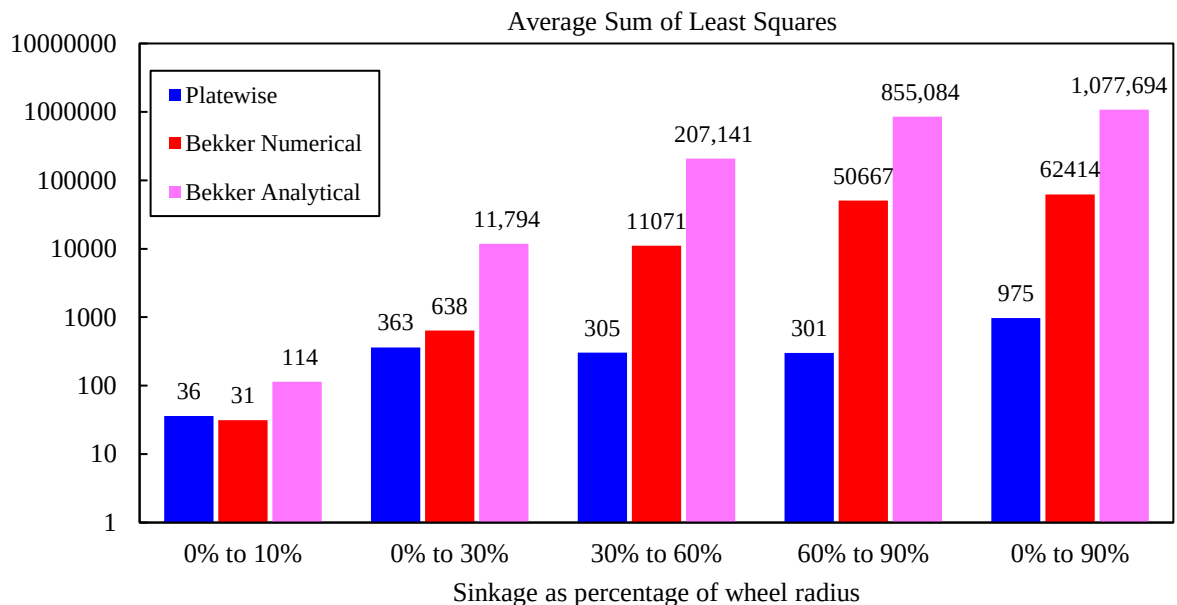


Figure 4.57 Average sum of least squares comparison between different Bekker methods for curved plates on fine sand.

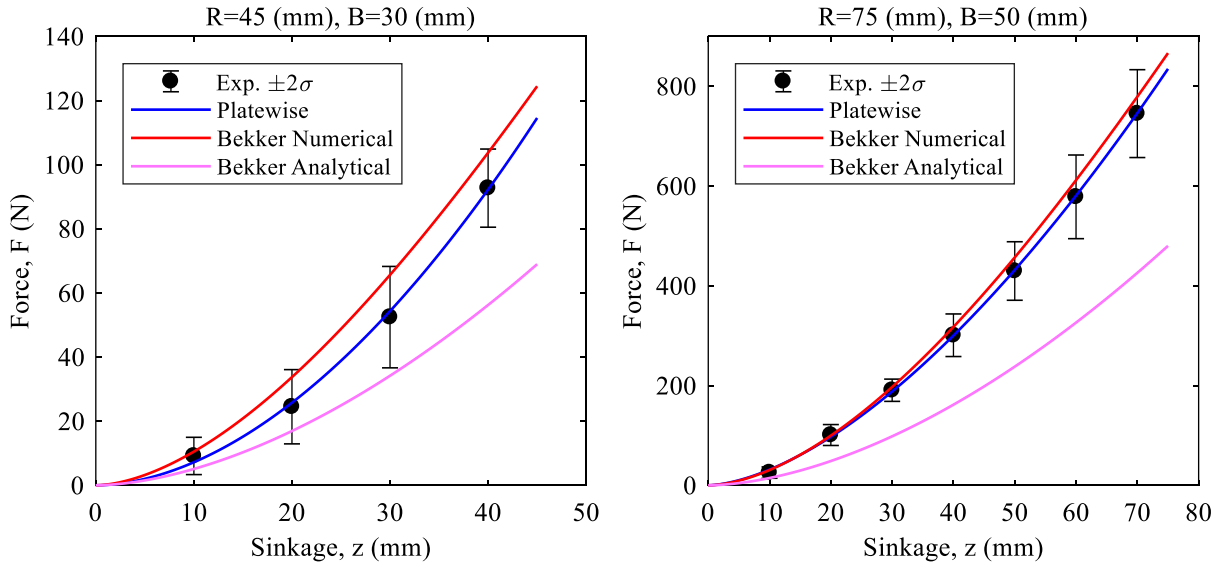


Figure 4.58 Comparison between Bekker-based methods for curved plates on fine sand.

Confirming the validity of Bekker's assumption regarding the generation of forces at the rigid wheel-soil interface, allowed the application to curved plates of the newly developed ADPS model by combining Equations 4.13 and 4.6 into Equation 4.16 below:

$$F_{V,T} = 2W = 2BR \int_0^{\theta_0} \left[a_k A^{(b_k+1)} \right] (R \cos \theta - R \cos \theta_0)^{\left[a_n A^{(b_n+1)} \right]} \cos \theta d\theta \quad 4.17$$

where the contact area is calculated as: $A = RB\theta$

Applying Equation 4.18 was found that it significantly underpredicts the forces on curved plates; an issue linked to the area calculation. If the contact area is assumed as $A = RB\theta$, then the overall equation's area dependency is effectively double accounted, as the $z = (R \cos \theta - R \cos \theta_0)$ already introduces an area dependency. The issue was resolved by simply removing parameter θ from the area equation, which is now defined as $A_c = RB$. Subsequently the total vertical force on a curved plate becomes:

$$F_{V,T} = 2W = 2BR \int_0^{\theta_0} \left[a_k A_c^{(b_k+1)} \right] (R \cos \theta - R \cos \theta_0)^{\left[a_n A_c^{(b_n+1)} \right]} \cos \theta d\theta \quad 4.18$$

Interestingly, a dependency of RB to the vertical force is observed in the test data. In Figure 4.59 the experimental forces of fine sand are plotted against the RB product, where the same shaped dots represent the forces of each plate at a particular sinkage level. As shown in Figure 4.59, a linear regression fits very well each group of equal sinkage forces, indicating the linear dependency between the geometric properties of the wheel and the soil forces for a given sinkage value.

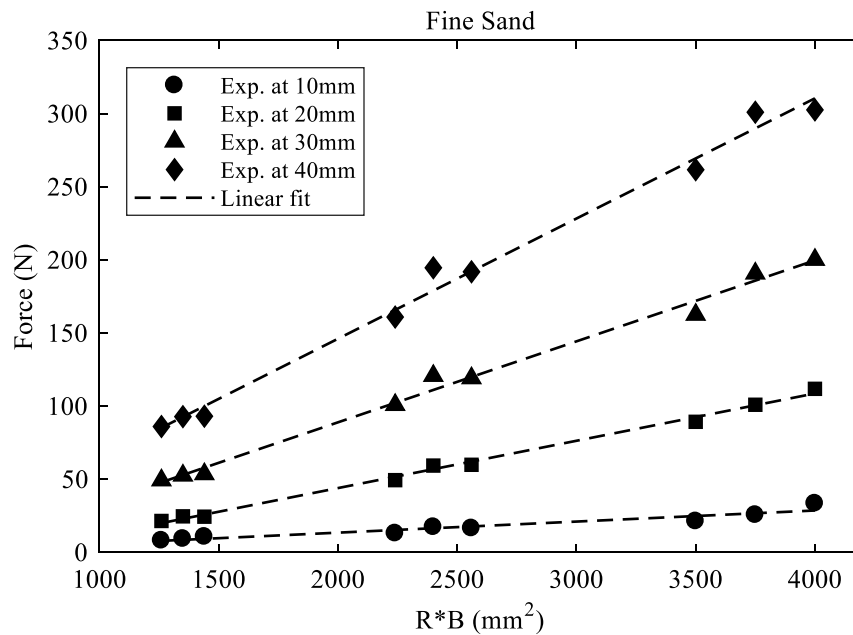


Figure 4.59 Vertical force dependency on the product of plate radius and width, fine sand.

The adapted ADPS model of Equation 4.18 applied to the tested curved plates of Table 4.6 on fine sand, using the ADPS soil parameters derived from the flat plate measurements as summarised in Table 4.3. As shown in the bar chart of Figure 4.60, the sum of least squares (average across all tested plates) using the ADPS model (green bars) is at all sinkage levels lower than Bekker's numerical method (red bars). Despite the reasonably good correlation, it is important to highlight though, that Bekker's methodology for calculating the total vertical forces does not consider any shearing forces. However, the curved plate measurements were performed at approximately 3 mm/s sinkage velocity, which means that some shearing effect is expected for this condition.

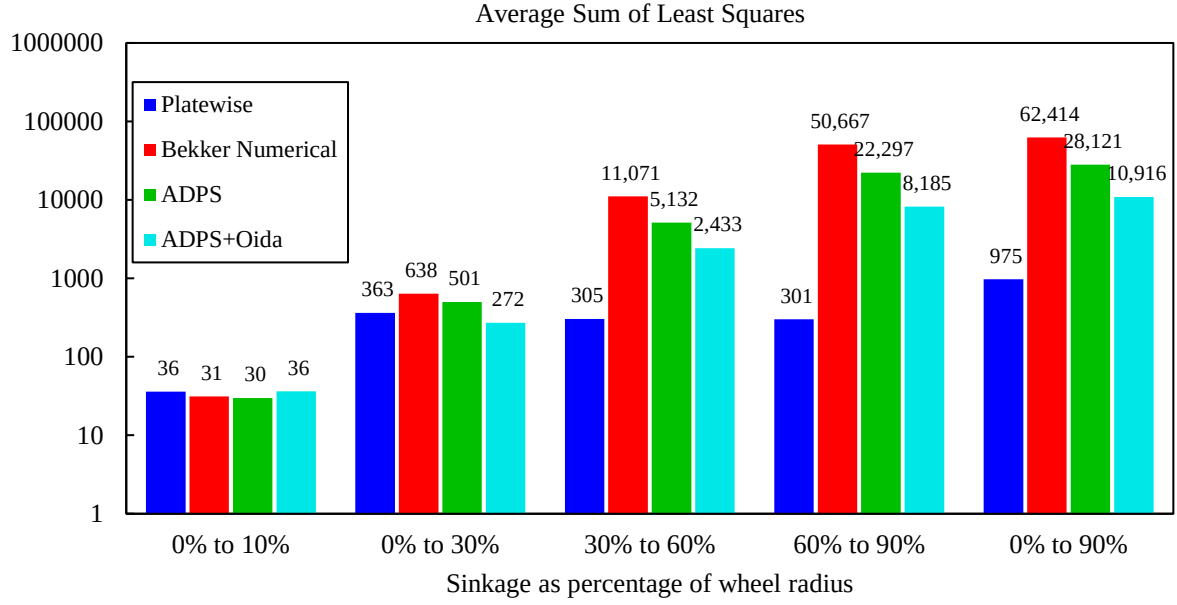


Figure 4.60 Average sum of least squares comparison between Bekker and ADPS models for curved plates on fine sand.

If the shear forces are considered for the calculation of the total vertical forces on curved plates, then the following Equation 4.19 shall be employed:

$$F_{V,T} = 2W = 2BR \left\{ \int_0^{\theta_0} \sigma \cos \theta d\theta + \int_0^{\theta_0} \tau \cos \theta d\theta \right\} \quad 4.19$$

where: σ is the normal stress as expressed by Equation 4.20, and τ the shear stress expressed by the extended Oida Equation 4.8 with the shear strain in this cases a function of contact angle θ .

$$\sigma = \left[a_k A_c^{(b_k+1)} \right] (R \cos \theta - R \cos \theta_0)^{\left[a_n A_c^{(b_n+1)} \right]} \quad 4.20$$

$$\tau = \tau_{\max} K'_r \left[1 - \frac{\sqrt{1-K'_r} \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}}}{\sqrt{1-K'_r} \left(1 - \frac{2}{K'_r} \right) + \frac{2}{K'_r} - 2} \right] \times \left[1 - \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}} \right] \quad 4.8$$

$$j = \dot{z} \sin \theta \quad 4.21$$

The cyan colour bars of Figure 4.60 present the average sum of least squares for the ADPS+Extended Oida model of Equation 4.19. As it can be seen, when including the shearing effect on the curved plates sinkage, the average fitting error reduces significantly at all sinkage levels. Furthermore, for up to 30 % wheel radius sinkage, the ADPS+Extended Oida produces smaller error even than the platewise fitting (blue bars), and above 30 % sinkage the average error is 6 times smaller than the Bekker numerical method (red bars).

In the graphs of Figure 4.61, two sample plates visually demonstrate the correlation quality between platewise, Bekker numerical, ADPS and ADPS+Oida methods. As shown in the graphs, the blue and cyan curves almost overlay up to moderate sinkage level for both plates, and in the rightmost graph overlay almost all across. In addition, it can be observed that the Bekker numerical method overestimates the vertical forces, which means that if shear forces were included the correlation would worsen. The above can be observed in all tested plates.

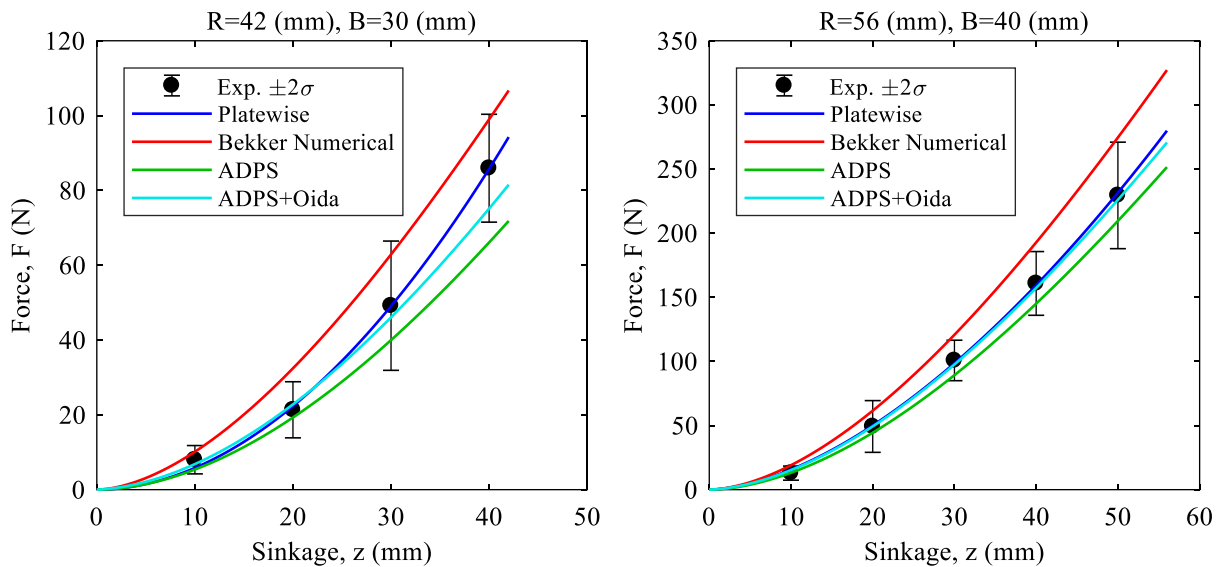


Figure 4.61 Comparison between various force-sinkage methods on fine sand.

In addition to the average sum of least squares and the visual force-sinkage graphs inspection, the radar chart of Figure 4.62 illustrate the relative error to two standard deviations for fine sand. For all tested plates, the results of all methods lie within the two standard deviations envelope. The ADPS+Oida method though demonstrates in all but one plate a far greater correlation, which is equivalent to the

platewise method for some plates. Additionally, the relative error to standard deviation for ADPS+Oida is approximately 25% for most plates, which is far smaller than one standard deviation from mean and further highlights its superior correlation accuracy.

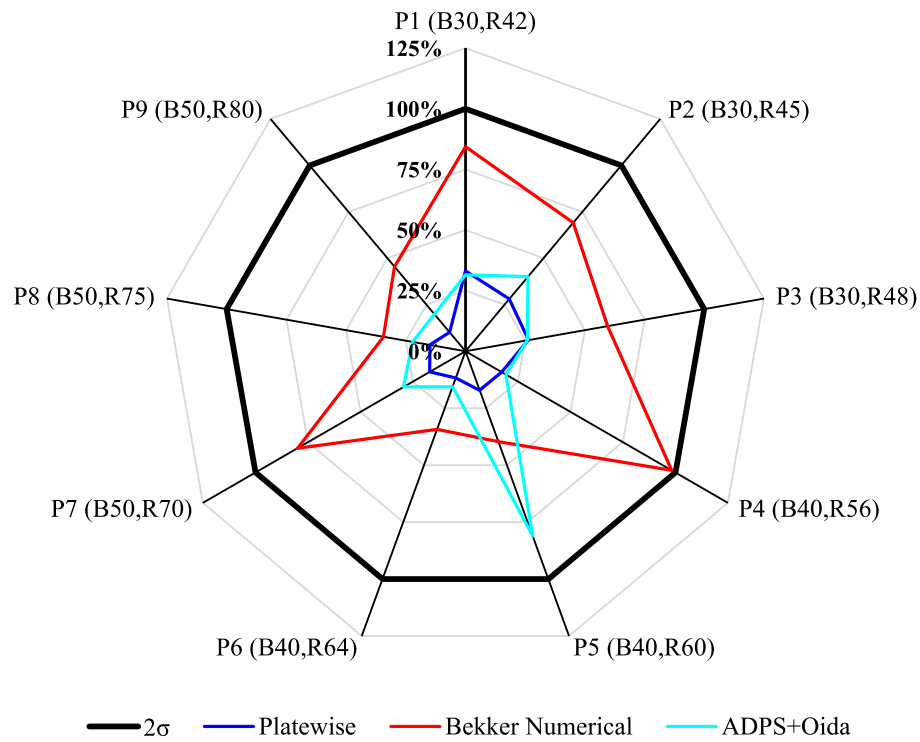


Figure 4.62 Relative error to standard deviation between fine sand test data on curved plates and fitting methods.

The methodology and models used for fine sand were then applied to sharp sand, with the soil parameters for the Bekker numerical method taken from Table 4.2, the ADPS soil parameters from Table 4.3 and the parameters for the extended Oida model from Table 4.5. Similarly to fine sand, a bar chart was created for the average sum of least squares and is presented in Figure 4.63, where the ADPS+Oida model demonstrating good correlation at shallow and moderate sinkage of up to 30 % of wheel radius. On the other hand, for higher sinkage levels the ADPS+Oida model correlation deteriorates significantly, as a result of underpredicting the soil forces of various plates, mostly the smaller of the list. This becomes more evident when looking into the sample of force-sinkage graphs of Figure 4.64.

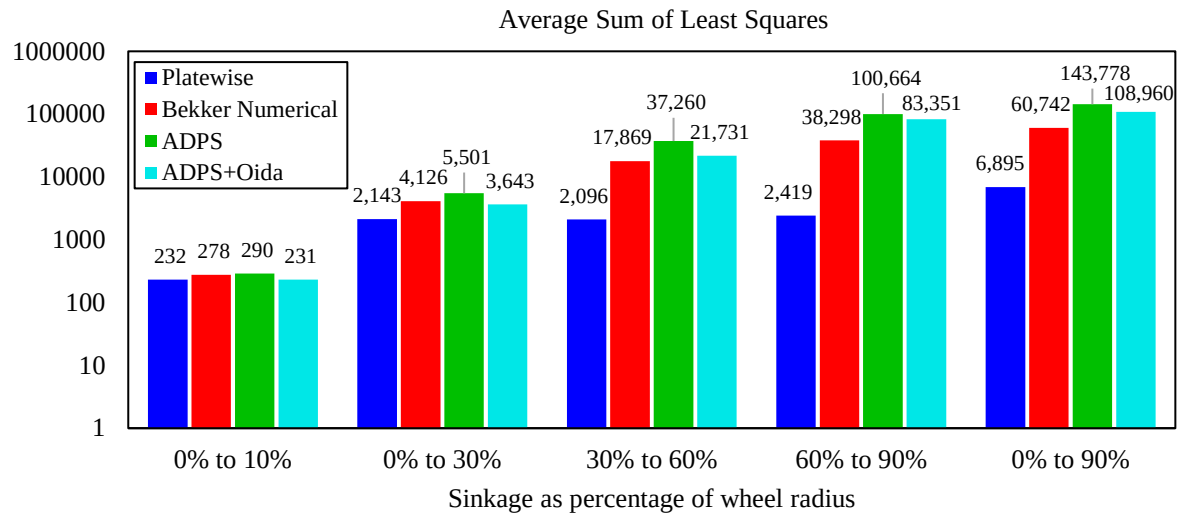


Figure 4.63 Average sum of least squares comparison between Bekker and ADPS models for curved plates on sharp sand.

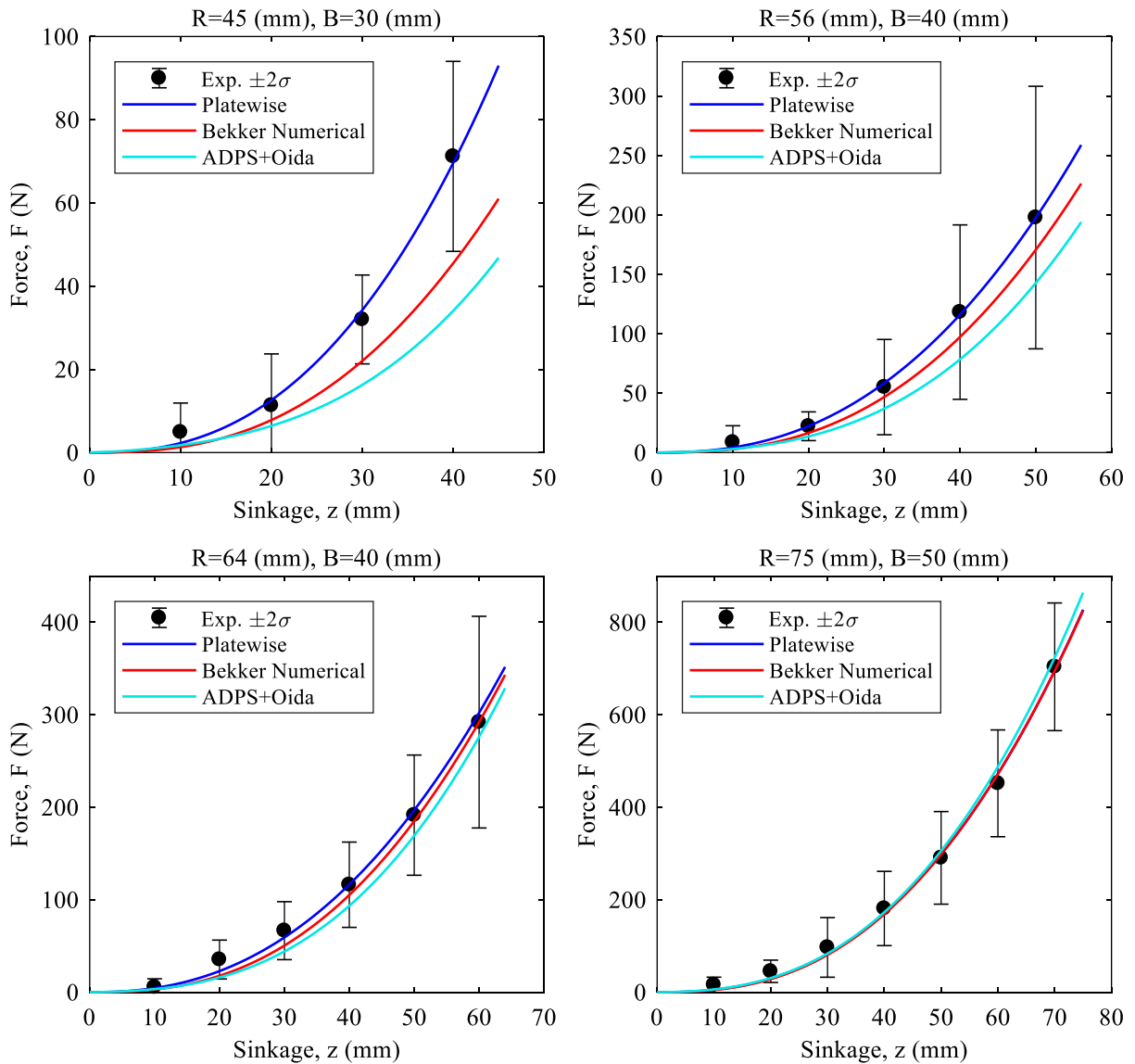


Figure 4.64 Examples of force-sinkage for four different plates on sharp sand.

The reason behind the lack of correlation quality seems to be linked to the soil itself. Figure 4.65 illustrates the normalised force-sinkage experimental results for fine and sharp sand, with the leftmost graph depicting the data of the 30×90 mm flat plate, and the rightmost graph a similar dimensions curved plate of $B = 30$ mm and $R = 48$ mm. In each graph, the curves are normalised with the maximum force of fine sand at 40 mm sinkage (in both cases produces the highest force), so that a relative force comparison can take place.

It can be clearly noted that the gap between the two curves for the flat plate is significant and widens as the plate sinks deeper into the soil. On the other hand, the curved plate results illustrate a different behaviour where the gap between the two curves remains quite small for up to 30 mm sinkage, and moderately widens at larger sinkage level. The actual difference between the fine and sharp sand curves is presented in Figure 4.66, where it becomes even more apparent that for flat plates the force difference between the two soils is much greater, reaching a maximum of approximately 55 % higher forces for fine sand. However, the force difference for the equivalent curved plate remains below 10 % for up to 30 mm of sinkage and even beyond that point it does not reach a value above 30 %. This means that there is an inconsistency in the measurement data between the two tested methods.

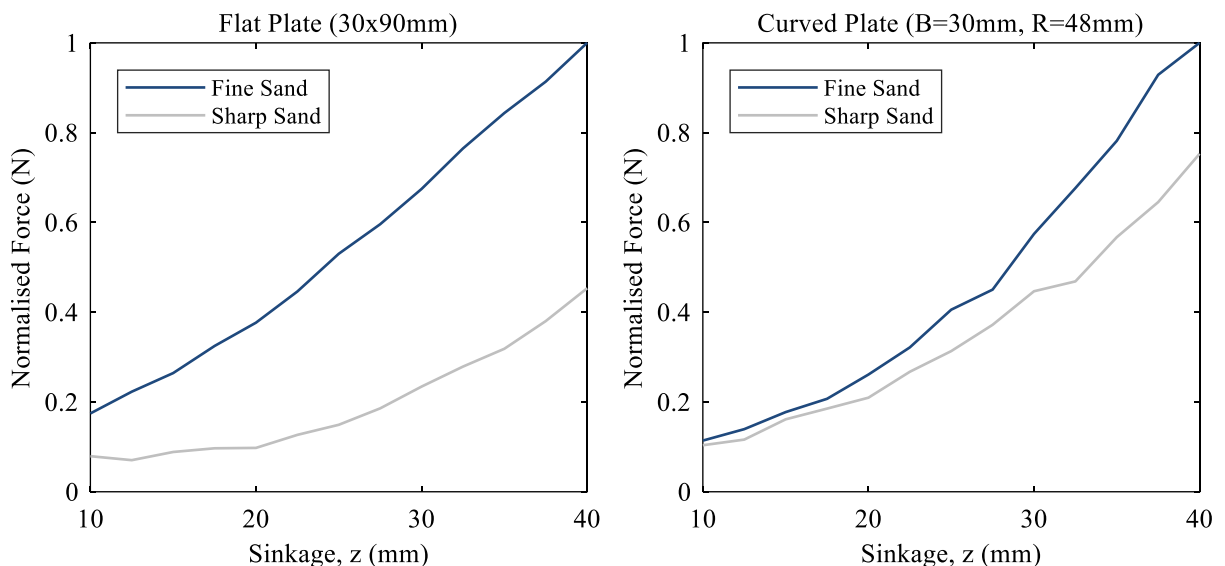


Figure 4.65 Comparison between fine and sharp normalised force-sinkage curves for flat and curved plates.

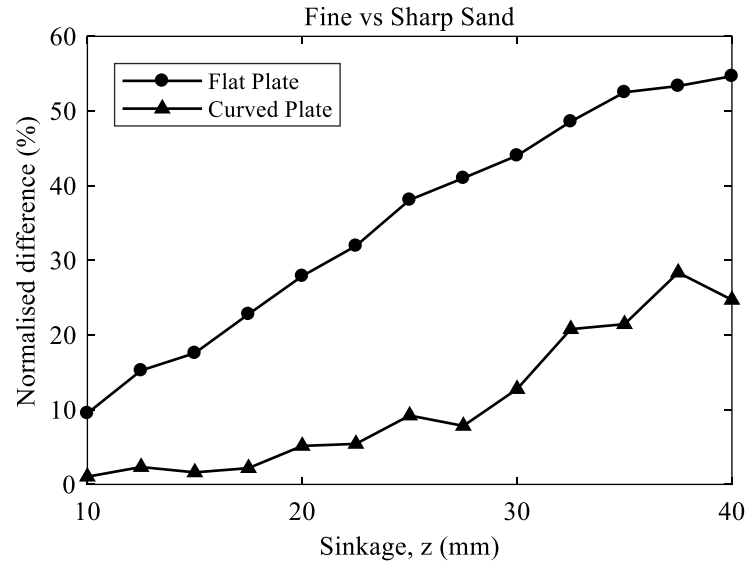


Figure 4.66 Percentage difference between fine and sharp sand for flat and curved plates.

Considering that the experimental methodology, data acquisition and data processing are the same between flat and curved plates, the only possible factor for the discrepancy is the soil conditions. As discussed in Chapter 4.2, the sharp sand contained an inherent moisture content of approximately 4 %, and it was later found that the soil can completely dry out in the lab in just one week. Moreover, the curved plate experiments were performed three days after completing with the flat plate, which is enough time for the soil conditions to change. Throughout the testing campaign, the author tried to maintain the soil's moisture content by adding moderate amount of water into soil box at frequent intervals, however since the soil could dry so quickly it is practically impossible achieve constant soil conditions, especially when the testing duration spans over several days.

In conclusion, it is believed that during the curved plate experiments the moisture content of sharp sand dropped below 4 %. As a consequence, there was a reduction of void ratio causing the soil to become denser and the particles to come closer; ultimately producing higher measured forces compared to flat plates. As expected, the smaller plates suffered the greater results discrepancy since any force variation accounts for larger portion of the total measured forces. The above described problem only affects the sharp sand measurements; since fine sand is completely dry provides better results repeatability as demonstrated by the coefficient of variance data presented earlier in this chapter (see Figure 4.7).

The radar chart of Figure 4.67 summarises the relative error to two standard deviations for the sharp sand data on curved plates. Despite the issues with the measurement results repeatability, the two thirds of measured plates (mainly the larger ones) are within the two standard deviation envelop for the ADPS+Oida model, with the results from Bekker's numerical method to provide a better correlation. However, as seen in the fine sand curved plate results, Bekker's numerical method tends to exaggerate the soil forces, but since all models underestimate the soil response for sharp sand, Bekker's results provide an answer closer to experimental data.

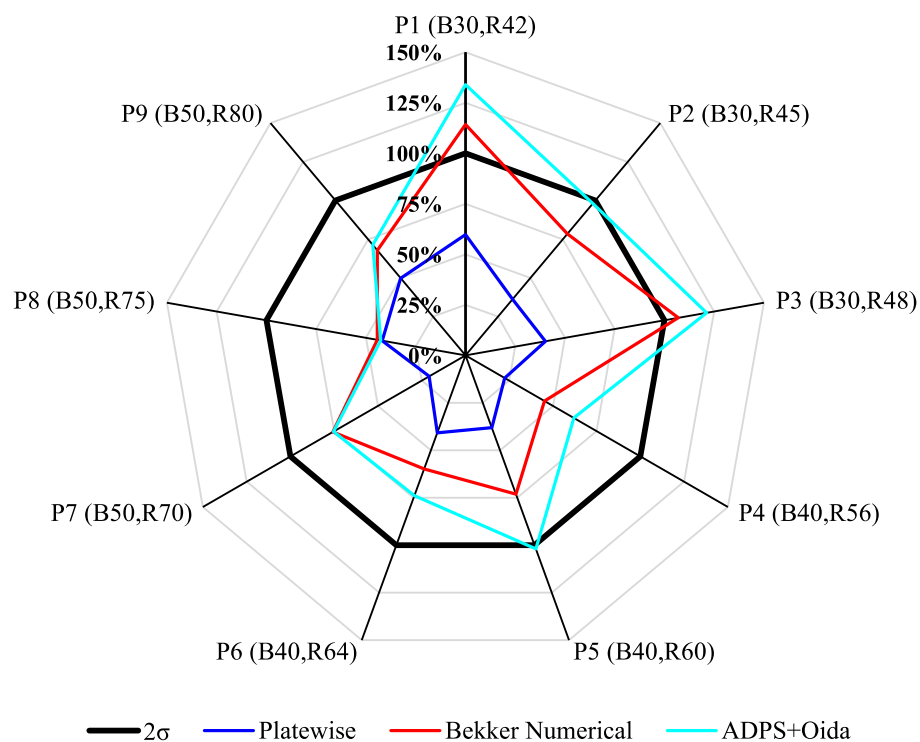


Figure 4.67 Relative error to standard deviation between sharp sand test data on curved plates and fitting methods.

4.5 Chapter Conclusions

In the current chapter, an extensive experimental campaign was illustrated and the results analysed for two different soil types. The experiments, which were conducted by the author using Loughborough University's Terramechanics scale rig, aimed at investigating three main ways of describing the soil behaviour through different test procedures, namely: pressure-sinkage on flat plates, pressure-sinkage on curved plates, and shear.

For pressure-sinkage on flat plates, it was found that the Bekker formulation cannot provide reasonable correlation to test data when the classical fitting method is used for deriving the model parameters. The Bekker's model prediction was significantly improved for both soil types when the author employed an optimisation algorithm for the model parameter identification process.

Analysis of the experimental data on flat plates revealed that the pressure-sinkage relationship is not only dependent on plate width as the Bekker model suggests, but rather the plate area. Based on that, the author developed a novel area-dependent pressure-sinkage model (ADPS), which provides a significant improvement in correlation compared to Bekker, and offers extensive extrapolation capability which is crucial for real life vehicle Terramechanics studies. The new model consists of just one additional parameter compared to Bekker's model, and it requires data from one additional plate in order to perform the parameter identification through an optimisation algorithm.

The soil measured responses displayed a 'hump' in shear stress as expected from the literature for a sandy soils. Three models available in the literature were evaluated and they could all successfully fit shear stress-strain curves. However, out of the three only the Oida model fully respects the crucial maximum shear stress input parameter, and for this reason was selected for further analysis. Analysing the shear data it was found that both the residual shear stress, as well as the strain at the maximum shear stress, are dependent on normal pressure. So, to enable the usage of the Oida model for road vehicle applications, the author extended the model to incorporate the normal stress dependency. In total, the extended Oida model requires six input parameters which are derived through a multi-step optimisation process developed by the author. The final set of parameters fit very well the experimental data.

When curved plates are used, the Bekker's assumption that the total soil reaction force is the sum of component forces along the soil-wheel interface proved valid. However, when Bekker simplified the

original integral equation into a simpler analytical form the vertical force prediction reduced its validity to shallow sinkage only. Instead, if the original integral equation is solved through numerical methods, the prediction accuracy significantly improves at moderate and deep sinkage.

Following the same principle, the author deployed the ADPS model for curved plates with the results providing an overall better correlation than both Bekker methods. Furthermore, because the pressure-sinkage tests were performed in a dynamic manner at an approximately 3 mm/s sinkage rate, the soil shearing effect cannot be ignored. For this reason, the author combined the ADPS model with the extended Oida model for shear, and the combined model further improved the correlation quality on fine sand. On the other hand, the correlation on sharp sand was inferior (but still reasonable) due to inconsistency of the soil material caused by variation in the moisture content.

Chapter 5: Soil-Wheel Interaction

The previous chapter focused on the procedure and analysis of in-house soil data acquisition, along with the creation of new models for predicting pressure-sinkage and shearing behaviour, which consist the elements necessary for an accurate soil-wheel interaction model. The current chapter concerns the data acquisition of rigid-towed wheels, post-processing methodologies for the estimation of wheel angular velocity and slip, as well as data analysis of all tested wheels. Finally, a newly developed soil-wheel model is described, and the results are compared to experimental data.

5.1 Rigid Wheel Experiments

For the tests on rotating rigid wheels the experimental apparatus is the same to the shear tests, with the addition of a video camera mounted on the rig's horizontal sliding platform. As shown in the schematic of Figure 5.1, the video camera is pointing to the wheel's side face with the purpose of recording the wheel rotation during the tests. The obtained video recordings are processed afterwards in order to extract the wheel angle and angular velocity, and to calculate the longitudinal slip.

The camera used for the data acquisition can record up to 120 fps, but during the experiments 60 fps were selected as a satisfactory frame rate. Usually, the file formats generated from video cameras use variable frame rate in order to reduce the final output file size, so the declared frame rate is essentially the maximum value the camera can record. However, for the purpose of calculating the wheel angle and angular velocity a constant frame rate is required. For this reason, all video output files were regenerated at a constant frame rate of 60 fps using a freeware video editing software. After converting the original video files to a constant 60 fps in *.mp4 format, they were imported into Matlab for calculating the wheel angle and angular velocity through various functions from Matlab's Image Processing Toolbox and Computer Vision System Toolbox.

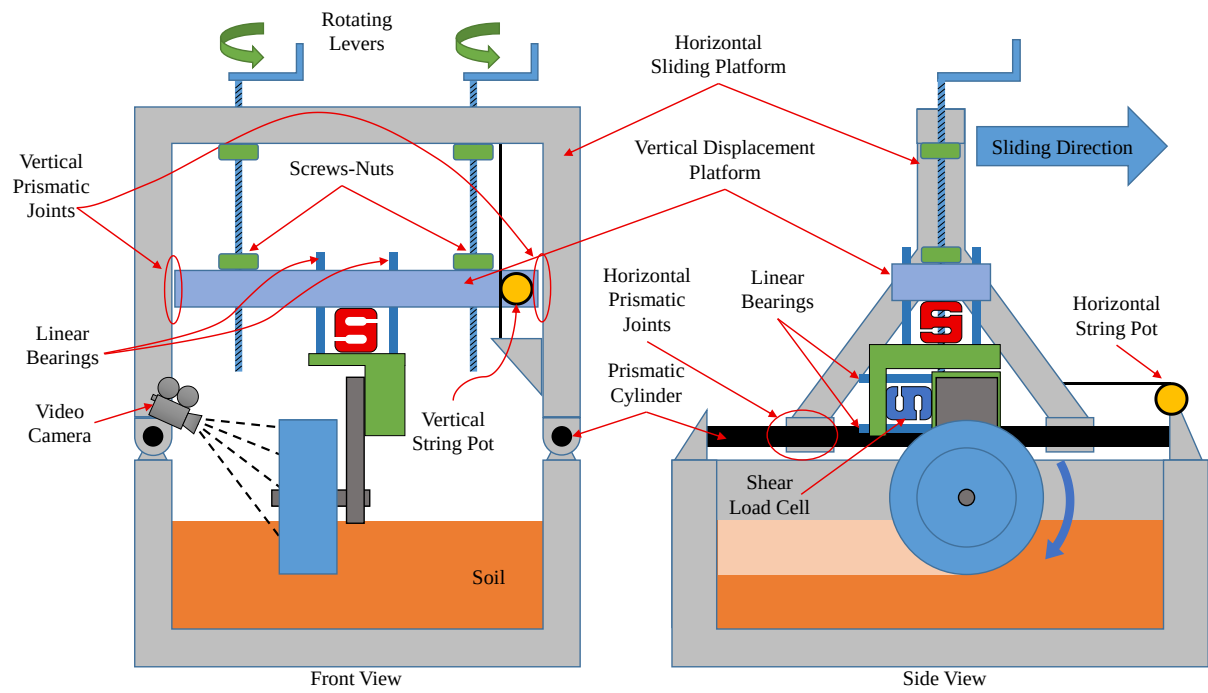


Figure 5.1 Terramechanics scale rig schematic setup for rotating wheels.

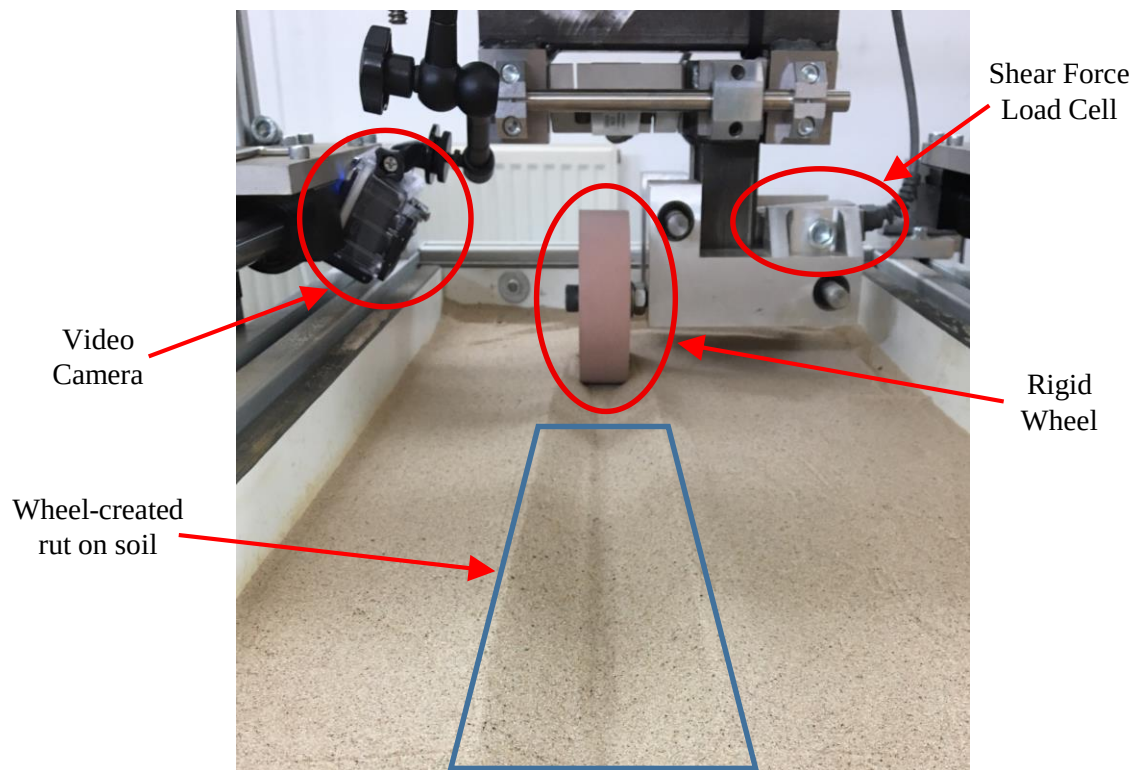


Figure 5.2 Rigid wheel rolling on fine sand using the Terramechanics scale rig.

Figure 5.4 illustrates the high level process, in the form of a flowchart, which was required for the extraction of wheel angle and angular velocity from the video recordings. After reading the *.mp4 files and extracting the total number of frames (steps 1 and 2), a loop starts (step 3) where in each iteration two consecutive frames are read and saved as images (step 4). In order for the preceding steps to successfully converge, some image pre-processing is required (steps 4 and 5) such as cropping out the image background, converting from RGB colour to grayscale and adjusting intensity for improved contrast. An example output image after step 4 is shown in Figure 5.3 (a). As it will be explained in more detail later in the text, it is necessary to remove the centre bolt from the image, which is done through the ‘regionfill’ Matlab function in step 5. This function removes objects from an image by applying an inward pixel interpolation in a specified region using the neighbouring pixels, with the example of Figure 5.3 (b) illustrating the result of removing the centre bolt in comparison to the original image of the same frame as shown by Figure 5.3 (a).

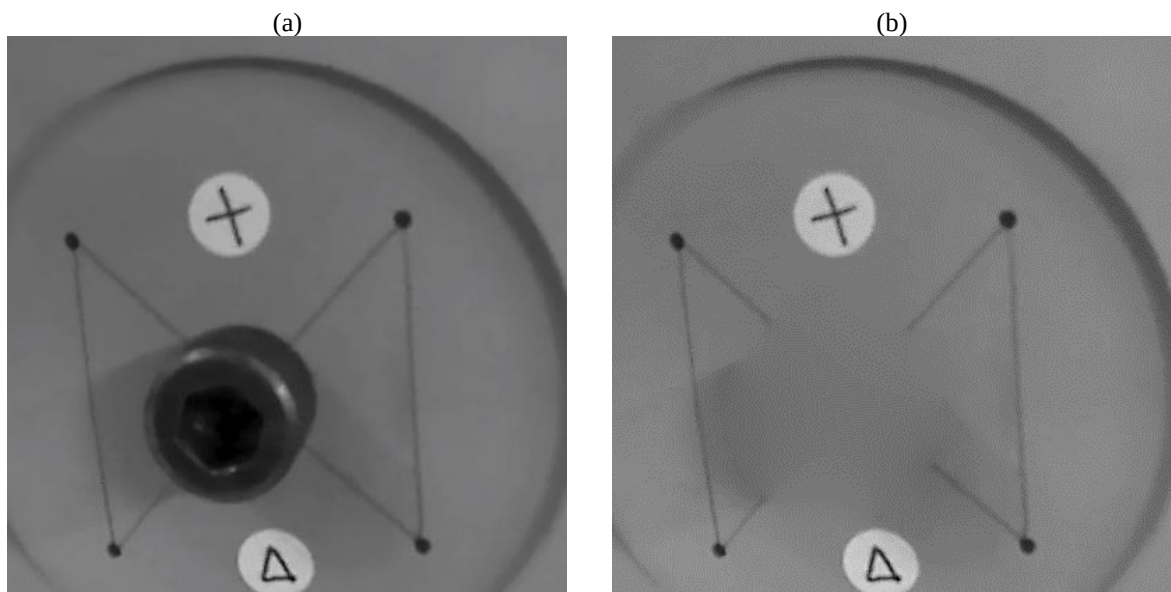


Figure 5.3 Original image of wheel (a) vs image with removed centre bolt (b).

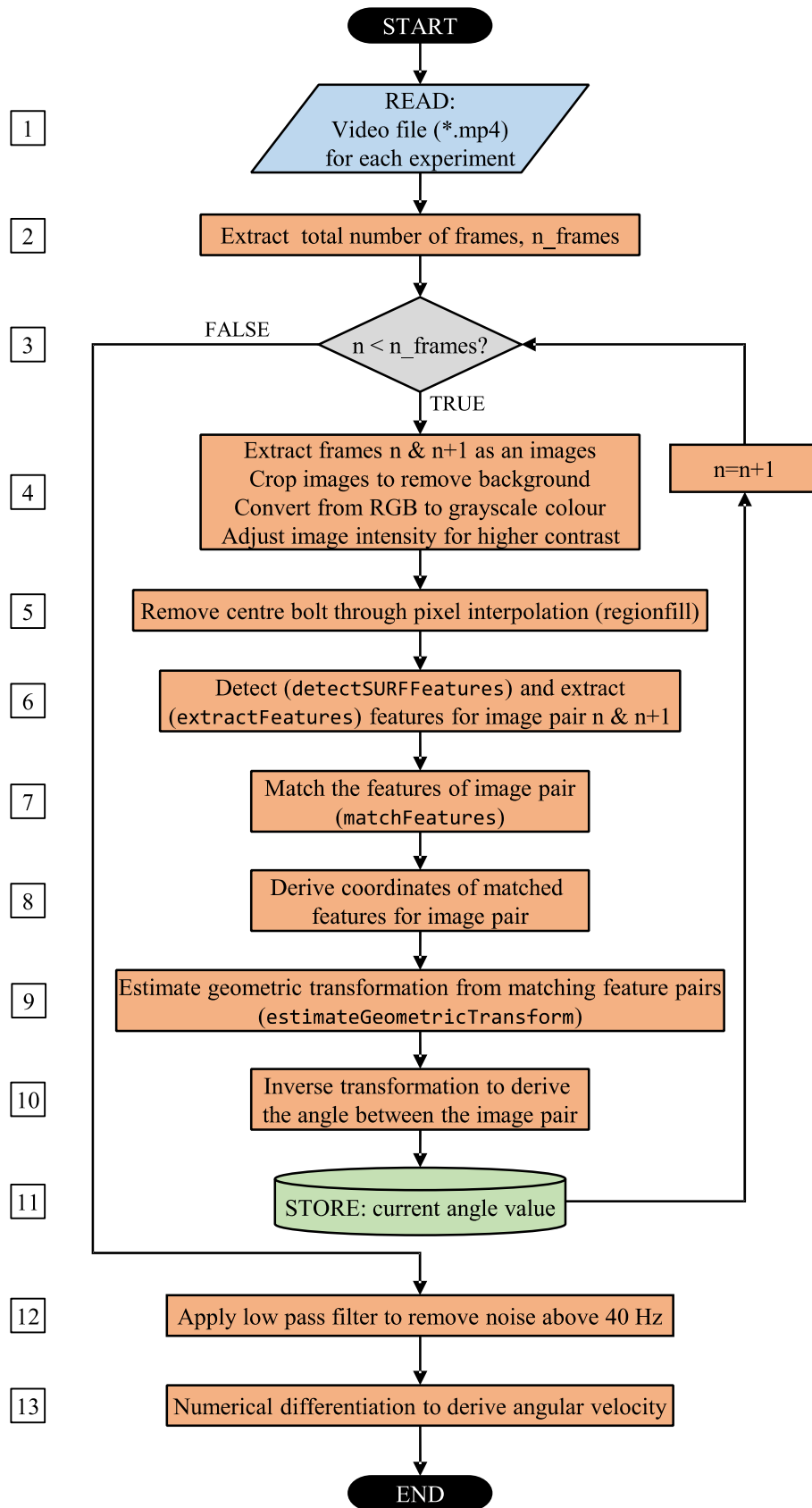


Figure 5.4 Flowchart of wheel angle and angular velocity calculation from video recordings.

After pre-processing, the image features of the two consecutive frames are detected and extracted (step 5), followed by a process of finding matching features between the two images (step 7) and deriving the matching feature pairs coordinates (step 8). Figure 5.5 presents an overlay of two consecutive frames, with the red circles and blue squares corresponding to the feature points of frame n and frame $n+1$ respectively as identified through steps 6-8, and the green lines corresponding to the magnitude of the distance between each feature pair. In step 9, the feature pairs are parsed into Matlab function 'estimateGeometricTransform' in order to calculate the transformation matrix defining the relative position and orientation of image $n+1$ with respect to image n . Through an inverse transformation (step 10), the relative angle of frame $n+1$ with respect to frame n can be derived. After the loop is completed, a low-pass Butterworth filter is applied to the angle vector in order to eliminate noise above 40 Hz (step 12). Finally, the wheel angular velocity is calculated via a numerical differentiation with time of the angle vector (step 13).

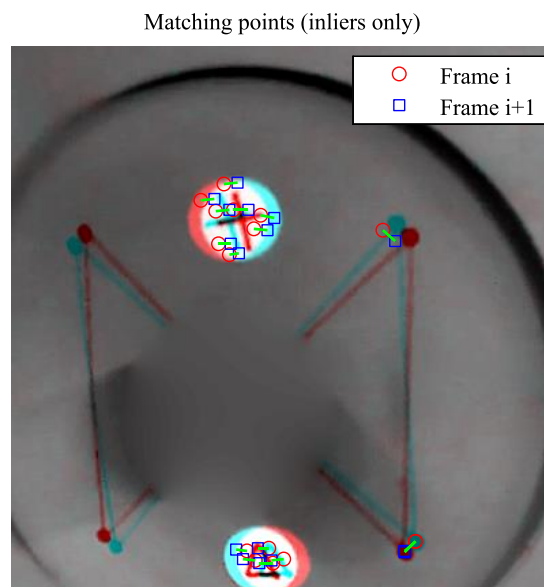


Figure 5.5 Two consecutive frames with matching features indicated by the red and blue dots, green lines correspond to distance between the feature point pairs.

As briefly described in the previous paragraph, in step 5 of the flowchart of Figure 5.4 the centre bolt is removed. This step is found necessary for improving the accuracy and robustness of the overall wheel angle calculation. Because the centre bolt – which attaches the wheel on the rig – is stationary throughout the test, often the identified features and feature pairs are concentrated on the bolt hex socket head. As a consequence, the transformation matrix estimation results in zero relative angle between two

consecutive frames, and since this error occurs multiple times for each video recording the overall wheel angle vector can result in significant error accumulation. In Figure 5.6 an example of a wheel angle identification process without removing the centre bolt ('Basic'), and with removing the centre bolt ('Advanced') are presented, where an obvious difference between the two curves can be observed. In this particular example, the final angle difference between the two curves is 6.4° , however differences of up to 20° have been recorded. The accuracy of the 'Advanced' method has been confirmed by the author via visual checks of the output curves versus the video recording at particular check points (e.g. every 90° wheel rotation). Furthermore, the 'Basic' method suffers from robustness issues such as the example of Figure 5.7, where between two time steps an angle as large as 90° or 180° can be calculated, or in other cases a negative angle is produced by the algorithm. Such issues are completely absent from the 'Advanced' method.

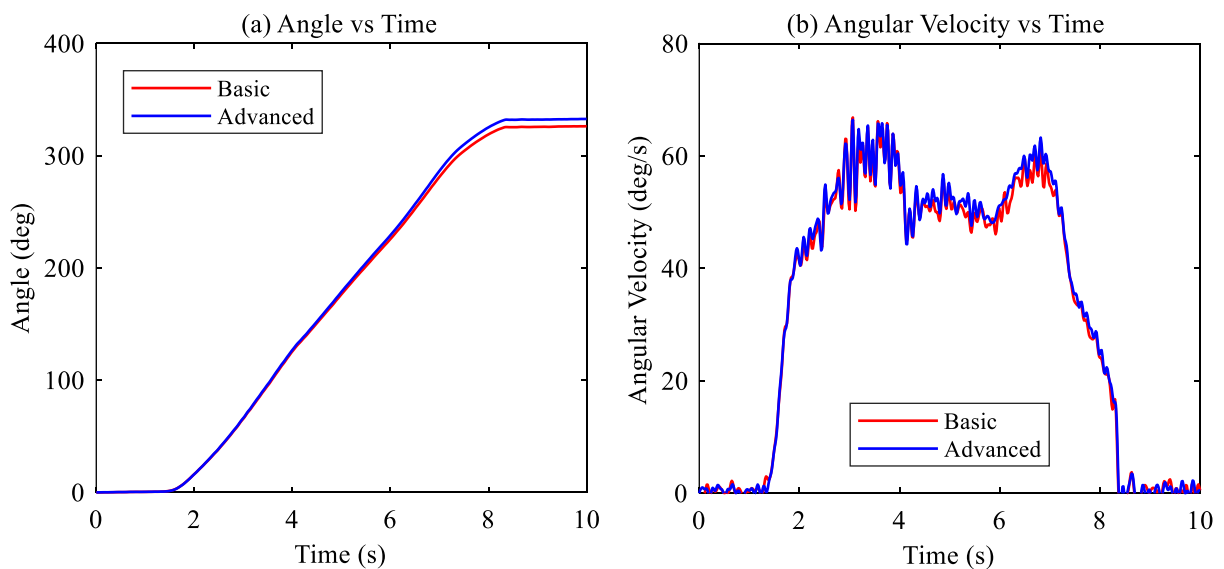


Figure 5.6 Wheel angle and angular velocity vs time, comparison between two calculation algorithms.

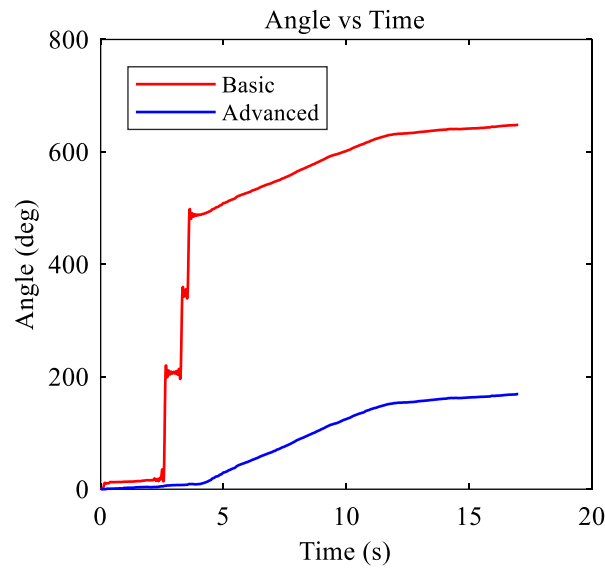


Figure 5.7 Example of robustness issue using the ‘Basic’ angle estimation algorithm.

In order to define appropriate dimensions for the scaled rigid wheels to be tested, a market research was conducted by the author, which analysed the wheel dimensions of sport utility vehicles (SUVs) sold in the UK market in 2016. The width/diameter ratios of more than sixty vehicle variants from a wide range of manufacturers are plotted in a histogram as shown in Figure 5.8. In order to cover the whole range of width/diameter ratios of Figure 5.8, it was decided to manufacture three rigid wheels; one for the smallest ratio ($B/D = 29.2$), one for the mean ratio ($B/D = 34$) and one for the largest ratio ($B/D = 44.6$), which from now on will be referred to as ‘Narrow’, ‘Nominal’ and ‘Wide’ respectively.

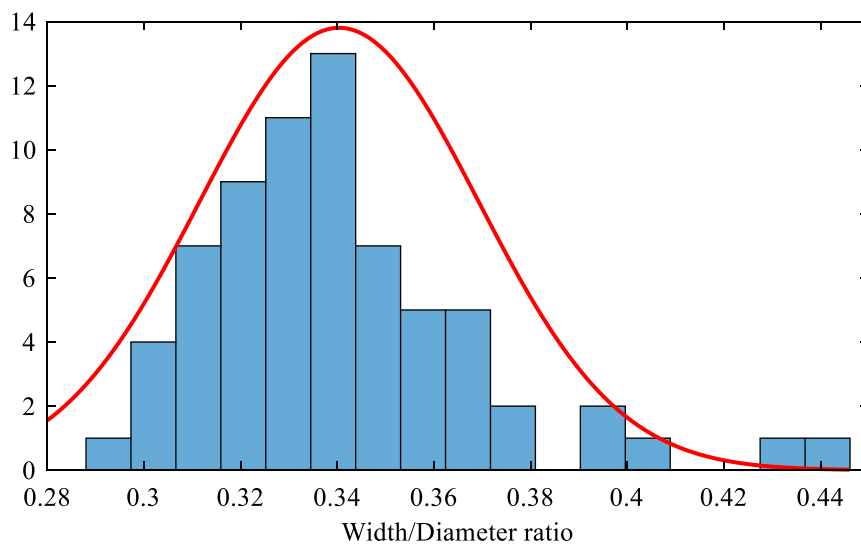


Figure 5.8 Histogram of the tyre width/diameter ratio using market data from passenger SUVs.

Considering the Terramechanics scale rig available space, it was decided to manufacture rigid wheels with a diameter of 100 mm, which corresponds to widths of 29.2 mm, 34 mm and 46 mm for the three wheels. These dimensions would allow up to two full rotations of the wheels (depending on slip ratio), as well as zero or negligible impact with the soil box inner walls. The wheel designs are presented in Figure 5.9 below.

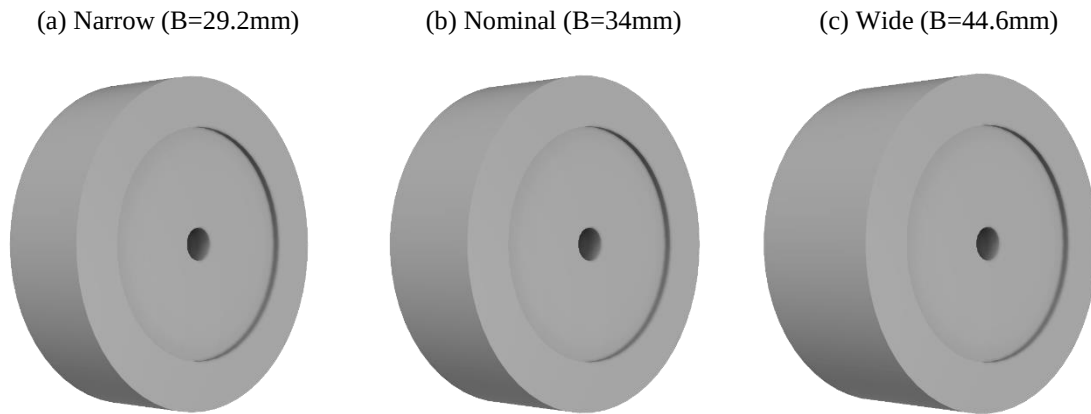


Figure 5.9 Tested rigid wheels with smooth surface.

5.2 Data Analysis

Apart from the experimental apparatus, the testing procedure and data post processing methodology of rigid towed wheels are almost identical to the shear tests as well. Each of the three rigid wheels of Figure 5.9 would be sunk into the soil at different depths at one end of the soil box, and then starting the horizontal sliding until reaching the other end of the soil box. During this process, forces and displacements in the vertical and horizontal directions are acquired, as well as video recordings of the wheel rotation. Since the video camera recording was not synchronised with the rest of the acquisition system, the start/end points had to be manually identified through triggers in the force/displacement channels and visual observation of the video footage.

The example plots of Figure 5.10 illustrate time-domain data after the post-processing was completed, which they cover the entire duration of one experimental repetition starting from the wheel at standstill, but already sunk into the soil, until reaching the end of the soil box. Looking at the plots it can be easily observed the presence of two distinct regions in the soil-wheel responses.

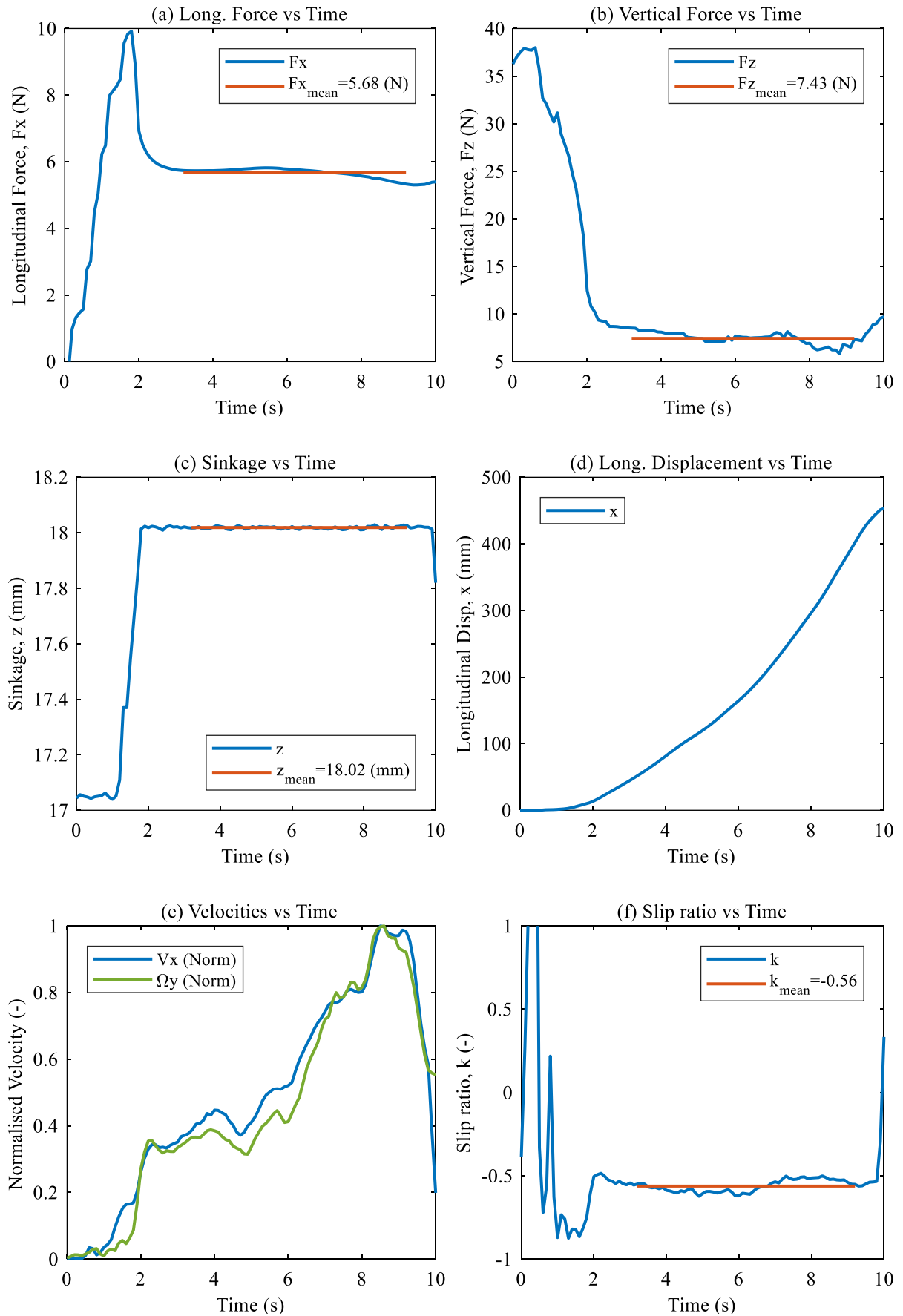


Figure 5.10 Example of experimental outputs from Narrow wheel on fine sand.

The first region (transient), which lasts for approximately 2 sec in this particular example, covers the initial part of the wheel movement where fluctuations are observed in the vertical and horizontal forces, vertical displacement and slip ratio. After entering the second region (steady-state), the responses in all plots are almost constant. The previous statement does not apply to plot (e), where the wheel normalised horizontal velocity (blue curve) and angular velocity (green curve) are overlaid, because the horizontal velocity is an input and not a response.

Despite the lack of consistency and inability to maintain completely constant velocity application by the rig operator, it is interesting to note that the wheel angular velocity follows almost the same longitudinal velocity pattern with the linear velocity. Such observation, which is confirmed in all test repetitions at different sinkage/wheel width/soil, indicates that the wheel angular velocity is not dependent on the longitudinal velocity; meaning that the slip ratio is also independent of the wheel longitudinal velocity.

As analysed in Section 4.3.2, both tested soils display a peak at the maximum shear stress which is present in the horizontal force of plot (a) in Figure 5.10. Since the horizontal force is directly affected by the shear stress of the soil-wheel interface, the peak force occurs at the transient region where the slip ratio is still small. After entering the steady-state condition, the longitudinal force settles at the residual shear stress area as the slip ratio (i.e. shear strain) has also settled at a fairly large value. This outcome also confirms that for sand-like soils, considering the Mohr-Coulomb failure criterion alone as normally done in the literature, would result in overestimation of longitudinal forces and rolling resistance.

The small sinkage difference between the transient and steady-state conditions, as shown in Figure 5.10 (c), is caused by the rig's prizing effect which was described in detail in Section 4.3.2. Furthermore, a significant vertical force drop when passing from transient to steady-state is caused by two factors. Firstly, the initial vertical force value is measured with both front and rear wheel surfaces in contact with the soil, since the wheel is still stationary and at equilibrium. Once the wheel starts the horizontal rolling, the majority of the rear surface is no longer in contact and a drop in force is expected. Secondly, according to literature ([34], [49]) for towed rigid wheels the maximum normal pressure shifts forward along the wheel surface generating rolling resistance (i.e. longitudinal force), which means that further vertical force reduction is expected.

For each of the three tested wheels and two soils (fine and sharp sand), the results demonstrate the same behaviour as in the example case of Figure 5.10. Since at steady-state condition the responses are almost constant, an average value is taken for each curve, with the red bars highlighting the magnitude and the range considered for the averaging. These average values are analysed and discussed later in the text.

It is known from the literature ([25], [33-35],[49]) that when rigid wheels traveling on deformable terrains, soil accumulation in front of the wheel's surface takes place called bulldozing. Hence, the actual sinkage depth in front of the wheel is higher than the original static sinkage level, which results in increase in rolling resistance. The bulldozing effect is not clearly understood [9], and with experimental methodologies required to measure being fairly complex its estimation is usually done through finite element simulations [57]. Since bulldozing is an essential part of the overall wheel response, especially for the prediction of rolling resistance, the author used the video recordings in order to estimate the total sinkage depth at steady-state condition. It is worth noting that after observing all video recordings, it is confirmed that the soil accumulation reaches its maximum value just a few centimetres after the wheel starts its horizontal travel, and that it remains constant thereafter. This statement is supported by the longitudinal force response which also remains effectively constant at the steady-state region.

For the total sinkage depth z_T , the author extracted snapshots from the video recordings with the wheels in steady-state condition, then marked various parts on the pictures (e.g. wheel top and centre, soil surface) and measured the pixel distance between them (see Figure 5.11). Using known geometric parameters, such as the wheel radius, it was possible to transform pixel lengths into actual distances and estimate the total sinkage depth. As the video camera was not installed completely normal to the wheel side surface due to space restrictions on the rig, a single angle transformation was also necessary in order to convert the projected lengths into the actual lengths, with the angle between the wheel side surface and the camera spanning between 30° and 45°.

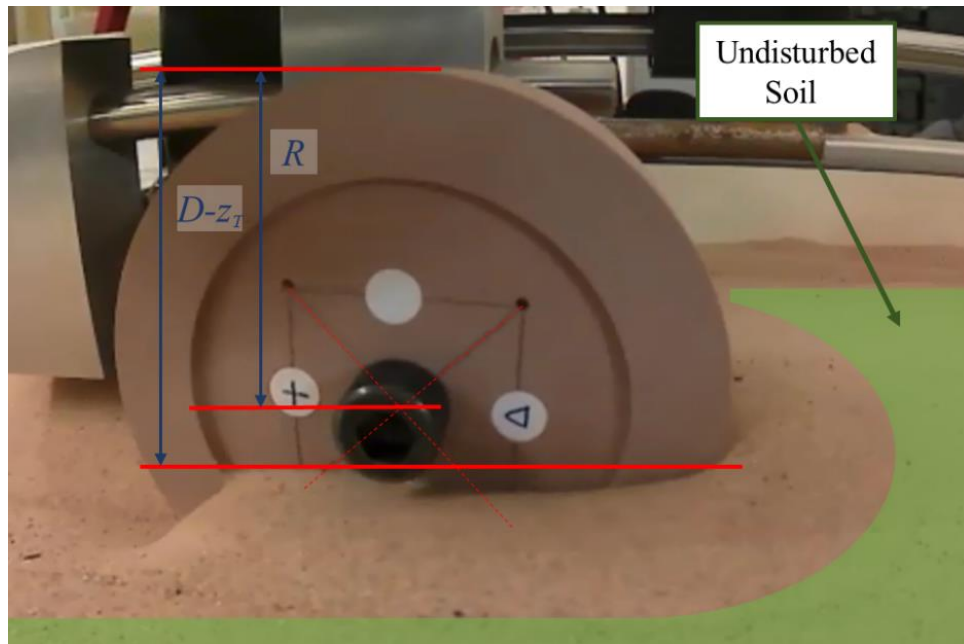


Figure 5.11 Method for obtaining bulldozing sinkage from snapshots.

The derived total sinkage depths are plotted against the static sinkage as shown in Figure 5.12 for the narrow, Figure 5.13 for the nominal and Figure 5.14 for the wide wheel respectively. In all plots, a linear regression line without y-intercept fits almost perfectly the test data, indicating that a simple linear relationship between the initial and total sinkage depth exist. The slope of each regression line increases with increase in wheel width, meaning that the larger the wheel width the higher the volume of soil accumulation, and consequently higher the total sinkage depth, which is in complete agreement with the literature [35].

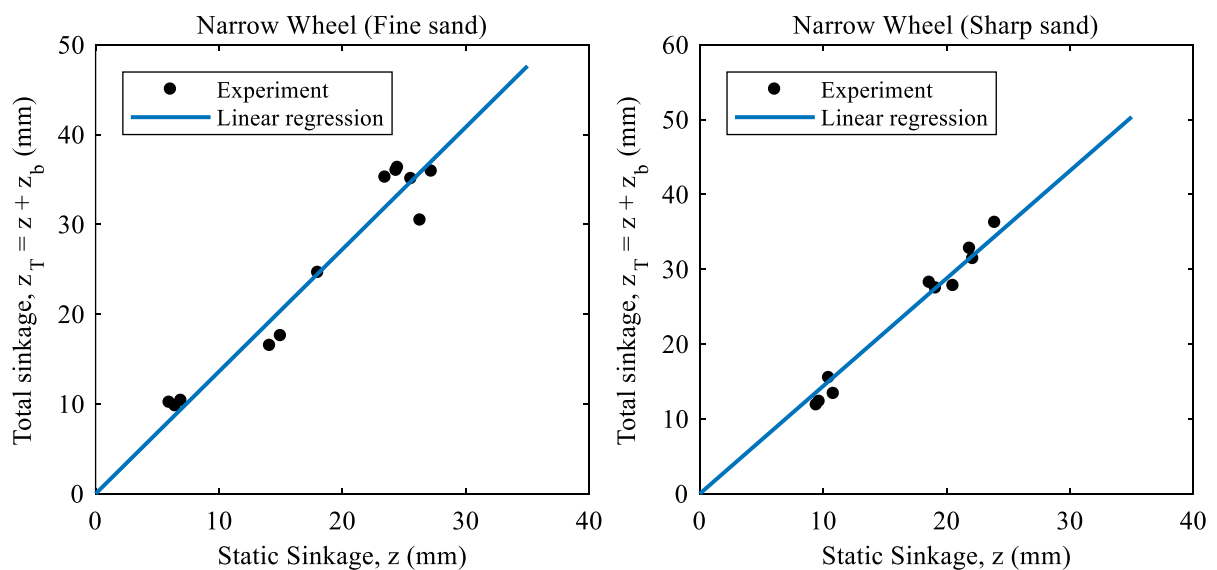


Figure 5.12 Bulldozing effect for the narrow rigid wheel on fine sand (left) and sharp sand (right).

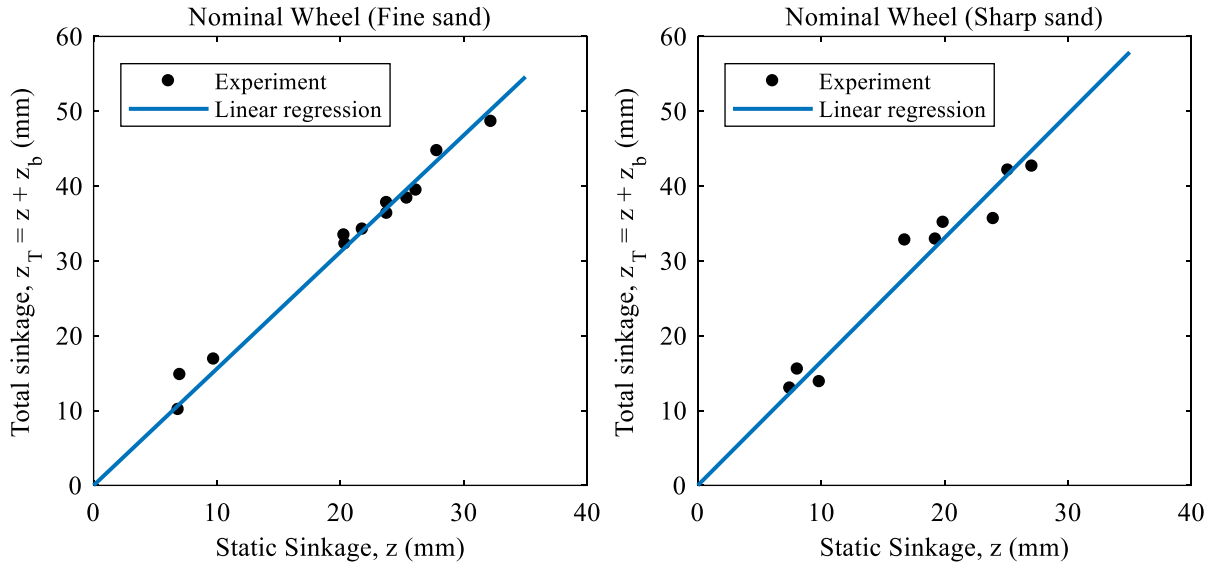


Figure 5.13 Bulldozing effect for the nominal rigid wheel on fine sand (left) and sharp sand (right).

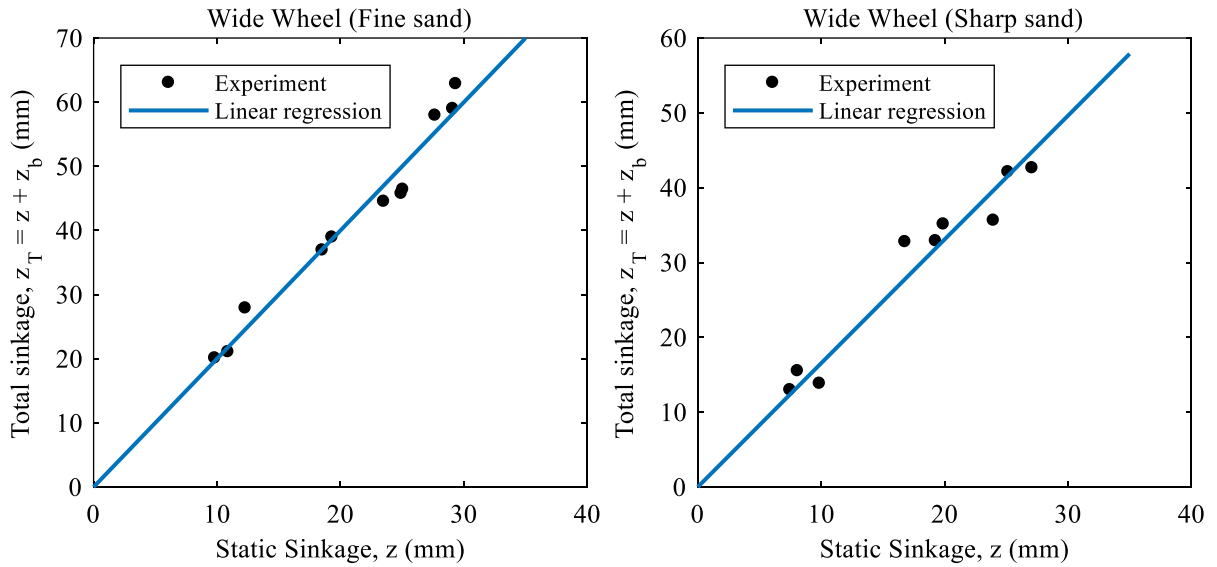


Figure 5.14 Bulldozing effect for the wide rigid wheel on fine sand (left) and sharp sand (right).

The total sinkage depth z_T is defined as the sum of the initial sinkage of the stationary wheel z and the sinkage due to bulldozing effect z_b (Equation 5.1). In addition, the total sinkage can also be characterised as a linear function (Equations 5.2) of the initial sinkage, as highlighted by the plots of Figures 5.12, 5.13 and 5.14 for narrow, nominal and wide wheels respectively.

$$z_T = z + z_b$$

5.1

$$z_T = \alpha z$$

5.2

The calculated slopes α of Equation 5.2 for all three wheels for fine and sharp sand are summarised in Table 5.1 below.

	α Narrow (-)	α Nominal (-)	α Wide (-)	β (1/m)
Fine Sand	1.36	1.56	2.00	45.51
Sharp Sand	1.44	1.65	2.14	48.46

Table 5.1 Summary of linear fitting slopes.

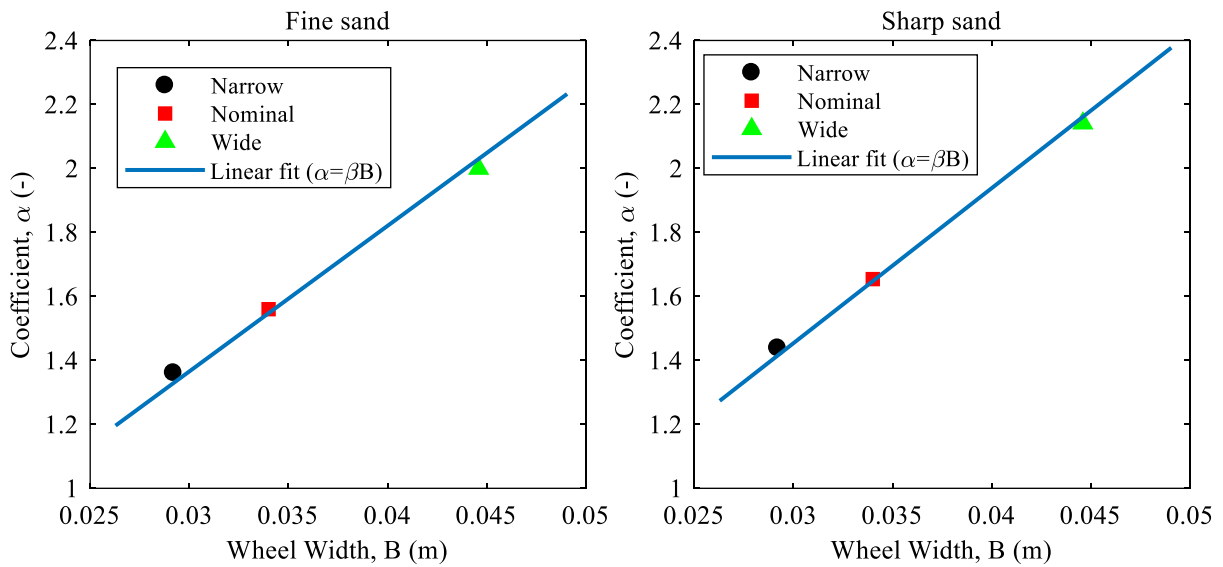


Figure 5.15 Wheel width dependency on bulldozing.

In Figure 5.15 the slopes α are plotted against their corresponding wheel widths for fine and sharp sand, where in both soils a linear regression line without y-intercept fits well the data. In both cases, as summarised in Table 5.1, the slopes β are approximately equal to the wheel radius B multiplied with 1000, leading to the following approximate relationship of Equation 5.3.

$$\alpha = \beta B \approx 1000RB$$

5.3

When combining Equations 5.1, 5.2 and 5.3, the new formulation of Equation 5.4 is created which describes the sinkage due to bulldozing as a function of the wheel geometric characteristics (radius and width) and the nominal static sinkage.

$$z_b = 1000RBz - z \quad 5.4$$

Applying Equation 5.4 on Equations 5.1, results in the following formulation for the total sinkage at steady state condition:

$$z_T = 1000RBz \quad 5.5$$

In an earlier paragraph of the current chapter, it was noted that the wheel slip and angular velocity is not affected by variations in the linear velocity. As shown in the Figures 5.16, 5.17 and 5.18 a direct relationship exist between the static sinkage and the wheel slip for both soil types. This relationship can be expressed quite well by a linear regression line. Furthermore, there is a difference between the behaviour of the two soils, where the initial y-intercept at fine sand is up to three times higher in absolute terms, compared to sharp sand. However, fine sand's slope is half of the one of sharp sand, and therefore smaller slip sensitivity to static sinkage compared to sharp sand.

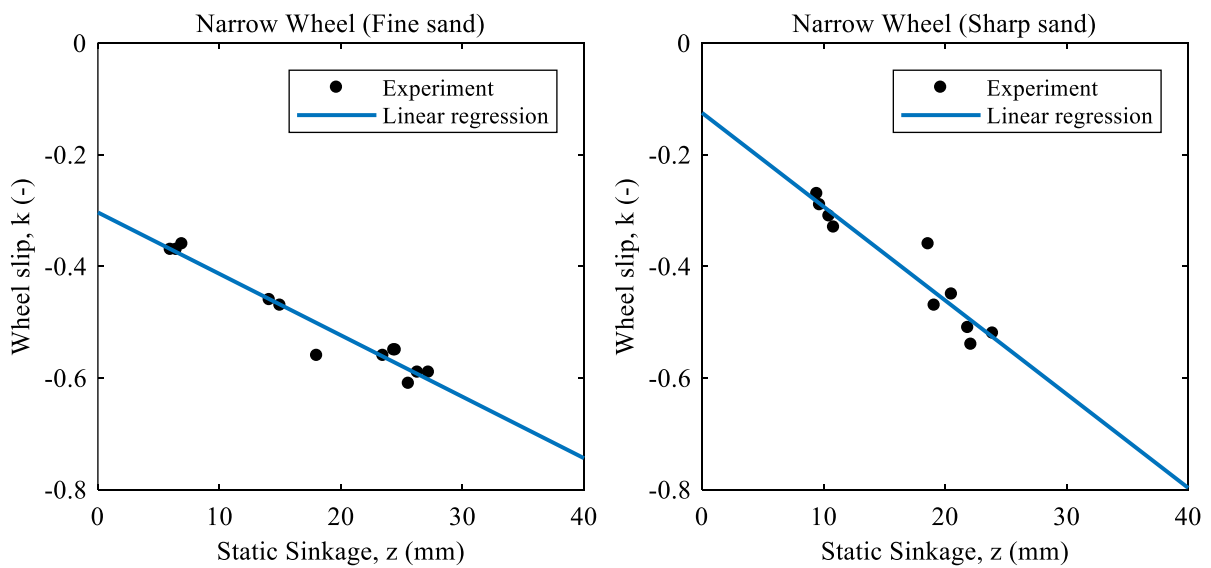


Figure 5.16 Slip vs static sinkage for the narrow rigid wheel on fine sand (left) and sharp sand (right).

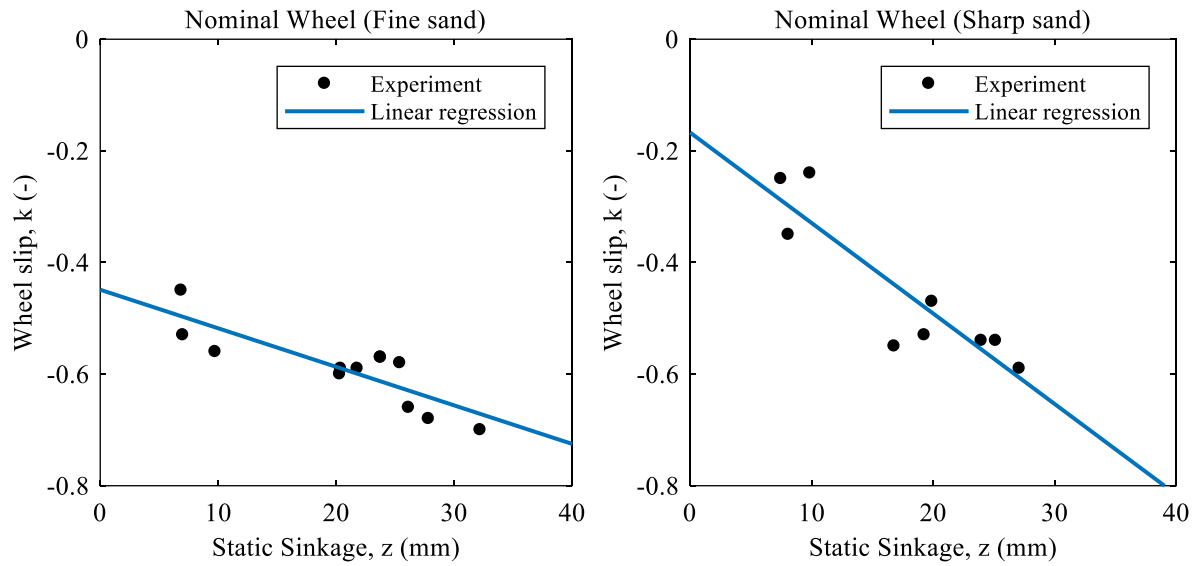


Figure 5.17 Slip vs static sinkage for the nominal rigid wheel on fine sand (left) and sharp sand (right).

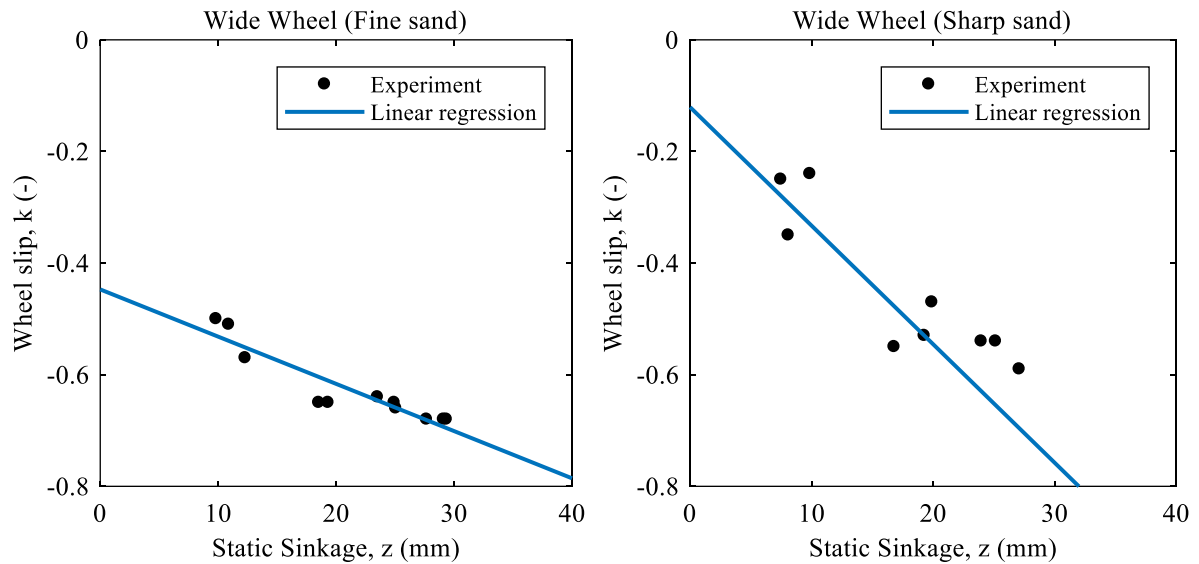


Figure 5.18 Wheel slip as a function of static sinkage for the wide rigid wheel on fine sand (left) and sharp sand (right).

The differences in and y-intercepts between the linear regression lines of the same soil indicate there are additional dependencies affecting the wheel slip other than the static sinkage, with the wheel width being the prime potential candidate. In order to gain a comprehensive understanding of the wheel slip mechanism, additional analysis and investigation is required which is not covered by the current text. The slope and y-intercepts of all wheels and soils are summarised in Table 5.2 below.

	Fine Sand		Sharp Sand	
	Slope (1/mm)	y-Intercept (-)	Slope (1/mm)	y-Intercept (-)
Narrow	-0.01	-0.31	-0.02	-0.13
Nominal	-0.01	-0.45	-0.02	-0.17
Wide	-0.01	-0.45	-0.02	-0.12

Table 5.2 Linear regression coefficients for slip vs static sinkage.

5.3 Soil-Wheel Model

As discussed in the literature review in Chapter 2, Wong and Reece [34] further investigated the findings of Onafeco and Reece [49] regarding soil flow patterns under rigid towed wheels on dry sand. According to Wong and Reece [34] the maximum normal stress occurs at an angle θ_M ahead of the wheel bottom-dead-centre, with the front region defined from θ_M until the contact angle θ_1 , and the rear region between θ_M and θ_2 as shown in Figure 5.19.

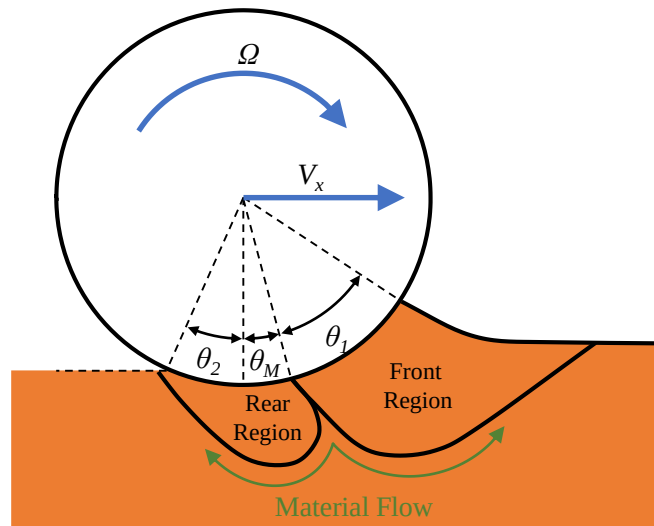


Figure 5.19 Material flow under rigid towed wheels on dry sand, as proposed by Wong and Reece [34].

The vertical (compaction) and horizontal (bulldozing) forces are calculated using Equations 5.6 and 5.7 respectively. Both equations consider the normal and shear stress distribution along the contact surface of the front and rear regions, and then transformed into the pure vertical and horizontal directions.

$$W = RB \left\{ \int_{\theta_M}^{\theta_1} \sigma_1(\theta) \cos \theta d\theta + \int_{\theta_2}^{\theta_M} \sigma_2(\theta) \cos \theta d\theta + \int_{\theta_M}^{\theta_1} \tau_1(\theta) \sin \theta d\theta - \int_{\theta_2}^{\theta_M} \tau_2(\theta) \sin \theta d\theta \right\} \quad 5.6$$

$$F_x = RB \left\{ \int_{\theta_M}^{\theta_1} \sigma_1(\theta) \sin \theta d\theta + \int_{\theta_2}^{\theta_M} \sigma_2(\theta) \sin \theta d\theta - \int_{\theta_M}^{\theta_1} \tau_1(\theta) \cos \theta d\theta + \int_{\theta_2}^{\theta_M} \tau_2(\theta) \cos \theta d\theta \right\} \quad 5.7$$

Also, Wong and Reece define the angle θ_M and through Equation 5.8 which is a function of wheel slip (denoted with the letter i here) and soil internal friction angle. The shear strain along the soil-wheel interface, which is necessary for the shear stress calculations, is defined by Equations 5.9 and 5.10 for the front and rear contact regions respectively.

$$\tan\left(45 - \frac{\phi}{2}\right) = \frac{\cos \theta_M - \frac{1}{1+i}}{\sin \theta_M} \quad 5.8$$

$$j_1 = R \left\{ (\theta_1 - \theta) \left[\frac{(1+i)(\sin \theta_1 - \sin \theta_M)}{(\theta_1 - \theta_M)} \right] - (1+i)(\sin \theta_1 - \sin \theta) \right\} \quad 5.9$$

$$j_2 = R[(\theta_M - \theta) - (1+i)(\sin \theta_M - \sin \theta)] \quad 5.10$$

The author has utilised the general equations provided by Wong and Reece in the creation of a dynamic soil-wheel model, but replaced the definitions for normal stress with the ADPS model (Equations 5.11 and 5.12) and for shear stress with the extended Oida model (Equations 4.8, 2.9, 4.9 and 4.10) which were both developed as part of the current research.

ADPS model equations:

$$\sigma_1(\theta) = \left[a_k A_c^{(b_k+1)} \right] (R \cos \theta - R \cos \theta_1) \quad \left[a_n A_c^{(b_n+1)} \right] \quad 5.11$$

$$\sigma_2(\theta) = \left[a_k A_c^{(b_k+1)} \right] \left\{ R \cos \left[\theta_1 - \left(\frac{\theta - \theta_2}{\theta_M - \theta_2} \right) (\theta_1 - \theta_M) \right] - R \cos \theta_1 \right\} \quad \left[a_n A_c^{(b_n+1)} \right] \quad 5.12$$

Extended Oida equations:

$$\tau = \tau_{\max} K'_r \left[1 - \frac{\sqrt{1-K'_r} \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}}}{\sqrt{1-K'_r} \left(1 - \frac{2}{K'_r} \right) + \frac{2}{K'_r} - 2} \right] \times \left[1 - \left(1 + \frac{\sqrt{1-K'_r}-1}{K'_r} \right)^{\frac{j}{K'_w}} \right] \quad 4.8$$

$$\tau_{\max} = c + \sigma \tan(\phi) \quad 2.9$$

$$\tau'_r = \tau_{r,a} \sigma + \tau_{r,b} \quad 4.9$$

$$K'_w = K_{w,a} \sigma + K_{w,b} \quad 4.10$$

$$K'_r = \frac{\tau'_r}{\tau_{\max}} = \frac{\tau_{r,a} \sigma + \tau_{r,b}}{c + \sigma \tan(\phi)} \quad 4.11$$

The implementation of the above equations was performed in Simulink environment; screenshots of the model are presented in Figure 5.19 (longitudinal force calculation) and Figure 5.20 (vertical force calculation). The red colour blocks contain Matlab function blocks of the actual equations presented above, with ramp blocks also present to ensure smooth transition between states for preventing numerical instabilities. For the vertical forces, as shown in Figure 5.20, the implemented model uses two formulations: (a) the adapted ADPS model for curved plates as developed in Section 4.4 which is used when there is no longitudinal velocity, and (b) the Wong and Reece approach – combined with ADPS and extended Oida models developed by the author – for the part where the wheel is travelling in the horizontal direction. Again, a smooth transition between one force law to another is ensured through ramp up/down blocks. Inputs for both vertical and longitudinal forces are: wheel slip, angle of maximum normal stress θ_M , front contact angle θ_1 , and rear contact angle θ_2 . The calculation of front

and rear contact angles are performed based on the sinkage level and as presented in the block diagram of Figure 5.22. The calculation of angle θ_M is performed iteratively using the Bisection method which ensures it always converges [61]. For correlation purposes, the model position and velocity states are provide by the virtual Terramechanics rig as presented in Figure 5.24.

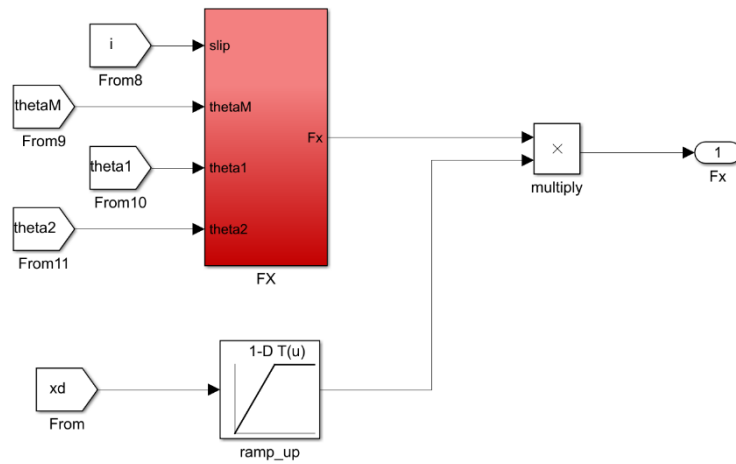


Figure 5.20 Longitudinal force calculation block diagram.

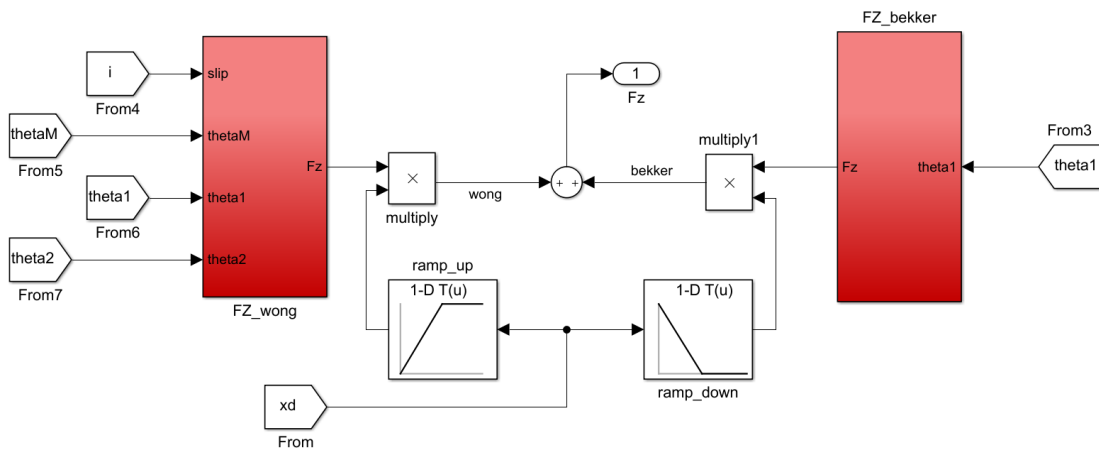


Figure 5.21 Vertical forces calculation block diagram with transition from static to rolling.

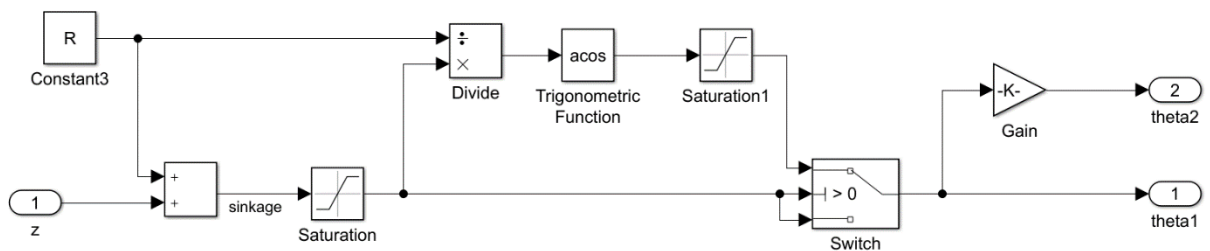


Figure 5.22 Contact angles θ_1 and θ_2 calculation block.

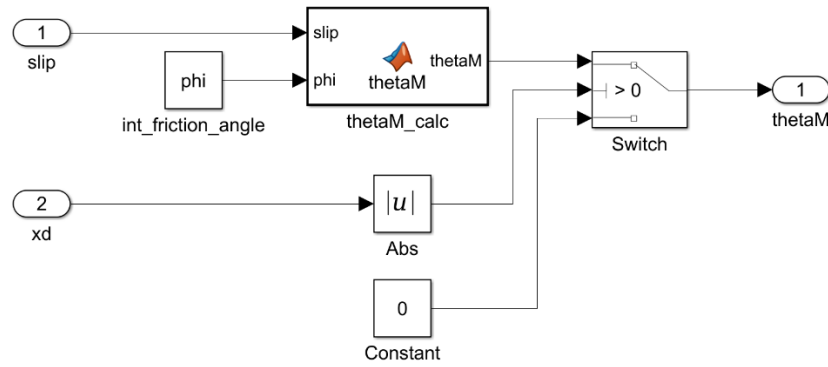


Figure 5.23 Calculation of angle at maximum normal stress θ_M .

As discussed earlier in the chapter, the wheel slip behaviour is not fully understood and further study is required. So, for the numerical simulations the linear relationships summarised in Table 5.2 were used. Also, the integrals along the soil-wheel interface are solved numerically using the trapezoidal method ('trapz' Matlab function) at 0.1° intervals. Finally, the model also considers the effect of bulldozing by adjusting the sinkage consideration based on the analytical formulas presented in the current chapter.

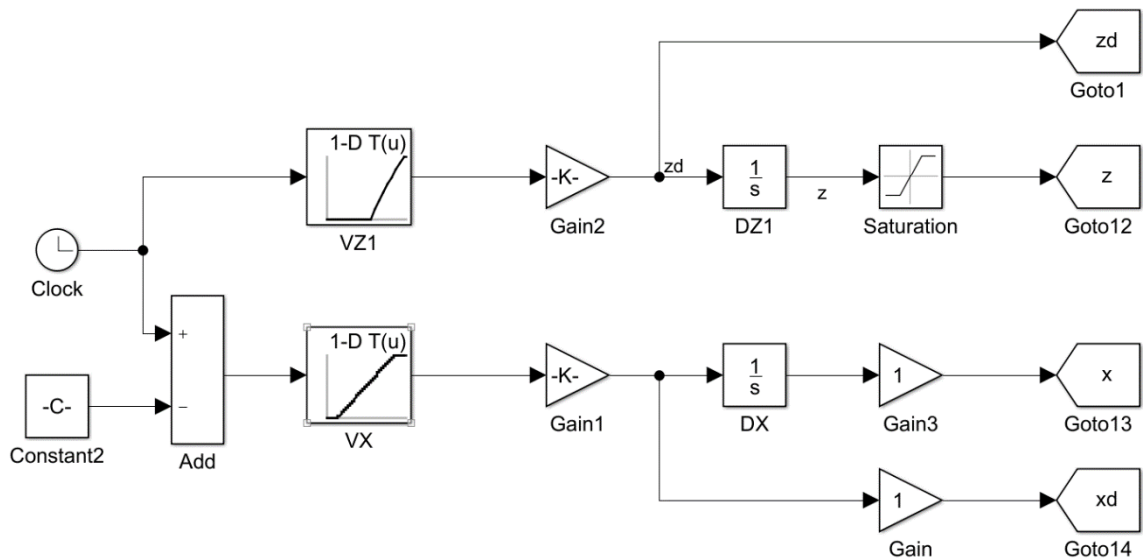


Figure 5.24 States generation from virtual Terramechanics scale rig.

Figures 5.25-5.30 presented in the following pages contain comparisons between steady-state test and simulation data for longitudinal and vertical forces at different sinkage levels. The results of both

longitudinal and vertical forces on fine sand are overestimated if 100% of the bulldozing sinkage is considered by the model. When the bulldozing sinkage reduces to 30% from maximum, then a good correlation for both longitudinal and vertical forces is achieved. Such outcome is reasonable though, because for bulldozing sinkage higher than 30% the soil volume in front of the wheel is not considerable. Since fine sand is a frictional soil, without normal pressure the shearing forces are low, and for the volume of soil in front of the wheel the only vertical load is provided by the soil weight itself.

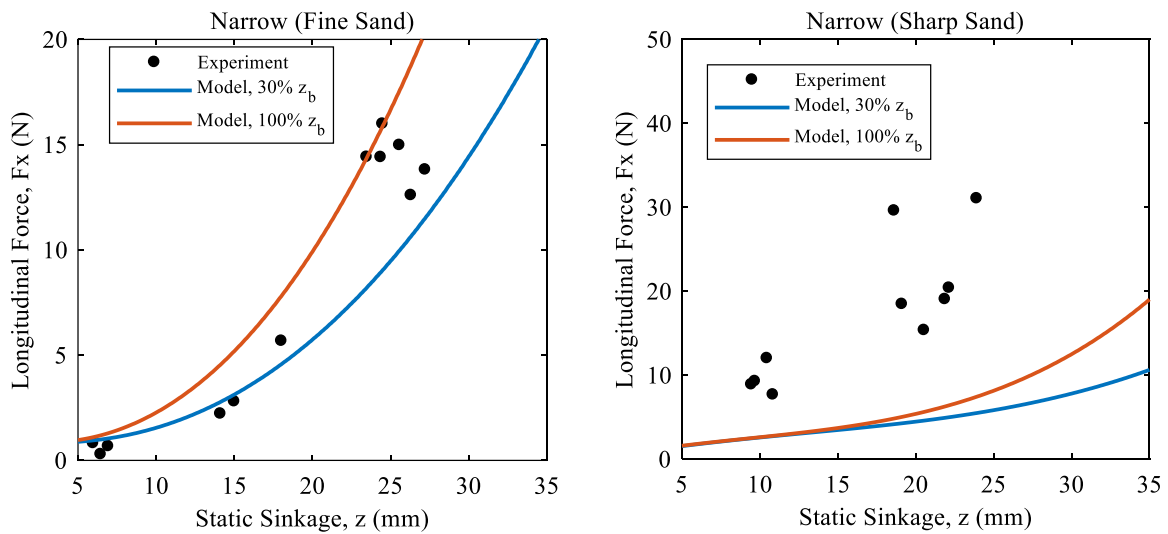


Figure 5.25 Longitudinal force vs sinkage for narrow rigid wheel on fine sand (left) and sharp sand (right).

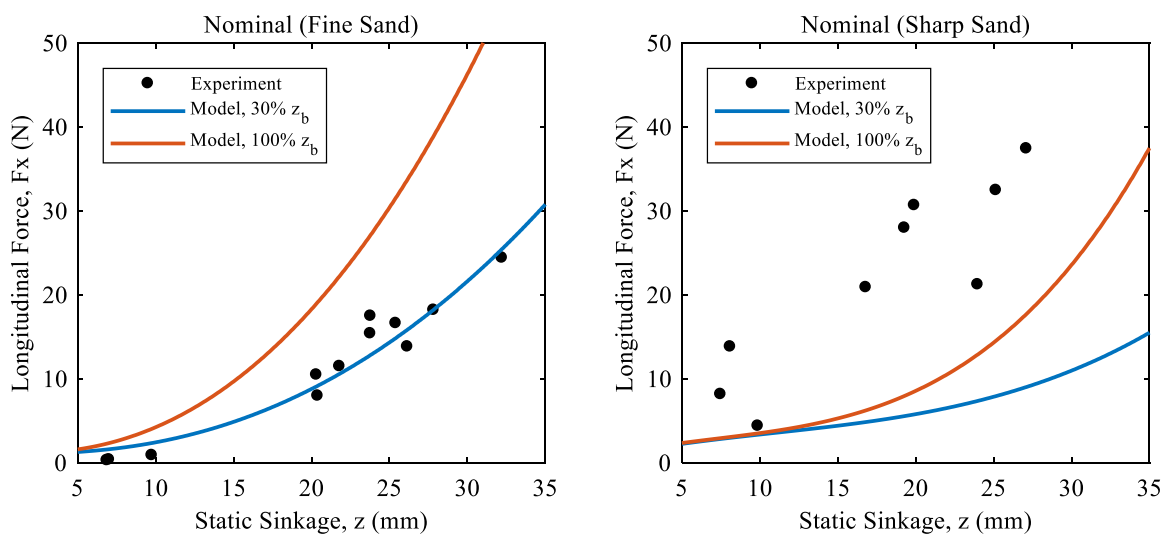


Figure 5.26 Longitudinal force vs sinkage for nominal rigid wheel on fine sand (left) and sharp sand (right).

When analysing the comparisons of sharp sand, the correlation is for narrow and nominal wheels is not good. In both wheels, the predicted model forces are underestimated even if 100% bulldozing height is considered. For the wide wheel, the correlation is reasonable with 100% bulldozing height. There are two potential reasons for this discrepancy. Firstly, the analysis from Wong and Reece describes the soil flow patterns of dry sand, however sharp sand contains a moisture content of approximately 4%, and the shear measurements have shown reasonably high cohesion value. Taking these into consideration, it is possible that the Wong and Reece proposal does not apply to sharp sand as the material may not follow the same flow pattern as a dry fine sand. Secondly, as discussed in Chapter 4, during the experiments it was difficult to maintain constant soil conditions for sharp sand as the moisture content would continuously change. Therefore, it is possible that the lack of correlation for narrow and nominal wheels is related to experimental errors. Interestingly, the experimental data of all three wheels overlay (see Figure 5.31), which may suggest that in sharp sand either the wheel forces hold little dependency to wheel width, or that the experimental discrepancy is so large that it is impossible to identify proper trends without a very large number of data points (i.e. experimental repetitions).

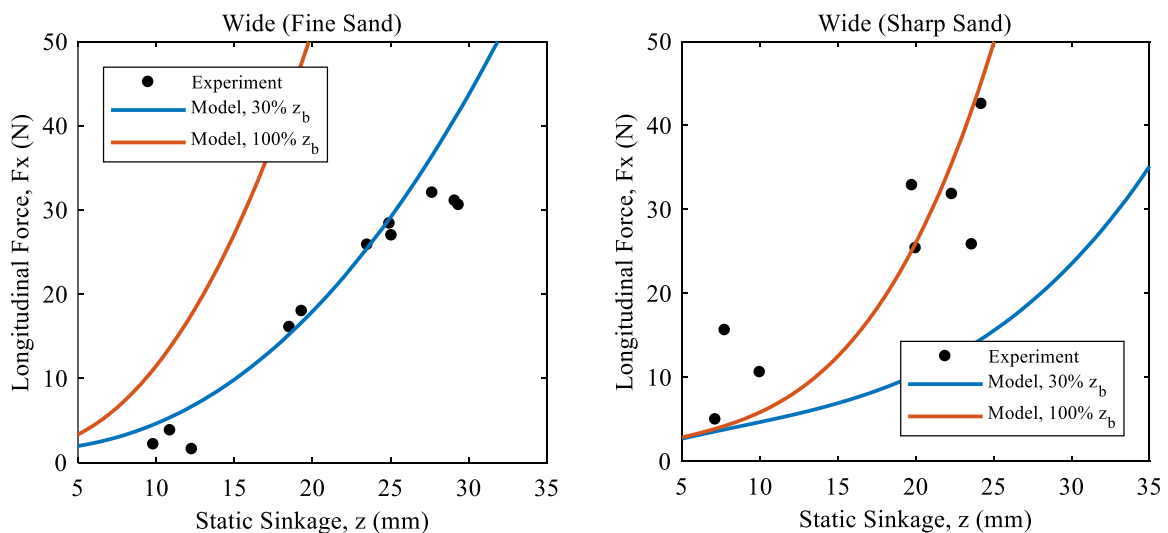


Figure 5.27 Longitudinal force vs sinkage for nominal rigid wheel on fine sand (left) and sharp sand (right).

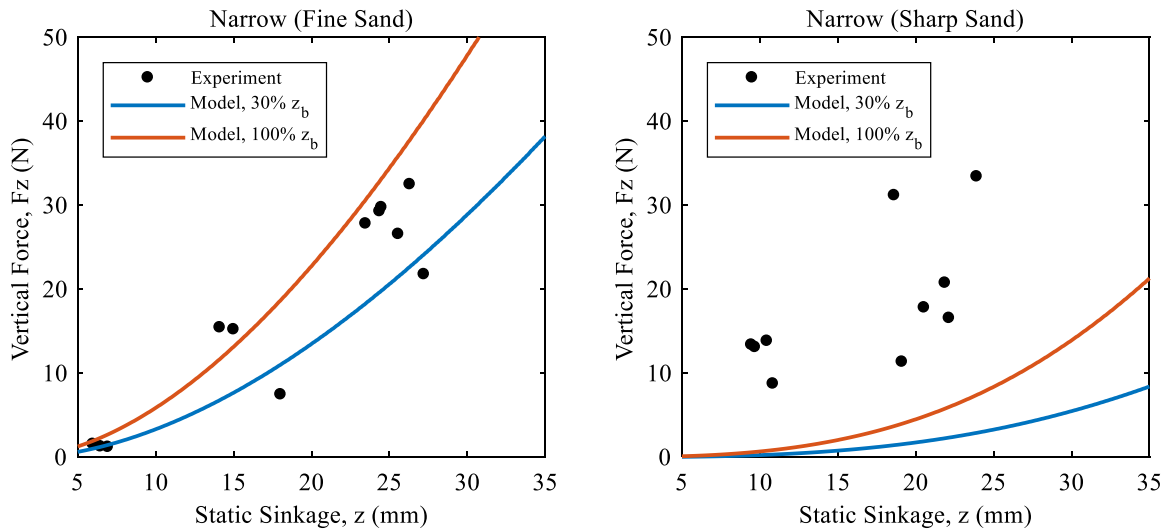


Figure 5.28 Vertical force vs sinkage for narrow rigid wheel on fine sand (left) and sharp sand (right).

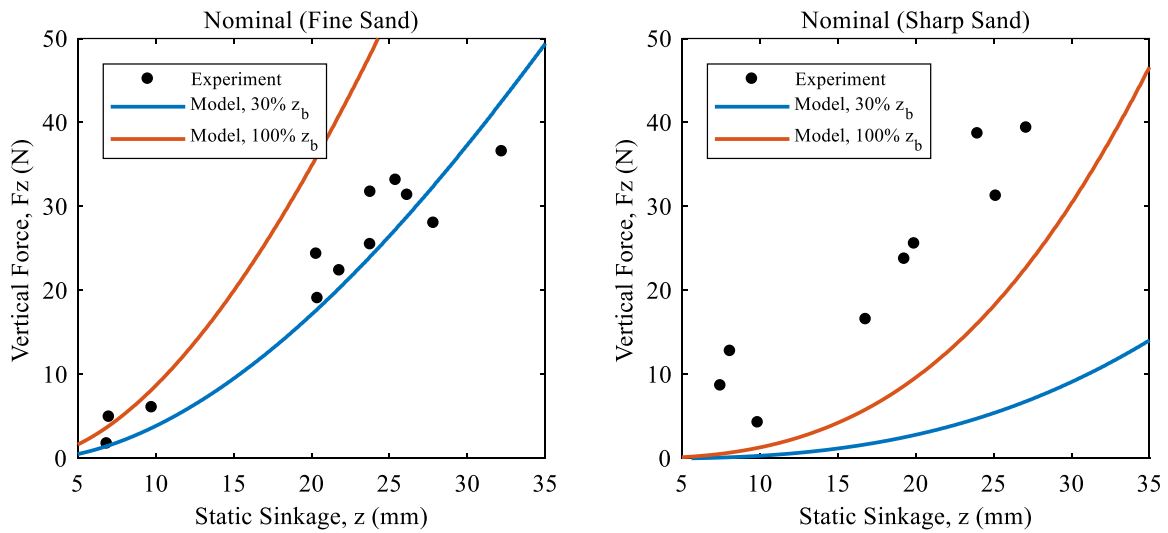


Figure 5.29 Vertical force vs sinkage for nominal rigid wheel on fine sand (left) and sharp sand (right).

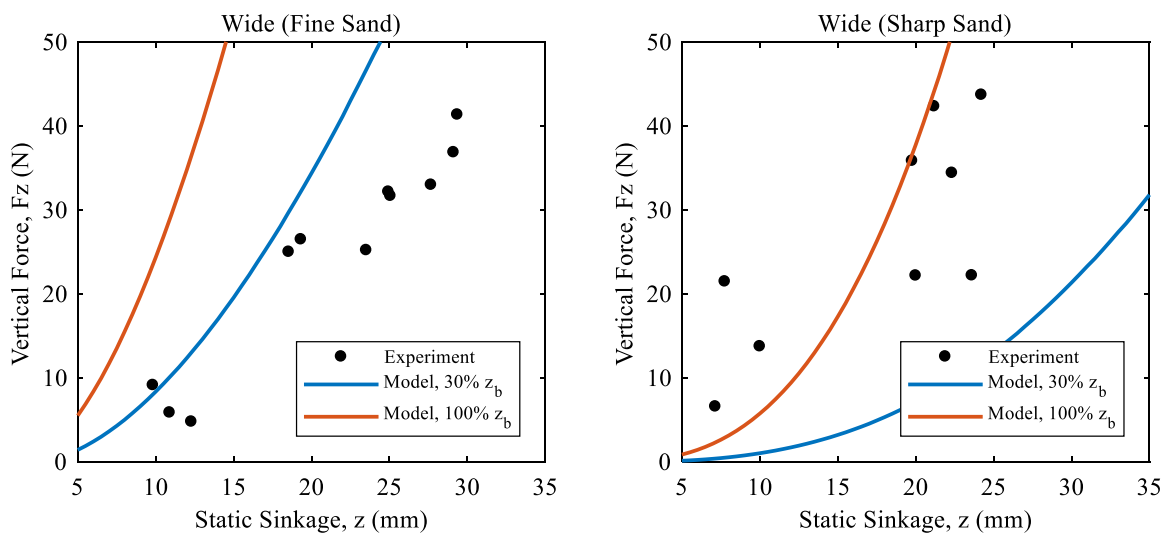


Figure 5.30 Vertical force vs sinkage for nominal rigid wheel on fine sand (left) and sharp sand (right).

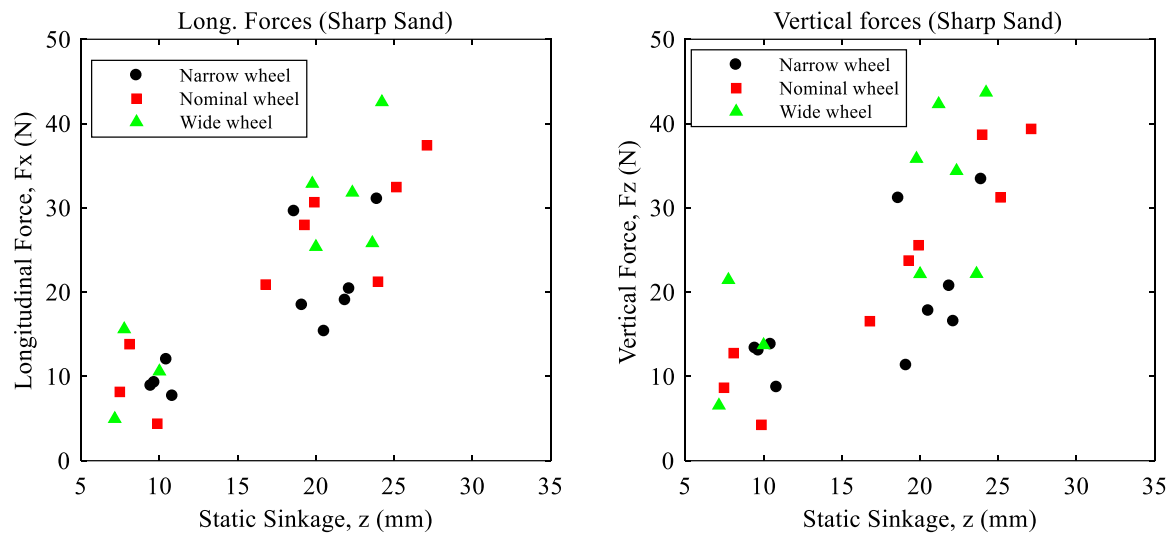


Figure 5.31 Overlay of experimental forces on sharp sand.

5.4 Chapter Conclusions

In this chapter a description of the experimental procedure and data processing for rigid wheels was presented along with analysis of the derived test data. It has been found that the wheel slip is independent of longitudinal velocity, and that a linear relationship with sinkage reasonably describes the experimental observations. In addition, a potential dependency of wheel width to the slip was highlighted. Analysis of the various wheel responses established the existence of two soil conditions; a transient condition which stops soon after the wheel starts rotating and where the longitudinal force peaks, and a steady-state condition where the forces settle at an almost constant value.

From the video recordings taken as part of the experimental procedure, the author would extract information about the bulldozing effect and could create a new and simple formulation. This formulation, along with the models for pressure-sinkage and shear developed in Chapter 4, created a new dynamic soil-wheel interaction model based on the Wong and Reece [34] theory for soil flow patterns under rigid wheels. The model can predict well the longitudinal and vertical forces exerted on rigid towed wheels on fine sand, but underestimates the forces on sharp sand. Further investigation is required to conclude if the proposed formulation is appropriate for non-dry sands.

Chapter 6: Final Conclusions

6.1 Overall Conclusions

The current research consists of a comprehensive analysis into the principles governing the soil-wheel interaction and emphasises on the concepts related to road vehicles. Today, the automotive industry is facing huge challenges regarding emissions and safety standards, as well as market trends for larger and heavier vehicles. Satisfying these demands require the utilisation of digital development methods and mathematical models that can accurately predict the component, system and overall vehicle behaviour. In the case of pneumatic tyres, vast research has been conducted over the past three decades, and various tyre models are currently available. However, these models are mostly focusing on the tyre behaviour on solid terrain such as dry asphalt, omitting the tyre performance under deformable terrains such as dry sand and saturated clay. Road vehicles are expected to perform equally well at all conditions, for example an anti-lock braking system must ensure that the vehicle can decelerate appropriately on dry and wet asphalt, ice/snow, mud and sand. But the absence of a reliable Terramechanics tyre model creates a gap in the overall virtual development process, and potentially compromises standards for safety, fuel economy and comfort at deformable terrains.

As highlighted by the literature review of Chapter 2, two of the most crucial aspects of the soil-wheel interaction is the rolling resistance and the tractive effort. The tyre rolling resistance, which directly affects the vehicle's fuel efficiency, is generated by a combination of vertical soil compaction (pressure-sinkage behaviour), horizontal soil displacement (bulldozing effect) and tyre deformation (when a flexible tyre is considered). On the other hand, the tractive effort is the result of the soil shearing behaviour at the soil-wheel interface.

For the prediction of soil compaction and bulldozing, two main approaches are employed by the researchers; (a) analytical formulations which they aim at describing the soil behaviour under the effect of an applied load from a macroscopic viewpoint, and (b) numerical methods such as FEM/DEM where the prediction of soil behaviour is conducted using material models and advanced numerical tools. The FEM/DEM models, although fairly accurate when correctly setup, suffer from numerical inefficiency

as it can take several hours to perform a few seconds of simulation, making such models unsuitable for overall vehicle performance and optimisation studies. In addition, their accuracy and applicability is usually limited to homogeneous soils. However, as emphasised in the literature review, the soil is a complex medium and far from homogeneous consisting of minerals, organic materials, water and gas. For these reasons, the author has decided to focus on analytical formulations for the development of a soil-wheel interaction model. The above paragraphs can confirm the achievement of objective (1).

In order to fulfil objective (2), in Chapter 3 the author conducted an evaluation of three existing pressure-sinkage analytical models proposed by Bekker, Reece and Lyasko respectively. The analysis was performed using numerical data found in the literature, and it concluded that the accuracy of all three models can range from reasonable to poor, but also being inconsistent when different soil types or size of penetration plate was changed. Furthermore, despite the appeal of the model proposed by Lyasko with its use of soil invariant parameters, the large number of complex equations included along with limited explanation of their origin and contribution, lead the author to finally disregard it from further research.

In addition, the analysis of Chapter 3 highlighted the difficulty to properly judge the models' prediction accuracy in the absence of statistical information in the soil data found in the literature. It has therefore confirmed the necessity of in-house soil data acquisition in order to enable a suitable evaluation of existing and newly developed predictive models.

An area that very rarely mentioned in the Terramechanics literature is the method for obtaining the necessary model parameters from test data. In this research, the author has shown that traditional fitting methods are inadequate, providing misleading model parameters and poor correlation to test data. When the fitting via optimisation was applied to the Bekker pressure-sinkage model, a vast correlation improvement was revealed. In addition, analysis has shown that optimisation fitting methods can converge to more reasonable model parameters, and that these methods should be used as common practice in Terramechanics related studies.

For objective (3), the author conducted hundreds of soil measurement experiments using Loughborough University's Terramechanics scale rig at two different soils (fine and sharp sand), using various sized flat and curved plates, and rotating rigid wheels. The pressure-sinkage soil responses confirmed, but also quantified, that test data dispersion is larger in less homogeneous soils such as sharp

sand, and that transient soil characteristics such as moisture content can greatly affect the soil behaviour and in all types of soil tests.

A broad assumption in the literature is that the pressure-sinkage behaviour is affected by the plate width but not its length. However, the assumption that two plates with vastly different length but the same width would sink equally into the soil under the effect of the same vertical load is questionable. Analysing the pressure-sinkage test data from flat plates, the author identified a dependency of the penetration plate total area on the soil responses and created a new pressure-sinkage model. The novel new model (Area-Dependent Pressure-Sinkage, ADPS) disregards the broad assumption in the literature that only plate width affects the sinkage and achieved significant correlation improvement to test data compared to the well-established and accepted Bekker model. ADPS demonstrated high prediction capability in the extrapolated region as well, with accurately estimating the sinkage of plates up to almost twice the size of the largest plate used for fitting. This is a significant achievement, because it can allow the soil testing using smaller, cheaper and more versatile data acquisition systems for field measurements without compromising the predictive quality of the actual vehicle wheel.

Analysis of the shear stress in-house experimental data, on both fine and sharp sands, was conducted and the parameters of the Mohr-Coulomb failure criterion obtained using an optimisation algorithm. Furthermore, it was found that the test data display a peak at the maximum shear stress; a response which is expected for brittle-type of soils. Existing models were analysed and evaluated (also linked to the 3rd objective), with the author deciding to use Oida's formulation and further expand it in the dimension of normal stress. The new model (extended Oida) combines the original Oida formulation with the Mohr-Coulomb failure criterion describing the maximum shear stress, and a newly developed dependency on normal stress for the residual shear stress and shear strain at maximum stress. Since forces due to shear develop at the soil-wheel interface as a result of wheel slip, for a dynamic model the ability to predict the shearing forces at different shear strain values (i.e. wheel slip) is crucial. Therefore, the accurate prediction of soil shear by the extended Oida model aids towards that scope.

For the prediction of sinking wheels, Bekker's proposal was used which suggests that the total vertical force on a rigid wheel is the sum of vertical components of radial forces equivalent to the ones produced by flat plates at the same sinkage. Initial results showed under-prediction of the ADPS model, however Bekker considers the forces with the wheel at equilibrium, i.e. static. For the pressure-sinkage

tests on curved plates, the penetration velocity was approximately constant at 3 mm/s, which means that shearing forces due to relative velocity at the soil-plate interface contributed to the overall vertical force. The author combined the ADPS and extended Oida in a dynamic model, which produced almost perfect force-sinkage responses compared to test data on fine sand. This finding further emphasises the improvement in the soil-wheel prediction when dynamic models are employed.

Finally, new soil-wheel dynamic model was developed, consisting of the also new ADPS model for pressure-sinkage and extended Oida model for shearing behaviour. The model can accurately predict the longitudinal and vertical forces for fine dry sand, but underestimates the forces in more cohesive type of soils. The creation of three new models (ADPS, extended Oida and soil-wheel interaction) achieved objective (4).

6.2 Suggestions for Future Work

Although the original objectives have been achieved, the current research has focused more on the analysis and interpretation of experimental data, and model validation of individual elements of soil behaviour than originally anticipated. Such was deemed necessary in order to gain appropriate understanding of the soil behaviour in order to obtain confidence about the overall soil-wheel model. It lead however to a reduction of the available effort towards the investigation of other phenomena regarding the soil-wheel interactions. The following provide a list of limitations of the existing research and suggestions for future work.

- The current soil-wheel model is validated for towed condition, meaning without externally applied torque. Similarly to towed rigid wheels, Wong and Reece [33] have analysed the flow patterns of dry sands in driven wheel conditions. The equations are similar to the towed wheels and the potential implementation into the dynamic soil-wheel model fairly easy. However, in order to validate the model additional experimental data are needed.
- As discussed in Chapter 5, the wheel slip behaviour is not fully understood and further analysis is required to establish the correct mechanism leading to the observed slip.
- The soil-wheel model could not accurately predict non-dry sand forces due to potential reasons discussed in Chapter 5. Additional investigation regarding this would be beneficial.

- The pressure used in modern tyres tends to increase for emissions reduction, which consequently increases the tyre stiffness. Considering this, the current research assumes the wheel as completely rigid. However, a comparison between a vehicle tyre and rigid wheel of the same dimensions is needed to validate or reject this assumption.
- The models developed in the current research are ultimately aiming at application on a vehicle development process. Therefore, confirmation of the models' behaviour and validity at full scale wheels should be conducted.

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65. G. Meirion-Griffith and M. Spenco, "A modified pressure-sinkage model for small, rigid wheels on deformable terrains," *Journal of Terramechanics*, vol. 48, pp. 149-155, 2011.

Appendix A: Soil Data from [57]

Sinkage [mm]	Pressure [kPa]				
	0.5×18"	1×18"	2×18"	3×18"	4×18"
12.7	53.7	53.6	45.9	56.9	57.4
25.4	61.3	59.3	56.5	72.7	76.2
38.1	72.7	67.0	65.0	81.7	88.3
50.8	80.3	78.6	79.3	92.4	100.0
63.5	99.4	92.0	87.9	103.4	111.7
76.2	114.8	107.6	99.6	115.5	122.7
88.9	134.1	122.7	114.1	128.2	134.1
101.6	149.6	140.0	126.2	142.0	145.5
114.3	168.6	157.2	141.3	156.9	157.2
127.0	187.5	174.4	156.9	173.7	170.3
139.7	206.8	193.7	172.0	191.7	182.7
152.4	226.1	212.4	188.9	215.8	209.6
165.1	244.8	233.7	203.4	228.2	224.1

Table A.1 Pressure as a function of sinkage for rectangular plates in wet sand, data from [57].

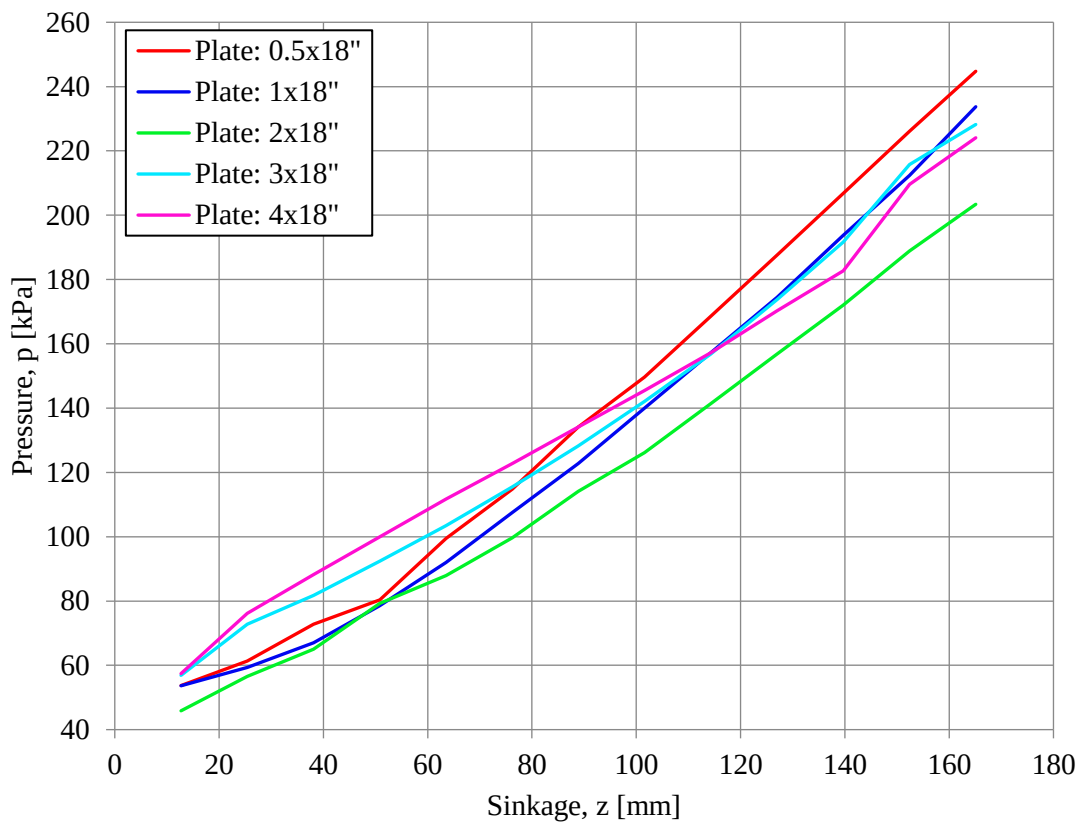


Figure A.1 Pressure vs sinkage for rectangular plates in wet sand, data from [57].

Sinkage [mm]	Pressure [kPa]					
	1" Dia	2" Dia	3" Dia	4" Dia	5" Dia	6" Dia
12.7	70.3	54.9	48.8	65.0	63.4	56.0
25.4	87.9	76.9	73.1	87.9	82.7	81.4
38.1	87.9	98.9	92.4	106.9	100.3	101.4
50.8	105.5	109.7	112.4	124.1	120.0	120.7
63.5	131.7	127.2	126.9	145.5	139.3	141.3
76.2	158.6	142.7	151.0	167.5	160.0	162.0
88.9	175.8	164.8	170.6	189.6	182.7	184.8
101.6	219.9	186.8	195.1	213.7	209.6	209.6
114.3	263.4	215.1	219.3	241.3	235.8	238.6
127.0	307.5	241.3	248.9	271.7	267.5	271.7
139.7	351.6	274.4	277.9	304.7	306.1	306.8
152.4	395.8	296.5	317.2	342.7	369.6	349.6

Table A.2 Pressure as a function of sinkage for circular plates in wet sand, data from [57].

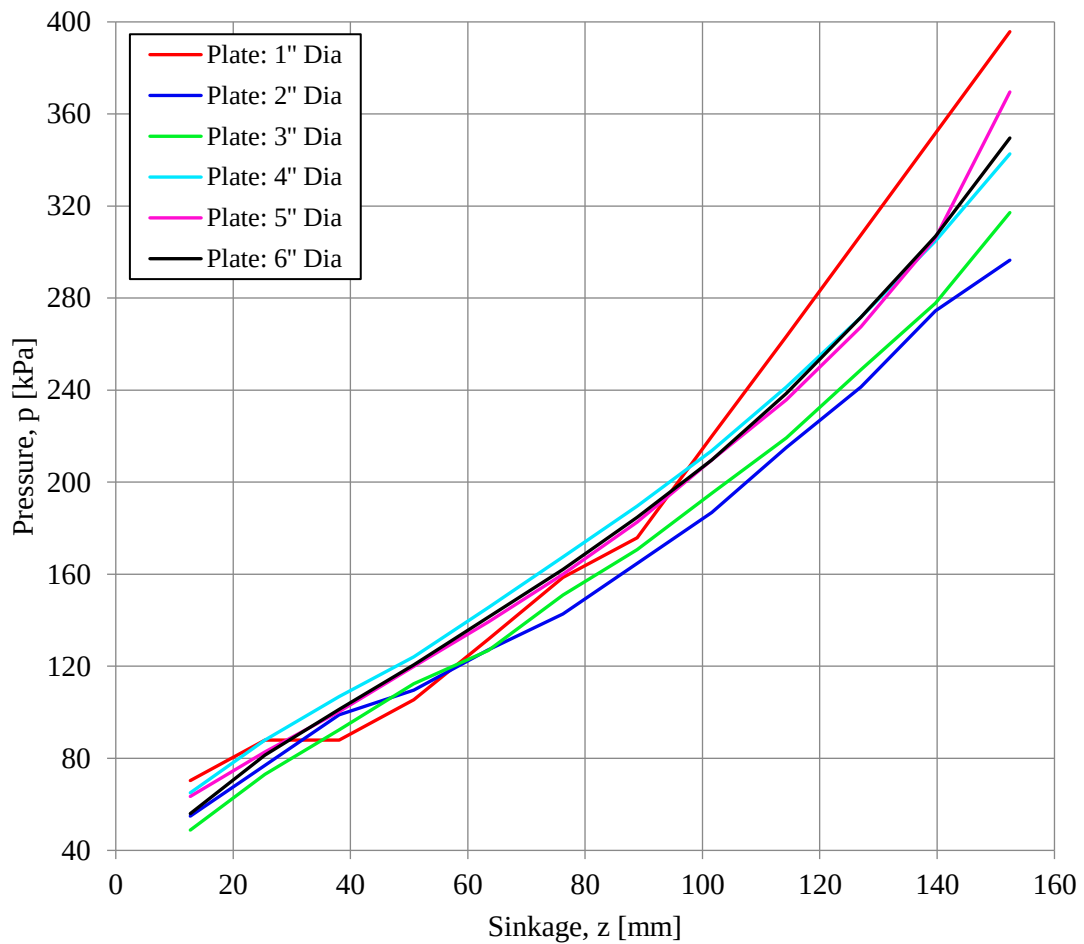


Figure A.2 Pressure as a function of sinkage for circular plates in wet sand, data from [57].

Sinkage [mm]	Pressure [kPa]				
	0.5×18"	1×18"	2×18"	3×18"	4×18"
12.7	9.2	11.5	19.2	23.6	29.6
25.4	19.2	21.1	26.9	33.1	42.1
38.1	30.6	28.7	34.5	43.4	50.7
50.8	39.8	38.3	43.1	51.7	58.6
63.5	49.7	46.1	51.7	61.4	68.3
76.2	61.2	55.6	61.4	71.7	76.5
88.9	75.8	65.2	71.0	81.4	86.2
101.6	84.1	76.5	81.4	93.1	97.9
114.3	99.3	86.2	92.4	104.8	109.6
127.0	114.5	100.0	105.5	117.2	122.7
139.7	133.8	113.1	118.6	131.7	137.9
152.4	148.9	126.9	131.0	146.9	155.1
165.1	164.8	140.0	149.6	164.8	176.5
177.8	183.4	155.1	166.9	181.3	

Table A.3 Pressure as a function of sinkage for rectangular plates in dry sand, data from [57].

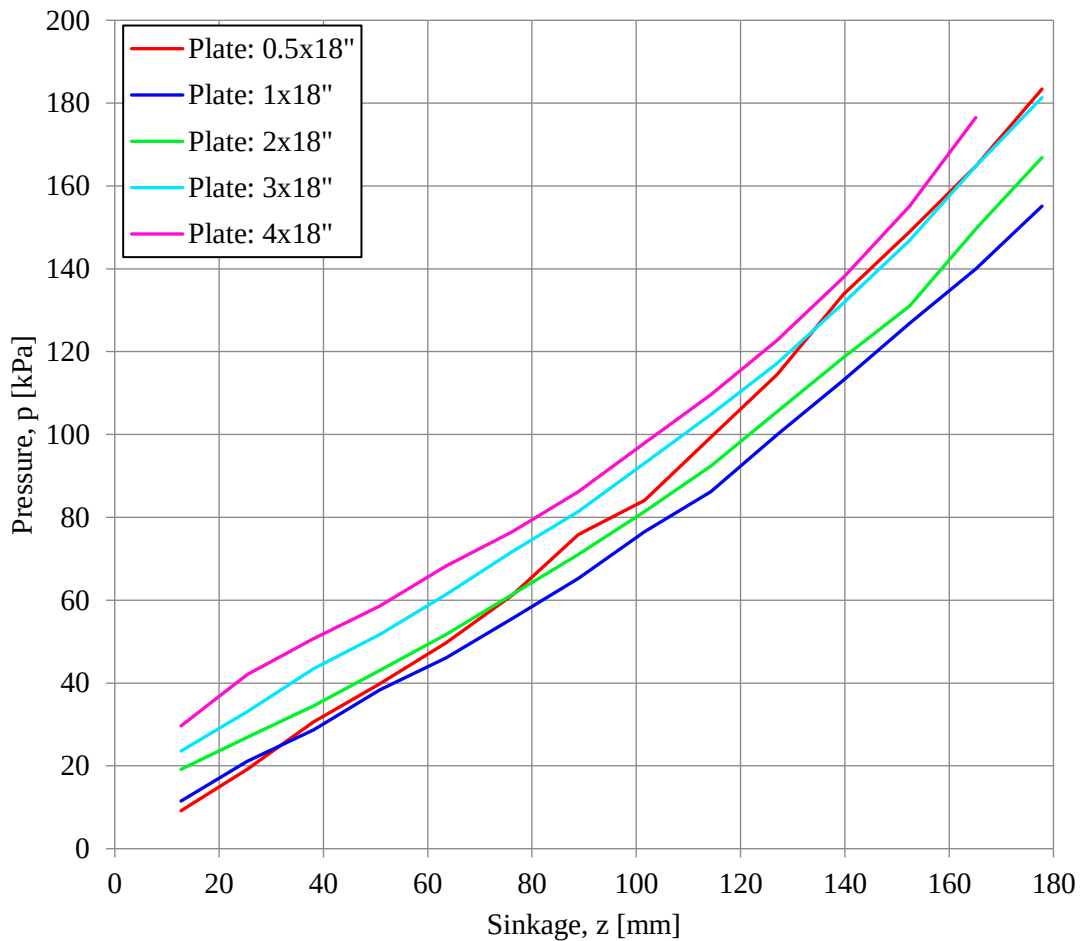


Figure A.3 Pressure as a function of sinkage for rectangular plates in dry sand, data from [57].

Sinkage [mm]	Pressure [kPa]					
	1" Dia	2" Dia	3" Dia	4" Dia	5" Dia	6" Dia
12.7	17.6	21.9	19.7	23.6	19.3	24.3
25.4	26.3	32.8	32.5	33.0	30.7	36.1
38.1	35.1	43.9	44.3	42.9	39.6	46.2
50.8	43.9	54.8	54.1	52.2	49.6	57.9
63.5	52.7	65.8	64.1	61.4	61.4	70.3
76.2	65.8	84.1	73.8	71.7	73.4	83.8
88.9	87.8	98.6	86.9	85.2	86.5	97.2
101.6	96.5	109.6	100.7	98.7	101.7	113.1
114.3	105.5	126.9	116.2	115.3	116.5	128.9
127.0	131.7	144.8	131.0	132.4	133.8	145.5
139.7	157.9	164.1	149.6	153.8	155.1	165.5
152.4	184.1	184.1	171.7	175.8	171.7	182.0
165.1	219.3	208.2	198.9	197.9	193.1	201.3
177.8	263.3	237.2	227.5	226.1	215.1	219.3

Table A.4 Pressure as a function of sinkage for circular plates in dry sand, data from [57].

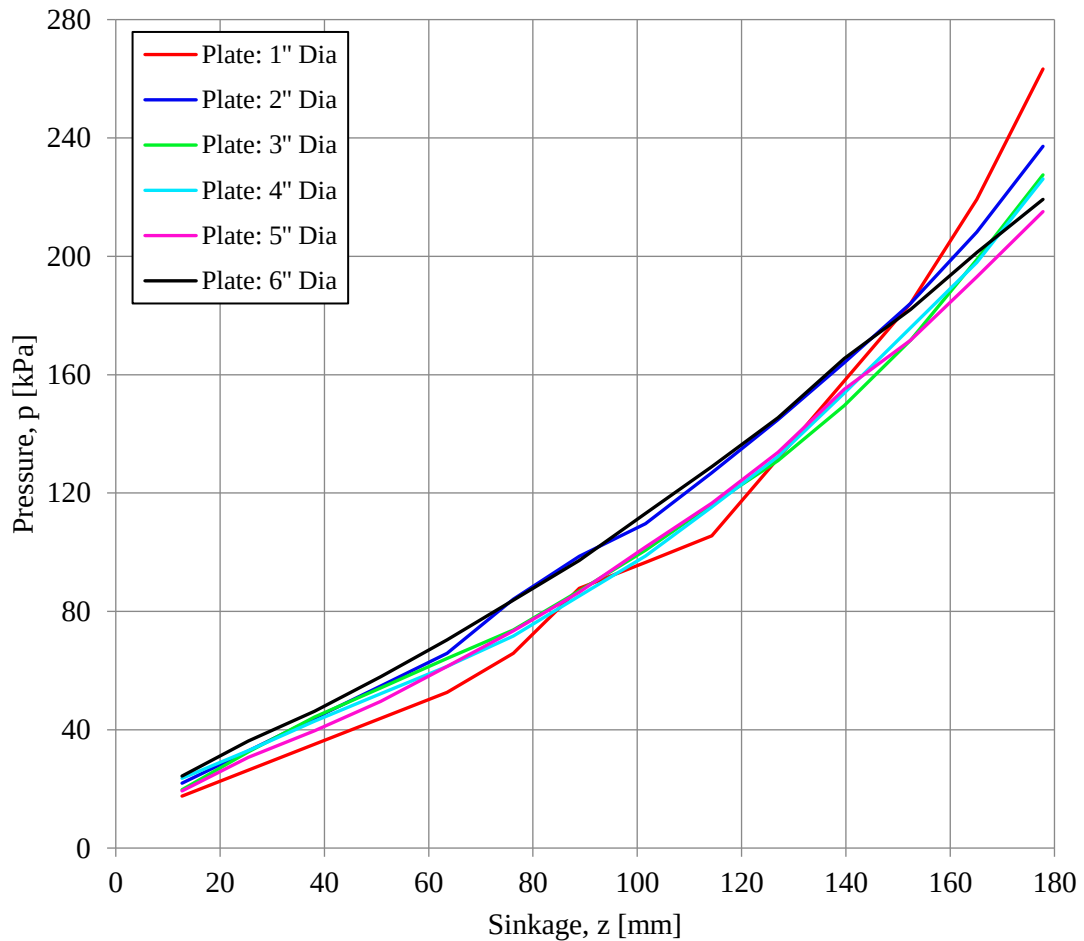


Figure A.4 Pressure as a function of sinkage for circular plates in dry sand, data from [57].

Sinkage [mm]	Pressure [kPa]			
	1×18"	2×18"	3×18"	4×18"
12.7	24.9	19.7	15.7	17.7
25.4	34.5	28.8	19.0	27.3
38.1	38.3	33.0	31.3	32.4
50.8	40.3	36.2	34.7	35.5
63.5	41.8	38.3	37.6	37.6
76.2	42.1	39.9	38.7	39.2
101.6	43.3	41.2	41.0	41.6
152.4	46.0	44.1	43.7	45.5
203.2	47.9	47.0	47.9	48.9
254.0	52.1	53.7	54.3	55.4

Table A.5 Pressure as a function of sinkage for rectangular plates in saturated clay, data from [57].

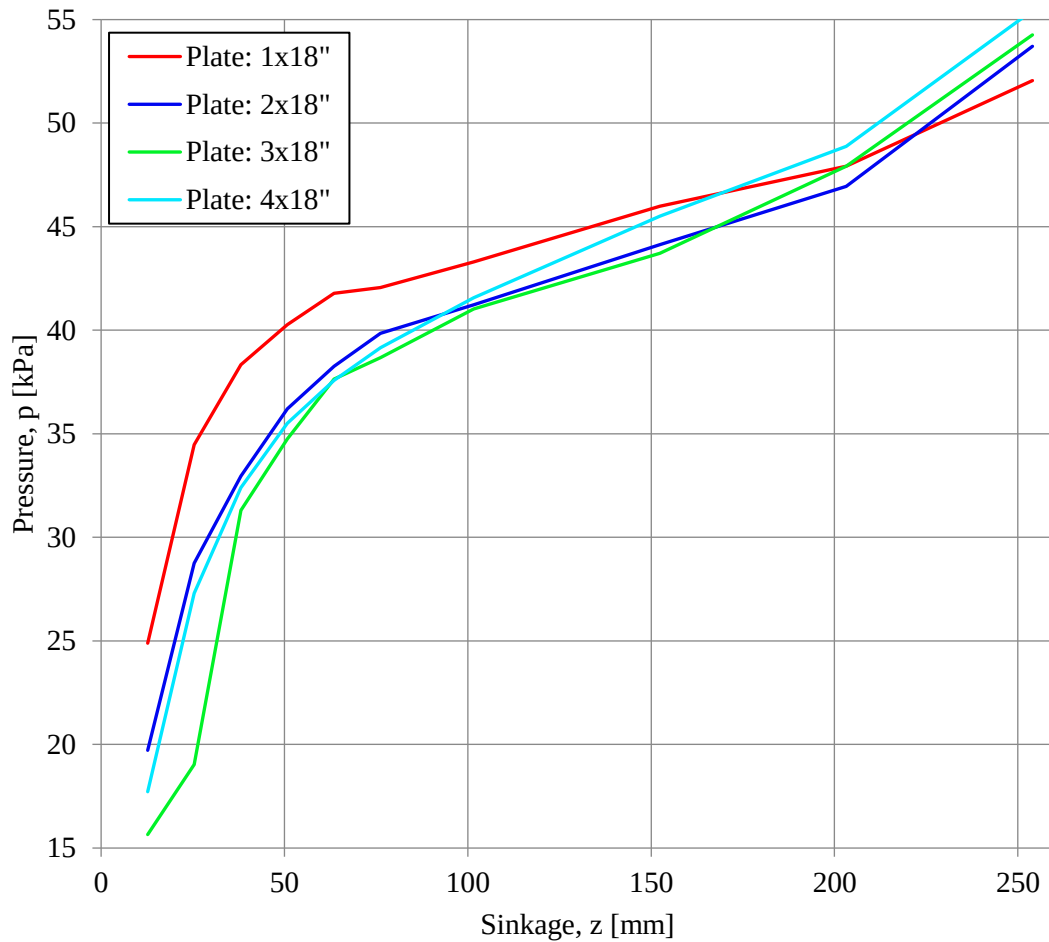


Figure A.5 Pressure as a function of sinkage for rectangular plates in saturated clay, data from [57].

Sinkage [mm]	Pressure [kPa]			
	1×4.5"	2×9"	3×13.5"	4×18"
12.7	29.2	20.7	19.7	19.4
25.4	36.8	29.9	26.4	22.8
38.1	39.9	35.6	30.3	27.8
50.8	41.4	39.0	33.2	31.2
63.5	44.5	41.4	35.4	33.5
76.2	47.6	43.0	37.2	35.5
101.6	50.7	44.1	40.0	38.7
152.4	53.7	45.2	43.4	43.6
203.2	62.9	46.0	46.9	48.9
254.0	81.2	51.0	53.8	57.5

Table A.6 Pressure as a function of sinkage for rectangular plates in saturated clay, data from [57].

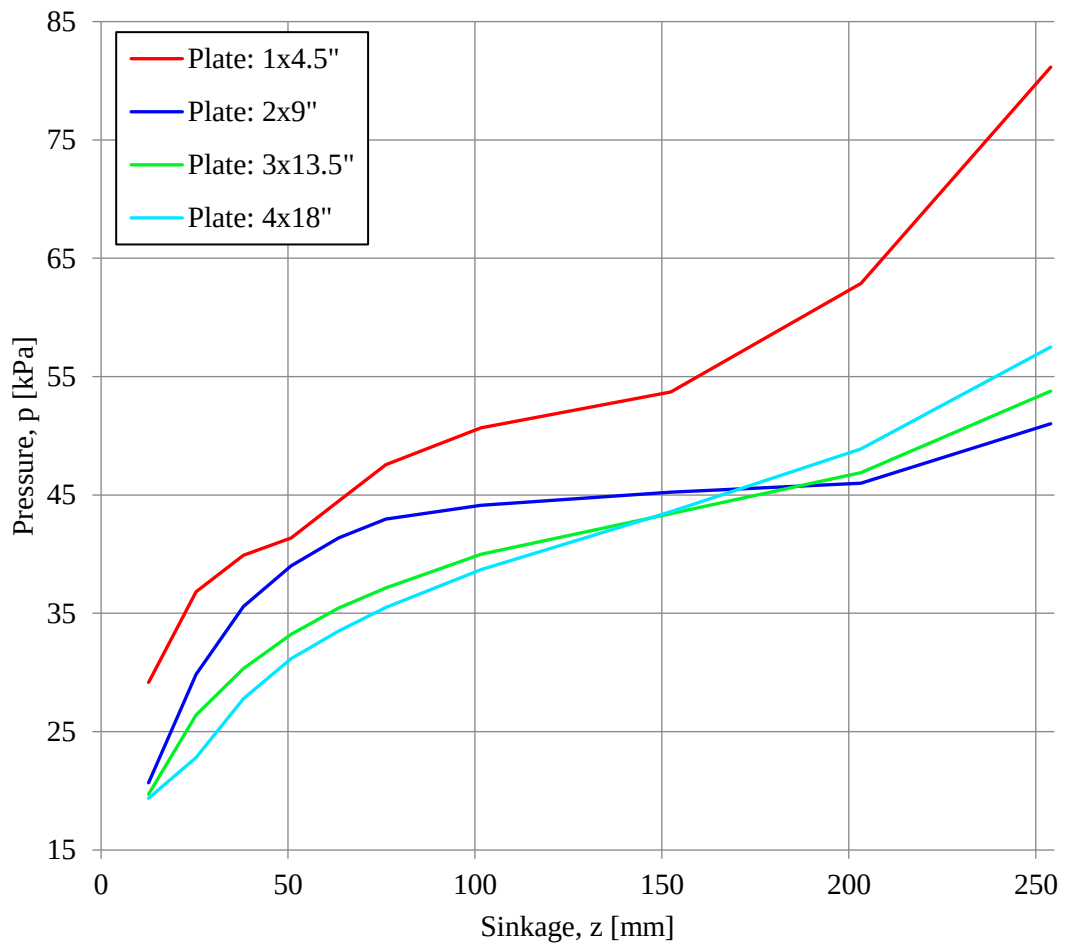


Figure A.6 Pressure as a function of sinkage for rectangular plates in saturated clay, data from [57].

Sinkage [mm]	Pressure [kPa]			
	1" Dia	2" Dia	4" Dia	6" Dia
12.70	39.51	28.54	22.48	17.17
25.40	44.82	37.37	33.51	26.54
38.10	44.82	42.82	41.16	31.72
50.80	44.82	46.19	43.99	34.61
76.20	44.82	47.44	45.51	37.30
101.60	44.82	48.75	46.61	38.96
152.40	52.74	49.37	45.99	41.44
203.20		49.37	49.44	45.85

Table A.7 Pressure as a function of sinkage for circular plates in saturated clay, data from [57].

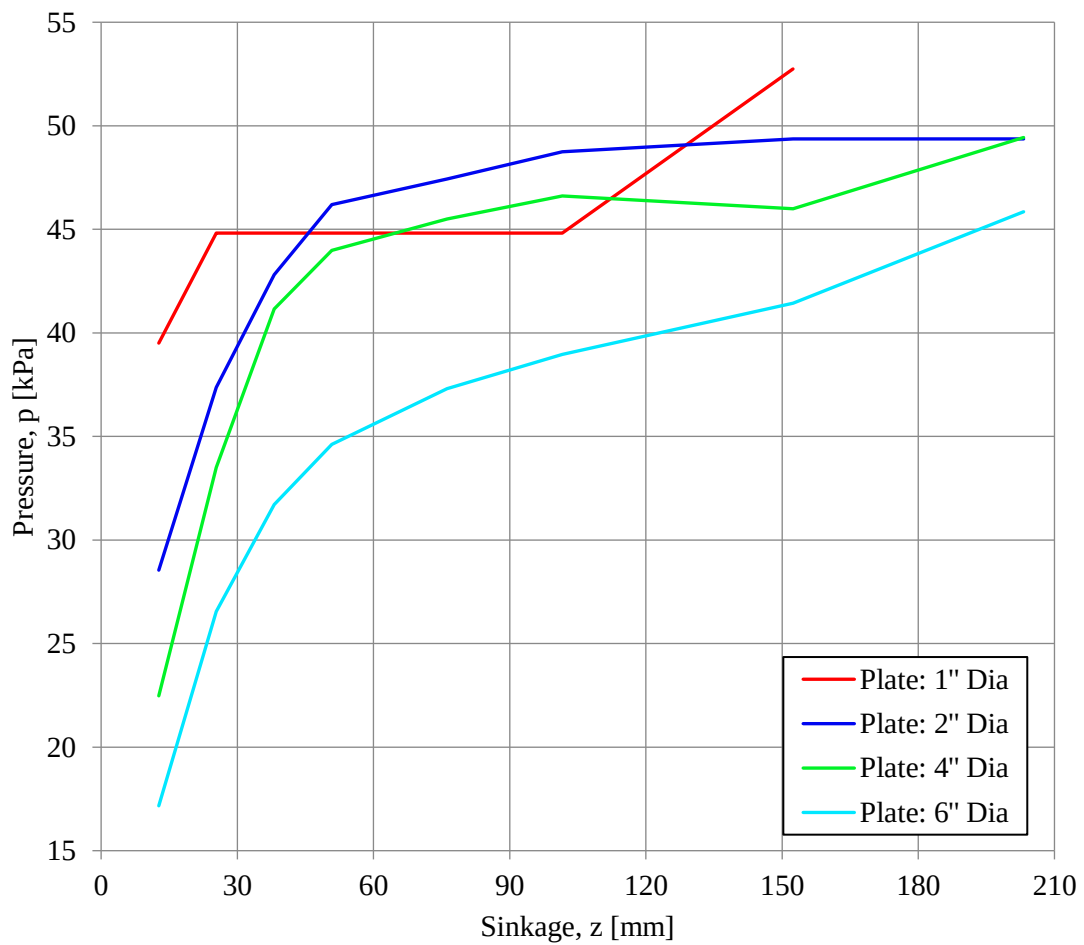


Figure A.7 Pressure as a function of sinkage for circular plates in saturated clay, data from [57].

	n	k_c	k_ϕ	k_c''	k_ϕ''	C	ϕ_0	w	γ	H
	-	N/m^{n+1}	N/m^{n+2}	N/m^2	N/m^3	N/m^2	deg	%	kg/m^3	m
Dry Sand Rectangular Plates	0.95	9012.7	508646.3	1861.9	1084546.7	0.0	32.0	0.0	1630.0	0.254
Dry Sand Circular Plates	1.07	16420.4	954438.0	-3446.9	1213239.4	0.0	32.0	0.0	1630.0	0.254
Wet Sand Rectangular Plates	0.68	5245.8	511361.7	24091.2	1375099.4	689.0	32.0	2.3	1770.0	0.254
Wet Sand Circular Plates	0.86	14698.9	1310837.9	31757.7	1909403.7	689.0	32.0	2.3	1770.0	0.254
Saturated Clay Rectangular Plates 1	0.22	-537.3	83936.9	33305.3	83583.2	6895.0	8.0	28.0	2000.0	0.600
Saturated Clay Rectangular Plates 2	0.34	504.5	85750.5	32317.5	61089.1	6895.0	8.0	28.0	2000.0	0.600
Saturated Clay Circular Plates	0.15	-355.1	69492.6	41589.7	-3085.3	6895.0	8.0	28.0	2000.0	0.600

Table A.8 Soil data used for comparison between Bekker, Reece and Lyasko models as found in [57]. Items highlighted with red font are assumptions.

Appendix B: Pressure-Sinkage Platewise Fitting Parameters

ID	Type	Width, B (mm)	Length, L (mm)	L/B (-)	Area, A (mm ²)	Fine Sand					
						Fitting			Optimisation		
						k (kN/m ⁿ)	n (-)	SSE (-)	k (kN/m ⁿ)	n (-)	SSE (-)
1	Rect.	30	60	2.0	1,800	6,576	1.52	104	6,374	1.51	101
2	Rect.	30	80	2.7	2,400	5,317	1.44	99	4,671	1.40	63
3	Rect.	30	90	3.0	2,700	3,452	1.31	68	3,280	1.29	63
4	Rect.	40	60	1.5	2,400	3,341	1.27	38	3,261	1.26	37
5	Rect.	40	100	2.5	4,000	2,134	1.10	84	2,220	1.11	79
6	Rect.	40	120	3.0	4,800	1,082	0.90	53	1,153	0.92	39
7	Rect.	50	75	1.5	3,750	2,108	1.08	150	2,385	1.12	92
8	Rect.	50	80	1.6	4,000	2,171	1.08	295	2,442	1.12	169
9	Rect.	50	120	2.4	6,000	1,481	0.96	42	1,567	0.97	27
10	Rect.	50	140	2.8	7,000	1,184	0.90	28	1,201	0.90	30
11	Rect.	60	90	1.5	5,400	2,815	1.12	166	3,087	1.15	123
12	Rect.	70	105	1.5	7,350	2,107	1.00	22	2,056	0.99	19
13	Rect.	80	120	1.5	9,600	2,653	1.02	45	2,447	1.00	34
14	Square	40	40	1.0	1,600	3,078	1.30	341	4,084	1.39	192
15	Square	50	50	1.0	2,500	4,352	1.38	58	4,595	1.40	48
16	Square	60	60	1.0	3,600	2,212	1.12	189	2,574	1.17	116
17	Square	70	70	1.0	4,900	2,155	1.07	366	2,618	1.13	198
18	Square	80	80	1.0	6,400	1,816	0.97	361	2,149	1.02	183

Table B.1 Summary of platewise Bekker pressure-sinkage model coefficients, comparison between fitting and optimisation methods, fine sand.

ID	Type	Width, <i>B</i> (mm)	Length, <i>L</i> (mm)	<i>L/B</i> (-)	Area, <i>A</i> (mm ²)	Sharp Sand					
						Fitting			Optimisation		
						<i>k</i> (kN/m ⁿ)	<i>n</i> (-)	SSE (-)	<i>k</i> (kN/m ⁿ)	<i>n</i> (-)	SSE (-)
1	Rect.	30	60	2.0	1,800	5,734	1.86	1484	97,492	2.76	164
2	Rect.	30	80	2.7	2,400	32,377	2.24	1926	93,016	2.56	250
3	Rect.	30	90	3.0	2,700	10,992	1.89	2736	89,730	2.54	157
4	Rect.	40	60	1.5	2,400	23,764	2.16	242	42,304	2.34	41
5	Rect.	40	100	2.5	4,000	12,735	1.73	1752	31,338	2.02	284
6	Rect.	40	120	3.0	4,800	17,546	1.81	1881	42,085	2.09	290
7	Rect.	50	75	1.5	3,750	21,861	1.87	809	36,951	2.04	227
8	Rect.	50	80	1.6	4,000	-	-	-	-	-	-
9	Rect.	50	120	2.4	6,000	9,331	1.59	2144	21,731	1.86	457
10	Rect.	50	140	2.8	7,000	8,659	1.58	2484	25,447	1.91	482
11	Rect.	60	90	1.5	5,400	10,178	1.64	2137	25,577	1.93	412
12	Rect.	70	105	1.5	7,350	8,920	1.58	2394	34,673	1.97	314
13	Rect.	80	120	1.5	9,600	17,165	1.73	718	46,755	2.01	116
14	Square	40	40	1.0	1,600	9,223	1.85	3351	123,017	2.67	342
15	Square	50	50	1.0	2,500	30,909	2.09	1117	72,863	2.36	59
16	Square	60	60	1.0	3,600	5,539	1.49	2616	17,086	1.84	596
17	Square	70	70	1.0	4,900	41,848	2.07	1321	84,564	2.30	395
18	Square	80	80	1.0	6,400	12,060	1.65	4042	44,755	2.06	832

Table B.2 Summary of platewise Bekker pressure-sinkage model coefficients, comparison between fitting and optimisation methods, sharp sand.

Appendix C: ADPS Correlation - Flat Plates

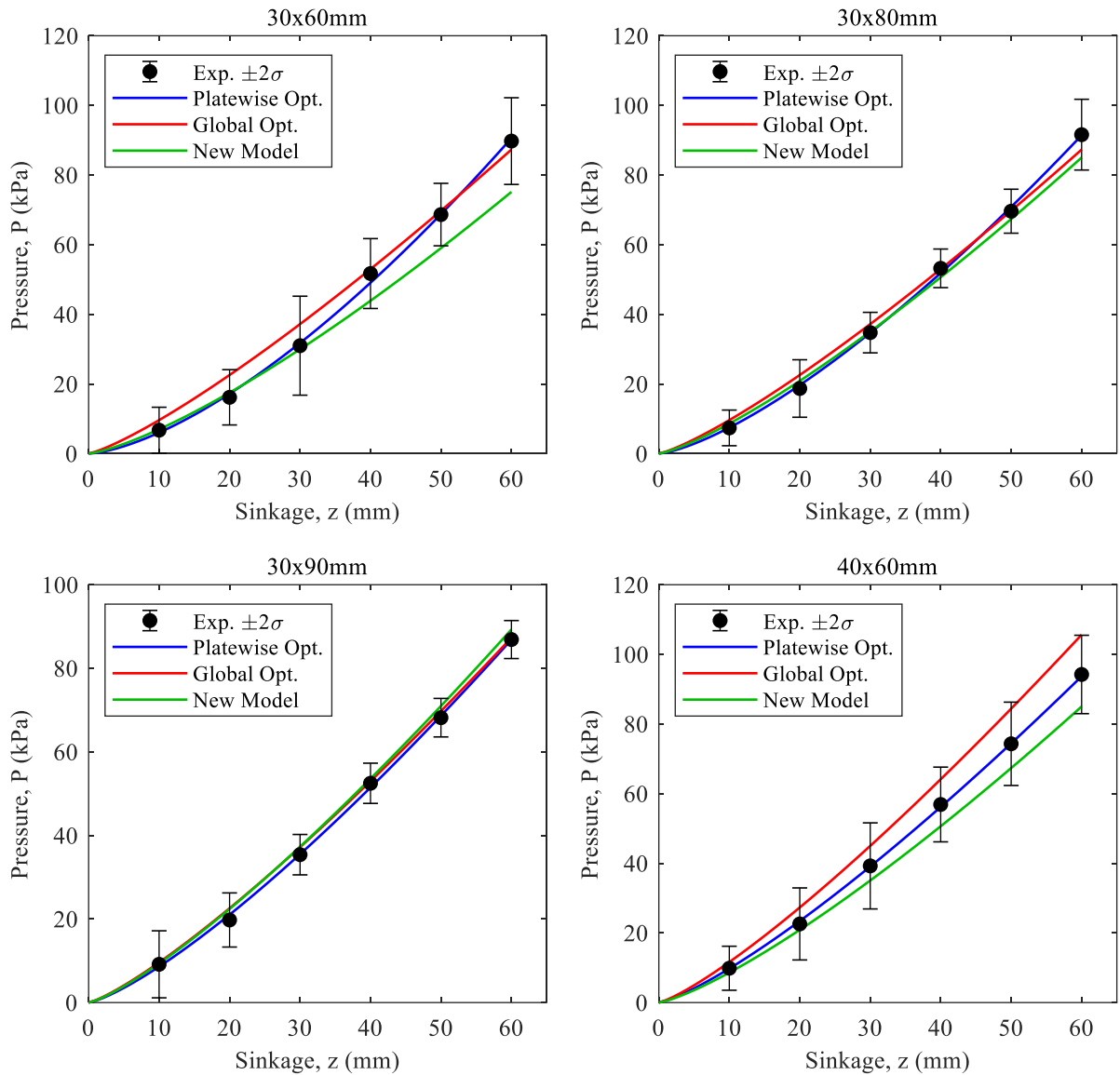


Figure C.1 Correlation of New pressure-sinkage model on flat plates, comparison with traditional Bekker formulation, fine sand.

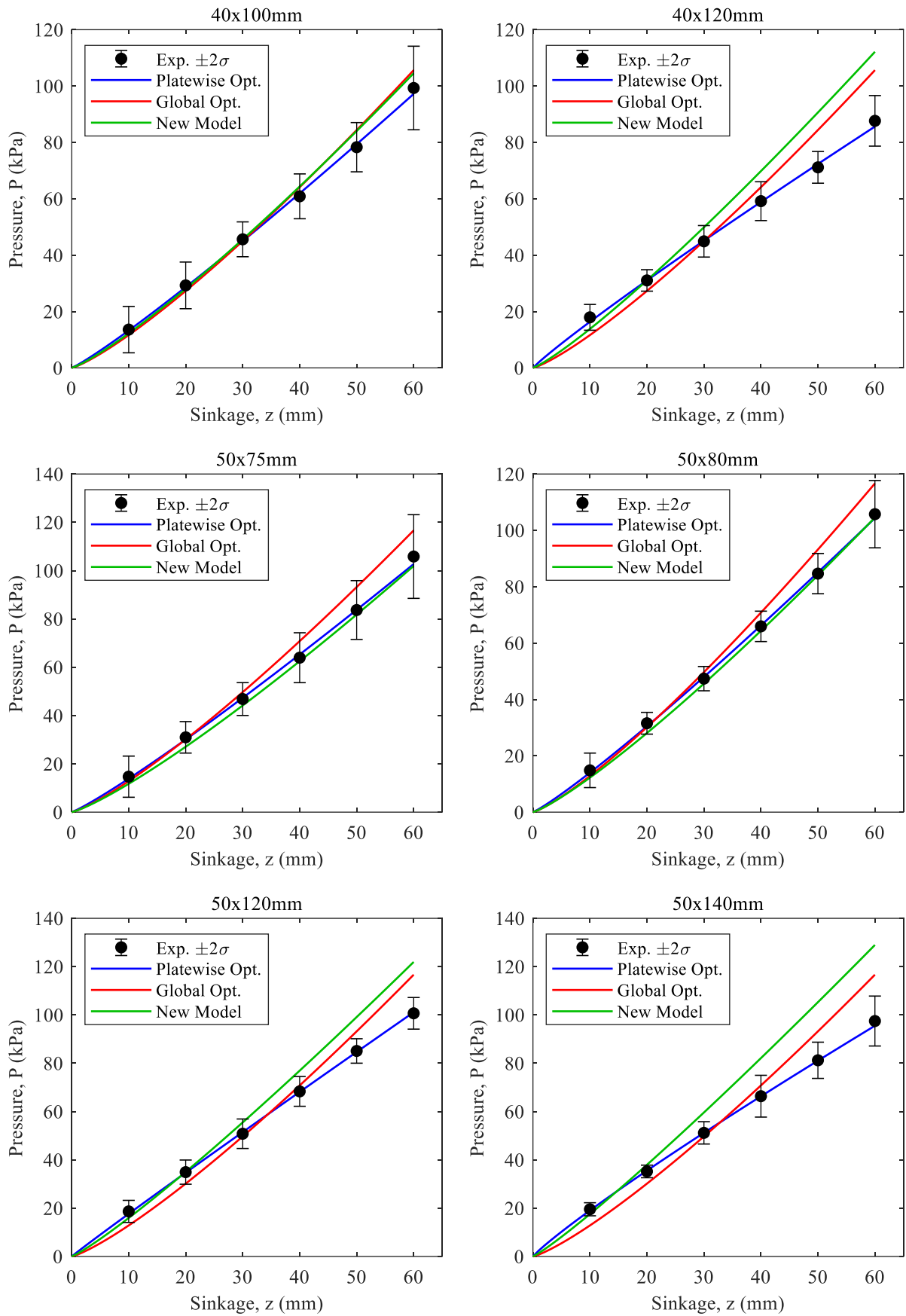


Figure C.1 (continued from previous page)

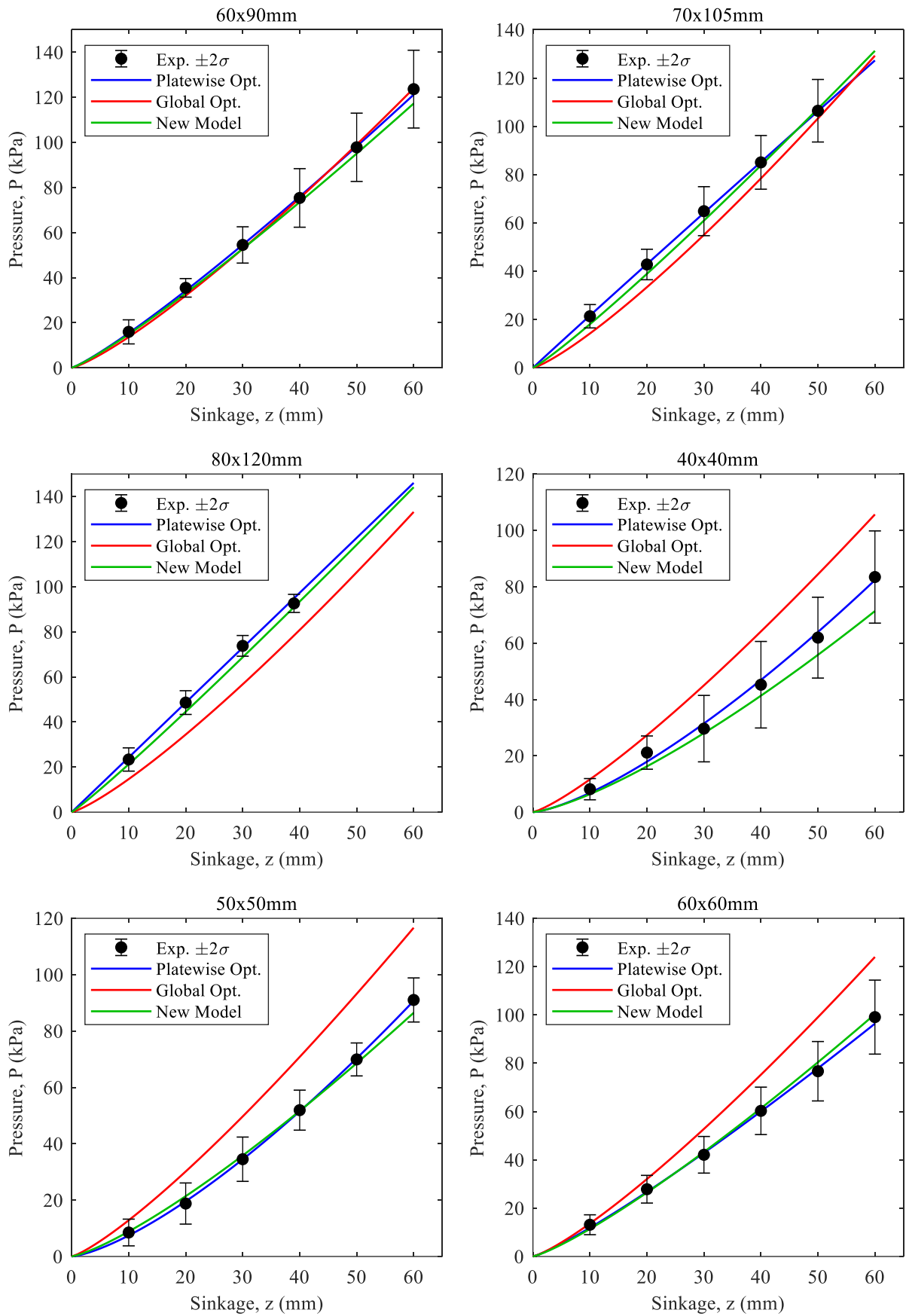


Figure C.1 (continued from previous page)

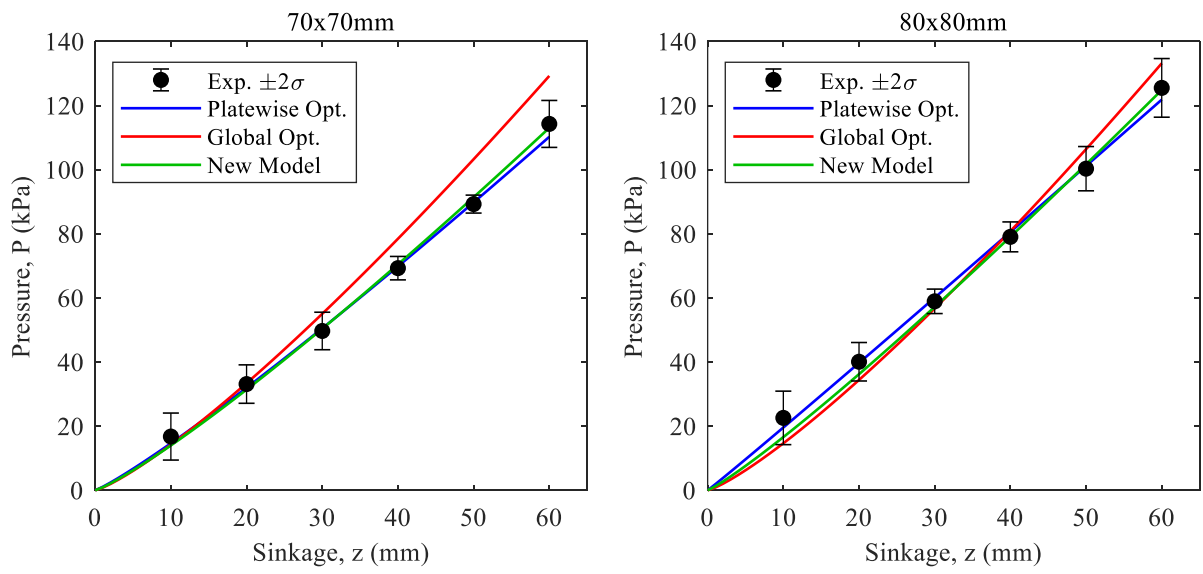


Figure C.1 (continued from previous page)

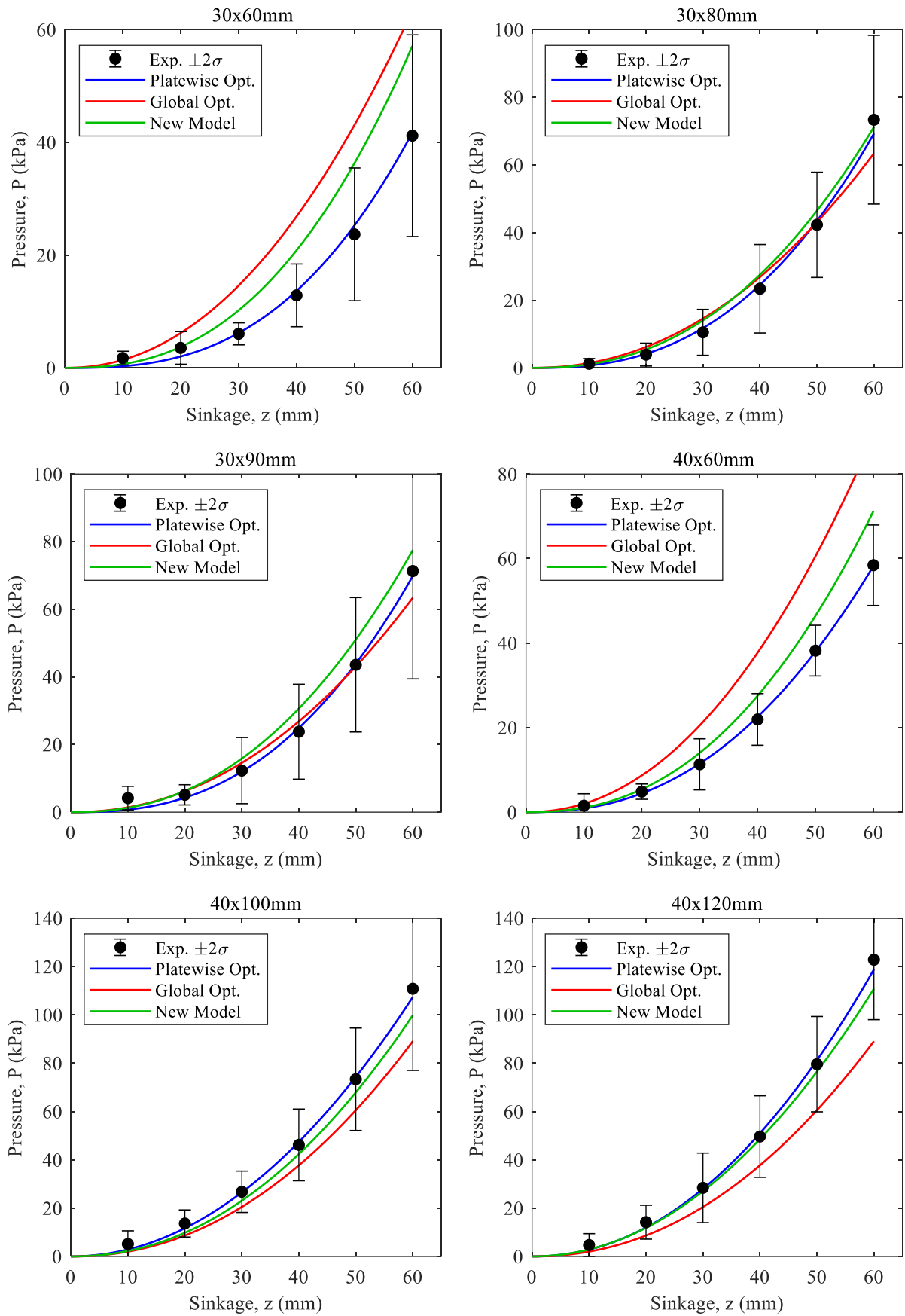


Figure C.2 Correlation of New pressure-sinkage model on flat plates, comparison with traditional Bekker formulation, sharp sand.

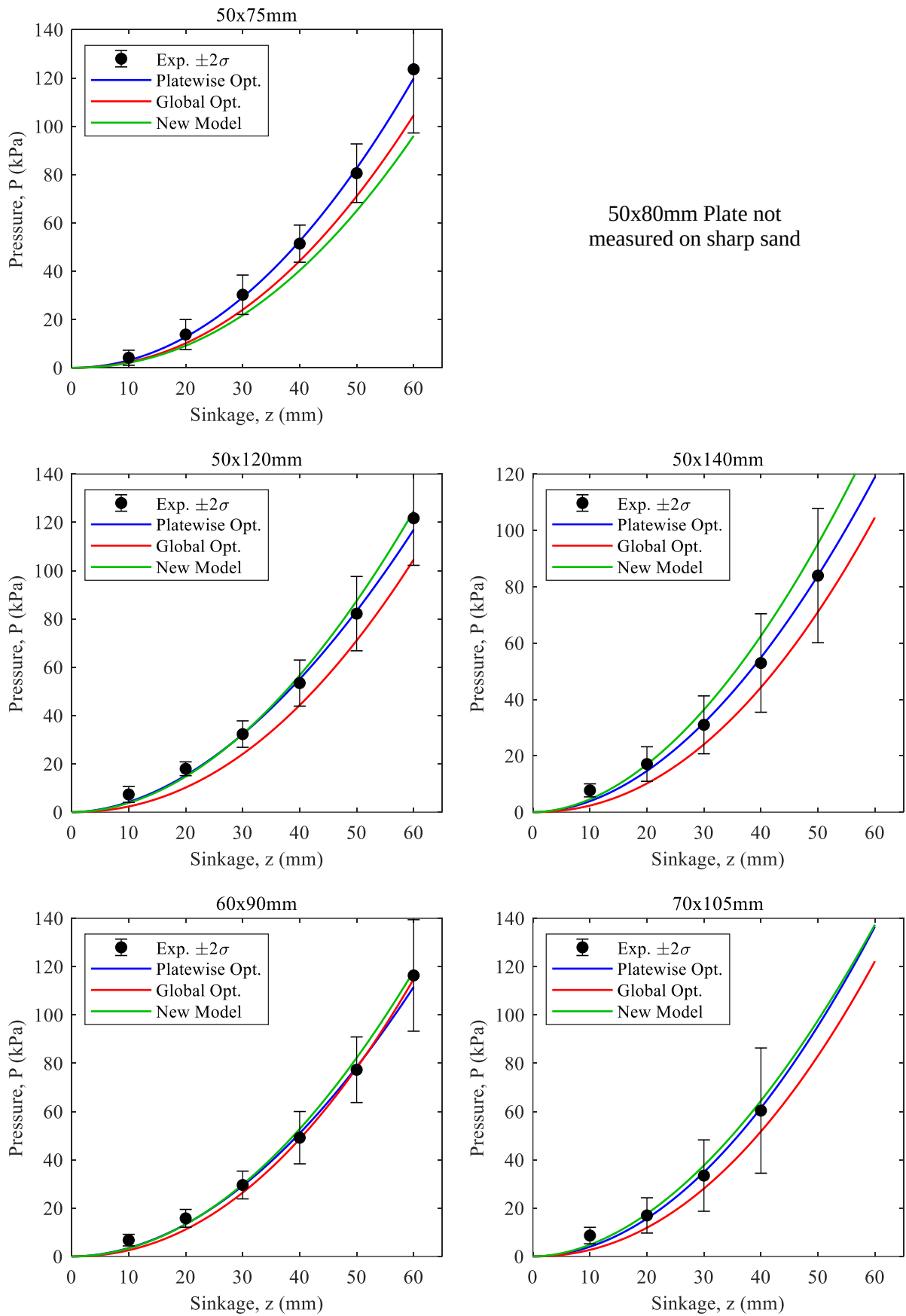


Figure C.2 (continued from previous page)

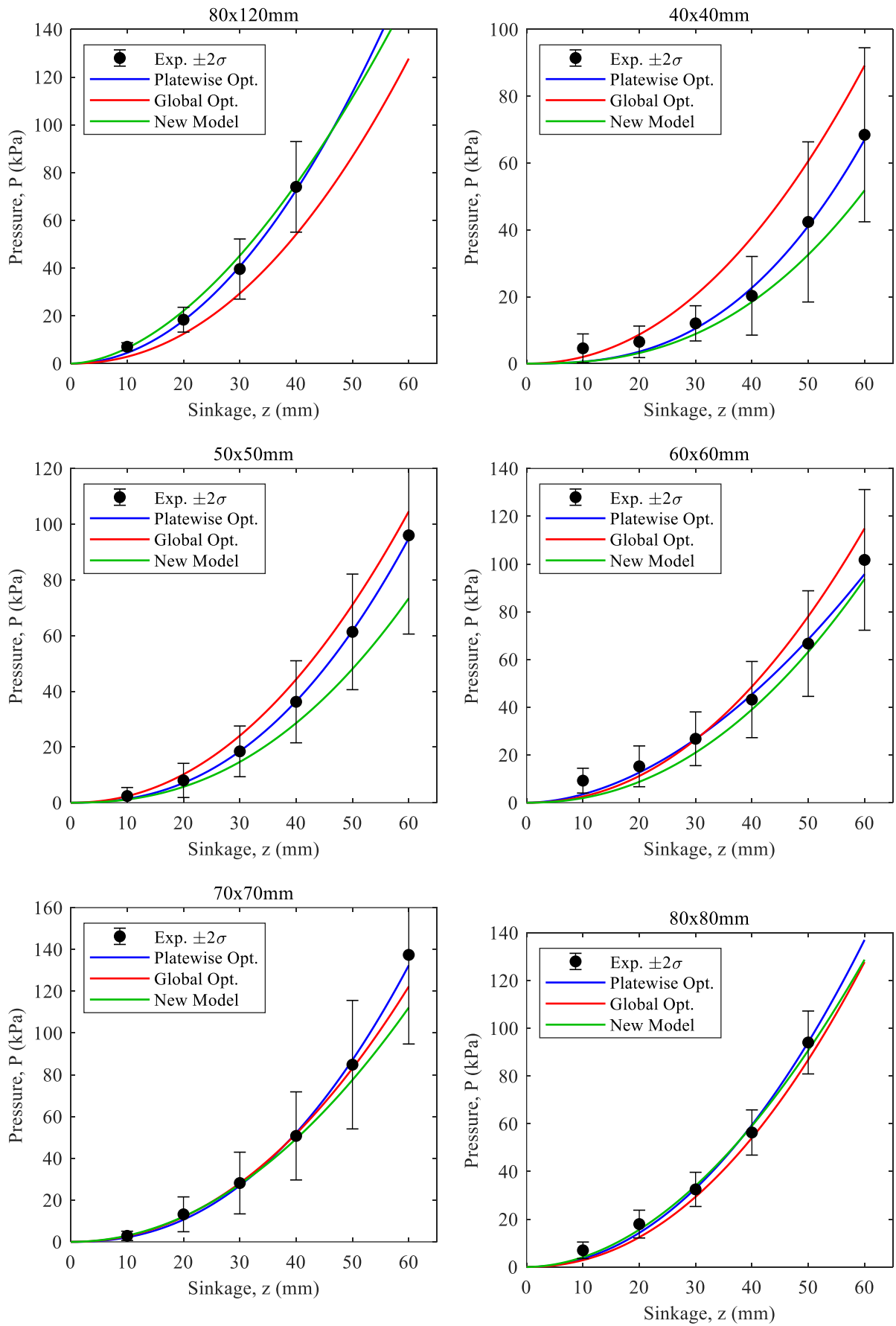


Figure C.2 (continued from previous page)

Appendix D: ADPS Correlation - Curved Plates

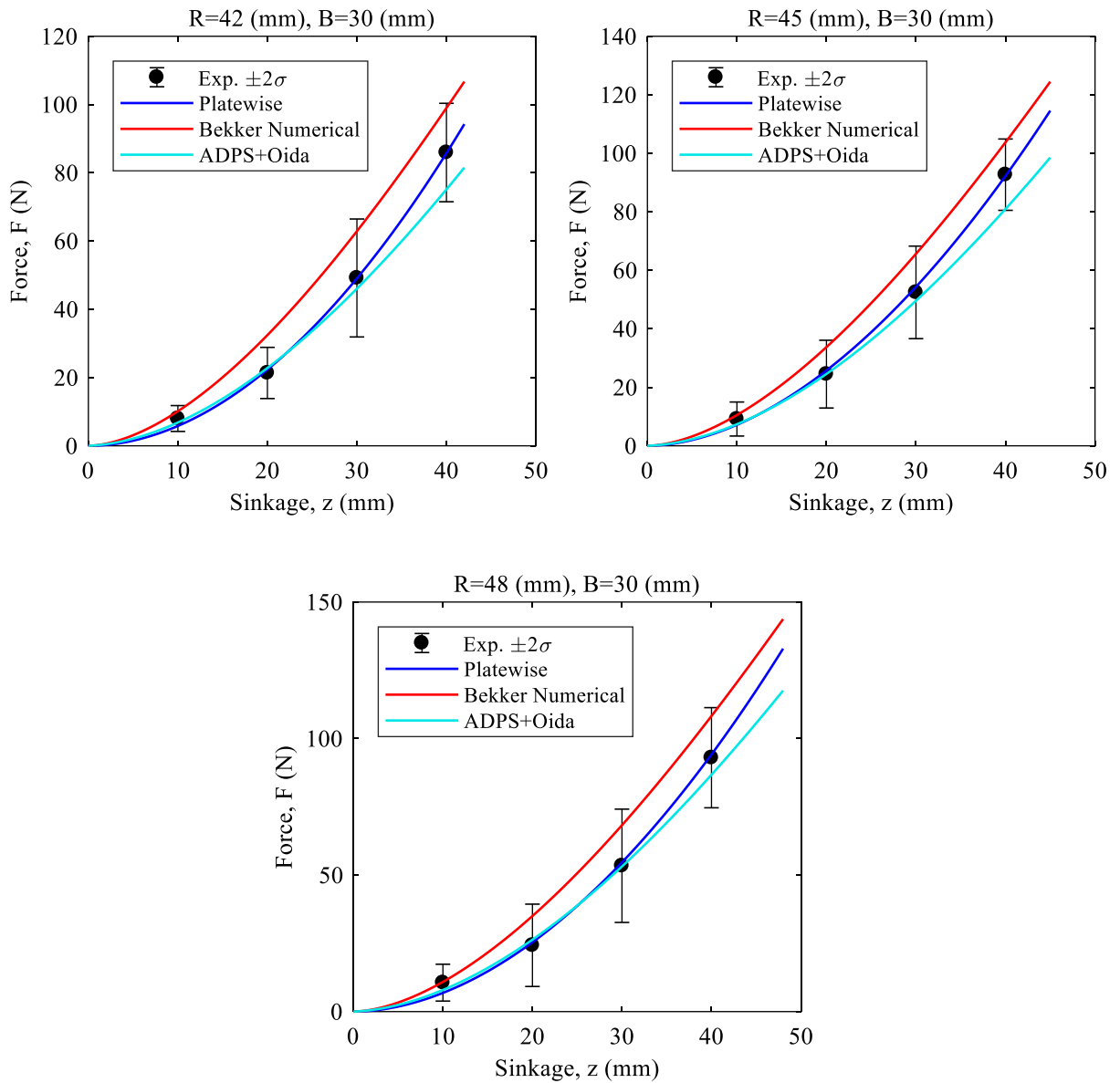


Figure D.1 Fine sand, plates with $B=30$ mm.

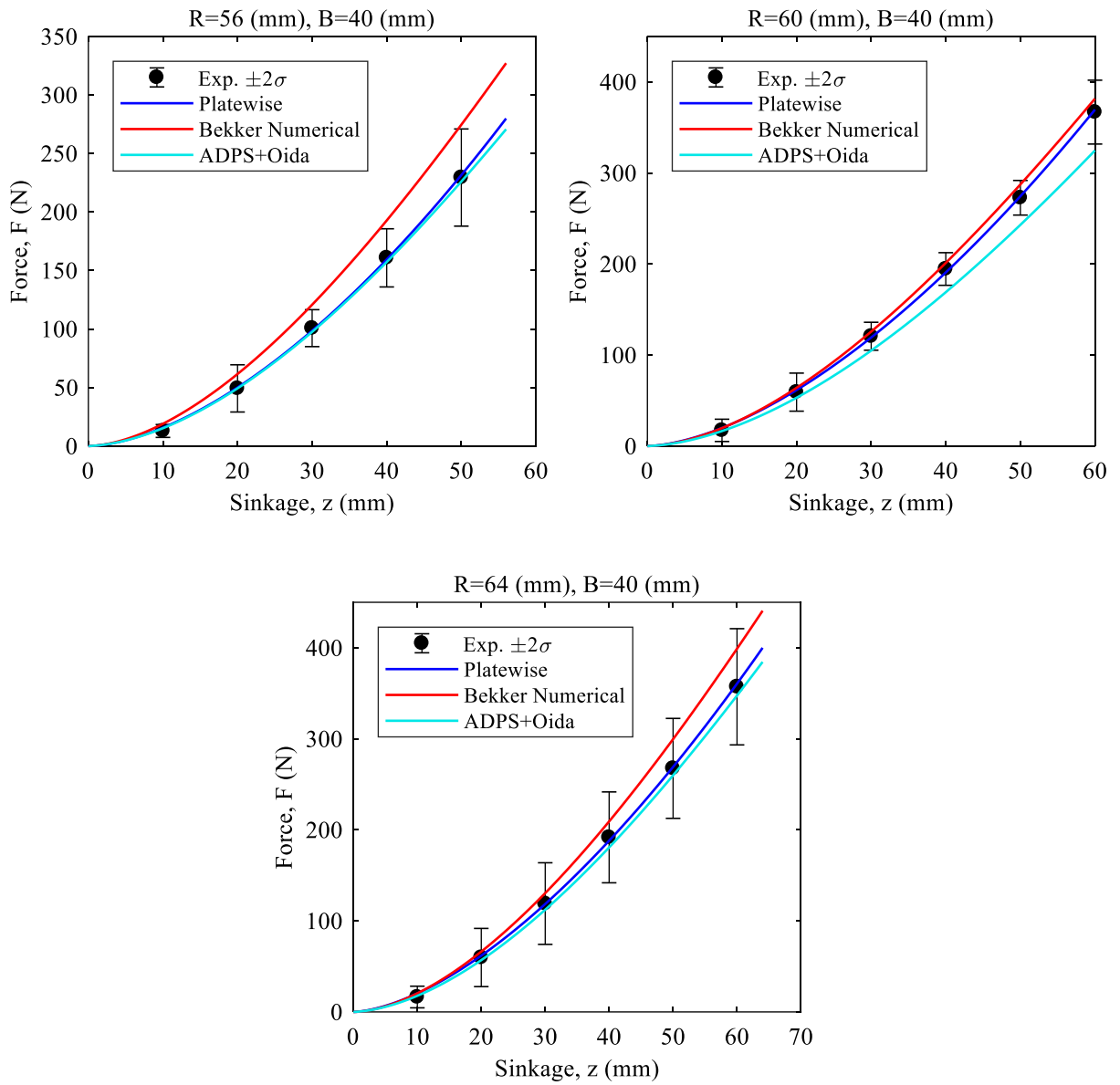


Figure D.2 Fine sand, plates with $B=40$ mm.

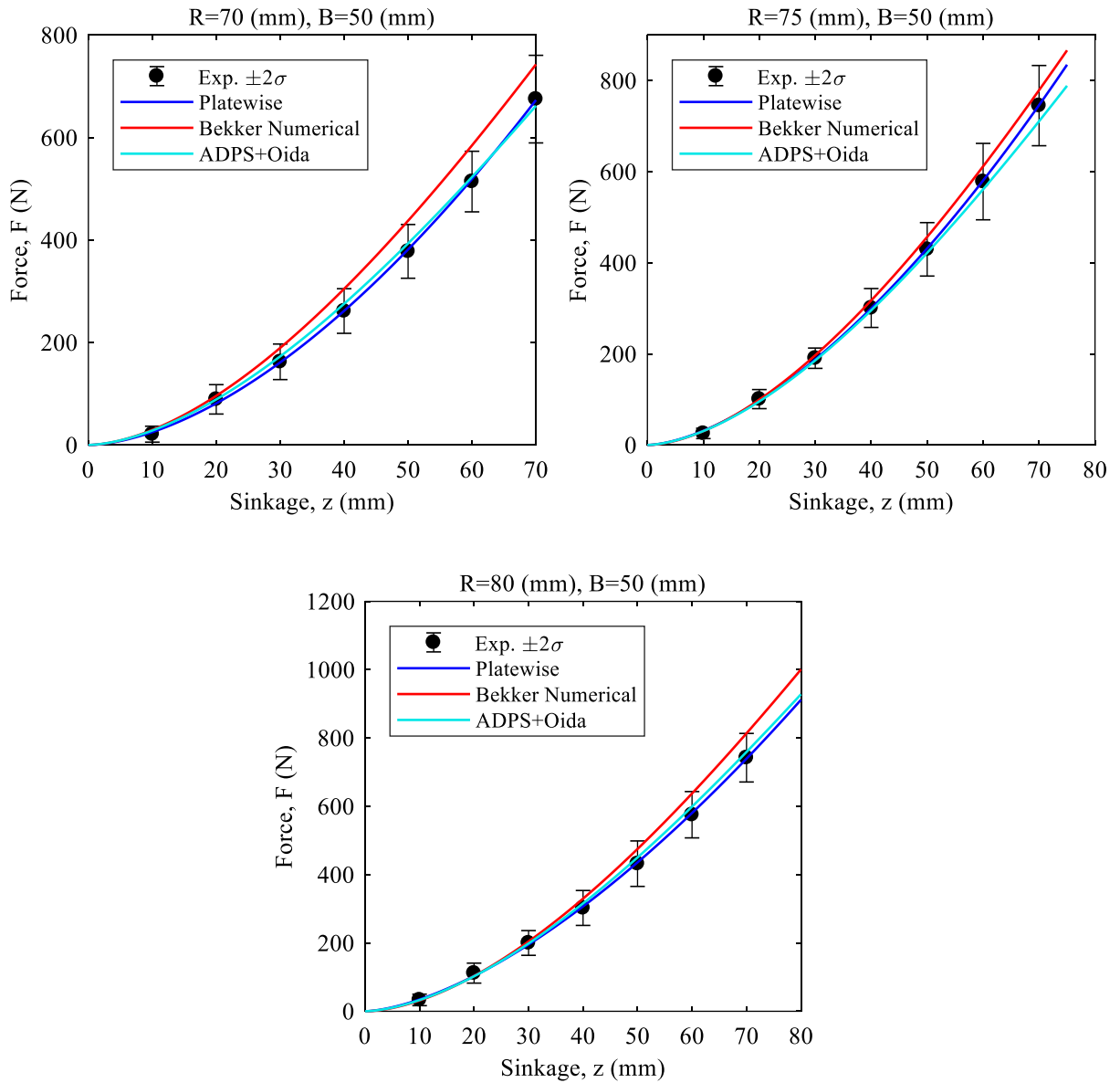


Figure D.3 Fine sand, plates with $B=50$ mm.

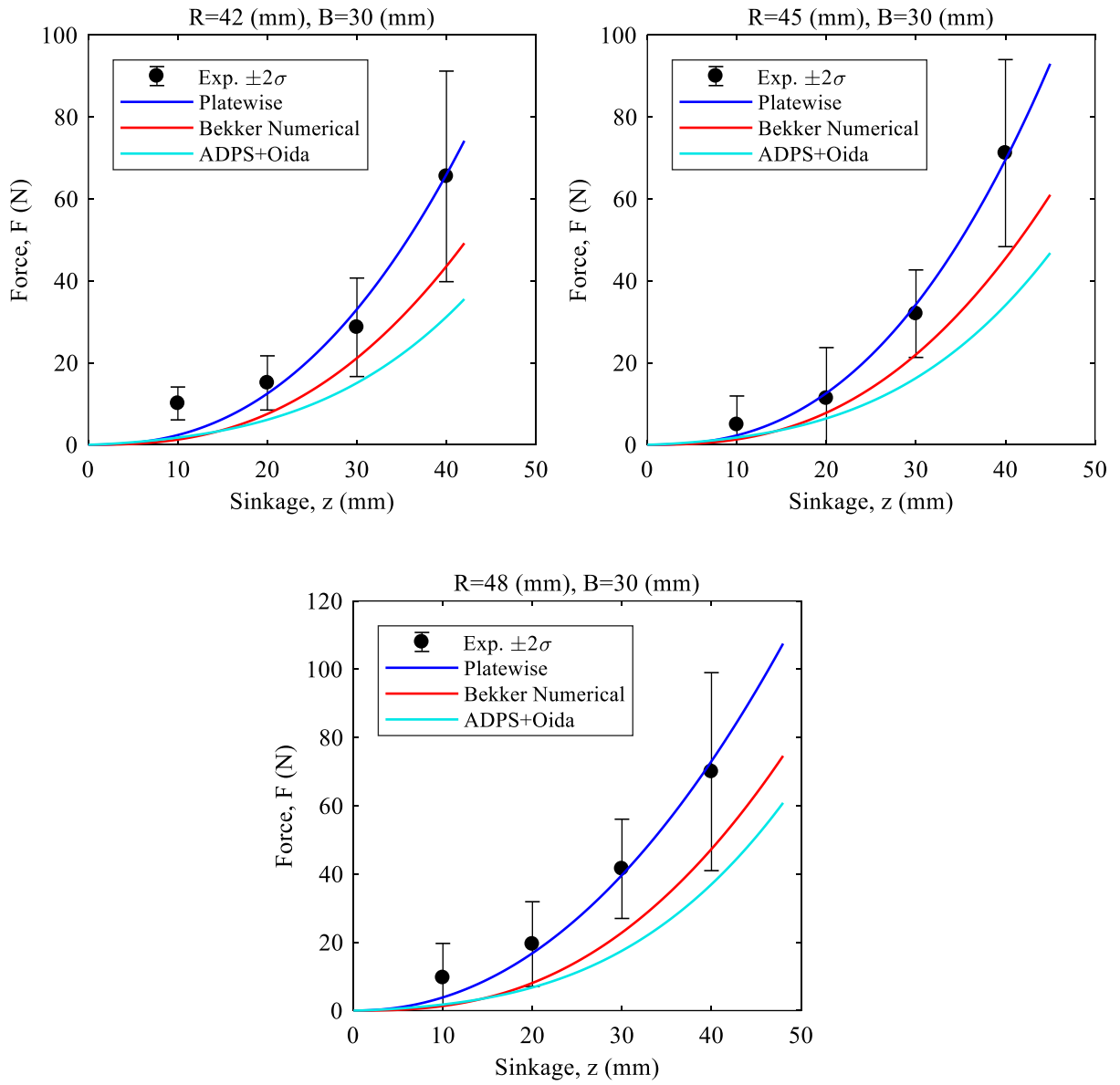


Figure D.4 Sharp sand, plates with $B=30$ mm.

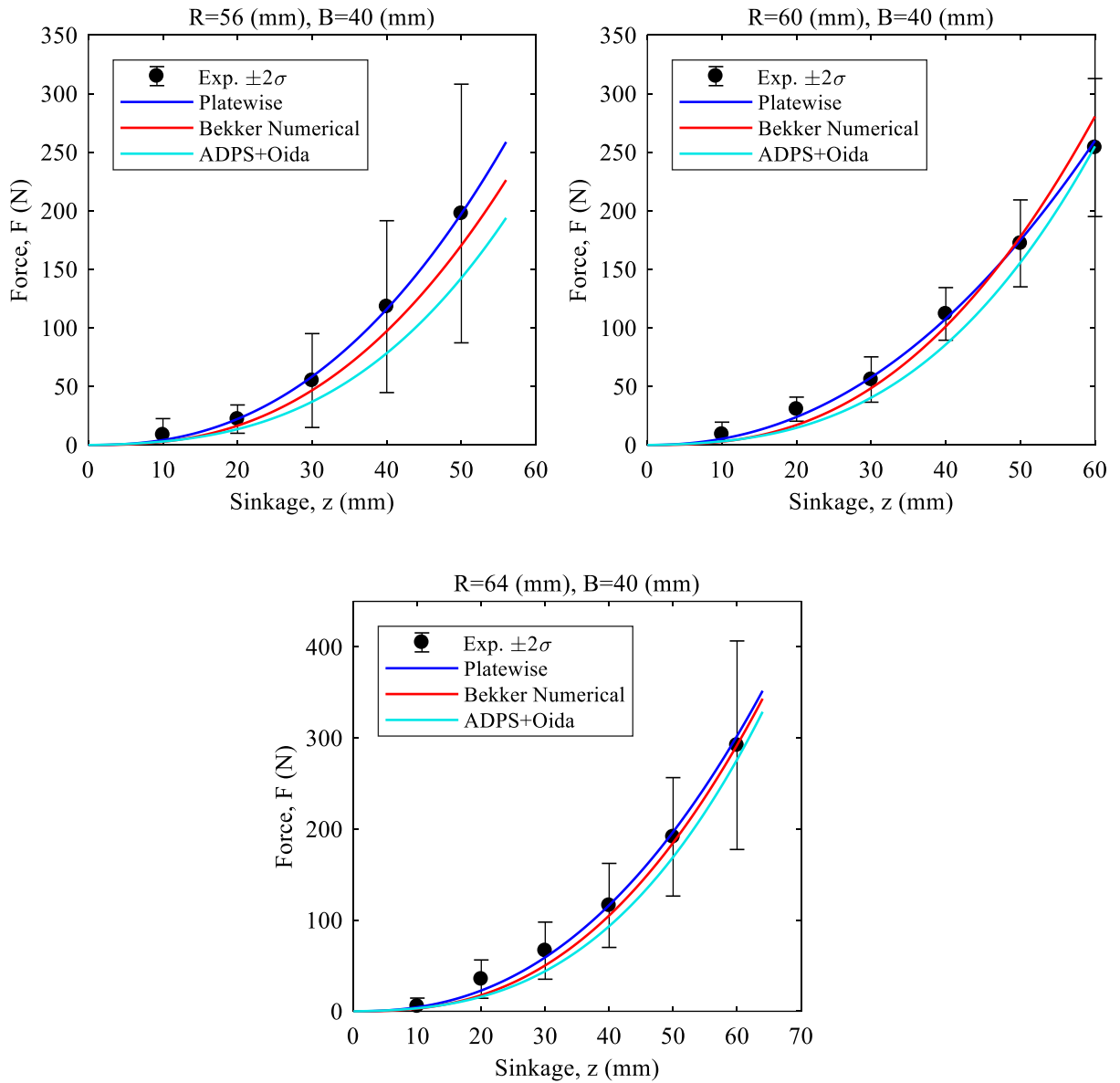


Figure D.5 Sharp sand, plates with $B=30$ mm.

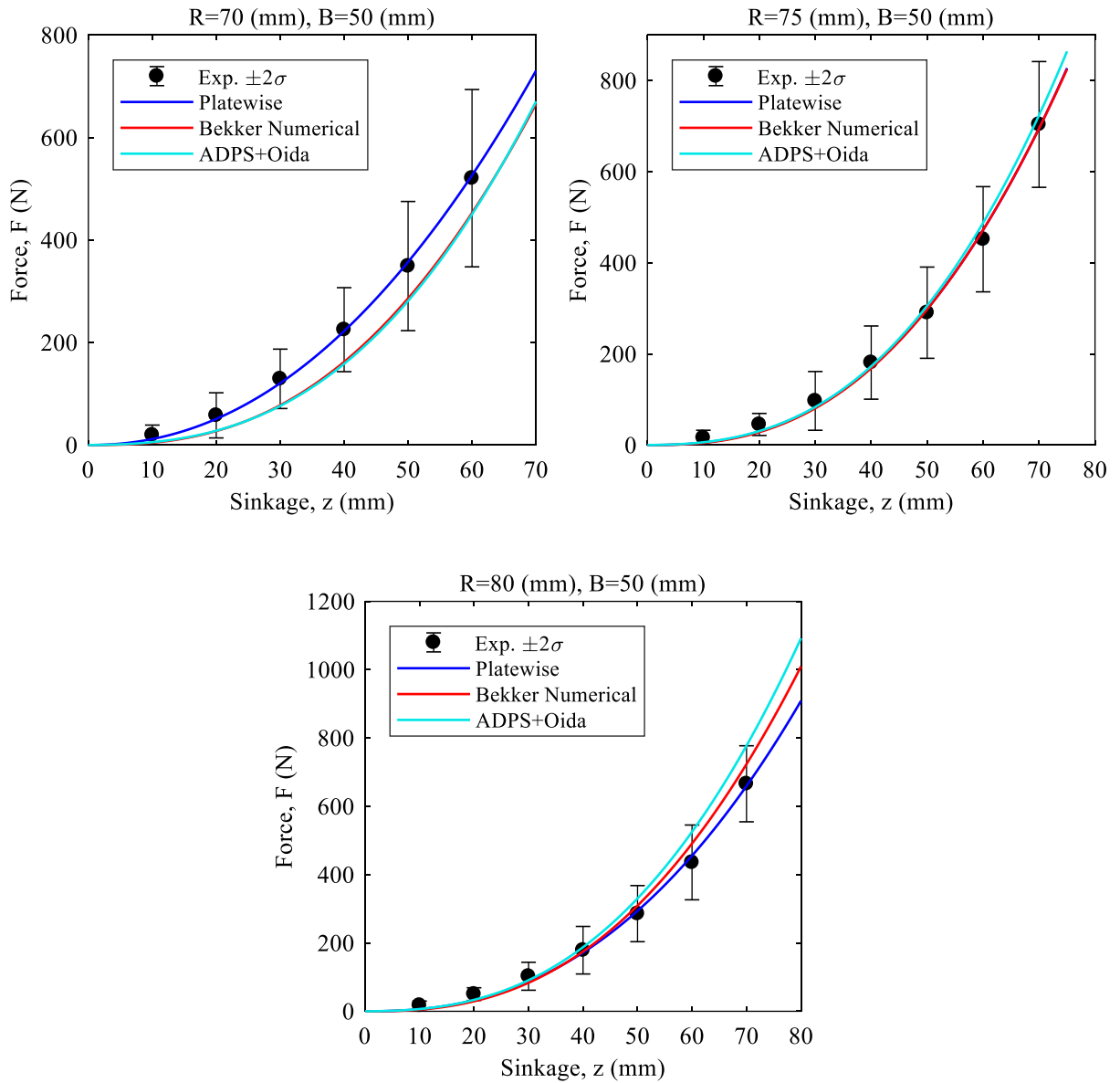


Figure D.6 Sharp sand, plates with $B=50$ mm.

Appendix E: Wheel Dimensions

Index	Make	Model	Tyre Width (mm)	Tyre Aspect ratio (-)	Rim diameter (in)	Tyre diameter (mm)	B/D
1	Land Rover	Range Rover	255	55	19	763.1	0.33
2			255	50	20	763.0	0.33
3		Discovery 4	235	70	17	760.8	0.31
4			255	60	18	763.2	0.33
5			255	55	19	763.1	0.33
6		Evoque	255	50	20	763.0	0.33
7			275	45	20	755.5	0.36
8			295	30	22	735.8	0.40
9			225	65	17	724.3	0.31
10			235	60	18	739.2	0.32
11			235	55	19	741.1	0.32
12			245	45	19	703.1	0.35
13			245	45	20	728.5	0.34
14		Range Rover Sport	275	50	19	757.6	0.36
15		Defender	235	85	16	805.9	0.29
16	Audi	Q3	215	65	16	685.9	0.31
17			235	55	17	690.3	0.34
18			235	50	18	692.2	0.34
19		Q5	255	40	19	686.6	0.37
20			235	65	17	737.3	0.32
21			235	60	18	739.2	0.32
22		Q7	235	55	19	741.1	0.32
23			255	45	20	737.5	0.35
24			235	60	18	739.2	0.32
25			255	55	18	737.7	0.35
26			265	55	18	748.7	0.35
27			275	45	20	755.5	0.36
28			295	35	21	739.9	0.40
29	BMW	X3	235	55	17	690.3	0.34
30			235	50	18	692.2	0.34
31		X5	255	55	18	737.7	0.35
32			275	40	20	728.0	0.38
33			325	30	21	728.4	0.45
34			315	35	20	728.5	0.43
35			255	55	18	737.7	0.35
36	Ford	Explorer	245	60	18	751.2	0.33
37			265	50	20	773.0	0.34
38		Kuga	265	35	22	744.3	0.36
39			225	50	18	682.2	0.33

Index	Make	Model	Tyre Width (mm)	Tyre Aspect ratio (-)	Rim diameter (in)	Tyre diameter (mm)	B/D
40		Ranger	235	55	17	690.3	0.34
41			215	70	16	707.4	0.30
42			235	85	15	780.5	0.30
43			265	65	17	776.3	0.34
44			265	60	18	775.2	0.34
45	Jeep	Cherokee	225	65	17	724.3	0.31
46		Wrangler	245	55	18	726.7	0.34
47			275	40	20	728.0	0.38
48			225	75	16	743.9	0.30
49			245	75	17	799.3	0.31
50			255	70	18	814.2	0.31
51			265	45	20	746.5	0.35
52	Mercedes	G-Class	265	60	18	775.2	0.34
53		M-Class	285	55	18	770.7	0.37
54			255	50	19	737.6	0.35
55			265	45	20	746.5	0.35
56			295	40	20	744.0	0.40
57	Mitsubishi	Outlander	215	70	16	707.4	0.30
58			255	55	18	737.7	0.35
59	Nissan	Juke	215	55	17	668.3	0.32
60		Navara	225	45	18	659.7	0.34
61			255	70	16	763.4	0.33
62			255	65	17	763.3	0.33
63			255	60	18	763.2	0.33
64		X-Trail	225	60	17	701.8	0.32
65			225	55	18	704.7	0.32
66			235	50	18	692.2	0.34
67	Toyota	Rav4	225	65	17	724.3	0.31
68		Land Cruiser	235	55	18	715.7	0.33
69			285	60	18	799.2	0.36

Table E.1 Summary of tyre dimensions for various SUVs available in the UK market in 2016.